

# spintent

SPANISH PARSE INTENTS

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©2026 by Pablo González L<sup>†</sup>

CTAN: <https://www.ctan.org/pkg/spintent>

 <https://github.com/pablgonz/spintent>

## Resumen

El paquete **spintent** proporciona una serie de utilidades para docentes de educación primaria y secundaria que necesiten crear documentos PDF *accesibles* (*tagged* PDF) en español usando Lua $\text{\LaTeX}$ .

## Índice

|       |                                             |   |       |                                            |     |
|-------|---------------------------------------------|---|-------|--------------------------------------------|-----|
| 1     | <b>Descripción</b>                          | 1 | 3.3   | El comando <code>\spmoney</code>           | 6   |
| 1.1   | Carga del paquete                           | 1 | 3.3.1 | Llaves para <code>\spmoney</code>          | 7   |
| 1.2   | El paquete <code>babel</code>               | 2 | 3.3.2 | Monedas y alias para <code>\spmoney</code> | 7   |
| 1.3   | Fuentes tipográficas                        | 2 | 3.4   | El comando <code>\spshort</code>           | 7   |
| 1.4   | El paquete <code>hyperref</code>            | 2 | 3.4.1 | Llaves para <code>\spshort</code>          | 9   |
| 2     | <b>El comando <code>\setspintent</code></b> | 3 | 3.4.2 | Argumentos para <code>\spshort</code>      | 9   |
| 3     | <b>The big four</b>                         | 3 | 4     | <b>Utilidades generales</b>                | 11  |
| 3.1   | El comando <code>\spnum</code>              | 3 | 5     | <b>Utilidades para símbolos</b>            | 11  |
| 3.1.1 | Llaves para <code>\spnum</code>             | 3 | 6     | <b>Utilidades para números</b>             | 13  |
| 3.2   | El comando <code>\spunit</code>             | 4 | 7     | <b>Utilidades para geometría</b>           | 17  |
| 3.2.1 | Llaves para <code>\spunit</code>            | 4 | 8     | <b>Referencias</b>                         | 32  |
| 3.2.2 | Argumentos para <code>\spunit</code>        | 4 | 9     | <b>Change history</b>                      | 33  |
| 3.2.3 | El comando <code>\spnewunit</code>          | 6 | 10    | <b>Índice de la Documentación</b>          | 34  |
| 3.2.4 | El comando <code>\spunitalias</code>        | 6 | 11    | <b>Implementation</b>                      | 37  |
|       |                                             |   | 12    | <b>Índice de la Implementación</b>         | 155 |

## Motivación y agradecimientos

Desde que desarrollé `scontents`[7] en 2019 y, más tarde, `enumext`[8] en 2024, me he ido interesando cada vez más por la accesibilidad en los documentos PDF. Al principio era un tema prácticamente desconocido para mí, pero a medida que fui aprendiendo sobre él comprendí la importancia que tiene, especialmente en el ámbito educativo.

El equipo del proyecto de  $\text{\LaTeX}$ [19] ha realizado un trabajo extraordinario en este campo. Hoy es posible generar documentos etiquetados, validarlos con herramientas como `veraPDF`[20] o inspeccionar su estructura con `ngpdf`[21]. Sin embargo, siempre me quedó una duda: cuando un estudiante utiliza NVDA[11] junto con `MathCAT`[12], ¿realmente escucha lo que el o la docente quiso expresar?

Basta con generar un documento accesible siguiendo las recomendaciones de `The LaTeX Tagged PDF Project` y escucharlo con NVDA/`MathCAT` para darse cuenta de que, en muchos casos, la *verbalización* no coincide con lo que uno espera. Esto no resulta extraño: las reglas del español, y en particular el lenguaje matemático, dependen en gran medida del contexto en que se utilizan.

El objetivo de **spintent** nace precisamente de esta necesidad: proporcionar herramientas para que el o la docente pueda indicar explícitamente el contexto de determinadas expresiones matemáticas desde el propio código fuente de  $\text{\LaTeX}$ , de modo que el documento PDF resultante no solo sea *accesible*, sino que también produzca una *verbalización más natural* y adecuada para el estudiante al utilizar NVDA/`MathCAT`.

El nuevo estándar PDF 2.0[14, 18, 16], las especificaciones de WTPDF[15] y la guía más reciente de la PDF Association, «Best Practice Guide: Math in PDF»[17], coinciden en que el camino hacia una accesibilidad matemática de calidad pasa por el uso de `MathML`. En ese contexto, Lua $\text{\LaTeX}$  ofrece una plataforma especialmente adecuada para incorporar esta tecnología.

<sup>\*</sup>Este archivo describe la documentación para la versión v0.99 revisada por última vez el 2026-09-21.

<sup>†</sup>E-mail: «[pablgonz@educarchile.cl](mailto:pablgonz@educarchile.cl)».

## Licencia y requerimientos

El paquete `spintent` carga y requiere el paquete `unicode-math`[1] y que se ejecute bajo Lua $\TeX$  por lo que es necesario contar con una distribución moderna de  $\TeX$  como  $\TeX$  Live 2026. Ha sido probado con las clases estándar proporcionadas por  $\LaTeX$ : `book`, `report`, `article` y `letter` en tamaños `10pt`, `11pt` y `12pt`.

Se otorga permiso para copiar, distribuir y/o modificar este paquete bajo los términos de  $\LaTeX$  Project Public License (LPPL), versión 1.3 o posterior (<https://www.latex-project.org/lppl.txt>). El paquete tiene la condición de mantenido al estilo «*A Work in Progress*»<sup>1</sup>.

- El requerimiento mínimo es  $\LaTeX$  release 2026-11-01.

Este paquete solo se puede utilizar solo con Lua $\TeX$  y *tagged* PDF activo. Está presente en  $\TeX$  Live y MiK $\TeX$ , para instalarlo, utilice el gestor de paquetes de su distribución. Si desea una instalación manual, descargue `spintent.zip` y descomprímalo, luego ejecute `luatex spintent.ins`, mueva todos los archivos a las ubicaciones correspondientes y luego ejecute `mktexlsr`. Para generar la documentación, ejecute `arara spintent.dtx`.

```
spintent.sty  ⇒ TDS:tex/lualatex/spintent/
spintent.lua  ⇒ TDS:tex/lualatex/spintent/
spintent.pdf  ⇒ TDS:doc/lualatex/spintent/
README.md    ⇒ TDS:doc/lualatex/spintent/
spintent.dtx ⇒ TDS:source/lualatex/spintent/
spintent.ins  ⇒ TDS:source/lualatex/spintent/
```

## 1 Descripción

El nombre `spintent` puede entenderse como «*Spanish Parse Intent*» o también como «*School Parse Intent*». Ambas interpretaciones reflejan el objetivo del paquete y el público para el que fue diseñado: docentes de educación primaria y secundaria.

Está pensado para crear «*apuntes*», «*hojas de ejercicios*», «*clases escritas*» enfocadas en el estudiante. Solo un docente sabe cual es el nivel de aprendizaje del estudiante. El paquete intenta sobrescribir muchas de las reglas dictadas por la heurística de NVDA/MathCAT teniendo esto presente.

- El paquete `spintent` usa y abusa del atributo `:intent` de MathML por medio del comando `\MathMLintent` definido en `latex-lab-mathintent`[30], un encapsulado de `\luamml_attribute:een` proveída por `luamml`[2]. ¡Gracias Marcel!

### 1.1 Carga del paquete

El paquete `spintent` trabaja solo con Lua $\TeX$  y *tagged* PDF activo usando `tagging=on` dentro del comando `\DocumentMetadata` al inicio del archivo. Las llaves («*keys*») aceptadas por `\DocumentMetadata` y en general para el código de *tagged* PDF están documentadas en los paquetes y `tagpdf`[29] `documentmetadata-support`[30], es deber del usuario revisar la documentación asociada a ellos.

```
\DocumentMetadata
{
  lang=es-CL,
  pdfversion=2.0,
  pdfstandard=ua-2,
  tagging=on,
  tagging-setup={math/setup={mathml-SE}},
}
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage{spintent}
```

### 1.2 El paquete babel

El paquete `babel-spanish`[10] está marcado como «parcialmente compatible» para *tagged* PDF, pero, es absolutamente necesario para la creación de documentos en español. En caso de encontrar problemas de compatibilidad con otros paquetes se puede desactivar usando:

```
\usepackage[spanish,shorthands=off]{babel}
```

Muchas de las «shorthands» proveídas por `babel-spanish` hoy no son necesarias, más allá de `\sptext` para las abreviaturas y ordinales el resto son solo atajos de teclas para escritura que hoy en día no se ocupan. El paquete `spintent` provee el comando `\spshort` (§3.4) que cubre estos casos, de esta manera podemos cargar `babel`[9] con todas sus funcionalidades sin problemas.

<sup>1</sup>A *Work in Progress* es un álbum en vivo de Neil Peart (\* 1952–† 2020), baterista y principal letrista de Rush. El nombre no pretende formar parte de la terminología de la LPPL; simplemente describe el estado permanente de este paquete mientras sigo aprendiendo sobre *accesibilidad* en PDF.

### 1.3 Fuentes tipográficas

Como `spintent` es un paquete Lua $\TeX$  debemos distinguir entre las fuentes tipográficas OpenType/TrueType versus las fuentes de 8bit estándar usadas por  $\TeX$  para el *modo texto* y el *modo matemático*. Para evitar problemas con *tagged* PDF preferiremos no combinar fuentes OpenType/TrueType con fuentes 8bit y trabajaremos solo con fuentes OpenType/TrueType que tengan soporte para matemáticas.

La carga de las fuentes tipográficas para Lua $\TeX$  se maneja por medio del paquete `fontspec`[3], si no se especifica ninguna se usarán las fuentes Latin Modern para el *modo texto* y Latin Modern Math para el *modo matemático*.

El tipo de fuente tipográfica escogida es preferencia de cada usuario, en lo personal utilizo las fuente Libertinus para el *modo texto* por medio del paquete `libertinus-otf`[4] y la fuente Erewhon-Math para el modo matemático cargada por el paquete `fourier-otf`[5]. Adicionalmente cargo la fuente Source Code Pro para el texto tipo «máquina de escribir» por medio del paquete `sourcecodepro`[6].

```
% \DocumentMetadata{...}
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage[osf,nomath,mono=false]{libertinus-otf}
\usepackage[osf,semibold]{sourcecodepro}
\usepackage[no-text]{fourier-otf}
\usepackage{spintent}
```

### 1.4 El paquete hyperref

El paquete `hyperref`[23] es «casi obligatorio» cuando se crean documentos *tagged* PDF, se encarga de las referencias cruzadas `\label` y `\ref`, las notas al pie `\footnote`, los hipervínculos `\href` y muchos otros elementos como marcadores, índices y metadatos.

```
\DocumentMetadata
{
  lang=es-CL,
  pdfversion=2.0,
  pdfstandard=ua-2,
  tagging=on,
  tagging-setup={math/setup={mathml-SE}},
}
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage[osf,nomath,mono=false]{libertinus-otf}
\usepackage[osf,semibold]{sourcecodepro}
\usepackage[no-text]{fourier-otf}
\setmathfont{Erewhon-Math}[range={cal,bfcal},StylisticSet=1]
\usepackage{spintent, hyperref}
\hypersetup
{
  colorlinks      = true,
  pdfstartview    = FitH,
  pdfdisplaydoctitle = true,
  pdftitle        = {Test para el paquete spintent},
  pdfinfo         =
  {
    Title   = {Test},
    Subject = {spintent},
  },
}
```

## 2 El comando \setspintent

---

`\setspintent` `\setspintent{⟨comando⟩}{⟨key-uno = valor, key-dos = valor, key-tres = valor, ...⟩}`

---

El comando `\setspintent` establece las llaves (*⟨keys⟩*) de forma global para los comandos provistos por `spintent`. Puede utilizarse tanto en el preámbulo como en el cuerpo del documento tantas veces como se desee. Las llaves (*⟨keys⟩*) definidas en el *argumento opcional* [*⟨key = val⟩*] de los comandos tienen prioridad sobre las establecidas por este, anulando las configuraciones globales de `\setspintent`.

Muchas de las utilidades que provee el paquete `spintent` implementan el sistema [*⟨key = val⟩*] y están pensadas para ser utilizadas en la generación de «*hojas de ejercicios*» o «*clases escritas*». Se recomienda usar `\setspintent` y establecerlas de manera global para cada documento, para mantener la coherencia en la verbalización de NVDA/MathCAT a lo largo del mismo.

El sistema [*⟨key = val⟩*] utilizado por `spintent` se implementa usando el módulo `l3keys` de `expl3`[24]. Por diseño, las llaves que reciben un *valor* después del signo '=' NO aceptan *valores vacíos* y es obligatorio pasar uno válido; las llaves documentadas como *⟨valor prohibido⟩* no reciben *ningun valor*. En caso de omitir un *valor* válido o pasar un *valor vacío* a una llave que acepte *valor* o pasar un *valor* a una llave del tipo *⟨valor prohibido⟩* se devolverá un "error" y se detendrá la compilación del documento.

## 3 The big four

El paquete `spintent` provee cuatro grandes comandos («*the big four*»): `\spnum` para el manejo de números, `\spunit` para el manejo de unidades, `\spmoney` para el manejo de dinero y `\spshort` para el manejo de abreviaturas, acrónimos y números ordinales.

- También se incluyen las utilidades `\spnewunit` y `\spunitalias` asociadas a al comando `\spunit`.

### 3.1 El comando \spnum

---

`\spnum` `\spnum{⟨número⟩}` `\spnum[⟨key = val⟩]{⟨número,número;número⟩}`  
`\spnum[⟨key = val⟩]{⟨+número⟩}` `\spnum[⟨key = val⟩]{⟨número.número;número⟩}`  
`\spnum[⟨key = val⟩]{⟨-número⟩}` `\spnum[⟨key = val⟩]{⟨número;número⟩}`  
`\spnum[⟨key = val⟩]{⟨número,número⟩}` `\spnum[⟨key = val⟩]{⟨número e exponente⟩}`  
`\spnum[⟨key = val⟩]{⟨número.número⟩}` `\spnum[⟨key = val⟩]{⟨número E exponente⟩}`

---

El comando `\spnum` trabaja en *modo matemático* y recibe el *argumento obligatorio* {*⟨número⟩*}, donde la «*parte entera*» puede iniciar con los signos '+' o '-', la «*parte decimal finita*» **debe** iniciar con una *coma* ',' o con un *punto* '.', y la «*parte decimal infinita (periódica)*» **debe** iniciar con un *punto* y *coma* ';'.

Si no se proporciona el *argumento opcional* [*⟨key = val⟩*] y se ejecuta `\$ \spnum{+23} \$` se imprimirá '+23' en la salida PDF y NVDA/MathCAT verbalizará «*más veintitrés*», si se ejecuta `\$ \spnum{3,14} \$` se imprimirá '3,14' en la salida PDF y NVDA/MathCAT verbalizará «*tres coma catorce*», si se ejecuta `\$ \spnum{3,1;4} \$` se imprimirá '3,14' en la salida PDF y NVDA/MathCAT verbalizará «*tres coma uno con cuatro periódico*».

Además, el *argumento obligatorio* {*⟨número⟩*} admite al final de este un sufijo de «*notación científica*». Este se indica mediante las letras 'e' o 'E', acompañadas de los signos opcionales '+' o '-' seguido por los dígitos del exponente. Por ejemplo `\$ \spnum{3,14e3} \$` imprimirá '3,14 · 10<sup>3</sup>' en la salida PDF y NVDA/MathCAT verbalizará «*tres coma catorce por diez al cubo*».

- La *coma* ',' o el *punto* '.' son *separadores decimales* y NO de millares, por lo tanto solo pueden aparecer una «única vez» dentro del {*⟨argumento⟩*}. El {*⟨argumento⟩*} puede contener espacios matemáticos válidos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{#1}` o `\mathbf{#1}` por ejemplo. Cualquier entrada fuera de esta sintaxis devolverá un "error" deteniendo la compilación del documento.

#### 3.1.1 Llaves para \spnum

`cifras-sep = {⟨true | false⟩}`

default: *true*

Establece el *espaciado* utilizado para agrupar los dígitos en bloques de tres, tanto en la parte entera como en la parte decimal finita. Si se establece en *true* se siguen las normas de la RAE y se utiliza un *espacio fino* '\,' entre las cifras; si se establece en *false*, se respetan los espacios matemáticos introducidos en el argumento. Por ejemplo `\$ \spnum{100000} \$` imprimirá '100 000' en la salida PDF y `\$ \spnum[cifras-sep=false]{10\,00\,00} \$` imprimirá '10 00 00' en la salida PDF.

`read-sign = {⟨terse | clear⟩}`

default: *terse*

Establece la *forma* en la que NVDA/MathCAT verbalizará los signos '+' o '-'. Si se establece en *terse*, NVDA/MathCAT verbalizará «*más*» o «*menos*» **antes** de leer el número; si se establece en *clear*, NVDA/MathCAT verbalizará «*positivo*» o «*negativo*» **después** de leer el número. Por ejemplo `\$ \spnum{-23} \$` imprimirá '-23' en la salida PDF y NVDA/MathCAT verbalizará «*menos veintitrés*» y `\$ \spnum[read-sign=clear]{-23} \$` imprimirá '-23' en la salida PDF y NVDA/MathCAT verbalizará «*veintitrés negativo*».

`read-period-sep = {⟨coma | punto⟩}`

default: *coma*

<sup>2</sup>Los «*cuatro grandes*» del Thrash Metal: Metallica, Megadeth, Slayer y Anthrax, que me acompañan mientras desarrollo `spintent`.

Establece la *forma* en la que NVDA/MathCAT verbalizará e imprimirá el signo de separación ‘;’ de la «*parte decimal infinita (periódica)*» cuando no se tiene la «*parte decimal finita*». Si se establece en *coma*, se imprimirá ‘,’ en la salida PDF y NVDA/MathCAT verbalizará «*coma*»; si se establece en *punto*, se imprimirá ‘.’ en la salida PDF y NVDA/MathCAT verbalizará «*punto*». Por ejemplo `\spnum{3;4}` imprimirá ‘3,4’ en la salida PDF y NVDA/MathCAT verbalizará «*tres coma cuatro periódico*» y `\spnum[read-period-sep=punto]{3;4}` imprimirá ‘3.4’ en la salida PDF y NVDA/MathCAT verbalizará «*tres punto cuatro periódico*»

- Esta llave es necesaria SOLO cuando se utilizan «*decimales periódicos puros*», es decir, cuando no se ha incluido en el `{(argumento)}` una *coma* ‘,’ o un *punto* ‘.’ antes del signo de separación ‘;’ periódica.

`notation-sym = {(times | cdot)}`

default: *cdot*

Establece el *símbolo* que se utilizará para imprimir la «*notación científica*» al usar ‘e’ o ‘E’. Si se establece en *times* imprimirá ‘x’ en la salida PDF y NVDA/MathCAT verbalizará «*por*»; si se establece en *cdot* imprimirá ‘.’ en la salida PDF y NVDA/MathCAT verbalizará «*por*». Por ejemplo `\spnum{3,14e3}` imprimirá ‘3,14 · 10<sup>3</sup>’ en la salida PDF y NVDA/MathCAT verbalizará «*tres coma catorce por diez al cubo*» y `\spnum[notation-sym=times]{3,14e3}` imprimirá ‘3,14 × 10<sup>3</sup>’ en la salida PDF y NVDA/MathCAT verbalizará «*tres coma catorce por diez al cubo*».

- Decir «*notación científica*» aquí es un poco exagerado, en realidad es «*multiplicación por potencias de 10*», no se verifica si la entrada cumple o no con la condición de estar escrita en «*notación científica*».

### 3.2 El comando \spunit

|                                                  |                                                                       |
|--------------------------------------------------|-----------------------------------------------------------------------|
| <code>\spunit{(unidad)}</code>                   | <code>\spunit[key = val]{(número unidad*unidad)}</code>               |
| <code>\spunit[key = val]{(unidad)}</code>        | <code>\spunit[key = val]{(número unidad*unidad/unidad)}</code>        |
| <code>\spunit[key = val]{(número unidad)}</code> | <code>\spunit[key = val]{(número unidad*unidad/unidad*unidad)}</code> |

El comando `\spunit` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{(unidad)}`, el cual puede contener una «*parte numérica*» opcional la cual se procesa internamente por el comando `\spnum` al inicio del `{(argumento)}` seguido de una `{(unidad)}` o un `{(alias)}` valido descritos en §3.2.2 o una `{(unidad)}` creada por el comando `\spnewunit` (§3.2.3) o un `{(alias)}` creado por el comando `\spunitalias` (§3.2.4).

Si no se proporciona el *argumento opcional* `[key = val]` y se ejecuta `\spunit{23 m}` se imprimirá ‘23 m’ en la salida PDF y NVDA/MathCAT verbalizará «*veintitrés metros*».

La multiplicación entre las `{(unidades)}` se realiza mediante un *asterisco* ‘\*’ y las potencias de estas se indican con un *circunflejo seguido del exponente entre llaves* ‘^`{(exponente)}`’. Por ejemplo si se ejecuta `\spunit{23 m*s^{2}}` se imprimirá ‘23 m s<sup>2</sup>’ en la salida PDF y NVDA/MathCAT verbalizará «*veintitrés metros segundos al cuadrado*».

Se pueden usar *fracciones lineales* con las `{(unidades)}` utilizando una «*única barra*» ‘/’ para la separación entre las `{(unidades)}` del numerador y el denominador, teniendo presente que el denominador NO puede contener números de ningún tipo. Por ejemplo si se ejecuta `\spunit{23 m/s^{2}}` se imprimirá ‘23 m/s<sup>2</sup>’ en la salida PDF y NVDA/MathCAT verbalizará «*veintitrés metros por segundos al cuadrado*».

- Solo imprimiremos *fracciones lineales*, si se requiere otra forma fraccionaria para explicaciones y otros usos se debe usar el comando por separado. El `{(argumento)}` puede contener espacios matemáticos validos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{#1}` o `\mathbf{#1}` por ejemplo, NO puede contener más de una «*única barra*» ‘/’ y NO puede contener números en el denominador. Cualquier entrada fuera de esta sintaxis devolverá un “error” deteniendo la compilación del documento.

#### 3.2.1 Llaves para \spunit

El comando `\spunit` posee las *meta-keys* *cifras-sep*, *read-period-sep* y *notation-sym* pasadas al comando `\spnum` (§3.1.1).

`print-cdot = {true | false}`

default: *false*

Establece si se *imprime* `\cdot` ‘·’ o un *espacio fino* ‘\,’ para la multiplicación de `{(unidades)}`. Si se establece en *true*, se imprimirá ‘·’ entre las `{(unidades)}` en la salida PDF; si se establece en *false* se imprimirá un *espacio fino* ‘\,’ entre las `{(unidades)}` en la salida PDF.

`read-cdot = {true | false}`

default: *false*

Establece la *forma* en la que NVDA/MathCAT verbalizará la multiplicación de `{(unidades)}`. Si se establece en *true*, NVDA/MathCAT verbalizará «*por*» entre las `{(unidades)}`; si se establece en *false*, NVDA/MathCAT no verbalizará nada entre las `{(unidades)}`.

`read-slash = {por | partido-por}`

default: *por*

Establece la *forma* en la que NVDA/MathCAT verbalizará la «*barra*» ‘/’ de separación entre el numerador y el denominador. Si se establece en *por*, NVDA/MathCAT verbalizará «*por*»; si se establece en *partido-por*, NVDA/MathCAT verbalizará «*partido por*».

#### 3.2.2 Argumentos para \spunit

Resumen de `{(unidades)}` y `{(alias)}` soportados por `\spunit`.

| Entrada | Alias soportados     | Nombre unidad       | Salida | NVDA/MathCAT          |
|---------|----------------------|---------------------|--------|-----------------------|
| m       | metro, metros        | Metro               | m      | «metro»               |
| g       | gramo, gramos        | Gramo               | g      | «gramo»               |
| s       | segundo, segundos    | Segundo (Tiempo)    | s      | «segundo (Tiempo)»    |
| l       | litro, l-litro       | Litro (minúscula)   | l      | «litro (minúscula)»   |
| h       | hora, horas          | Hora                | h      | «hora»                |
| d       | dia, dias            | Día                 | d      | «día»                 |
| A       | amperio, ampere      | Amperio             | A      | «amperio»             |
| V       | voltio, volt         | Voltio              | V      | «voltio»              |
| W       | vatio, watt          | Vatio               | W      | «vatio»               |
| J       | julio, joule         | Julio               | J      | «julio»               |
| N       | newton               | Newton              | N      | «newton»              |
| Pa      | pascal               | Pascal              | Pa     | «pascal»              |
| Hz      | hercio, hertz        | Hercio              | Hz     | «hercio»              |
| Ω       | ohmio, ohm, Omega    | Ohmio               | Ω      | «Ohmio»               |
| F       | faradio              | Faradio             | F      | «Faradio»             |
| C       | culombio             | Culombio            | C      | «Culombio»            |
| K       | kelvin, Kelvin       | Kelvin              | K      | «Kelvin»              |
| °C      | celsius, degC, °C    | Grados Celsius      | °C     | «Grados Celsius»      |
| °F      | fahrenheit, degF, °F | Grados Fahrenheit   | °F     | «Grados Fahrenheit»   |
| cal     | caloria, calorías    | Caloría             | cal    | «Caloría»             |
| b       | bit                  | Bit                 | b      | «Bit»                 |
| B       | byte, bytes          | Byte                | B      | «Byte»                |
| in      | pulgada, pulgadas    | Pulgada             | in     | «Pulgada»             |
| ft      | pie, pies            | Pie                 | ft     | «Pie»                 |
| yd      | yarda, yardas        | Yarda               | yd     | «Yarda»               |
| mi      | milla, millas        | Milla               | mi     | «Milla»               |
| lb      | libra, libras        | Libra               | lb     | «Libra»               |
| oz      | onza, onzas          | Onza                | oz     | «Onza»                |
| gal     | galon, galones       | Galón               | gal    | «Galón»               |
| Å       | angstrom             | Angstrom            | Å      | «Angstrom»            |
| μ       | micro                | Micro (Prefijo)     | μ      | «Micro (Prefijo)»     |
| Ω       | mho                  | Mho / Siemens       | Ω      | «Mho / Siemens»       |
| '       | minmin               | Minuto (Arco)       | '      | «Minuto (Arco)»       |
| "       | segseg               | Segundo (Arco)      | "      | «Segundo (Arco)»      |
| u       | u-masa               | Unidad masa atómica | u      | «Unidad masa atómica» |

Unidades directas (sin alias):

|     |   |                    |     |                      |
|-----|---|--------------------|-----|----------------------|
| L   | - | Litro (Mayúscula)  | L   | «Litro (Mayúscula)»  |
| ℓ   | - | Litro (Cursiva)    | ℓ   | «Litro (Cursiva)»    |
| w   | - | Vatio (Minúscula)  | w   | «Vatio (Minúscula)»  |
| °K  | - | Grado Kelvin       | °K  | «Grado Kelvin»       |
| Bq  | - | Becquerel          | Bq  | «Becquerel»          |
| Da  | - | Dalton             | Da  | «Dalton»             |
| Gy  | - | Gray               | Gy  | «Gray»               |
| S   | - | Siemens            | S   | «Siemens»            |
| Sv  | - | Sievert            | Sv  | «Sievert»            |
| T   | - | Tesla              | T   | «Tesla»              |
| Wb  | - | Weber              | Wb  | «Weber»              |
| atm | - | Atmósfera          | atm | «Atmósfera»          |
| au  | - | Unidad astronómica | au  | «Unidad astronómica» |
| bar | - | Bar                | bar | «Bar»                |
| cd  | - | Candela            | cd  | «Candela»            |
| dB  | - | Decibelio          | dB  | «Decibelio»          |
| dyn | - | Dina               | dyn | «Dina»               |
| eV  | - | Electronvoltio     | eV  | «Electronvoltio»     |
| erg | - | Ergio              | erg | «Ergio»              |
| ha  | - | Hectárea           | ha  | «Hectárea»           |
| kat | - | Katal              | kat | «Katal»              |
| kg  | - | Kilogramo          | kg  | «Kilogramo»          |
| km  | - | Kilómetro          | km  | «Kilómetro»          |
| lm  | - | Lumen              | lm  | «Lumen»              |

Continúa en la siguiente página...

| Continuación de la tabla 1 |                  |                  |        |                    |
|----------------------------|------------------|------------------|--------|--------------------|
| Entrada                    | Alias soportados | Nombre unidad    | Salida | MathCAT            |
| lx                         | -                | Lux              | lx     | «Lux»              |
| cm                         | -                | Centímetro       | cm     | «Centímetro»       |
| min                        | -                | Minuto (Tiempo)  | min    | «Minuto (Tiempo)»  |
| mm                         | -                | Milímetro        | mm     | «Milímetro»        |
| mol                        | -                | Mol              | mol    | «Mol»              |
| pc                         | -                | Pársec           | pc     | «Pársec»           |
| rad                        | -                | Radián           | rad    | «Radián»           |
| rpm                        | -                | Rev. por minuto  | rpm    | «Rev. por minuto»  |
| sr                         | -                | Estereorradián   | sr     | «Estereorradián»   |
| t                          | -                | Tonelada         | t      | «Tonelada»         |
| a                          | -                | Año / Área (are) | a      | «Año / Área (are)» |
| y                          | -                | Año (Year)       | y      | «Año (Year)»       |
| °                          | -                | Grado (Arco)     | °      | «Grado (Arco)»     |

3.2.3 El comando \spnewunit

\spnewunit

```
\spnewunit{<nueva unidad>}{<verbalización NVDA>}
```

El comando `\spnewunit` trabaja en *modo texto* y recibe dos *argumentos obligatorios*: `{<nueva unidad>}` que establece lo que se imprimirá en la salida PDF y `{<verbalización NVDA>}` que establece como NVDA/MathCAT verbalizará la `{<nueva unidad>}`.

La `{<nueva unidad>}` no debe estar registrada como `{<unidad>}` o `{<alias>}` para `\spunit` descritos en §3.2.2. Si la `{<nueva unidad>}` ya está registrada se devolverá un “error” y se detendrá la compilación del documento.

La declaración de la `{<nueva unidad>}` es «global» y se almacena en el módulo `spintent.lua` como una `{<unidad>}` valida para pasar como `{<argumento>}` a `\spunit`. Por ejemplo: `\spnewunit{px}{píxeles}` creará la `{<nueva unidad>}` ‘px’ y cuando se utilice `\spunit{100 px}` se imprimirá ‘100 px’ en la salida PDF y NVDA/MathCAT verbalizará «cien píxeles».

3.2.4 El comando \spunitalias

\spunitalias

```
\spunitalias{<nuevo alias>}{<expresión de unidades>}
```

El comando `\spunitalias` trabaja en *modo texto* y recibe dos *argumentos obligatorios*: `{<nuevo alias>}` que establece es el nuevo `{<alias>}` de `{<argumento>}` para `\spunit` y `{<expresión de unidades>}` que puede contener multiplicación o división de `{<unidades>}` ya registradas que se imprimirán en la salida PDF.

El `{<nuevo alias>}` no debe estar registrado como `{<unidad>}` o `{<alias>}` para `\spunit` descritos en §3.2.2 o estar definido como `{<unidad>}` por medio del comando `\spnewunit`. Si el `{<nuevo alias>}` ya está registrado se devolverá un “error” y se detendrá la compilación del documento.

La declaración de un `{<nuevo alias>}` es «global» y se almacena en el módulo `spintent.lua` como una `{<alias>}` valido para pasar como `{<argumento>}` a `\spunit`. Por ejemplo: `\spunitalias{caudal}{m^3/s}` creará el `{<alias>}` ‘caudal’ y cuando se utilice `$\spunit{50 caudal}$` se imprimirá ‘50 m³/s’ en la salida PDF y NVDA/MathCAT verbalizará «cincuenta metros al cubo por segundos».

3.3 El comando \spmoney

\spmoney

```
\spmoney{<cifra>}
\spmoney{<cifra moneda>}
\spmoney[<key = val>]{<cifra moneda>}
```

El comando `\spmoney` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<cifra>}` que contiene la «cifra numérica» positiva o negativa la se cual procesa internamente por el comando `\spnum` al inicio del `{<argumento>}` acompañada de un «símbolo de moneda» o «alias de moneda» opcional descritos en §3.3.2.

Si no se proporciona el *argumento opcional* `[<key = val>]` y se ejecuta `$\spmoney{500}$` se imprimirá ‘\$500’ en la salida PDF y NVDA/MathCAT verbalizará «quinientos pesos».

El comportamiento de `\spmoney` depende de como se pase el *argumento obligatorio* que tiene prioridad por sobre los valores por defecto. Por ejemplo: `$\spmoney{500 dolares}$` imprimirá ‘\$500’ en la salida PDF y NVDA/MathCAT verbalizará «quinientos dolares» aunque la moneda por defecto sean pesos.

Si se pasa un «alias de moneda» del tipo ISO (código de divisas) como CLP o USD (mayúsculas) se modifica la verbalización en NVDA/MathCAT agregando el país y se ajusta la salida PDF al estandar ISO. Por ejemplo: `$\spmoney{500 CLP}$` imprimirá ‘CLP500’ en la salida PDF y NVDA/MathCAT verbalizará «quinientos pesos chilenos».

Si se pasa un «alias de moneda» del tipo ISO (código de divisas) como clp o usd (minúsculas) se modifica la verbalización en NVDA/MathCAT agregando el país, pero, NO se modifica la salida PDF imprimiendo el

«*símbolo de moneda*» correspondiente. Por ejemplo: `\$ \spmoney{500 c\p}` imprimirá '\$500' en la salida PDF y NVDA/MathCAT verbalizará «quinientos pesos chilenos»

Pasar de manera directa en el *argumento obligatorio* un «*símbolo de moneda*» como '\$', '€', '£' o un «*alias de moneda*» como peso, euro o libra produce la salida PDF y verbalización estándar esperada para NVDA/MathCAT.

- La posición del «*símbolo de moneda*» a la izquierda o a la derecha de la «*cifra numérica*» y la verbalización para NVDA/MathCAT se manejan desde el módulo Lua `spintent.lua` y no se proveen llaves para modificar este comportamiento.
- El «*símbolo de moneda*» es tomado de la fuente tipográfica definida para el *modo matemático* al momento de ejecutar el comando, se debe tener presente en caso de querer usar un «*símbolo de moneda*» poco habitual que esté soportada por `spintent`. El `{\argumento}` puede contener espacios matemáticos válidos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{\#1}` o `\mathbf{\#1}` por ejemplo. La coma ',' o el punto '.' como *separador decimal* solo pueden aparecer una «única vez». Cualquier entrada fuera de esta sintaxis devolverá un "error" deteniendo la compilación del documento.

### 3.3.1 Llaves para \spmoney

El comando `\spmoney` posee la *meta-key* `cifras-sep` pasada al comando `\spnum` (§3.1.1).

`currency = {\moneda | alias}` default: peso

Establece el *símbolo de moneda* por defecto que se imprimirá en la salida PDF y cómo lo verbalizará NVDA/MathCAT cuando SOLO se pasa una «*cifra numérica*» como `{\argumento}` al comando `\spmoney`. El «*símbolo de moneda*» o el «*alias de moneda*» pasado a esta llave debe ser alguno de los descritos en §3.3.2.

`sub-units = {\true | false}` default: false

Establece si se usarán *subunidades* de la moneda en curso para la verbalización en NVDA/MathCAT. La moneda en curso debe aceptar *subunidades*. Por ejemplo: `\$ \spmoney[sub-units=true]{100,5 dolares}` imprimirá '\$100,5' en la salida PDF y NVDA/MathCAT verbalizará «*cient dólares con cincuenta centavos*».

`dolar-math = {\true | false}` default: true

Establece si el *símbolo* '\$' se imprimirá en *modo texto* o en *modo matemático*. Si se establece en `true`, se usará la fuente tipográfica definida para el *modo matemático* en curso para imprimir '\$' en la salida PDF; si se establece en `false`, se usará la fuente tipográfica definida para el *modo texto* para imprimir '\$' en la salida PDF.

### 3.3.2 Monedas y alias para \spmoney

Resumen de los `{\símbolos de moneda}` y `{\alias de moneda}` soportados como `{\argumento}` para `\spmoney`.

## 3.4 El comando \spshort

---

`\spshort` `\spshort{\abreviaturas | acrónimos | números ordinales}`  
`\spshort[key = val]{\abreviaturas | acrónimos | números ordinales}`  
`\spshort[read-arg={\verbalización NVDA}]{\comandos LATEX}`

El comando `\spshort` trabaja *solo en modo texto* y recibe un *argumento obligatorio* del tipo `{\abreviaturas}`, `{\acrónimos}` o `{\números ordinales}` descritos en §3.4.2.

Si no se proporciona el *argumento opcional* `[key = val]` y se ejecuta `\spshort{dr. a}` se imprimirá 'Dr.<sup>a</sup>' en la salida PDF y NVDA/MathCAT verbalizará «*doctora*»; si se ejecuta `\spshort{onu}` se imprimirá 'ONU' en la salida PDF y NVDA/MathCAT verbalizará «*organización de las naciones unidas*» y si se ejecuta `\spshort{mineduc}` se imprimirá 'Mineduc' en la salida PDF y NVDA/MathCAT verbalizará «*ministerio de educación*».

Para la escritura de `{\números ordinales}` se puede ejecutar `\spshort{1. a}` que imprimirá '1.<sup>a</sup>' en la salida PDF y NVDA/MathCAT verbalizará «*primera*»; si se ejecuta `\spshort{1. o}` imprimirá '1.<sup>o</sup>' en la salida PDF y NVDA/MathCAT verbalizará «*primero*».

Para crear una nueva `{\abreviatura}`, `{\acrónimo}` o `{\número ordinal}` que no se encuentre registrada por `spintent` dentro del módulo Lua `spintent.lua` descritas en §3.4.2 se debe utilizar la llave `read-arg` y definir esta dentro del *argumento obligatorio* que puede contener `{\comandos LATEX}` en *modo texto* los cuales se imprimirán en salida PDF.

- Para la escritura de `{\números ordinales}` se debe tener cuidado y no confundir la «*letra o voladita*» 'o' con el símbolo de grado '°'. Si la fuente tipográfica del *modo texto* está utilizando `Numbers=OldStyle` los números pueden quedar muy separados de la «*letra voladita*», la solución para esto es agregar `\addfontfeatures{Numbers=Lining}` dentro de un grupo junto al comando: `{\addfontfeatures{Numbers=Lining}\spshort{1. o}}`.
- El comando `\spshort` está pensado como un reemplazo para *shorthands* y el comando `\sptext` provistos por el paquete `babel` en español compatible con *tagged* PDF. Es de las pocas utilidades que provee el paquete `spintent` que trabaja *solo en modo texto*. Internamente utiliza la llave `E` (*expansion text*) proveída por el paquete `tagpdf`[29].

| Entrada | Alias                             | Nombre  | Salida | NVDA/MathCAT         |
|---------|-----------------------------------|---------|--------|----------------------|
| \$      | peso, pesos                       | peso    | \$     | «peso / pesos»       |
| €       | euro, euros                       | euro    | €      | «euro / euros»       |
| £       | libra, libras                     | libra   | £      | «libra / libras»     |
| ¥       | yen, yenes                        | yen     | ¥      | «yen / yenes»        |
| ¥       | yuan, yuanes                      | yuan    | ¥      | «yuan / yuanes»      |
| ₩       | won, wons                         | won     | ₩      | «won / wons»         |
| ₪       | shekel, shekels, sequel, sequeles | séquel  | ₪      | «séquel / séqueles»  |
| ₹       | rupia, rupias                     | rupia   | ₹      | «rupia / rupias»     |
| ₽       | rublo, rublos                     | rublo   | ₽      | «rublo / rublos»     |
| ₺       | lira, liras                       | lira    | ₺      | «lira / liras»       |
| ₴       | grivna, grivnas                   | grivna  | ₴      | «grivna / grivnas»   |
| ¢       | centavo, centavos                 | centavo | ¢      | «centavo / centavos» |

Divisas en formato ISO (Mayúsculas)

|     |   |               |     |                                                  |
|-----|---|---------------|-----|--------------------------------------------------|
| CLP | — | peso chileno  | CLP | «peso chileno / pesos chilenos»                  |
| MXN | — | peso mexicano | MXN | «peso mexicano / pesos mexicanos»                |
| USD | — | dólar         | USD | «dólar estadounidense / dólares estadounidenses» |
| EUR | — | euro          | EUR | «euro / euros»                                   |
| GBP | — | libra         | GBP | «libra / libras»                                 |
| JPY | — | yen           | JPY | «yen / yenes»                                    |
| CNY | — | yuan          | CNY | «yuan / yuanes»                                  |
| KRW | — | won           | KRW | «won / wons»                                     |
| ILS | — | séquel        | ILS | «séquel / séqueles»                              |
| INR | — | rupia         | INR | «rupia / rupias»                                 |
| RUB | — | rublo         | RUB | «rublo / rublos»                                 |
| TRY | — | lira          | TRY | «lira / liras»                                   |
| UAH | — | grivna        | UAH | «grivna / grivnas»                               |

Códigos ISO en minúsculas

|     |                |               |    |                                                  |
|-----|----------------|---------------|----|--------------------------------------------------|
| clp | —              | peso chileno  | \$ | «peso chileno / pesos chilenos»                  |
| mxn | —              | peso mexicano | \$ | «peso mexicano / pesos mexicanos»                |
| usd | dolar, dolares | dólar         | \$ | «dólar estadounidense / dólares estadounidenses» |
| eur | —              | euro          | €  | «euro / euros»                                   |
| gbp | —              | libra         | £  | «libra / libras»                                 |
| jpy | —              | yen           | ¥  | «yen / yenes»                                    |
| cny | —              | yuan          | ¥  | «yuan / yuanes»                                  |
| krw | —              | won           | ₩  | «won / wons»                                     |
| ils | —              | séquel        | ₪  | «séquel / séqueles»                              |
| inr | —              | rupia         | ₹  | «rupia / rupias»                                 |
| rub | —              | rublo         | ₽  | «rublo / rublos»                                 |
| try | —              | lira          | ₺  | «lira / liras»                                   |
| uah | —              | grivna        | ₴  | «grivna / grivnas»                               |

3.4.1 Llaves para \spshort

`read-arg = {⟨verbalización NVDA⟩}` default: no usado

Establece la verbalización del argumento {⟨comandos  $\LaTeX$ ⟩} de la nueva {⟨abreviatura⟩}, {⟨acrónimo⟩} o {⟨número ordinal⟩} para NVDA. El valor de esta llave siempre debe pasarse entre '{ }'. Por ejemplo: `\spshort[read-arg={octavo}]{8.\textsuperscript{\emph{avo}}}` creará una nueva {⟨abreviatura⟩} que imprimirá '8.<sup>avo</sup>' en la salida PDF y NVDA verbalizará «octavo».

`small-caps = {⟨true | false⟩}` default: false

Establece si se usarán versalitas para imprimir {⟨acrónimos⟩} y para la «letra volada» en {⟨abreviaturas⟩} y {⟨números ordinales⟩} respetando las reglas de la RAE para el uso de estas. Si se establece en `true`, las abreviaturas con mayúscula como EE. UU. y los acrónimos de cinco o más letras como Unicef no se verán afectados; si se establece en `false`, no se aplicarán versalitas. Por ejemplo: `\spshort[small-caps=true]{dr.a}` imprimirá 'Dr.<sup>a</sup>' en la salida PDF y NVDA verbalizará «doctora».

3.4.2 Argumentos para \spshort

El cuadro 2 resume las abreviaturas, expresiones latinas y títulos registrados en el módulo Lua `spintent.lua`.

Cuadro 2: Abreviaturas soportadas

| Entrada                                    | Alias / Variantes | Salida PDF         | Lectura NVDA       |
|--------------------------------------------|-------------------|--------------------|--------------------|
| Abreviaturas con letras voladas            |                   |                    |                    |
| dr.a                                       | dr. <sup>a</sup>  | Dr. <sup>a</sup>   | «doctora»          |
| dr.as                                      | —                 | Dr. <sup>as</sup>  | «doctoras»         |
| prof.a                                     | —                 | Prof. <sup>a</sup> | «profesora»        |
| d.a                                        | d. <sup>a</sup>   | D. <sup>a</sup>    | «doña»             |
| m.a                                        | —                 | M. <sup>a</sup>    | «maría»            |
| n.o                                        | n. <sup>o</sup>   | N. <sup>o</sup>    | «número»           |
| n.os                                       | —                 | N. <sup>os</sup>   | «números»          |
| c.ia                                       | —                 | C. <sup>ia</sup>   | «compañía»         |
| Abreviaturas comunes y expresiones latinas |                   |                    |                    |
| pág.                                       | pag.              | pág.               | «página»           |
| vol.                                       | —                 | vol.               | «volumen»          |
| etc.                                       | —                 | etc.               | «etcétera»         |
| et al.                                     | et. al.           | et al.             | «y otros»          |
| ibíd.                                      | ibid.             | ibíd.              | «ibídem»           |
| op. cit.                                   | op.cit.           | op. cit.           | «obra citada»      |
| loc. cit.                                  | loc.cit.          | loc. cit.          | «lugar citado»     |
| v. gr.                                     | v.gr.             | v. gr.             | «verbigracia»      |
| i. e.                                      | i.e.              | i. e.              | «esto es»          |
| e. g.                                      | e.g.              | e. g.              | «por ejemplo»      |
| p. ej.                                     | p.ej.             | p. ej.             | «por ejemplo»      |
| Títulos, profesiones y cargos              |                   |                    |                    |
| sr.                                        | —                 | Sr.                | «señor»            |
| sra.                                       | —                 | Sra.               | «señora»           |
| srta.                                      | —                 | Srta.              | «señorita»         |
| dr.                                        | —                 | Dr.                | «doctor»           |
| dra.                                       | —                 | Dra.               | «doctora»          |
| dres.                                      | —                 | Dres.              | «doctores»         |
| dras.                                      | —                 | Dras.              | «doctoras»         |
| prof.                                      | —                 | Prof.              | «profesor»         |
| profa.                                     | —                 | Profa.             | «profesora»        |
| profs.                                     | —                 | Profs.             | «profesores»       |
| ing.                                       | —                 | Ing.               | «ingeniero»        |
| ings.                                      | —                 | Ings.              | «ingenieros»       |
| lic.                                       | —                 | Lic.               | «licenciado»       |
| v. b.                                      | v.b.              | V. B.              | «visto bueno»      |
| Abreviaturas compuestas e institucionales  |                   |                    |                    |
| s. a.                                      | s.a.              | S. A.              | «sociedad anónima» |
| ee. uu.                                    | ee.uu.            | EE. UU.            | «estados unidos»   |
| dd. hh.                                    | dd.hh.            | DD. HH.            | «derechos humanos» |
| jj. oo.                                    | jj.oo.            | JJ. OO.            | «juegos olímpicos» |

Continúa en la siguiente página...

Cuadro 2: (Continuación - Abreviaturas soportadas)

| Entrada  | Alias / Variantes | Salida PDF | Lectura NVDA                      |
|----------|-------------------|------------|-----------------------------------|
| ff. aa.  | ff.aa.            | FF. AA.    | «fuerzas armadas»                 |
| rr. ee.  | rr.ee.            | RR. EE.    | «relaciones exteriores»           |
| cc. aa.  | cc.aa.            | CC. AA.    | «comunidades autónomas»           |
| p. d.    | p.d.              | P. D.      | «posdata»                         |
| a. c.    | a.c.              | a. C.      | «antes de Cristo»                 |
| d. c.    | d.c.              | d. C.      | «después de Cristo»               |
| d. n. i. | d.n.i.            | D. N. I.   | «documento nacional de identidad» |
| r. u. t. | r.u.t.            | R. U. T.   | «rol único tributario»            |
| r. u. n. | r.u.n.            | R. U. N.   | «rol único nacional»              |

El cuadro 3 detalla las siglas y acrónimos registrados en el módulo Lua `spintent.lua`.

Cuadro 3: Siglas y acrónimos soportados

| Entrada                               | Alias / Variantes | Salida PDF | Lectura NVDA                                                                     |
|---------------------------------------|-------------------|------------|----------------------------------------------------------------------------------|
| Siglas (Deletreo o lectura expandida) |                   |            |                                                                                  |
| dni                                   | —                 | DNI        | «documento nacional de identidad»                                                |
| rut                                   | —                 | RUT        | «rol único tributario»                                                           |
| run                                   | —                 | RUN        | «rol único nacional»                                                             |
| onu                                   | —                 | ONU        | «organización de las naciones unidas»                                            |
| rae                                   | —                 | RAE        | «real academia española»                                                         |
| ong                                   | ongs              | ONG        | «organización no gubernamental»                                                  |
| urss                                  | —                 | URSS       | «unión de repúblicas socialistas soviéticas»                                     |
| oea                                   | —                 | OEA        | «organización de los estados americanos»                                         |
| oms                                   | —                 | OMS        | «organización mundial de la salud»                                               |
| fmi                                   | —                 | FMI        | «fondo monetario internacional»                                                  |
| bid                                   | —                 | BID        | «banco interamericano de desarrollo»                                             |
| otan                                  | —                 | OTAN       | «organización del tratado del atlántico norte»                                   |
| ue                                    | —                 | UE         | «unión europea»                                                                  |
| ru                                    | —                 | RU         | «reino unido»                                                                    |
| Acrónimos silábicos                   |                   |            |                                                                                  |
| unicef                                | —                 | Unicef     | «fondo de las naciones unidas para la infancia»                                  |
| unesco                                | —                 | Unesco     | «organización de las naciones unidas para la educación, la ciencia y la cultura» |
| mercosur                              | —                 | Mercosur   | «mercado común del sur»                                                          |
| cepal                                 | —                 | Cepal      | «comisión económica para américa latina»                                         |
| mineduc                               | —                 | Mineduc    | «ministerio de educación»                                                        |
| conicet                               | —                 | Conicet    | «consejo nacional de investigaciones»                                            |

El cuadro 4 muestra ejemplos de números ordinales registrados en el módulo Lua `spintent.lua`.

Cuadro 4: Números ordinales soportados

| Entrada                           | Alias / Variantes | Salida PDF       | Lectura NVDA |
|-----------------------------------|-------------------|------------------|--------------|
| Sufijos para ordinales (Ejemplos) |                   |                  |              |
| 1.a                               | 1. <sup>a</sup>   | 1. <sup>a</sup>  | «primera»    |
| 2.o                               | 2. <sup>o</sup>   | 2. <sup>o</sup>  | «segundo»    |
| 3.er                              | —                 | 3. <sup>er</sup> | «tercer»     |
| 4.os                              | —                 | 4. <sup>os</sup> | «cuartos»    |
| 5.as                              | —                 | 5. <sup>as</sup> | «quintas»    |

## 4 Utilidades generales

Además de los «cuatro grandes» y sus asociados, el paquete `spintent` provee tres utilidades de uso general: `\spsiglo` para la escritura y lectura de siglos, `\sptime` para la escritura y lectura de tiempo y `\spdate` para la escritura y lectura de fechas.

### El comando `\spsiglo`

`\spsiglo`

`\spsiglo{<siglo>}`  
`\spsiglo[<key = val>]{<siglo>}`

El comando `\spsiglo` trabaja en *modo texto* y en *modo matemático*, recibe el *argumento obligatorio* `{<siglo>}`, el cual puede ser un *número arábigo* ‘21’ o un *número romano* ‘XXI’ mayor que ‘0’ y menor que ‘4000’.

Si no se proporciona el *argumento opcional* `[<key = val>]` y se ejecuta `\spsiglo{21}` o `\spsiglo{XXI}` se imprimirá ‘xxi’ en la salida PDF y NVDA/MathCAT verbalizará «veintiuno».

- El *argumento obligatorio* `{<siglo>}` «siempre» es convertido a *número romano en versalitas* y es verbalizado por NVDA/MathCAT como *número arábigo*. Esta es la única utilidad que provee el paquete `spintent` que trabaja en *modo texto* y en *modo matemático*. Internamente utiliza la llave `alt` proveída por el paquete `tagpdf` en *modo texto* y usa `\MathMLintent` para el *modo matemático*.

### Llaves para `\spsiglo`

`print-era = {<ac | dc | none>}`

default: none

Establece si se *imprime* o no la abreviatura después de imprimir el argumento `{<siglo>}`. Si se establece en `ac`, `\spsiglo[print-era=ac]{XXI}` imprimirá ‘xi a. C.’ en la salida PDF y NVDA/MathCAT verbalizará «veintiuno antes de cristo»; si se establece en `dc`, `\spsiglo[print-era=dc]{XXI}` imprimirá ‘xi d. C.’ en la salida PDF y NVDA/MathCAT verbalizará «veintiuno después de cristo»; si se establece en `none`, `\spsiglo[print-era=none]{XXI}` imprimirá ‘xi’ en la salida PDF y NVDA/MathCAT verbalizará «veintiuno».

### El comando `\sptime`

`\sptime`

`\sptime{<horas : minutos>}`  
`\sptime{<horas : minutos : segundos>}`

El comando `\sptime` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<horas : minutos>}` o `{<horas : minutos : segundos>}`. Por ejemplo: `$\sptime{23:45}$` imprimirá ‘23:45’ en *modo texto* la salida PDF y NVDA/MathCAT verbalizará «las veintitrés y cuarenta y cinco minutos».

### El comando `\spdate`

`\spdate`

`\spdate{<DD/ MM/YYYY>}`  
`\spdate{<YYYY/ MM/DD>}`

`\spdate{<DD- MM-YYYY>}`  
`\spdate{<YYYY- MM-DD>}`

El comando `\spdate` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<DD/MM/YYYY>}`, `{<DD-MM-YYYY>}` o `{<YYYY-MM-DD>}`. Por ejemplo: `$\spdate{1981-01-02}$` y `$\spdate{02/01/1981}$` imprimirán en *modo texto* la salida PDF ‘1981-01-02’ y ‘02/01/1981’ respectivamente y en ambos casos, NVDA/MathCAT verbalizará «dos de enero de mil novecientos ochenta y uno».

- Si se pasa el argumento `{<YYYY/MM/DD>}`, internamente se convertirá en `{<DD/MM/YYYY>}` y se imprimirá de esa forma por compatibilidad con NVDA/MathCAT.

## 5 Utilidades para símbolos

El paquete `spintent` provee tres comandos `\spread`, `\spdsym` y `\spmsym` para trabajar con «símbolos matemáticos» descritos en esta sección. Están pensando desde el punto de vista docente, para situaciones donde la lectura de un «símbolo» (acorde al contexto) es relevante.

- A diferencia de las otras utilidades implementados por `spintent` (que analizan y dan formato a la salida), estos no procesan los «símbolos matemáticos», solo afectan la *forma* (semántica) en que serán verbalizados por NVDA/MathCAT.

### El comando `\spnewsym`

`\spnewsym`

`\spnewsym{<símbolo>}{<verbalización NVDA>}`

El comando `\spnewsym` trabaja en *modo texto* y recibe dos *argumentos obligatorios*: `{<símbolo>}` y `{<verbalización NVDA>}` que establece como NVDA/MathCAT verbalizará ese `{<símbolo>}` al utilizarlo con `\spread`.

El `{<símbolo>}` no debe estar registrado. Si ya está registrado se devolverá un “error” y se detendrá la compilación del documento.

La declaración es «global» y se almacena en el módulo `spintent.lua`. Por ejemplo: `\spnewsym{\cong}{es-congruente-}` registrará ‘\cong’ y cuando se utilice `$\spread{\cong}$` se imprimirá ‘≅’ en la salida PDF y NVDA/MathCAT verbalizará «es congruente con».

## El comando `\spread`

---

```
\spread <{símbolo}>
\spread[<key = val>]{<símbolo>}
```

---

El comando `\spread` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<símbolo>}`, buscando su verbalización registrada por medio de `\spnewsym`.

Si el `{<símbolo>}` está registrado, `\spread{<cong>}` imprimirá ‘ $\cong$ ’ en la salida PDF y NVDA/MathCAT verbalizará la `{<verbalización NVDA>}` establecida por `\spnewsym`.

Si el `{<símbolo>}` *no* está registrado y no se proporciona la llave `read-arg`, se devolverá un “error” y se detendrá la compilación del documento.

### Llaves para `\spread`

`read-arg = {<verbalización NVDA>}` default: *no usado*

Sobrescribe, *solo para esta ejecución*, la verbalización registrada por `\spnewsym` para el `{<símbolo>}` sin modificar la impresión de este. No requiere que el `{<símbolo>}` esté registrado. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’. Por ejemplo: `\spread[read-arg={es-paralela-a}]{\parallel}` imprimirá ‘ $\parallel$ ’ en la salida PDF y NVDA/MathCAT verbalizará «*es paralela a*».

`silent-cmd = {<true | false>}` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del `{<símbolo>}` pasado al comando sin modificar la impresión de este, sin necesitar que esté registrado ni pasar `read-arg`. Si se establece en `true`, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en `false`, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spusep`

---

```
\spusep <{separador}>
\spusep[<key = val>]{<separador>}
```

---

El comando `\spusep` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<separador>}`, el cual debe ser uno de los «delimitadores» soportados: ‘(’, ‘)’, ‘[’, ‘]’, ‘{’, ‘}’ o ‘\’. Imprime el *delimitador* indicado con el tamaño establecido por la llave `size` y controla si NVDA/MathCAT lo verbaliza o no mediante la llave `silent`.

Si no se proporciona el *argumento opcional* `[<key = val>]`, `\spusep{< >}` imprimirá ‘(’ en la salida PDF y NVDA/MathCAT verbalizará «*paréntesis izquierdo*».

- El motivo de existencia de `\spusep` es que los comandos `\left`, `\right`, `\Uleft` y `\Uright` provistos por LuaTeX crean un *grupo* implícito que dificulta el código para *tagged* PDF cuando se necesita interrumpir algo entre ellos, por ejemplo al construir expresiones complejas con `\spnZ`, `\spmQ` o `\spdiv`. El comando `\spusep` imprime el *delimitador* sin crear ningún grupo.

### Llaves para `\spusep`

`size = {<none | big | Big | bigg | Bigg>}` default: *none*

Establece el *tamaño* del `{<delimitador>}` que se imprimirá en la salida PDF. Los valores corresponden a los cuatro tamaños fijos provistos por  $\TeX$ : si se establece en `none`, el `{<delimitador>}` se imprime en el tamaño por defecto del modo matemático en curso; si se establece en `big`, se usa `\bigl\bigl`; si se establece en `Big`, se usa `\Bigl\Bigl`; si se establece en `bigg`, se usa `\biggl\biggr`; y si se establece en `Bigg`, se usa `\Biggl\Biggr`. El tamaño se aplica automáticamente en el lado correcto (izquierdo o derecho) según el `{<delimitador>}` pasado.

`silent = {<true | false>}` default: *false*

Establece si NVDA/MathCAT verbalizará el `{<delimitador>}` impreso por `\spusep`. Si se establece en `true`, el `{<delimitador>}` se marcará con el *intent* `:silent` y NVDA/MathCAT *no* lo verbalizará; si se establece en `false`, NVDA/MathCAT verbalizará el `{<delimitador>}` de forma normal.

## El comando `\spdsym`

---

```
\spdsym
\spdsym[<key = val>]
```

---

El comando `\spdsym` trabaja en *modo matemático* y recibe el *argumento opcional* `[<key = val>]` mediante el cual se establece la «*forma*» en la que se imprimirá el *símbolo de división* y como se verbalizará en NVDA/MathCAT.

Si no se proporciona el *argumento opcional* `[<key = val>]`, `\spdsym` imprimirá ‘:’ en la salida PDF y NVDA/MathCAT verbalizará «*dividido*».

### Llaves para `\spdsym`

|                                                                                                                                                                                                                                                                                                                                                                     |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <code>prefix = {⟨prefijo⟩}</code>                                                                                                                                                                                                                                                                                                                                   | default: <i>no usado</i> |
| Establece el <i>prefijo</i> que será <i>verbalizado</i> por NVDA/MathCAT «antes» de <i>verbalizar</i> el {⟨valor⟩} establecido por la llave <code>read-sym</code> . El valor de esta llave <i>siempre</i> debe pasarse entre llaves '{ }'.                                                                                                                          |                          |
| <code>suffix = {⟨sufijo⟩}</code>                                                                                                                                                                                                                                                                                                                                    | default: <i>no usado</i> |
| Establece el <i>sufijo</i> que será <i>verbalizado</i> por NVDA/MathCAT «después» de <i>verbalizar</i> el {⟨valor⟩} establecido por la llave <code>read-sym</code> . El valor de esta llave <i>siempre</i> debe pasarse entre llaves '{ }'.                                                                                                                         |                          |
| <code>set-sym = {⟨old-style   two-dots   slash⟩}</code>                                                                                                                                                                                                                                                                                                             | default: <i>two-dots</i> |
| Establece el <i>símbolo de división</i> que se imprimirá en la salida PDF. Si se establece en <code>old-style</code> , imprimirá '÷'; si se establece en <code>two-dots</code> , imprimirá '⋅' y si se establece <code>slash</code> , imprimirá '/'.                                                                                                                |                          |
| <code>read-sym = {⟨dividido   dividido-por   dividido-entre⟩}</code>                                                                                                                                                                                                                                                                                                | default: <i>dividido</i> |
| Establece la <i>forma</i> con la cual NVDA/MathCAT verbalizará el <i>símbolo de división</i> . Si se establece en <code>dividido</code> , NVDA/MathCAT verbalizará «dividido»; si se establece en <code>dividido-por</code> , NVDA/MathCAT verbalizará «dividido por» y si se establece en <code>dividido-entre</code> , NVDA/MathCAT verbalizará «dividido entre». |                          |

### El comando `\spmsym`

---

```
\spmsym \spmsym
\spmsym[⟨key = val⟩]
```

---

El comando `\spmsym` trabaja en *modo matemático* y recibe el *argumento opcional* [⟨key = val⟩] mediante el cual se establece la «forma» en la que se imprimirá el *símbolo de multiplicación* y como se verbalizará en NVDA/MathCAT.

Si no se proporciona el *argumento opcional* [⟨key = val⟩], `\spmsym` imprimirá '·' en la salida PDF y NVDA/MathCAT verbalizará «por».

#### Llaves para `\spmsym`

|                                                                                                                                                                                                                                                                                                                                                       |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <code>prefix = {⟨prefijo⟩}</code>                                                                                                                                                                                                                                                                                                                     | default: <i>no usado</i> |
| Establece el <i>prefijo</i> que será <i>verbalizado</i> por NVDA/MathCAT «antes» de <i>verbalizar</i> el {⟨valor⟩} establecido por la llave <code>read-sym</code> . El valor de esta llave <i>siempre</i> debe pasarse entre llaves '{ }'.                                                                                                            |                          |
| <code>suffix = {⟨sufijo⟩}</code>                                                                                                                                                                                                                                                                                                                      | default: <i>no usado</i> |
| Establece el <i>sufijo</i> que será <i>verbalizado</i> por NVDA/MathCAT «después» de <i>verbalizar</i> el {⟨valor⟩} establecido por la llave <code>read-sym</code> . El valor de esta llave <i>siempre</i> debe pasarse entre llaves '{ }'.                                                                                                           |                          |
| <code>set-sym = {⟨cross   dot   asterisk⟩}</code>                                                                                                                                                                                                                                                                                                     | default: <i>dot</i>      |
| Establece el <i>símbolo de multiplicación</i> que se imprimirá en la salida PDF. Si se establece en <code>cross</code> , imprimirá '×'; si se establece en <code>dot</code> , imprimirá '·' y si se establece en <code>asterisk</code> , imprimirá '*'.                                                                                               |                          |
| <code>read-sym = {⟨por   multiplicado-por   veces⟩}</code>                                                                                                                                                                                                                                                                                            | default: <i>por</i>      |
| Establece la <i>forma</i> con la cual NVDA/MathCAT verbalizará el <i>símbolo de multiplicación</i> . Si se establece en <code>por</code> , NVDA/MathCAT verbalizará «por»; si se establece en <code>multiplicado-por</code> , NVDA/MathCAT verbalizará «multiplicado por» y si se establece en <code>veces</code> , NVDA/MathCAT verbalizará «veces». |                          |
| <code>not-print = {⟨true   false⟩}</code>                                                                                                                                                                                                                                                                                                             | default: <i>false</i>    |
| Deshabilita la <i>impresión</i> del <i>símbolo de multiplicación</i> . Cuando se establece esta llave, NVDA/MathCAT verbalizará el <i>símbolo de multiplicación</i> acorde al valor establecido por la llave <code>read-sym</code> , pero, no se imprimirá este en la salida PDF.                                                                     |                          |

## 6 Utilidades para números

El paquete `spintent` provee utilidades para trabajar con números descritas en esta sección.

### El comando `\spdiv`

---

```
\spdiv \spdiv{⟨dividendo⟩}{⟨divisor⟩}
\spdiv[⟨key = val⟩]{⟨dividendo⟩}{⟨divisor⟩}
```

---

El comando `\spdiv` trabaja en *modo matemático* y recibe dos *argumentos obligatorios* {⟨dividendo⟩} y {⟨divisor⟩}.

Si no se proporciona el *argumento opcional* [⟨key = val⟩], `\spdiv{20}{26}` imprimirá '20 : 26' en la salida PDF y NVDA/MathCAT verbalizará «veinte dividido veinte y seis»

### Llaves para `\spdiv`

El comando `\spdiv` posee las *meta-keys* `cifras-sep`, `read-sign`, `read-period-sep` y `notation-sym` pasadas al comando `\spnum` (§3.1.1) junto a las *meta-keys* `set-sym` y `read-sym` pasadas al comando `\spdsym` (pág. 13).

`wrap-parens = {⟨true | false⟩}` default: *false*

Establece si se *imprimen* paréntesis redondos ‘(...)’ alrededor de los *argumentos* {⟨*dividendo*⟩} y {⟨*divisor*⟩} pasados a `\spdiv`. Si se establece en *true*, se imprimirán paréntesis redondos alrededor de ambos *argumentos*; si se establece en *false*, no se imprimirán los paréntesis.

`read-parens = {⟨true | false⟩}` default: *true*

Establece si NVDA/MathCAT *verbalizará* los paréntesis redondos ‘(...)’ alrededor de los *argumentos* {⟨*dividendo*⟩} y {⟨*divisor*⟩} pasados a `\spdiv`. Si se establece en *true*, NVDA/MathCAT verbalizará la lectura de los paréntesis; si se establece en *false*, NVDA/MathCAT no los verbalizará.

### El comando `\spmul`

---

```
\spmul {⟨factor⟩}{⟨factor⟩}
\spmul[⟨key = val⟩]{⟨factor⟩}{⟨factor⟩}
```

---

El comando `\spmul` trabaja en *modo matemático* y recibe dos *argumentos obligatorios* {⟨*factor*⟩} y {⟨*factor*⟩}. Si no se proporciona el *argumento opcional* [⟨*key = val*⟩], `\spmul{20}{26}` imprimirá ‘20 · 26’ en la salida PDF y NVDA/MathCAT verbalizará «veinte por veinte y seis».

### Llaves para `\spmul`

El comando `\spmul` posee las *meta-keys* `cifras-sep`, `read-sign`, `read-period-sep` y `notation-sym` pasadas al comando `\spnum` (§3.1.1) junto a las *meta-keys* `set-sym` y `read-sym` pasadas al comando `\spmsym` (pág. 13).

`wrap-parens = {⟨true | false⟩}` default: *false*

Establece si se *imprimen* paréntesis redondos ‘(...)’ alrededor de los *argumentos* {⟨*factor*⟩} y {⟨*factor*⟩} pasados a `\spmul`. Si se establece en *true*, se imprimirán paréntesis redondos alrededor de ambos *argumentos*; si se establece en *false*, no se imprimirán los paréntesis.

`read-parens = {⟨true | false⟩}` default: *true*

Establece si NVDA/MathCAT *verbalizará* los paréntesis redondos ‘(...)’ alrededor de los *argumentos* {⟨*factor*⟩} y {⟨*factor*⟩} pasados a `\spmul`. Si se establece en *true*, NVDA/MathCAT verbalizará la lectura de los paréntesis; si se establece en *false*, NVDA/MathCAT no los verbalizará.

### El comando `\spnD`

---

```
\spnD \spnD
\spnD(⟨número natural⟩)
\spnD[⟨key = val⟩](⟨número natural⟩)
```

---

El comando `\spnD` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* (⟨*número natural*⟩), por ejemplo: ‘(6)’. Si se usa sin (⟨*argumento*⟩), `\spnD` imprimirá ‘D(*n*)’ en la salida PDF y NVDA/MathCAT verbalizará «divisores de *n*».

Si se usa con (⟨*argumento*⟩), por ejemplo, `\spnD(6)`, imprimirá ‘D(6)’ en la salida PDF y NVDA/MathCAT verbalizará «divisores de seis».

### Llaves para `\spnD`

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT «antes» de verbalizar `\spnD(⟨argumento⟩)`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`result = {⟨true | false⟩}` default: *false*

Establece si se *imprime* el resultado de calcular los *divisores* del (⟨*número natural*⟩) pasado a `\spnD`. Si se establece en *true*, `\spnD[result=true](6)` imprimirá ‘D(6) = {1, 2, 3, 6}’ en la salida PDF y NVDA/MathCAT verbalizará «los divisores de seis son uno, dos, tres y seis»; si se establece en *false*, no se imprimirá en la salida PDF el resultado de calcular los *divisores*.

`result-num = {⟨número natural⟩}` default: *5*

Establece la *cantidad* de números que se mostrarán en la salida PDF al activar `result=true`. Si se establece en *2*, solo se imprimirán dos valores en la salida PDF; si no se establece, se imprimirán como máximo *5* valores y luego se añadirá ‘...’ en la salida PDF.

`sample-arg = {⟨true | false⟩}` default: *false*

Desactiva la *validación* interna del (⟨*argumento*⟩) pasado a `\spnD`, permitiendo usar ejemplos «no numéricos» como entrada. Si se establece en *true*, `\spnD[sample-arg=true](m)` imprimirá ‘D(*m*)’ en la salida PDF y NVDA/MathCAT verbalizará «divisores de *m*»; si se establece en *false*, el (⟨*argumento*⟩) debe ser un número natural, en caso contrario se devolverá un “error”.

`silent-cmd = { <true | false> }` default: *false*

Establece si se *silencia* la verbalización para NVDA/MathCAT de `\spnD(<argumento>)` cuando se establece `sample-arg=true` o cuando `\spnD` se usa sin (`<argumento>`). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

**El comando `\spnM`**

`\spnM` `\spnM`  
`\spnM(<número natural>)`  
`\spnM[<key = val>](<número natural>)`

El comando `\spnM` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* (`<número natural>`), por ejemplo: `'(6)'`. Si se usa sin (`<argumento>`), `\spnM` imprimirá `'M(n)'` en la salida PDF y NVDA/MathCAT verbalizará *«múltiplos de n»*.  
Si se usa con (`<argumento>`), por ejemplo, `\spnM(6)`, imprimirá `'M(6)'` en la salida PDF y NVDA/MathCAT verbalizará *«múltiplos de seis»*.

**Llaves para `\spnM`**

`prefix = { <prefijo> }` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *«antes»* de verbalizar `\spnM(<argumento>)`. El valor de esta llave *siempre* debe pasarse entre llaves `{ }`.

`result = { <true | false> }` default: *false*

Establece si se *imprime* el resultado de calcular los *múltiplos* del (`<número natural>`) pasado a `\spnM`. Si se establece en `true`, `\spnM[result=true](6)` imprimirá `'M(6) = {6, 12, 18, 24, 30, ...}'` en la salida PDF y NVDA/MathCAT verbalizará *«los múltiplos de seis son seis, doce, dieciocho, veinticuatro, treinta puntos suspensivos»*; si se establece en `false`, no se imprimirá en la salida PDF el resultado de calcular los *múltiplos*.

`result-num = { <número natural> }` default: *5*

Establece la *cantidad* de números que se mostrarán en la salida PDF al activar `result=true`. Si se establece en `2`, solo se imprimirán dos valores en la salida PDF seguidos de `'...'`; si no se establece, se imprimirán por defecto los `5` primeros valores y luego se añadirá `'...'` en la salida PDF.

`sample-arg = { <true | false> }` default: *false*

Desactiva la *validación* interna del (`<argumento>`) pasado a `\spnM`, permitiendo usar ejemplos *«no numéricos»* como entrada. Si se establece en `true`, `\spnM[sample-arg=true](m)` imprimirá `'M(m)'` en la salida PDF y NVDA/MathCAT verbalizará *«múltiplos de m»*; si se establece en `false`, el (`<argumento>`) debe ser un número natural, en caso contrario se devolverá un “error”.

`silent-cmd = { <true | false> }` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spnM(<argumento>)` cuando se establece `sample-arg=true` o cuando `\spnM` se usa sin (`<argumento>`). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false` NVDA/MathCAT lo verbalizará.

**El comando `\spmc`**

`\spmc` `\spmc`  
`\spmc(<números separados por coma>)`  
`\spmc[<key = val>](<números separados por coma>)`

El comando `\spmc` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* (`<números separados por coma>`), el cual debe ser una lista de *números naturales*, por ejemplo: `'(2, 3, 6)'`. Si se usa sin (`<argumento>`) `\spmc` imprimirá `'mcd'` en la salida PDF, y NVDA/MathCAT verbalizará *«máximo común divisor»* o *«m c d»*, acorde al `{<valor>}` establecido por la llave `read-cmd`.  
Si se usa con (`<argumento>`) `\spmc(2, 3, 6)` imprimirá `'mcd(2, 3, 6)'` en la salida PDF, y NVDA/MathCAT verbalizará *«máximo común divisor entre dos, tres y seis»* o *«m c d entre dos, tres y seis»* acorde al `{<valor>}` establecido por la llave `read-cmd`.

**Llaves para `\spmc`**

`upper-case = { <true | false> }` default: *false*

Establece si se usarán *mayúsculas* o *minúsculas* en la salida PDF para el comando `\spmc`. Si se establece en `true`, se imprimirá `'MCD'` en la salida PDF; si se establece en `false`, se imprimirá `'mcd'` en la salida PDF.

`read-cmd = { <terse | clear> }` default: *clear*

Establece la *forma* en la que NVDA/MathCAT verbalizará el comando `\spmc`. Si se establece en `terse`, NVDA/MathCAT verbalizará *«m c d»*; si se establece en `clear`, NVDA/MathCAT verbalizará *«máximo común divisor»*.

`prefix = { <prefijo> }` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *«antes»* de verbalizar el valor establecido por la llave `read-cmd`. El valor de esta llave *siempre* debe pasarse entre llaves `{ }`.

`result = { <true | false> }` default: *false*

Establece si se *imprime* el resultado de calcular el «máximo común divisor» del (*argumento*) pasado a `\spgcd`. Si se establece en `true`: `\spgcd[result=true](2, 3, 6)` imprimirá ‘mcd(2, 3, 6) = 1’ en la salida PDF y NVDA/MathCAT verbalizará «máximo común divisor entre dos, tres y seis es igual a uno»; si se establece en `false` no se imprimirá en la salida PDF el resultado de calcular el «máximo común divisor».

`sample-arg = { (true | false) }` default: false

Desactiva la *validación* interna del (*argumento*) pasado a `\spgcd`, permitiendo usar ejemplos «no numéricos» como entrada. Si se establece en `true`: `\spgcd[sample-arg=true](m, n)`, imprimirá ‘mcd(*m*, *n*)’ en la salida PDF y NVDA/MathCAT verbalizará «máximo común divisor entre *m* y *n*»; si se establece en `false` el (*argumento*) debe ser una lista de *números naturales*, en caso contrario se devolverá un “error”.

`silent-cmd = { (true | false) }` default: false

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spgcd`(*argumento*) cuando se establece `sample-arg=true` o cuando `\spgcd` se usa sin (*argumento*). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

## El comando `\spmcn`

---

```
\spmcn \spmcn
\spmcn(<números separados por coma>)
\spmcn[<key = val>](<números separados por coma>)
```

---

El comando `\spmcn` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* (*números separados por coma*), el cual debe ser una lista de *números naturales*, por ejemplo: ‘(2, 3, 6)’. Si se usa sin (*argumento*) `\spmcn` imprimirá ‘mcm’ en la salida PDF, y NVDA/MathCAT verbalizará «mínimo común múltiplo» o «*m c m*», acorde al {*valor*} establecido por la llave `read-cmd`.

Si se usa con (*argumento*) `\spmcn(2, 3, 6)` imprimirá ‘mcm(2, 3, 6)’ en la salida PDF, y NVDA/MathCAT verbalizará «mínimo común múltiplo entre dos, tres y seis» o «*m c m entre dos, tres y seis*» acorde al {*valor*} establecido por la llave `read-cmd`.

### Llaves para `\spmcn`

`upper-case = { (true | false) }` default: false

Establece si se usarán *mayúsculas* o *minúsculas* en la salida PDF para el comando `\spmcn`. Si se establece en `true` se imprimirá ‘MCM’ en la salida PDF; si se establece en `false` se imprimirá ‘mcm’ en la salida PDF.

`read-cmd = { (terse | clear) }` default: clear

Establece la *forma* en la que NVDA/MathCAT verbalizará el comando `\spmcn`. Si se establece en `terse`, NVDA/MathCAT verbalizará «*m c m*»; si se establece en `clear`, NVDA/MathCAT verbalizará «mínimo común múltiplo».

`prefix = { (prefijo) }` default: no usado

Establece el *prefijo* que será verbalizado por NVDA/MathCAT «antes» de verbalizar el valor establecido por la llave `read-cmd`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`result = { (true | false) }` default: false

Establece si se *imprime* el resultado de calcular el «mínimo común múltiplo» del (*argumento*) pasado a `\spmcn`. Si se establece en `true`: `\spmcn[result=true](2, 3, 6)` imprimirá ‘mcm(2, 3, 6) = 6’ en la salida PDF y NVDA/MathCAT verbalizará «mínimo común múltiplo entre dos, tres y seis es igual a seis»; si se establece en `false` no se imprimirá en la salida PDF el resultado de calcular el «mínimo común múltiplo».

`sample-arg = { (true | false) }` default: false

Desactiva la *validación* interna del (*argumento*) pasado a `\spmcn`, permitiendo usar ejemplos «no numéricos» como entrada. Si se establece en `true`: `\spmcn[sample-arg=true](m, n)`, imprimirá ‘mcm(*m*, *n*)’ en la salida PDF y NVDA/MathCAT verbalizará «mínimo común múltiplo entre *m* y *n*»; si se establece en `false` el (*argumento*) debe ser una lista de *números naturales*, en caso contrario se devolverá un “error”.

`silent-cmd = { (true | false) }` default: false

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spmcn`(*argumento*) cuando se establece `sample-arg=true` o cuando `\spmcn` se usa sin (*argumento*). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false` NVDA/MathCAT SÍ lo verbalizará.

## El comando `\spnZ`

---

```
\spnZ \spnZ{<número entero>}
\spnZ[<key = val>]{<número entero>}
```

---

El comando `\spnZ` trabaja en *modo matemático* y recibe el *argumento obligatorio* {*número entero*} el cual se procesa internamente bajo el comando `\spnum`. Este comando está dedicado exclusivamente al manejo de «*números enteros*», y añade algunas utilidades prácticas para el demarcado e interacción con NVDA/MathCAT.

### Llaves para `\spnZ`

El comando `\spnZ` posee las *meta-keys* `cifras-sep` y `read-sign` pasadas al comando `\spnum` (§3.1.1).

`wrap-parens = {\langle true | false \rangle}` default: *false*

Establece si se *imprimen* paréntesis redondos ‘(...)’ alrededor del argumento  $\{\langle \text{número entero} \rangle\}$  pasado a `\spnZ`. Si se establece en *true*, se imprimirán paréntesis redondos alrededor del  $\{\langle \text{número entero} \rangle\}$ ; si se establece en *false*, no se imprimirán los paréntesis.

`read-parens = {\langle true | false \rangle}` default: *false*

Establece si NVDA/MathCAT *verbalizará* los paréntesis redondos ‘(...)’ alrededor del *argumento*  $\{\langle \text{número entero} \rangle\}$  pasado a `\spnZ`. Si se establece en *true*, NVDA/MathCAT verbalizará la lectura del paréntesis ‘(...)’; si se establece en *false*, NVDA/MathCAT no los verbalizará.

### El comando `\spmQ`

---

```
\spmQ{\langle número entero \rangle}{\langle numerador \rangle}{\langle denominador \rangle}
\spmQ[key = val]{\langle número entero \rangle}{\langle numerador \rangle}{\langle denominador \rangle}
```

---

El comando `\spmQ` trabaja en *modo matemático* y recibe *tres argumentos obligatorios*:  $\{\langle \text{número entero} \rangle\}$ ,  $\{\langle \text{numerador} \rangle\}$  y  $\{\langle \text{denominador} \rangle\}$ , los procesa internamente y devuelve un «*número mixto*».

Si no se proporciona el *argumento opcional*  $[\langle key = val \rangle]$ , `\spmQ{2}{5}{7}` imprimirá ‘ $2\frac{5}{7}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «*dos y cinco séptimos*».

- En el caso de las fracciones, y en especial con los «*números mixtos*», no se pueden manejar muchas cosas desde el lado de **spintent**. Por ejemplo, la entrada `\$3\frac{5}{7}` con *tagged* PDF se verbaliza por NVDA/MathCAT como «*tres más cinco séptimos*», lo cual no está mal, pero pedagógicamente hablando no es del todo correcto.

### Llaves para `\spmQ`

`join-word = {\langle entero | enteros | y \rangle}` default: *y*

Establece la *forma* en la que NVDA/MathCAT verbaliza la conexión entre el entero y la fracción que forman el *número mixto*. Si se establece en *entero*, al usar `\$ \spmQ[join-word=entero]{1}{5}{7}` imprimirá ‘ $1\frac{5}{7}$ ’ en la salida PDF, y NVDA/MathCAT verbalizará «*un entero cinco séptimos*»; si se establece en *enteros*, al usar `\$ \spmQ[join-word=enteros]{2}{5}{7}` imprimirá ‘ $2\frac{5}{7}$ ’ en la salida PDF, y NVDA/MathCAT verbalizará «*dos enteros cinco séptimos*»; si se establece en *y*, al usar `\$ \spmQ[join-word=y]{2}{5}{7}` imprimirá ‘ $2\frac{5}{7}$ ’ en la salida PDF, y NVDA/MathCAT verbalizará «*dos y cinco séptimos*».

`use-spnZ = {\langle true | false \rangle}` default: *true*

Establece si el  $\{\langle \text{numerador} \rangle\}$  y el  $\{\langle \text{denominador} \rangle\}$  que forman la fracción del *número mixto* se procesan internamente bajo el comando `\spnZ`, añadiendo así *semántica de entero* en la estructura MathML de la salida PDF. Si se establece en *true*,  $\{\langle \text{numerador} \rangle\}$  y  $\{\langle \text{denominador} \rangle\}$  se pasarán a `\spnZ` y NVDA/MathCAT los verbalizará correctamente como *números ordinales*; si se establece en *false*, los argumentos se tipografían directamente sin procesamiento semántico adicional.

`print-dfrac = {\langle true | false \rangle}` default: *false*

Establece si la fracción del *número mixto* se imprime usando `\dfrac` en lugar de `\frac`. Si se establece en *true*, al usar `\$ \spmQ[print-dfrac=true]{2}{5}{7}` imprimirá la fracción en tamaño *display* ‘ $2\frac{5}{7}$ ’ en la salida PDF; si se establece en *false*, la fracción se imprime con `\frac` siguiendo el tamaño matemático del contexto en curso.

`wrap-parens = {\langle true | false \rangle}` default: *false*

Establece si se *imprimen* paréntesis redondos ‘(...)’ alrededor del *número mixto* completo devuelto por `\spmQ`. Si se establece en *true*, se imprimirán paréntesis redondos cuyo tamaño se ajusta automáticamente al estilo matemático en curso; si se establece en *false*, no se imprimirán los paréntesis.

`read-parens = {\langle true | false \rangle}` default: *true*

Establece si NVDA/MathCAT *verbalizará* los paréntesis redondos ‘(...)’ alrededor del *número mixto* cuando la llave `wrap-parens` está activa. Si se establece en *true*, NVDA/MathCAT verbalizará la lectura de los paréntesis; si se establece en *false*, NVDA/MathCAT no los verbalizará.

## 7 Utilidades para geometría

El paquete **spintent** provee comandos para trabajar con *geometría* descritos en esta sección. Todos los comandos trabajan en *modo matemático* y reciben como  $\{\langle \text{argumento} \rangle\}$  base una *única* letra (mayúscula en el caso de puntos) con un modificador del tipo subíndice/prima, un número o comando.

Algunos comandos reciben  $\{\langle \text{argumento} \rangle\}$  único (modo “*single*”) o más de uno (modo “*batch*”) donde opcionalmente se pueden separar por comas ‘,’.

- Las primas se debe pasar de forma directa ‘’ y los subíndices siempre deben escribirse usando llaves ‘A\_{...}’ y nunca sin estas ‘A\_n’ que funciona solo por coincidencia. .

## El comando `\spcoord`

---

|                       |                                            |                                                         |
|-----------------------|--------------------------------------------|---------------------------------------------------------|
| <code>\spcoord</code> | <code>\spcoord{⟨abscisa, ordenada⟩}</code> | <code>\spcoord[⟨key = val⟩]{⟨abscisa, ordenada⟩}</code> |
|                       | <code>\spcoord{⟨abscisa; ordenada⟩}</code> | <code>\spcoord[⟨key = val⟩]{⟨abscisa; ordenada⟩}</code> |

---

El comando `\spcoord` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{⟨abscisa, ordenada⟩}` el cual se procesa internamente bajo el comando `\spnum` cuando es numérico no fraccionario.

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `$\spcoord{-3, 4}$` imprimirá ‘(-3, 4)’ en la salida PDF y NVDA/MathCAT verbalizará «menos tres coma cuatro»; `$\spcoord{x_{1}, y_{1}}$` imprimirá ‘(x<sub>1</sub>, y<sub>1</sub>)’ y verbalizará «exis subíndice uno y subíndice uno».

- El comando `\spcoord` solo acepta un *separador* ‘;’ o ‘,’ para la abscisa y la ordenada cuando son valores numéricos no fraccionarios.

### Llaves para `\spcoord`

`silent-cmd = {⟨true | false⟩}` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del `{⟨argumento⟩}` pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spPoint`

---

|                       |                                                     |                                                                  |
|-----------------------|-----------------------------------------------------|------------------------------------------------------------------|
| <code>\spPoint</code> | <code>\spPoint{⟨punto⟩}</code>                      | <code>\spPoint[⟨key = val⟩]{⟨punto⟩}</code>                      |
|                       | <code>\spPoint{⟨punto⟩}(⟨abscisa, ordenada⟩)</code> | <code>\spPoint[⟨key = val⟩]{⟨punto⟩}(⟨abscisa, ordenada⟩)</code> |

---

El comando `\spPoint` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{⟨punto⟩}` el cual es una *única letra mayúscula* con prima ‘’ o subíndice ‘\_...’ y el *argumento opcional delimitado por paréntesis* `(⟨abscisa, ordenada⟩)` el cual se procesa internamente bajo el comando `\spcoord`.

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `$\spPoint{A}$` imprimirá ‘A’ en la salida PDF y NVDA/MathCAT verbalizará «punto A»; `$\spPoint{A_{1}}$` imprimirá ‘A<sub>1</sub>’ y verbalizará «punto A sub uno».

Si se proporciona el *argumento opcional delimitado por paréntesis*, `$\spPoint{A}(-3, 4)$` imprimirá ‘A(-3, 4)’ en la salida PDF y NVDA/MathCAT verbalizará «punto A menos tres coma cuatro»; `$\spPoint{A_{1}}(-3, 4)$` imprimirá ‘A<sub>1</sub>(-3, 4)’ y verbalizará «punto A sub uno menos tres coma cuatro».

### Llaves para `\spPoint`

`read-cmd = {⟨school | mathcat⟩}` default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento `{⟨punto⟩}`. Si se establece en *school*, NVDA/MathCAT verbalizará el argumento `{⟨punto⟩}` sin el prefijo «mayúscula», `$\spPoint{A}$` se verbaliza «punto A»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el concepto «punto». El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`concept = {⟨true | false⟩}` default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «punto» al ejecutar el comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará el concepto y el `{⟨argumento⟩}`; si se establece en *false*, NVDA/MathCAT solo verbalizará el `{⟨argumento⟩}` acorde al valor establecido por la llave `read-cmd`.

`silent-cmd = {⟨true | false⟩}` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «punto» y del `{⟨argumento⟩}` pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spvec`

|                     |                                                                 |                                                                        |
|---------------------|-----------------------------------------------------------------|------------------------------------------------------------------------|
| <code>\spvec</code> | <code>\spvec{⟨letra⟩}</code>                                    | <code>\spvec{⟨punto, punto⟩}</code>                                    |
|                     | <code>\spvec{⟨letra⟩} (⟨abscisa, ordenada⟩)</code>              | <code>\spvec{⟨punto, punto⟩} (⟨abscisa, ordenada⟩)</code>              |
|                     | <code>\spvec[⟨key = val⟩]{⟨letra⟩} (⟨abscisa, ordenada⟩)</code> | <code>\spvec[⟨key = val⟩]{⟨punto, punto⟩} (⟨abscisa, ordenada⟩)</code> |

El comando `\spvec` trabaja en *modo matemático* y recibe un *argumento obligatorio* de la forma `{⟨letra⟩}` (minúscula) o `{⟨punto, punto⟩}` junto al *argumento opcional delimitado por paréntesis* `(⟨abscisa, ordenada⟩)` el cual se procesa internamente bajo el comando `\spcoord`.

Si el *argumento* es de la forma `{⟨punto, punto⟩}` con prima(s) ‘`’` o subíndice(s) ‘`_{}...`’ opcionales se aplica `\overrightarrow` en la impresión; si el *argumento* es de la forma `{⟨letra⟩}` (minúscula) con subíndice ‘`_{}...`’ opcional se aplica `\vec` en la impresión.

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `\spvec{a}` imprimirá ‘ $\vec{a}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «vector  $a$ »; `\spvec{AB}` imprimirá ‘ $\overrightarrow{AB}$ ’ y verbalizará «vector  $A B$ ».

Si se proporciona el *argumento opcional delimitado por paréntesis*, `\spvec{a}(-3,4)` imprimirá ‘ $\vec{a} = (-3, 4)$ ’ en la salida PDF y NVDA/MathCAT verbalizará «vector  $a$  igual a menos tres coma cuatro»; `\spvec{AB}(-3,4)` imprimirá ‘ $\overrightarrow{AB} = (-3, 4)$ ’ y verbalizará «vector  $A B$  igual a menos tres coma cuatro».

### Llaves para `\spvec`

`read-cmd = {⟨school | mathcat⟩}` default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento `{⟨punto, punto⟩}`. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spvec{AB}` se verbaliza «vector  $A B$ »; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el concepto «vector». El valor de esta llave *siempre* debe pasarse entre llaves ‘`{ }`’.

`autofit = {⟨true | false⟩}` default: *false*

Establece si se *utiliza* `\vphantom{b}` al ejecutar el comando cuando se utiliza el argumento `{⟨letra⟩}`. Si se establece en *true*, se aplicará `\vphantom{b}` al inicio del argumento `{⟨letra⟩}`; si se establece en *false*, no se aplicará `\vphantom{b}`.

`concept = {⟨true | false⟩}` default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «vector» al ejecutar el comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará el concepto y el `{⟨argumento⟩}`; si se establece en *false*, NVDA/MathCAT solo verbalizará el `{⟨argumento⟩}` acorde al valor establecido por la llave `read-cmd`.

`silent-cmd = {⟨true | false⟩}` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «vector» y del `{⟨argumento⟩}` pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\sprecta`

|                       |                                                    |                                             |
|-----------------------|----------------------------------------------------|---------------------------------------------|
| <code>\sprecta</code> | <code>\sprecta{⟨punto, punto⟩}</code>              | <code>\sprecta{⟨letra⟩}</code>              |
|                       | <code>\sprecta[⟨key = val⟩]{⟨punto, punto⟩}</code> | <code>\sprecta[⟨key = val⟩]{⟨letra⟩}</code> |

El comando `\sprecta` trabaja en *modo matemático* y recibe un *argumento obligatorio* de la forma `{⟨letra⟩}` o `{⟨punto, punto⟩}` con prima(s) ‘`’` o subíndice(s) ‘`_{}...`’ opcionales. Si el *argumento* es de la forma `{⟨punto, punto⟩}` se aplica `\overleftarrow` en la impresión; si el *argumento* es de la forma `{⟨letra⟩}` (mayúscula o minúscula), se imprimirá solo la `{⟨letra⟩}` sin aplicar `\overleftarrow`.

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `\sprecta{AB}` imprimirá ‘ $\overleftarrow{AB}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «recta  $A B$ »; `\sprecta{m}` imprimirá ‘ $m$ ’ en la salida PDF y NVDA/MathCAT verbalizará «recta  $m$ ».

### Llaves para `\sprecta`

`read-cmd = {⟨school | mathcat⟩}` default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento `{⟨punto, punto⟩}` y el argumento `{⟨letra⟩}` cuando está en mayúscula. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\sprecta{AB}` se verbaliza «recta  $A B$ »; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el concepto «recta». El valor de esta llave *siempre* debe pasarse entre llaves ‘`{ }`’.

`concept = {⟨true | false⟩}`

default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «*recta*» al ejecutar el comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará el concepto y el {⟨*argumento*⟩}; si se establece en *false*, NVDA/MathCAT solo verbalizará el {⟨*argumento*⟩} acorde al valor establecido por la llave `read-cmd`.

`silent-cmd = {⟨true | false⟩}`

default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «*recta*» y del {⟨*argumento*⟩} pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\sprayo`

---

`\sprayo` `\sprayo{⟨punto, punto⟩}`  
`\sprayo[⟨key = val⟩]{⟨punto, punto⟩}`

---

El comando `\sprayo` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*punto, punto*⟩} con prima(s) ‘*’* o subíndice(s) ‘*\_*{...}’ opcionales. Representa un *rayo* o *semirrecta* con origen en el primer punto y dirección hacia el segundo, aplicando `\overrightarrow` en la impresión.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩], `\sprayo{AB}` imprimirá ‘ $\overrightarrow{AB}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «*rayo A B*».

### Llaves para `\sprayo`

`read-cmd = {⟨school | mathcat⟩}`

default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento {⟨*punto, punto*⟩}. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\sprayo{AB}` se verbaliza «*rayo A B*»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`read-sty = {⟨rayo | semirrecta⟩}`

default: *rayo*

Establece el *concepto* que verbalizará NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. Si se utiliza `\sprayo[read-sty=rayo]{AB}`, NVDA/MathCAT verbalizará «*rayo A B*»; si se utiliza `\sprayo[read-sty=semirrecta]{AB}`, NVDA/MathCAT verbalizará «*semirrecta A B*».

`prefix = {⟨prefijo⟩}`

default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el «*concepto*» establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`concept = {⟨true | false⟩}`

default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del «*concepto*» establecido por la llave `read-sty` al ejecutar el comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará el concepto y el {⟨*argumento*⟩}; si se establece en *false*, NVDA/MathCAT solo verbalizará el {⟨*argumento*⟩} acorde al valor establecido por la llave `read-cmd`.

`silent-cmd = {⟨true | false⟩}`

default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del «*concepto*» establecido por la llave `read-sty` y del {⟨*argumento*⟩} pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

- La impresión ‘ $\overrightarrow{AB}$ ’ en la salida PDF es idéntica en ambos casos. La elección depende de la terminología del libro de texto o del currículo nacional: en Chile y España se prefiere «semirrecta»; en México y otros países de América Latina se usa «rayo».

## El comando `\spside`

---

`\spside` `\spside{⟨punto, punto⟩}`  
`\spside[⟨key = val⟩]{⟨punto, punto⟩}`

---

El comando `\spside` trabaja en *modo matemático* y recibe el *argumento obligatorio* {⟨*punto, punto*⟩} con prima(s) ‘*’* o subíndice(s) ‘*\_*{...}’ opcionales. Representa un *lado*, *segmento* o *trazo* determinado por dos puntos, aplicando `\overline` en la impresión.

Si no se proporciona el *argumento opcional* [⟨*key = val*⟩], `\spside{AB}` imprimirá ‘ $\overline{AB}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «*lado A B*».

### Llaves para `\spside`

`read-cmd` = { `<school | mathcat>` } default: `school`

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento { `<punto, punto>` }. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spside{AB}` se verbaliza «lado A B»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`read-sty` = { `<lado | segmento | trazo>` } default: `lado`

Establece el *concepto* que verbalizará NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. Si se utiliza `\spside[read-sty=lado]{AB}`, NVDA/MathCAT verbalizará «lado A B»; si se utiliza `\spside[read-sty=segmento]{AB}`, NVDA/MathCAT verbalizará «segmento A B» y si se utiliza `\spside[read-sty=trazo]{AB}`, NVDA/MathCAT verbalizará «trazo A B»

`prefix` = { `<prefijo>` } default: `no usado`

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el «concepto» establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`concept` = { `<true | false>` } default: `true`

Establece si se *silencia* la verbalización en NVDA/MathCAT del «concepto» establecido por la llave `read-sty` al ejecutar el comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT verbalizará el concepto y el { `<argumento>` }; si se establece en `false`, NVDA/MathCAT solo verbalizará el { `<argumento>` } acorde al valor establecido por la llave `read-cmd`.

`read-arg` = { `<verbalización NVDA>` } default: `no usado`

Sobrescribe el *concepto* establecido por la llave `read-sty` que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'. Por ejemplo: `\spside[read-arg={hipotenusa}]{AB}` imprimirá  $\overline{AB}$  en la salida PDF y NVDA/MathCAT verbalizará «hipotenusa A B».

`silent-cmd` = { `<true | false>` } default: `false`

Establece si se *silencia* la verbalización en NVDA/MathCAT del «concepto» establecido por la llave `read-sty` y del { `<argumento>` } pasado al comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en `false`, NVDA/MathCAT sí verbalizará la salida del comando.

### El comando `\spmside`

---

`\spmside`    `\spmside{<punto, punto>}`  
               `\spmside[<key = val>]{<punto, punto>}`

El comando `\spmside` trabaja en *modo matemático* y recibe el *argumento obligatorio* { `<punto, punto>` } con prima(s) ' ' o subíndice(s) ' \_{...}' opcionales. Representa la *medida* del *lado*, *segmento* o *trazo* determinado por dos puntos. La distinción entre `\spside` y `\spmside` es semántica: `\spside` representa el *objeto geométrico*  $\overline{AB}$ ; `\spmside` representa su *medida* (un número real positivo).

Si no se proporciona el *argumento opcional* [ `<key = val>` ], `\spmside{AB}` imprimirá  $\overline{AB}$  en la salida PDF y NVDA/MathCAT verbalizará «medida del lado A B».

#### Llaves para `\spmside`

`read-cmd` = { `<school | mathcat>` } default: `school`

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento { `<punto, punto>` }. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spmside{AB}` se verbaliza «medida del lado A B»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`read-sty` = { `<lado | segmento | trazo>` } default: `lado`

Establece el *concepto* que verbalizará NVDA/MathCAT después de verbalizar el término «medida del» al ejecutar el comando sin modificar la impresión de este. Si se utiliza `\spmside[read-sty=lado]{AB}`, NVDA/MathCAT verbalizará «medida del lado A B»; si se utiliza `\spmside[read-sty=segmento]{AB}`, NVDA/MathCAT verbalizará «medida del segmento A B» y si se utiliza `\spmside[read-sty=trazo]{AB}`, NVDA/MathCAT verbalizará «medida del trazo A B»

`prefix` = { `<prefijo>` } default: `no usado`

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «medida del» seguido por el «concepto» establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`concept` = { `<true | false>` } default: `true`

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «medida del» y del «concepto» establecido por la llave `read-sty` al ejecutar el comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT verbalizará el término «medida del», el concepto y el { `<argumento>` }; si se establece en `false`, NVDA/MathCAT solo verbalizará el { `<argumento>` } acorde al valor establecido por la llave `read-cmd`.

`read-arg = {⟨verbalización NVDA⟩}`

default: *no usado*

Sobrescribe el *concepto* establecido por la llave `read-sty` que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’ y se debe usar en conjunto con la llave `concept`.

Por ejemplo: `$\spmside[concept=false,read-arg={medida de la cuerda}]{AB}$` imprimirá ‘ $\overline{AB}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida de la cuerda A B».

`print-m = {⟨true | false⟩}`

default: *false*

Establece si se *imprime* la notación visual ‘m’ antes aplicar `\overline` al {⟨argumento⟩}. Si se establece en *true*, `$\spmside[print-m=true]{AB}$` imprimirá ‘ $m\overline{AB}$ ’ en la salida PDF, si se establece en *false* imprimirá ‘ $\overline{AB}$ ’ en la salida PDF. En ambos casos NVDA/MathCAT verbalizará «medida del lado A B».

`silent-cmd = {⟨true | false⟩}`

default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «medida del», del «concepto» establecido por la llave `read-sty` y del {⟨argumento⟩} pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spangle`

`\spangle {⟨punto | letra griega | número⟩}`

`\spangle {⟨punto, punto, punto⟩}`

`\spangle [⟨key = val⟩] {⟨punto | letra griega | número⟩}`

`\spangle [⟨key = val⟩] {⟨punto, punto, punto⟩}`

El comando `\spangle` trabaja en *modo matemático* y recibe un *argumento obligatorio* de la forma {⟨punto | letra griega | número⟩} o {⟨punto, punto, punto⟩} aplicando `\angle` o `\rightanglesqr` en la impresión. Solo los argumentos de la forma {⟨punto⟩} o {⟨punto, punto, punto⟩} admiten prima(s) ‘’ o subíndice(s) ‘\_ { . . . }’ opcionales.

Si no se proporciona el *argumento opcional* [⟨key = val⟩], `$\spangle{A}$` imprimirá ‘ $\angle A$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo en A»; `$\spangle{\alpha}$` imprimirá ‘ $\angle \alpha$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo alfa»; `$\spangle{1}$` imprimirá ‘ $\angle 1$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo uno» y `$\spangle{AOB}$` imprimirá ‘ $\angle AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo A O B».

- Si no se tiene instalada la fuente matemática NewCMath-Regular.otf parte del paquete `newcomputermodern` en el sistema o no se provee una fuente matemática con soporte para `\rightanglesqr` se usará `\rightangle` en su reemplazo.

### Llaves para `\spangle`

`read-cmd = {⟨school | mathcat⟩}`

default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento {⟨punto⟩} o {⟨punto, punto, punto⟩}. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `$\spangle{AOB}$` se verbaliza «ángulo A O B»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}`

default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el concepto «ángulo». El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`recto = {⟨true | false⟩}`

default: *false*

Establece si se *usará* `\rightanglesqr` en la impresión añadiendo el término «recto» a la verbalización en NVDA/MathCAT. Si se establece en *true*, `$\spangle[recto=true]{A}$` imprimirá ‘ $\perp A$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo recto en A», `$\spangle[recto=true]{AOB}$` imprimirá ‘ $\perp AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo recto A O B»; si se establece en *false*, imprimirá usando `\angle` en la salida PDF y NVDA/MathCAT omitirá el término «recto».

`concept = {⟨true | false⟩}`

default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «ángulo» al ejecutar el comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará el concepto «ángulo» y el {⟨argumento⟩}; si se establece en *false*, NVDA/MathCAT solo verbalizará el {⟨argumento⟩} acorde al valor establecido por la llave `read-cmd`.

`silent-cmd = {⟨true | false⟩}`

default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «ángulo» y del {⟨argumento⟩} pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spmangle`

---

```
\spmangle <{punto | letra griega | número}> <{punto, punto, punto}>
\spmangle[key = val]{<punto | letra griega | número>} \spmangle[key = val]{<punto, punto, punto>}
```

---

El comando `\spmangle` trabaja en *modo matemático* y recibe un *argumento obligatorio* de la forma `{<punto | letra griega | número>}` o `{<punto, punto, punto>}` aplicando `\measuredangle` o `\measuredrightangle` en la impresión. Solo los argumentos de la forma `{<punto>}` o `{<punto, punto, punto>}` admiten prima(s) ‘`’` o subíndice(s) ‘`_`’ opcionales.

Si no se proporciona el *argumento opcional* `[key = val]`, `\spmangle{A}` imprimirá ‘ $\angle A$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo en A»; `\spmangle{\alpha}` imprimirá ‘ $\angle \alpha$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo alfa»; `\spmangle{1}` imprimirá ‘ $\angle 1$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo uno» y `\spmangle{AOB}` imprimirá ‘ $\angle AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo A O B».

- La distinción entre `\spangle` y `\spmangle` es semántica: `\spangle` representa el *objeto geométrico* ‘ $\angle AOB$ ’; `\spmangle` representa su *medida* (un valor en grados).

### Llaves para `\spmangle`

```
read-cmd = {<school | mathcat>} default: school
```

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento `{<punto>}` o `{<punto, punto, punto>}`. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spmangle{AOB}` se verbaliza «medida del ángulo A O B»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

```
prefix = {<prefijo>} default: no usado
```

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «medida del» seguido por el concepto «ángulo». El valor de esta llave *siempre* debe pasarse entre llaves ‘`{ }`’.

```
recto = {<true | false>} default: false
```

Establece si se *usará* `\measuredrightangle` en la impresión añadiendo el término «recto» a la verbalización en NVDA/MathCAT. Si se establece en `true`, `\spmangle[recto=true]{A}` imprimirá ‘ $\angle A$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo recto en A»; `\spmangle[recto=true]{AOB}` imprimirá ‘ $\angle AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo recto A O B»; si se establece en `false`, imprimirá usando `\measuredangle` en la salida PDF y NVDA/MathCAT omitirá el término «recto».

```
print-m = {<true | false>} default: false
```

Establece si se *imprime* la notación visual ‘m’ antes del argumento pasado al comando sin modificar la verbalización de este. Si se establece en `true`, `\spmangle[print-m=true]{AOB}` imprimirá ‘m  $\angle AOB$ ’ en la salida PDF, si se establece en `false` imprimirá ‘ $\angle AOB$ ’ en la salida PDF. En ambos casos NVDA/MathCAT verbalizará «medida del ángulo A O B».

```
concept = {<true | false>} default: true
```

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «medida del» seguido por el concepto «ángulo» al ejecutar el comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT verbalizará «medida del ángulo» y el `{<argumento>}`; si se establece en `false`, NVDA/MathCAT solo verbalizará el `{<argumento>}` acorde al valor establecido por la llave `read-cmd`.

```
silent-cmd = {<true | false>} default: false
```

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «medida del», del concepto «ángulo» y del `{<argumento>}` pasado al comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en `false`, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spang`

---

```
\spang <{grados}>
\spang <{grados : minutos}>
\spang <{grados : minutos : segundos}>
```

---

El comando `\spang` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<grados>}`, `{<grados : minutos>}` o `{<grados : minutos : segundos>}`. Los procesa internamente bajo los comandos `\spnum` y `\spunit` para imprimir y verbalizar. Por ejemplo, `\spang{45:30:15}` imprimirá ‘45°30’15” en la salida PDF y NVDA/MathCAT verbalizará «cuarenta y cinco grados treinta minutos quince segundos».

## El comando `\sparc`

---

|                     |                                                  |                                                         |
|---------------------|--------------------------------------------------|---------------------------------------------------------|
| <code>\sparc</code> | <code>\sparc{⟨punto, punto⟩}</code>              | <code>\sparc{⟨punto, punto, punto⟩}</code>              |
|                     | <code>\sparc[⟨key = val⟩]{⟨punto, punto⟩}</code> | <code>\sparc[⟨key = val⟩]{⟨punto, punto, punto⟩}</code> |

---

El comando `\sparc` trabaja en *modo matemático* y recibe un *argumento obligatorio* de la forma `{⟨punto, punto⟩}` o `{⟨punto, punto, punto⟩}` con prima(s) ‘`’` o subíndice(s) ‘`_{}...`’ opcionales aplicando `\widearc` en la impresión.

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `\sparc{AB}` imprimirá ‘ $\widehat{AB}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «arco A B».

- El acento `\widearc` es proporcionado por algunos paquetes como `fourier-otf` o `kpfonts-otf`, en caso de no estar disponible se utilizará una versión interna para este.

### Llaves para `\sparc`

`read-cmd = {⟨school | mathcat⟩}` default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento `{⟨punto, punto⟩}` o `{⟨punto, punto, punto⟩}`. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\sparc{AB}` se verbaliza «arco A B»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el concepto «arco». El valor de esta llave *siempre* debe pasarse entre llaves ‘`{ }`’.

`concept = {⟨true | false⟩}` default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «arco» al ejecutar el comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT verbalizará «arco» y el `{⟨argumento⟩}`; si se establece en `false`, NVDA/MathCAT solo verbalizará el `{⟨argumento⟩}` acorde al valor establecido por la llave `read-cmd`.

`silent-cmd = {⟨true | false⟩}` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «arco» y del `{⟨argumento⟩}` pasado al comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en `false`, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spmarc`

---

|                      |                                                   |                                                          |
|----------------------|---------------------------------------------------|----------------------------------------------------------|
| <code>\spmarc</code> | <code>\spmarc{⟨punto, punto⟩}</code>              | <code>\spmarc{⟨punto, punto, punto⟩}</code>              |
|                      | <code>\spmarc[⟨key = val⟩]{⟨punto, punto⟩}</code> | <code>\spmarc[⟨key = val⟩]{⟨punto, punto, punto⟩}</code> |

---

El comando `\spmarc` trabaja en *modo matemático* y recibe un *argumento obligatorio* de la forma `{⟨punto, punto⟩}` o `{⟨punto, punto, punto⟩}` con prima(s) ‘`’` o subíndice(s) ‘`_{}...`’ opcionales colocando ‘`m`’ antes de aplicar `\widearc` en la impresión.

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `\spmarc{AB}` imprimirá ‘ $m\widehat{AB}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del arco A B».

- La distinción entre `\sparc` y `\spmarc` es semántica: `\sparc` representa el *objeto geométrico* ‘ $\widehat{AB}$ ’; `\spmarc` representa su *medida* (un valor en grados).

### Llaves para `\spmarc`

`read-cmd = {⟨school | mathcat⟩}` default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento `{⟨punto, punto⟩}` o `{⟨punto, punto, punto⟩}`. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spmarc{AB}` se verbaliza «medida del arco A B»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «medida del» seguido por el concepto «arco». El valor de esta llave *siempre* debe pasarse entre llaves ‘`{ }`’.

`concept = {⟨true | false⟩}` default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «medida del» y el concepto «arco» al ejecutar el comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT verbalizará «medida del arco» y el `{⟨argumento⟩}`; si se establece en `false`, NVDA/MathCAT solo verbalizará el `{⟨argumento⟩}` acorde al valor establecido por la llave `read-cmd`.

`silent-cmd = {⟨true | false⟩}` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «medida del», el concepto «arco» y el `{⟨argumento⟩}` pasado al comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en `false`, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spPoly`

---

```
\spPoly \spPoly{⟨punto, punto, punto, ...⟩}
\spPoly[⟨key = val⟩]{⟨punto, punto, punto, ...⟩}
```

---

El comando `\spPoly` trabaja en *modo matemático* y recibe el *argumento obligatorio*  $\{\langle \text{punto, punto, punto, ...} \rangle\}$ , una lista de al menos tres  $\{\langle \text{puntos} \rangle\}$  con prima(s) `'` o subíndice(s) `_{...}` opcionales. El *símbolo* usado en la impresión se maneja por medio de la llave `print-sym` y la verbalización del concepto asociado se determinan acorde al número de puntos en el argumento y la llave `concept`.

Si son tres puntos se aplica `\triangle` y añade el concepto «*triángulo*» a la verbalización; si son cuatro puntos se aplica `\mdlgwhtsquare` y añade el concepto «*cuadrado*» a la verbalización; si son cinco o más puntos no se aplica ningún símbolo ni se añade verbalización.

Si no se proporciona el *argumento opcional*  $[\langle \text{key} = \text{val} \rangle]$ ,  $\$ \backslash \text{spPoly}\{ABC\} \$$  imprimirá  $\triangle ABC$  en la salida PDF y NVDA/MathCAT verbalizará «*triángulo A B C*»;  $\$ \backslash \text{spPoly}\{ABCD\} \$$  imprimirá  $\square ABCD$  y verbalizará «*cuadrado A B C D*»;  $\$ \backslash \text{spPoly}\{ABCDE\} \$$  imprimirá  $ABCDE$  y verbalizará «*A B C D E*».

### Llaves para `\spPoly`

`read-cmd` =  $\{\langle \text{school} \mid \text{mathcat} \rangle\}$  default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento  $\{\langle \text{punto, punto, punto, ...} \rangle\}$ . Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»:  $\$ \backslash \text{spPoly}\{ABC\} \$$  se verbaliza «*triángulo A B C*»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` =  $\{\langle \text{prefijo} \rangle\}$  default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el  $\{\langle \text{argumento} \rangle\}$  o los conceptos «*triángulo*» y «*cuadrado*» establecido por la llave `concept`. El valor de esta llave *siempre* debe pasarse entre llaves `{ }`.

`concept` =  $\{\langle \text{true} \mid \text{false} \rangle\}$  default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT de los conceptos «*triángulo*» y «*cuadrado*» acorde al número de puntos en el  $\{\langle \text{argumento} \rangle\}$  al ejecutar el comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará «*triángulo*» o «*cuadrado*» y el  $\{\langle \text{argumento} \rangle\}$ ; si se establece en *false*, NVDA/MathCAT solo verbalizará el  $\{\langle \text{argumento} \rangle\}$  acorde al valor establecido por la llave `read-cmd`.

`print-sym` =  $\{\langle \text{true} \mid \text{false} \rangle\}$  default: *true*

Establece si se *usará* `\triangle` o `\mdlgwhtsquare` en la impresión sin modificar la verbalización de este. Si se establece en *true*,  $\$ \backslash \text{spPoly}[\text{print-sym}=\text{true}]\{ABC\} \$$  imprimirá  $\triangle ABC$  en la salida PDF y NVDA/MathCAT verbalizará «*triángulo A B C*»;  $\$ \backslash \text{spPoly}[\text{print-sym}=\text{true}]\{ABCD\} \$$  imprimirá  $\square ABCD$  en la salida PDF y NVDA/MathCAT verbalizará «*cuadrado A B C*»; si se establece en *false*, solo se imprimirá el  $\{\langle \text{argumento} \rangle\}$  en la salida PDF y la verbalización será la misma.

`silent-cmd` =  $\{\langle \text{true} \mid \text{false} \rangle\}$  default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT de los conceptos establecido por la llave `concept` y el  $\{\langle \text{argumento} \rangle\}$  pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spAfig`

---

```
\spAfig \spAfig(⟨punto, punto, punto, ...⟩)
\spAfig[⟨key = val⟩] \spAfig[⟨key = val⟩](⟨punto, punto, punto, ...⟩)
\spAfig(⟨1 | 2 | 3 | ... | n⟩) \spAfig(F | P | B | L | T)
\spAfig[⟨key = val⟩](⟨1 | 2 | 3 | ... | n⟩) \spAfig[⟨key = val⟩](F | P | B | L | T)
```

---

El comando `\spAfig` trabaja en *modo matemático* y recibe el *argumento opcional delimitado por paréntesis* de la forma  $(\langle 1 \mid 2 \mid 3 \mid \dots \mid n \rangle)$ ,  $(\langle \text{punto, punto, punto, ...} \rangle)$  o  $(\langle F \mid P \mid B \mid L \mid T \rangle)$  aplicando `\mathcal{A}` a la impresión. El *símbolo* usado en la impresión se maneja por medio de la llave `print-sym` y la verbalización del concepto asociado se determinan acorde al  $(\langle \text{argumento} \rangle)$  y la llave `concept`.

Si se proporciona el *argumento*  $(\langle 1 \mid 2 \mid 3 \mid \dots \mid n \rangle)$ ,  $\$ \backslash \text{spAfig}(1) \$$  imprimirá  $\mathcal{A}_1$  en la salida PDF y NVDA/MathCAT verbalizará «*área uno*»;  $\$ \backslash \text{spAfig}(n) \$$  imprimirá  $\mathcal{A}_n$  en la salida PDF y NVDA/MathCAT verbalizará «*área ene*».

Si se proporciona el *argumento*  $(\langle \text{punto, punto, punto, ...} \rangle)$ ,  $\$ \backslash \text{spAfig}(ABC) \$$  imprimirá  $\mathcal{A}_{\triangle ABC}$  en la salida PDF y NVDA/MathCAT verbalizará «*área del triángulo A B C*»;  $\$ \backslash \text{spAfig}(ABCD) \$$  imprimirá  $\mathcal{A}_{\square ABCD}$  en la salida PDF y NVDA/MathCAT verbalizará «*área del cuadrado A B C D*»;

Si se proporciona el *argumento*  $(\langle F \mid P \mid B \mid L \mid T \rangle)$ ,  $\$ \backslash \text{spAfig}(F) \$$  imprimirá  $\mathcal{A}_F$  en la salida PDF y NVDA/MathCAT verbalizará «*área de la figura*»;  $\$ \backslash \text{spAfig}(F_{\{1\}}) \$$  imprimirá  $\mathcal{A}_{F_1}$  en la salida PDF y NVDA/MathCAT verbalizará «*área de la figura uno*».

Si no se proporciona el *argumento opcional* [*(key = val)*] ni el *argumento opcional delimitado por paréntesis*,  $\$ \backslash \text{spAfig} \$$  imprimirá ‘A’ en la salida PDF y NVDA/MathCAT verbalizará «área».

- El *argumento* (*(punto, punto, punto, ...)*) acepta prima(s) ‘\_’ o subíndice(s) ‘\_{...}’. El *argumento* (*(F|P|B|L|T)*) solo acepta subíndices ‘\_{...}’ y tiene una impresión y verbalización fija para cada letra: F «figura», P «polígono», B «basal», L «lateral» y T «total».

### Llaves para $\backslash \text{spAfig}$

$\text{read-cmd} = \{ \langle \text{school} \mid \text{mathcat} \rangle \}$  default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»:  $\$ \backslash \text{spAfig} (\text{ABC}) \$$  se verbaliza «área del triángulo A B C»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

$\text{prefix} = \{ \langle \text{prefijo} \rangle \}$  default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «área del» y el concepto «nombre de la figura» acorde al (*argumento*) pasado al comando. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

$\text{concept} = \{ \langle \text{true} \mid \text{false} \rangle \}$  default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «área del» y el concepto «nombre de la figura» acorde al (*argumento*) pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará el término «área del», el concepto «nombre de la figura» y el (*argumento*); si se establece en *false*, NVDA/MathCAT solo verbalizará el (*argumento*) acorde al valor establecido por la llave *read-cmd*.

$\text{read-arg} = \{ \langle \text{nombre} \rangle \}$  default: *no usado*

Sobrescribe el término «área del» y el concepto «nombre de la figura» cuando se utiliza el argumento (*(punto, punto, punto, ...)*) que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’ y se debe usar en conjunto con la llave *concept*. Por ejemplo:  $\$ \backslash \text{spAfig} [\text{read-arg}=\{\text{trapezio}\}] (\text{ABCD}) \$$  imprimirá ‘ $\mathcal{A}_{\square \text{ABCD}}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «área del trapezio A B C D».

- Esta llave no se puede utilizar con los argumentos (*(F|P|B|L|T)*) y (*(1|2|3|...|n)*).

$\text{concept}^* = \langle \text{valor prohibido} \rangle$  default: *no usado*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «nombre de la figura» acorde al (*argumento*) pasado al comando sin modificar la impresión de este. Si se establece, NVDA/MathCAT verbalizará el término «área» y el (*argumento*) acorde al valor establecido por la llave *read-cmd*.

$\text{read-sub} = \{ \langle \text{true} \mid \text{false} \rangle \}$  default: *false*

Establece si se *antepone* la palabra «sub» al verbalizar los argumentos (*(1|2|3|...|n)*) y (*(F|P|B|L|T)*) para NVDA/MathCAT. Si se utiliza (*(1|2|3|...|n)*) y se establece en *true*,  $\$ \backslash \text{spAfig} [\text{read-sub}=\text{true}] (1) \$$  imprimirá ‘ $\mathcal{A}_1$ ’ en la salida PDF y NVDA/MathCAT verbalizará «área sub uno»; si se establece en *false*, la impresión será la misma y se verbaliza «área uno».

Si se utiliza (*(F|P|B|L|T)*) y se establece en *true*,  $\$ \backslash \text{spAfig} [\text{read-sub}=\text{true}] (\text{F}_{\{1\}}) \$$  imprimirá ‘ $\mathcal{A}_{F_1}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «área de la figura sub uno»; si se establece en *false*, la impresión será la misma y se verbaliza «área de la figura uno».

$\text{print-sym} = \{ \langle \text{auto} \mid \text{triangle} \mid \text{square} \mid \text{none} \rangle \}$  default: *auto*

Establece si se *usará*  $\backslash \text{triangle}$  o  $\backslash \text{mdlgwhtsquare}$  en la impresión para los argumentos (*(punto, punto, punto, ...)*) y (*(1|2|3|...|n)*) pasado al comando sin modificar la verbalización de este. Si se utiliza (*(punto, punto, punto, ...)*) se infiere el *símbolo* acorde al número de puntos; si establece en *auto* se imprime y si se establece en *none* se omite:  $\$ \backslash \text{spAfig} [\text{print-sym}=\text{auto}] (\text{ABC}) \$$  y  $\$ \backslash \text{spAfig} [\text{print-sym}=\text{none}] (\text{ABC}) \$$  imprimirán ‘ $\mathcal{A}_{\triangle \text{ABC}}$ ’ y ‘ $\mathcal{A}_{\text{ABC}}$ ’ en la salida PDF y en ambos casos NVDA/MathCAT verbalizará «área del triángulo A B C».

Si se utiliza (*(1|2|3|...|n)*) los valores *auto* y *none* quedan sin efecto; si se establece en *triangle* o *square* anida el argumento bajo el *símbolo*:  $\$ \backslash \text{spAfig} [\text{print-sym}=\text{triangle}] (1) \$$  imprimirá ‘ $\mathcal{A}_{\triangle 1}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «área del triángulo uno»;  $\$ \backslash \text{spAfig} [\text{print-sym}=\text{square}] (1) \$$  imprimirá ‘ $\mathcal{A}_{\square 1}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «área del cuadrado uno».

- Esta llave no se puede utilizar con el argumento (*(F|P|B|L|T)*).

$\text{silent-cmd} = \{ \langle \text{true} \mid \text{false} \rangle \}$  default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «área del», el concepto «nombre de la figura» y el (*argumento*) pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spPfig`

|                                                                                        |                                                                                             |                                                                                    |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <code>\spPfig</code>                                                                   | <code>\spPfig</code>                                                                        | <code>\spPfig(\langle punto, punto, punto, \dots \rangle)</code>                   |
| <code>\spPfig[\langle key = val \rangle]</code>                                        | <code>\spPfig[\langle key = val \rangle](\langle punto, punto, punto, \dots \rangle)</code> | <code>\spPfig[\langle key = val \rangle](\langle F   P   B   L   T \rangle)</code> |
| <code>\spPfig(\langle 1   2   3   \dots   n \rangle)</code>                            | <code>\spPfig(\langle F   P   B   L   T \rangle)</code>                                     |                                                                                    |
| <code>\spPfig[\langle key = val \rangle](\langle 1   2   3   \dots   n \rangle)</code> | <code>\spPfig[\langle key = val \rangle](\langle F   P   B   L   T \rangle)</code>          |                                                                                    |

El comando `\spPfig` trabaja en *modo matemático* y recibe el *argumento opcional delimitado por paréntesis* de la forma  $(\langle 1 | 2 | 3 | \dots | n \rangle)$ ,  $(\langle punto, punto, punto, \dots \rangle)$  o  $(\langle F | P | B | L | T \rangle)$  aplicando `\mathcal{P}` a la impresión. El *símbolo* usado en la impresión se maneja por medio de la llave `print-sym` y la verbalización del concepto asociado se determinan acorde al  $(\langle argumento \rangle)$  y la llave `concept`.

Si se proporciona el *argumento*  $(\langle 1 | 2 | 3 | \dots | n \rangle)$ , `\spPfig(1)` imprimirá  $\mathcal{P}_1$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro uno»; `\spPfig(n)` imprimirá  $\mathcal{P}_n$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro ene».

Si se proporciona el *argumento*  $(\langle punto, punto, punto, \dots \rangle)$ , `\spPfig(ABC)` imprimirá  $\mathcal{P}_{\triangle ABC}$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro del triángulo A B C»; `\spPfig(ABCD)` imprimirá  $\mathcal{P}_{\square ABCD}$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro del cuadrado A B C D»;

Si se proporciona el *argumento*  $(\langle F | P | B | L | T \rangle)$ , `\spPfig(F)` imprimirá  $\mathcal{P}_F$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro de la figura»; `\spPfig(F_{1})` imprimirá  $\mathcal{P}_{F_1}$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro de la figura uno».

Si no se proporciona el *argumento opcional*  $[\langle key = val \rangle]$  ni el *argumento opcional delimitado por paréntesis*, `\spPfig` imprimirá  $\mathcal{P}$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro».

- El *argumento*  $(\langle punto, punto, punto, \dots \rangle)$  acepta prima(s)  $'_{'$  o subíndice(s)  $'_{\{...\}}'$ . El *argumento*  $(\langle F | P | B | L | T \rangle)$  solo acepta subíndices  $'_{\{...\}}'$  y tiene una impresión y verbalización fija para cada letra: F «figura», P «polígono», B «basal», L «lateral» y T «total».

### Llaves para `\spPfig`

`read-cmd` =  $\{\langle school | mathcat \rangle\}$  default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spPfig(ABC)` se verbaliza «perímetro del triángulo A B C»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` =  $\{\langle prefijo \rangle\}$  default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT antes de verbalizar el término «perímetro del» y el concepto «nombre de la figura» acorde al  $(\langle argumento \rangle)$  pasado al comando. El valor de esta llave *siempre* debe pasarse entre llaves  $\{\}$ .

`concept` =  $\{\langle true | false \rangle\}$  default: *true*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «perímetro del» y el concepto «nombre de la figura» acorde al  $(\langle argumento \rangle)$  pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará el término «perímetro del», el concepto «nombre de la figura» y el  $(\langle argumento \rangle)$ ; si se establece en *false*, NVDA/MathCAT solo verbalizará el  $(\langle argumento \rangle)$  acorde al valor establecido por la llave `read-cmd`.

`read-arg` =  $\{\langle nombre \rangle\}$  default: *no usado*

Sobrescribe el término «perímetro del» y el concepto «nombre de la figura» cuando se utiliza el argumento  $(\langle punto, punto, punto, \dots \rangle)$  que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves  $\{\}$  y se debe usar en conjunto con la llave `concept`. Por ejemplo: `\spPfig[read-arg=\langle trapecio \rangle](ABCD)` imprimirá  $\mathcal{P}_{\square ABCD}$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro del trapecio A B C D».

- Esta llave no se puede utilizar con los argumentos  $(\langle F | P | B | L | T \rangle)$  y  $(\langle 1 | 2 | 3 | \dots | n \rangle)$ .

`concept*`  $\langle valor prohibido \rangle$  default: *no usado*

Establece si se *silencia* la verbalización en NVDA/MathCAT del concepto «nombre de la figura» acorde al  $(\langle argumento \rangle)$  pasado al comando sin modificar la impresión de este. Si se establece, NVDA/MathCAT verbalizará el término «perímetro» y el  $(\langle argumento \rangle)$  acorde al valor establecido por la llave `read-cmd`.

`read-sub` =  $\{\langle true | false \rangle\}$  default: *false*

Establece si se *antepone* la palabra «sub» al verbalizar los argumentos  $(\langle 1 | 2 | 3 | \dots | n \rangle)$  y  $(\langle F | P | B | L | T \rangle)$  para NVDA/MathCAT. Si se utiliza  $(\langle 1 | 2 | 3 | \dots | n \rangle)$  y se establece en *true*, `\spPfig[read-sub=true](1)` imprimirá  $\mathcal{P}_1$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro sub uno»; si se establece en *false*, la impresión será la misma y se verbaliza «perímetro uno».

Si se utiliza  $(\langle F | P | B | L | T \rangle)$  y se establece en *true*, `\spPfig[read-sub=true](F_{1})` imprimirá  $\mathcal{P}_{F_1}$  en la salida PDF y NVDA/MathCAT verbalizará «perímetro de la figura sub uno»; si se establece en *false*, la impresión será la misma y se verbaliza «perímetro de la figura uno».

`print-sym` =  $\{\langle auto | triangle | square | none \rangle\}$  default: *auto*

Establece si se usará `\triangle` o `\mdlgwhtsquare` en la impresión para los argumentos (`\punto`, `punto`, `punto`, ...) y (`\1` | `2` | `3` | ... | `n`) pasado al comando sin modificar la verbalización de este. Si se utiliza (`\punto`, `punto`, `punto`, ...) se infiere el *símbolo* acorde al número de puntos; si establece en `auto` se imprime y si se establece en `none` se omite: `\spPfig[print-sym=auto](ABC)$` y `\spPfig[print-sym=none](ABC)$` imprimirán ' $\mathcal{P}_{\triangle ABC}$ ' y ' $\mathcal{P}_{ABC}$ ' en la salida PDF y en ambos casos NVDA/MathCAT verbalizará «*perímetro del triángulo A B C*»

Si se utiliza (`\1` | `2` | `3` | ... | `n`) los valores `auto` y `none` quedan sin efecto; si se establece en `triangle` o `square` anida el argumento bajo el *símbolo*: `\spPfig[print-sym=triangle](1)$` imprimirá ' $\mathcal{P}_{\triangle_1}$ ' en la salida PDF y NVDA/MathCAT verbalizará «*perímetro del triángulo uno*»; `\spPfig[print-sym=square](1)$` imprimirá ' $\mathcal{P}_{\square_1}$ ' en la salida PDF y NVDA/MathCAT verbalizará «*perímetro del cuadrado uno*».

🔑 Esta llave no se puede utilizar con el argumento (`\F` | `\P` | `\B` | `\L` | `\T`).

`silent-cmd` = {`\true` | `false`}

default: `false`

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «*perímetro del*», el concepto «*nombre de la figura*» y el {`\argumento`} pasado al comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en `false`, NVDA/MathCAT sí verbalizará la salida del comando.

## Los comandos `\sprad` y `\spdiam`

|                      |                      |                                                 |
|----------------------|----------------------|-------------------------------------------------|
| <code>\sprad</code>  | <code>\sprad</code>  | <code>\sprad[\langle key = val \rangle]</code>  |
| <code>\spdiam</code> | <code>\spdiam</code> | <code>\spdiam[\langle key = val \rangle]</code> |

Los comandos `\sprad` y `\spdiam` trabajan en *modo matemático* y no reciben ningún argumento: imprimen y verbalizan la letra 'r'/'d' (o 'R'/'D', según `upper-case`) que representa el radio o el diámetro de una circunferencia.

Si no se proporciona el *argumento opcional* [`\langle key = val \rangle`], `\sprad$` imprimirá 'r' en la salida PDF y NVDA/MathCAT verbalizará «*radio*»; `\spdiam$` imprimirá 'd' en la salida PDF y NVDA/MathCAT verbalizará «*diámetro*».

### Llaves para `\sprad` y `\spdiam`

`read-cmd` = {`\school` | `mathcat`}

default: `school`

Establece la *forma* en la que NVDA/MathCAT verbalizará la letra cuando `concept` es `false`. Si se establece en `school`, NVDA/MathCAT verbaliza la letra deletreada; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas.

`upper-case` = {`\true` | `false`}

default: `false`

Establece si se imprime la letra en mayúscula sin modificar la verbalización de este. Si se establece en `true`, `\sprad` [upper-case=true]\$ imprimirá 'R' en la salida PDF; `\spdiam` [upper-case=true]\$ imprimirá 'D'. En ambos casos la verbalización no cambia.

`print-sym` = {`\true` | `false`}

default: `false`

Establece si se usará `\diameter` antes de la letra en la impresión sin modificar la verbalización de este. Si se establece en `true`, `\spdiam` [print-sym=true]\$ imprimirá ' $\varnothing d$ ' en la salida PDF y NVDA/MathCAT verbalizará «*diámetro*».

🔑 La llave `print-sym` no tiene ningún efecto en `\sprad`.

`sub-index` = {`\subíndice`}

default: `no usado`

Establece un *subíndice* sobre la letra. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`read-sub` = {`\true` | `false`}

default: `false`

Establece si se *antepone* la palabra «sub» al verbalizar el `sub-index` para NVDA/MathCAT. Si se establece en `true`, `\sprad` [sub-index={1}, read-sub=true]\$ imprimirá ' $r_1$ ' en la salida PDF y NVDA/MathCAT verbalizará «*radio sub uno*»; si se establece en `false`, se verbaliza «*radio uno*».

`concept` = {`\true` | `false`}

default: `true`

Establece si NVDA/MathCAT verbaliza «*radio*» (o «*diámetro*») en lugar de la letra misma. Si se establece en `true`, `\sprad$` se verbaliza «*radio*»; si se establece en `false`, se verbaliza la letra acorde al valor establecido por la llave `read-cmd`.

`read-arg` = {`\verbalización NVDA`}

default: `no usado`

Sobrescribe «*radio*» (o «*diámetro*») que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '{ }' y se debe usar en conjunto con la llave `concept`. Por ejemplo: `\sprad` [read-arg={medida del radio}]\$ imprimirá 'r' en la salida PDF y NVDA/MathCAT verbalizará «*medida del radio*».

`silent-cmd` = {`\true` | `false`}

default: `false`

Establece si se *silencia* la verbalización en NVDA/MathCAT de «*radio*» (o «*diámetro*») y de la letra al ejecutar el comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en `false`, NVDA/MathCAT sí verbalizará la salida del comando.

## El comando `\spcirc`

```
\spcirc \spcirc \spcirc[⟨key = val⟩](⟨centro⟩)
\spcirc[⟨key = val⟩] \spcirc(⟨centro, radio⟩)
\spcirc(⟨centro⟩) \spcirc[⟨key = val⟩](⟨centro, radio⟩)
```

El comando `\spcirc` trabaja en *modo matemático* y recibe el *argumento opcional delimitado por paréntesis* de la forma  $(\langle \text{centro} \rangle)$  o  $(\langle \text{centro}, \text{radio} \rangle)$ , donde  $\langle \text{radio} \rangle$  es opcional y debe pasarse junto al argumento  $(\langle \text{centro} \rangle)$  separado por una coma ‘,’ obligatoria.

La forma en la que se procesa el  $\langle \text{radio} \rangle$  junto a su impresión y verbalización se establece de manera interna: Si es un  $\langle \text{número} \rangle$  se usa `\spnum`; si es una  $\langle \text{medida} \rangle$  se usa `\spunit` y si es un segmento  $\langle \text{punto punto} \rangle$  se usa `\spside` o `\spmside` acorde al valor establecido por la llave `radio-sty`. En cualquier otro caso no se procesa de manera interna.

Si no se proporciona el *argumento opcional*  $[\langle \text{key} = \text{val} \rangle]$  ni el *argumento opcional delimitado por paréntesis*, `\spcirc` imprimirá ‘ $\odot$ ’ en la salida PDF y NVDA/MathCAT verbalizará «circunferencia».

Si se proporciona el *argumento*  $(\langle \text{centro} \rangle)$ , `\spcirc(O)` imprimirá ‘ $\odot(O)$ ’ en la salida PDF y NVDA/MathCAT verbalizará «circunferencia de centro en O».

Si se proporciona el *argumento*  $(\langle \text{centro}, \text{radio} \rangle)$ , `\spcirc(O, 3)` imprimirá ‘ $\odot(O, 3)$ ’ en la salida PDF y NVDA/MathCAT verbalizará «circunferencia de centro en O y radio tres».

- El argumento  $(\langle \text{centro} \rangle)$  acepta prima ‘ $\prime$ ’ o subíndice ‘ $_{\dots}$ ’.

### Llaves para `\spcirc`

`read-cmd = {⟨school | mathcat⟩}` default: school

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento  $(\langle \text{centro} \rangle)$ . Si se establece en `school`, NVDA/MathCAT verbaliza el argumento  $(\langle \text{centro} \rangle)$  sin el prefijo «mayúscula»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

- Esta llave no tiene efecto sobre el argumento  $\langle \text{radio} \rangle$ .

`prefix = {⟨prefijo⟩}` default: no usado

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el «concepto» establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`print-sym = {⟨usual | odot⟩}` default: odot

Establece si se usará `\bigodot` o la letra ‘C’ en la impresión del comando sin modificar la verbalización de este. Si se establece en `usual`, `\spcirc[print-sym=usual](O)` imprimirá ‘C(O)’ en la salida PDF; si se establece en `odot`, `\spcirc[print-sym=odot](O)` imprimirá ‘ $\odot(O)$ ’ en la salida PDF.

`read-sty = {⟨circun | circle⟩}` default: circun

Establece el *concepto* que verbalizará NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. Si se establece en `circun`, `\spcirc[read-sty=circun](O)`, NVDA/MathCAT verbalizará «circunferencia de centro en O»; si se establece en `circle`, `\spcirc[read-sty=circle](O)`, NVDA/MathCAT verbalizará «círculo de centro en O».

`diameter = {⟨true | false⟩}` default: false

Establece *como* se verbaliza el argumento  $\langle \text{radio} \rangle$  al ejecutar el comando sin modificar la impresión de este. Si se establece en `true`, `\spcirc[diameter=true](O, 3)`, NVDA/MathCAT verbalizará «circunferencia de centro en O y diámetro tres»; si se establece en `false`, `\spcirc[diameter=false](O, 3)`, NVDA/MathCAT verbalizará «circunferencia de centro en O y radio tres».

`radio-sty = {⟨side | mside⟩}` default: mside

Establece el *comando* que procesa internamente el argumento  $\langle \text{radio} \rangle$  cuando tiene la forma  $\langle \text{punto punto} \rangle$ . Si se establece en `side` se usará `\spside{⟨punto punto⟩}`, `\spcirc[radio-sty=side](O, AB)` imprimirá ‘ $\odot(O, \overline{AB})$ ’ en la salida PDF y NVDA/MathCAT verbalizará «circunferencia de centro en O y radio lado A B»; si se establece en `mside` se usará `\spmside{⟨punto punto⟩}`, `\spcirc[radio-sty=mside](O, AB)` imprimirá ‘ $\odot(O, AB)$ ’ en la salida PDF y NVDA/MathCAT verbalizará «circunferencia de centro en O y radio medida del lado A B».

`concept = {⟨true | false⟩}` default: true

Establece si se *silencia* la verbalización en NVDA/MathCAT del «concepto» establecido por la llave `read-sty` al ejecutar el comando sin modificar la impresión de este. Si se establece en `true`, NVDA/MathCAT verbalizará el «concepto» y el argumento  $(\langle \text{centro} \rangle)$  o  $(\langle \text{centro}, \text{radio} \rangle)$ ; si se establece en `false`, NVDA/MathCAT solo verbalizará el argumento  $(\langle \text{centro} \rangle)$  o  $(\langle \text{centro}, \text{radio} \rangle)$  acorde al valor establecido por la llave `read-cmd`.

`sub-index = {⟨subíndice⟩}` default: no usado

Establece un *subíndice* sobre `\bigodot` o la letra ‘C’ establecido por la llave `print-sym`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’. Por ejemplo: `\spcirc[sub-index={1}, radio-sty=mside](O, AB)`

imprimirá  $\odot_1(O, AB)$  en la salida PDF y NVDA/MathCAT verbalizará «circunferencia uno de centro en O y radio medida del lado A B».

`read-sub = {⟨true | false⟩}` default: false

Establece si se *antepone* la palabra «sub» al verbalizar el valor establecido por la llave `sub-index`. para NVDA/MathCAT.

`read-arg = {⟨verbalización NVDA⟩}` default: no usado

Sobrescribe el *concepto* establecido por la llave `read-sty` que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`silent-cmd = {⟨true | false⟩}` default: false

Establece si se *silencia* la verbalización en NVDA/MathCAT del «concepto» establecido por la llave `read-sty`, del argumento (`⟨centro⟩`) o (`⟨centro, radio⟩`) pasados al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

- Las llaves `spside` y `spmside` reenvían sus valores directamente a `\spside` y `\spmside`, respectivamente, cuando el radio es un segmento de dos letras.

## El comando `\spAcirc`

|                              |                                              |                                                           |
|------------------------------|----------------------------------------------|-----------------------------------------------------------|
| <u><code>\spAcirc</code></u> | <code>\spAcirc</code>                        | <code>\spAcirc[⟨key = val⟩](⟨1   2   3   ...   n⟩)</code> |
|                              | <code>\spAcirc[⟨key = val⟩]</code>           | <code>\spAcirc(⟨centro⟩)</code>                           |
|                              | <code>\spAcirc(⟨1   2   3   ...   n⟩)</code> | <code>\spAcirc[⟨key = val⟩](⟨centro⟩)</code>              |

El comando `\spAcirc` trabaja en *modo matemático* y recibe el *argumento opcional delimitado por paréntesis* de la forma (`⟨1 | 2 | 3 | ... | n⟩`) o (`⟨centro⟩`), aplicando `\mathcal{A}` a la impresión.

Si se proporciona el *argumento* (`⟨1 | 2 | 3 | ... | n⟩`), `\spAcirc(1)` imprimirá ' $A_1$ ' en la salida PDF y NVDA/MathCAT verbalizará «área de la circunferencia uno».

Si se proporciona el *argumento* (`⟨centro⟩`), `\spAcirc(0)` imprimirá ' $A_{\odot(O)}$ ' en la salida PDF y NVDA/MathCAT verbalizará «área de la circunferencia con centro en O».

Si no se proporciona el *argumento opcional* [`⟨key = val⟩`] ni el *argumento opcional delimitado por paréntesis*, `\spAcirc` imprimirá ' $A$ ' en la salida PDF y NVDA/MathCAT verbalizará «área de la circunferencia».

- El *argumento* (`⟨centro⟩`) acepta prima(s) '`'`' o subíndice(s) '`_{\dots}`' y no acepta un radio, una coma '`,`' en el argumento siempre produce un error.

### Llaves para `\spAcirc`

`read-cmd = {⟨school | mathcat⟩}` default: school

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}` default: no usado

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «área de la» seguido por el «concepto» establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`print-sym = {⟨usual | odot⟩}` default: odot

Establece si se *usará* `\bigodot` o la letra 'C' en la impresión sin modificar la verbalización del comando. Si se establece en *usual*, `\spAcirc[print-sym=usual](0)` imprimirá ' $A_{C(O)}$ ' en la salida PDF y NVDA/MathCAT verbalizará «área de la circunferencia con centro en O»; si se establece en *odot*, `\spAcirc[print-sym=odot](0)` imprimirá ' $A_{\odot(O)}$ ' en la salida PDF y NVDA/MathCAT verbalizará «área de la circunferencia con centro en O».

`read-sty = {⟨circun | circle⟩}` default: circun

Establece el *concepto* que verbalizará NVDA/MathCAT después de verbalizar el término «área de la» al ejecutar el comando sin modificar la impresión de este. Si se utiliza `\spAcirc[read-sty=circun](0)`, NVDA/MathCAT verbalizará «área de la circunferencia con centro en O»; si se utiliza `\spAcirc[read-sty=circle](0)`, NVDA/MathCAT verbalizará «área del círculo con centro en O». Solo tiene efecto cuando `read-arg` no se proporciona.

`concept = {⟨true | false⟩}` default: true

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «área de la» y del «concepto» establecido por la llave `read-sty` al ejecutar el comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT verbalizará el término, el concepto y el (`⟨argumento⟩`); si se establece en *false*, NVDA/MathCAT solo verbalizará el (`⟨argumento⟩`) (y, si corresponde, el centro) acorde al valor establecido por la llave `read-cmd`.

`sub-index = {⟨subíndice⟩}` default: no usado

Establece un *subíndice* sobre el símbolo. Solo tiene efecto cuando se proporciona el *argumento* (`⟨centro⟩`). El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`read-sub = {⟨true | false⟩}`

default: *false*

Establece si se *antepone* la palabra «sub» al verbalizar el argumento ( $\langle 1 | 2 | 3 | \dots | n \rangle$ ). Si se establece en *true*,  $\$ \backslash \text{spAcirc}[read-sub=true](1) \$$  se verbaliza «área de la circunferencia sub uno»; si se establece en *false*, se verbaliza «área de la circunferencia uno».

`read-arg = {⟨verbalización NVDA⟩}`

default: *no usado*

Sobrescribe el *concepto* establecido por la llave `read-sty` que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '{ }' y funciona con cualquier forma del argumento. El artículo sigue siempre `read-sty`, incluso cuando `read-arg` reemplaza el concepto. Por ejemplo:  $\$ \backslash \text{spAcirc}[read-arg=\{\text{esfera}\}](O) \$$  imprimirá ' $\mathcal{A}_{\odot(O)}$ ' en la salida PDF y NVDA/MathCAT verbalizará «área de la esfera con centro en O».

`silent-cmd = {⟨true | false⟩}`

default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «área de la», del concepto establecido por la llave `read-sty` y del ( $\langle \text{argumento} \rangle$ ) pasado al comando sin modificar la impresión de este. Si se establece en *true*, NVDA/MathCAT NO verbalizará la salida del comando; si se establece en *false*, NVDA/MathCAT sí verbalizará la salida del comando.

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## 9 Change history

**v0.99 (ctan), 2026-09-21** – First public release.

10 Índice de la Documentación

Los números en cursiva indican las páginas donde se describe la entrada correspondiente.

|                                                |    |                                                 |    |
|------------------------------------------------|----|-------------------------------------------------|----|
| C                                              |    | Keys for \spAfig provide by <b>spintent</b> :   |    |
| Document class:                                |    | concept <sup>*</sup>                            | 26 |
| article                                        | 1  | concept                                         | 26 |
| book                                           | 1  | prefix                                          | 26 |
| letter                                         | 1  | print-sym                                       | 26 |
| report                                         | 1  | read-arg                                        | 26 |
| Commands provide by <b>spintent</b> :          |    | read-cmd                                        | 26 |
| \setspintent                                   | 3  | read-sub                                        | 26 |
| \spAcirc                                       | 30 | silent-cmd                                      | 26 |
| \spAfig                                        | 25 | Keys for \spangle provide by <b>spintent</b> :  |    |
| \spPfig                                        | 27 | concept                                         | 22 |
| \spPoint                                       | 18 | prefix                                          | 22 |
| \spPoly                                        | 25 | read-cmd                                        | 22 |
| \spangle                                       | 22 | recto                                           | 22 |
| \spang                                         | 23 | silent-cmd                                      | 22 |
| \sparc                                         | 24 | Keys for \sparc provide by <b>spintent</b> :    |    |
| \spcirc                                        | 29 | concept                                         | 24 |
| \spcoord                                       | 18 | prefix                                          | 24 |
| \spdate                                        | 11 | read-cmd                                        | 24 |
| \spdiam                                        | 28 | silent-cmd                                      | 24 |
| \spdiv                                         | 13 | Keys for \spcirc provide by <b>spintent</b> :   |    |
| \spdsym                                        | 12 | concept                                         | 29 |
| \spmQ                                          | 17 | diameter                                        | 29 |
| \spmangle                                      | 23 | prefix                                          | 29 |
| \spmarc                                        | 24 | print-sym                                       | 29 |
| \spmcd                                         | 15 | radio-sty                                       | 29 |
| \spmcm                                         | 16 | read-arg                                        | 30 |
| \spmoney                                       | 6  | read-cmd                                        | 29 |
| \spmside                                       | 21 | read-sty                                        | 29 |
| \spmsym                                        | 13 | read-sub                                        | 30 |
| \spmul                                         | 14 | silent-cmd                                      | 30 |
| \spnD                                          | 14 | sub-index                                       | 29 |
| \spnM                                          | 15 | Keys for \spcoord provide by <b>spintent</b> :  |    |
| \spnZ                                          | 16 | silent-cmd                                      | 18 |
| \spnewsym                                      | 11 | Keys for \spdiv provide by <b>spintent</b> :    |    |
| \spnewunit                                     | 6  | read-parens                                     | 14 |
| \spnum                                         | 3  | wrap-parens                                     | 14 |
| \sprad                                         | 28 | Keys for \spdsym provide by <b>spintent</b> :   |    |
| \sprayo                                        | 20 | prefix                                          | 13 |
| \spread                                        | 12 | read-sym                                        | 13 |
| \sprecta                                       | 19 | set-sym                                         | 13 |
| \spshort                                       | 7  | suffix                                          | 13 |
| \spside                                        | 20 | Keys for \spmangle provide by <b>spintent</b> : |    |
| \spsiglo                                       | 11 | concept                                         | 23 |
| \sptime                                        | 11 | prefix                                          | 23 |
| \spunitalias                                   | 6  | print-m                                         | 23 |
| \spunit                                        | 4  | read-cmd                                        | 23 |
| \spusep                                        | 12 | recto                                           | 23 |
| \spvec                                         | 19 | silent-cmd                                      | 23 |
| K                                              |    | Keys for \spmarc provide by <b>spintent</b> :   |    |
| Keys for \spAcirc provide by <b>spintent</b> : |    | concept                                         | 24 |
| concept                                        | 30 | prefix                                          | 24 |
| prefix                                         | 30 | read-cmd                                        | 24 |
| print-sym                                      | 30 | silent-cmd                                      | 24 |
| read-arg                                       | 31 | Keys for \spmcd provide by <b>spintent</b> :    |    |
| read-cmd                                       | 30 | prefix                                          | 15 |
| read-sty                                       | 30 | read-cmd                                        | 15 |
| read-sub                                       | 31 | result                                          | 15 |
| silent-cmd                                     | 31 | sample-arg                                      | 16 |
| sub-index                                      | 30 | silent-cmd                                      | 16 |
|                                                |    | upper-case                                      | 15 |

Keys for \spmcn provide by **spintent**:

prefix

read-cmd

result

sample-arg

silent-cmd

upper-case

16

16

16

16

16

16

Keys for \spmoney provide by **spintent**:

currency

dolar-math

sub-units

7

7

7

Keys for \spmQ provide by **spintent**:

join-word

print-dfrac

read-parens

use-spnZ

wrap-parens

17

17

17

17

17

Keys for \spmside provide by **spintent**:

concept

prefix

print-m

read-arg

read-cmd

read-sty

silent-cmd

21

21

22

22

21

21

22

Keys for \spmsym provide by **spintent**:

not-print

prefix

read-sym

set-sym

suffix

13

13

13

13

13

Keys for \spm\ provide by **spintent**:

read-parens

wrap-parens

14

14

Keys for \spnD provide by **spintent**:

prefix

result-num

result

sample-arg

silent-cmd

14

14

14

14

15

Keys for \spnM provide by **spintent**:

prefix

result-num

result

sample-arg

silent-cmd

15

15

15

15

15

Keys for \spnum provide by **spintent**:

cifras-sep

notation-sym

read-period-sep

read-sign

3

4

3

3

Keys for \spnZ provide by **spintent**:

read-parens

wrap-parens

17

17

Keys for \spPfig provide by **spintent**:

concept\*

concept

prefix

print-sym

read-arg

read-cmd

read-sub

silent-cmd

27

27

27

27

27

27

27

28

Keys for \spPoint provide by **spintent**:

concept

18

prefix

read-cmd

silent-cmd

18

18

18

Keys for \spPoly provide by **spintent**:

concept

prefix

print-sym

read-cmd

silent-cmd

25

25

25

25

25

Keys for \sprad provide by **spintent**:

concept

print-sym

read-arg

read-cmd

read-sub

silent-cmd

sub-index

upper-case

28

28

28

28

28

28

28

28

Keys for \sprayo provide by **spintent**:

concept

prefix

read-cmd

read-sty

silent-cmd

20

20

20

20

20

Keys for \spread provide by **spintent**:

read-arg

silent-cmd

12

12

Keys for \sprecta provide by **spintent**:

concept

prefix

read-cmd

silent-cmd

20

19

19

20

Keys for \spshort provide by **spintent**:

read-arg

small-caps

9

9

Keys for \spside provide by **spintent**:

concept

prefix

read-arg

read-cmd

read-sty

silent-cmd

21

21

21

21

21

21

Keys for \spsiglo provide by **spintent**:

print-era

11

Keys for \spunit provide by **spintent**:

print-cdot

read-cdot

read-slash

4

4

4

Keys for \spusep provide by **spintent**:

silent

size

12

12

Keys for \spvec provide by **spintent**:

autofit

concept

prefix

read-cmd

silent-cmd

19

19

19

19

19

Keys for \spdiv provide by **spintent**:

cifras-sep

notation-sym

read-period-sep

read-sign

read-sym

set-sym

14

14

14

14

14

14

Keys for \spdsym provide by **spintent**:  
    read-sym ..... 13

Keys for \spmcd provide by **spintent**:  
    read-cmd ..... 15

Keys for \spmcm provide by **spintent**:  
    read-cmd ..... 16

Keys for \spmoney provide by **spintent**:  
    cifras-sep ..... 7

Keys for \spmQ provide by **spintent**:  
    wrap-parens ..... 17

Keys for \spmsym provide by **spintent**:  
    read-sym ..... 13

Keys for \spm\ provide by **spintent**:  
    cifras-sep ..... 14  
    notation-sym ..... 14  
    read-period-sep ..... 14  
    read-sign ..... 14  
    read-sym ..... 14  
    set-sym ..... 14

Keys for \spnZ provide by **spintent**:  
    cifras-sep ..... 17  
    read-sign ..... 17

Keys for \spside provide by **spintent**:  
    read-sty ..... 21, 22

Keys for \spunit provide by **spintent**:  
    cifras-sep ..... 4  
    notation-sym ..... 4  
    read-period-sep ..... 4

Keys for \spusep provide by **spintent**:  
    silent ..... 12  
    size ..... 12

**M**  
Lua module provide by **spintent**:  
    spintent.lua ..... 6, 7, 9–11

**P**  
Packages :  
    babel-spanish ..... 2  
    babel ..... 2, 7  
    documentmetadata-support ..... 1  
    enumext ..... 1  
    expl3 ..... 3  
    fontspec ..... 2  
    fourier-otf ..... 2  
    hyperref ..... 2  
    l3keys ..... 3  
    latex-lab-mathintent ..... 1  
    libertinus-otf ..... 2  
    luamml ..... 1  
    scontents ..... 1  
    sourcecodepro ..... 2  
    tagpdf ..... 1, 7, 11  
    unicode-math ..... 1

**S**  
\sprad ..... 28

## 11 Implementation

The most recent publicly released version of `spintent` is available at CTAN: <https://www.ctan.org/pkg/spintent>. While general feedback via email is welcomed, specific bugs or feature requests should be reported through the issue tracker: <https://github.com/pablgonz/spintent/issues>.

- The documentation presented here is far from professional, it contains a lot of obvious information that to the eye of a TeXpert are superfluous, but, after so many years developing this project is the only way to remember what does what.

### 11.1 General conventions

Functions of the form `\__spintent_luafun_some:n` or `\__spintent_luafun_some:nn` are defined (created) in the Lua module `spintent.lua` and only their variants are declared in `expl3`.

Similarly, variables of the form `\l__spintent_luaset_some_tl` or `\l__spintent_luaset_some_str` are defined in the Lua module `spintent.lua` and only declared in `expl3`.

- All variables and functions defined in this package are private and are NOT intended to work or be used by another package or module.

### 11.2 Initial set up

Start the DocStrip guards.

```
1 (*package)
```

Identify the internal prefix (L<sup>A</sup>T<sub>E</sub>X3 DocStrip convention) for `l3doc` class.

```
2 (@@=spintent)
```

### 11.3 Declaration of the package

First we will make sure we have a minimum (super updated) release version of L<sup>A</sup>T<sub>E</sub>X to work correctly, otherwise, we will return an “error”.

```
3 \NeedsTeXFormat{LaTeX2e}[2026-11-01]
4 \IfFormatAtLeastF { 2026-11-01 }
5 {
6   \PackageError { spintent }
7     { LaTeX~kernel~too~old~need~2026-11-01~or~later }
8     {
9       spintent~requires~a~LaTeX2e~release~2026-11-01~or~
10      later.~Update~your~TeX~distribution.
11     }
12 }
```

Now declare the `spintent` package and add only two keys: `mathcal` and `AFghost`, which can be passed as package options.

```
13 \ProvidesExplPackage {spintent} {2026-09-21} {0.99} {Spanish parse intents}
14 \keys_define:nn { spintent }
15 {
16   mathcal .usage:n    = load,
17   mathcal .bool_set:N = \l__spintent_mathcal_bool,
18   mathcal .initial:n  = true,
19   mathcal .value_required:n = true,
20   AFghost .usage:n    = load,
21   AFghost .bool_set:N = \l__spintent_debug_ghost_bool,
22   AFghost .initial:n  = false,
23   AFghost .value_required:n = true,
24 }
25 \ProcessKeyOptions [ spintent ]
```

We check if the requirements for running the `spintent` package are met; that is: `\DocumentMetadata` must be active, Lua<sub>L</sub>TeX must be running, and the `unicode-math` or `lua-unicode-math` package must be loaded. If neither is present, we will load `unicode-math`. If the conditions cannot be met, we will return a “fatal error”.

```
26 \IfDocumentMetadataTF
27 {
28   \sys_if_engine luatex:TF
29   {
30     \msg_new:nnn { spintent } { package-not-load }
31     {
32       The~'#1'~package~will~be~loaded~as~a~dependency.
33     }
34     \IfPackageLoadedF { unicode-math }
35     {
36       \IfPackageLoadedF { lua-unicode-math }
37       {
38         \msg_info:nnn { spintent } { package-not-load } { unicode-math }
39         \RequirePackage{unicode-math}

```

```

40     }
41   }
42   \msg_new:nnn { spintent } { env-success }
43   { Everything~looks~correct,~spintent~will~make~the~intent! }
44   \msg_info:nn { spintent } { env-success }
45 }
46 {
47   \msg_new:nnnn { spintent } { fatal-engine-requires-luatex }
48   { The~spintent~package~requires~LuaLaTeX. }
49   { This~package~relies~on~Lua~module.~Compile~with~lualatex. }
50   \msg_fatal:nn { spintent } { fatal-engine-requires-luatex }
51 }
52 }
53 {
54   \msg_new:nnnn { spintent } { fatal-metadata-missing }
55   { \token_to_str:N \DocumentMetadata \c_space_tl is-missing. }
56   {
57     The~spintent~package~requires~tagged~PDF~active.\\
58     You~must~declare~\token_to_str:N \DocumentMetadata { tagging=on } ~
59   }
60   \msg_fatal:nn { spintent } { fatal-metadata-missing }
61 }

```

## 11.4 Some kernel variants

```

\str_if_eq:nVTF \str_if_eq:nVF
\str_uppercase:V \str_lowercase:V
\seq_set_from_clist:Ne
\cs_generate_variant:Nn \str_if_eq:nnTF { V }
\cs_generate_variant:Nn \str_if_eq:nnF { V }
\cs_generate_variant:Nn \str_uppercase:n { V }
\cs_generate_variant:Nn \str_lowercase:n { V }
\cs_generate_variant:Nn \seq_set_from_clist:Nn { Ne }

```

Non-standard kernel variants.

(End of definition for `\str_if_eq:nVTF` and others.)

## 11.5 Variables for the key `\mathcal`, `\rightanglesqr` and `\attribute`

First, we declare the string variable `\l__spintent_luaset_cal_fam_ok_str` used by the function `\__spintent_luafun_classic_mathcal:nn` (controlled by the key `mathcal`), the string variable `\l__spintent_luafun_right_angle_sqr` used by the function `\__spintent_luafun_right_angle_sqr:` to copy `\rightanglesqr` alongside the string variable `\l__spintent_luaset_scope_attr_str` used by `\attribute` before loading the Lua module `spintent.lua`. This is necessary due to the implementation of the functions within the Lua module; otherwise, `expl3` would issue a warning regarding duplicate variable declarations.

```

67 \str_new:N \l__spintent_luaset_cal_fam_ok_str
68 \str_new:N \l__spintent_luaset_sym_fam_ok_str
69 \str_new:N \l__spintent_luaset_scope_attr_str

```

(End of definition for `\l__spintent_luaset_cal_fam_ok_str`, `\l__spintent_luaset_sym_fam_ok_str`, and `\l__spintent_luaset_scope_attr_str`.)

## 11.6 Loading the Lua module and functions

`spintent.lua` Loads the Lua module `spintent.lua`, where every `_luafun_` function used throughout the package is registered.

```

70 \lua_load_module:n { spintent.lua }

```

(End of definition for `spintent.lua`.)

`\__spintent_luafun_set_debug_ghost:n` Passes `AFghost` to Lua as a plain string. When `true`, both ghost characters are marked `sptmp="outer"` or `sptmp="inner"` in the file `-luamml-mathml.html`, so an external script can identify and strip them before generating the Associated~Files structure for `tagged PDF` — `math-AF` mode does not want either ghost present.

```

71 \bool_if:NTF \l__spintent_debug_ghost_bool
72 { \__spintent_luafun_set_debug_ghost:n { true } }
73 { \__spintent_luafun_set_debug_ghost:n { false } }

```

(End of definition for `\__spintent_luafun_set_debug_ghost:n`.)

`\l__spintent_scope_attr_int` A LuaTeX attribute number, allocated collision-free from Lua via `luatexbase.new_attribute` and exposed here as an integer. Every command that uses the shared function `\__spintent_luafun_mrow_intent_arg:nn` activates it, via `\attribute \l__spintent_scope_attr_int = 1`, inside its own group — marking every node typeset within as eligible for that queue to claim. Without this mark, the queue could claim content from an unrelated command sitting in the same formula.

```

74 \int_new:N \l__spintent_scope_attr_int
75 \int_set:Nn \l__spintent_scope_attr_int { \l__spintent_luaset_scope_attr_str }

```

(End of definition for `\__spintent_scope_attr_int`.)

`\__spintent_luafun_spunit_lookup_alias:n` (used by `\spunit`), `\__spintent_luafun_spmoney_lookup_data:n` (used by `\spmoney`), `\__spintent_luafun_spmQ_scale_int:nn` (used by `\spmQ`) and `\__spintent_luafun_clean_split_and_set:n` (used by `\spnum`).

```

76 \cs_generate_variant:Nn \__spintent_luafun_spunit_lookup_alias:n { V }
77 \cs_generate_variant:Nn \__spintent_luafun_spmoney_lookup_data:n { V }
78 \cs_generate_variant:Nn \__spintent_luafun_spmQ_scale_int:nn { Vn }
79 \cs_generate_variant:Nn \__spintent_luafun_clean_split_and_set:n { V }

```

(End of definition for `\__spintent_luafun_spunit_lookup_alias:n` and others.)

## 11.7 Internal implementation for `\mathcal` and `\rightanglesqr`

To simplify user configuration and obtain the expected output style pdf<sub>TeX</sub> for `\mathcal`, we will provide a fallback to the font NewCMMath-Regular.otf, which contains the glyphs for these characters. Using this same font, we will obtain the symbol `\rightanglesqr` for the key `recto`.

`mathcal-fallback`  
`right-angle-sqr-fallback`

First, we'll define two error messages in case our fallback fails.

```

80 \hook_gput_code:nnn { begindocument } { spintent }
81 {
82   \__spintent_luafun_init_math_fallbacks:
83 }
84 \msg_new:nnn { spintent } { mathcal-fallback }
85 {
86   Using~classic~\token_to_str:N \mathcal \c_space_tl as~fallback~
87   (font~not~found~or~no~free~math~family~available).~\msg_line_context:
88 }
89 \msg_new:nnn { spintent } { right-angle-sqr-fallback }
90 {
91   Using~\token_to_str:N \rightangle \c_space_tl as~fallback~
92   for~the~right-angle~ glyph.~\msg_line_context:
93 }

```

(End of definition for `mathcal-fallback` and `right-angle-sqr-fallback`.)

`\__spintent_mathcal:n`

Now we define the internal function `\__spintent_mathcal:n` for `\mathcal` which checks the state of the variable `\__spintent_mathcal_bool` set by the package key `mathcal`; if its state is true, we check the string variable `\__spintent_luaset_cal_fam_ok_str` and if the latter has the value true, we call the function `\__spintent_luafun_classic_mathcal:n` and otherwise (we cannot find the source in the system) we return a warning message and directly use `\mathcal`. Otherwise, that is, `mathcal=false` we simply use `\mathcal`.

```

94 \cs_new_protected:Nn \__spintent_mathcal:n
95 {
96   \bool_if:NTF \__spintent_mathcal_bool
97   {
98     \str_if_eq:VnTF \__spintent_luaset_cal_fam_ok_str { true }
99     { \__spintent_luafun_classic_mathcal:n {#1} }
100     {
101       \msg_warning:nn { spintent } { mathcal-fallback }
102       \mathcal {#1}
103     }
104   }
105   { \mathcal {#1} }
106 }

```

(End of definition for `\__spintent_mathcal:n`.)

`\__spintent_right_angle_sqr:`

The internal function `\__spintent_right_angle_sqr:` establishes the internal copy of the command `\rightanglesqr`. First, we check the string variable `\__spintent_luaset_sym_fam_ok_str`. If it is true, we call the function `\__spintent__luafun_right_angle_sqr:`; otherwise (we do not find the font in the system), we return a warning message and use `\rightangle` instead.

```

107 \cs_new_protected:Nn \__spintent_right_angle_sqr:
108 {
109   \str_if_eq:VnTF \__spintent_luaset_sym_fam_ok_str { true }
110   { \__spintent_luafun_right_angle_sqr: }
111   {
112     \msg_warning:nn { spintent } { right-angle-sqr-fallback }
113     \rightangle
114   }
115 }

```

(End of definition for `\__spintent_right_angle_sqr:`.)

## 11.8 Shared annotation mechanisms: the queue and the ghosts

Almost all commands in the `spintent` package share three annotation functions, the mechanism of which is defined in the Lua module `spintent.lua`.

`\__spintent_luafun_mrow_intent_arg:nn`

The internal function `\__spintent_luafun_mrow_intent_arg:nn` — called directly, with no `expl3` wrapper — pushes one entry onto a queue: a concept `intent` (`#1`, possibly empty) and a Lua table literal `arg=/  
intent=` (`#2`). A `pre_mlist_to_hlist_filter` callback (installed once, before `luamml[2]`’s own) walks the formula’s top-level noads in order and, for each one whose nucleus or subscript was typeset inside a `spintent` command’s own group (marked via a LuaTeX `\attribute` activated by every command that uses this queue), claims it for the next pending entry: `arg`/a literal `intent` land on that leaf, and, when the concept `intent` is non-empty, the leaf is wrapped in a native `<mrow>` carrying it — no token, no ghost involved. Entries are consumed strictly in order; any left unclaimed when the formula ends are discarded rather than carried into the next one. A native `\MathMLarg/\MathMLintent` claim is invisible to this queue: mixing the two for the same content risks the queue claiming something already tagged natively, or never being claimed at all.

(End of definition for `\__spintent_luafun_mrow_intent_arg:nn`.)

The other two are “ghost” characters (an invisible U+2063, self-tagged `:silent` through their own `mathml_filter` — so the queue above never mistakes either for one of its own free targets) dropped in as an extra sibling so the resulting `<mrow>` is not the formula’s sole top-level content, which `luamml` would otherwise promote straight to `<math intent=...>` instead of leaving it nested (issue #26).

`\__spintent_invisible_sep:`

Placed once, before opening the formula’s own outer `\MathMLintent` — protects the *whole formula* from collapsing when that outer group is its only top-level content. Fires at most once per document: past the first call, in any command, it does nothing, since one sibling anywhere in the formula is already enough.

```
116 \cs_new_protected:Nn \__spintent_invisible_sep:
117 {
118   \__spintent_luafun_invisible_sep:
119 }
```

🔗 Executing this function writes the following line directly to the output file `-luamml-mathml.html`:

```
<mi intent=":silent" mathvariant="normal"></mi>
```

and if the package key `AFghost=true` is used, the output is:

```
<mi intent=":silent" mathvariant="normal" sptmp="outer"></mi>
```

If you wish to use math-AF mode, you can remove the line generated by `AFghost=true` using the Lua script `spiafcls`.

(End of definition for `\__spintent_invisible_sep:`.)

`\__spintent_inner_sep:`

Placed just inside a native `\MathMLintent` group — protects *that specific group* when it too would otherwise hold only one child. Never deduplicates: called once per such group, however many appear, nested or sibling, in the same formula.

```
120 \cs_new_protected:Nn \__spintent_inner_sep:
121 {
122   \__spintent_luafun_inner_sep:
123 }
```

🔗 Executing this function writes the following line directly to the output file `-luamml-mathml.html`:

```
<mi intent=":silent" mathvariant="normal"></mi>
```

and if the package key `AFghost=true` is used, the output is:

```
<mi intent=":silent" mathvariant="normal" sptmp="inner"></mi>
```

If you wish to use math-AF mode, you can remove the line generated by `AFghost=true` using the Lua script `spiafcls`.

(End of definition for `\__spintent_inner_sep:`.)

`\c__spintent_invisible_sep_int`

A raw, untagged copy of the same U+2063 character, via the LuaTeX primitive `\Umathchardef` — for use as ordinary content *inside* an explicit `\MathMLintent{<intent>}` call (typically `:silent`), not as a sibling on its own.

```
124 \hook_gput_code:nnn { begindocument } { spintent }
125 {
126   \Umathchardef \c__spintent_invisible_sep_int = "0" "0" "2063
127 }
```

🔗 Documented in `source3` on page 887 for `\int_const:Nn`: “As stated, most constants can be defined as `\chardef` or `\mathchardef` but that’s engine dependent.” —so this appears to be the correct option here.

(End of definition for `\c__spintent_invisible_sep_int`.)

## 11.9 Normalize argument and affixes

We normalize the `{⟨argument⟩}` to support affixes (prefixes/suffixes) by trimming leading and trailing spaces, converting internal spaces to hyphens, collapsing consecutive hyphens, and removing leading or trailing hyphens.

`\__spintent_collapse_hyphens:N` A recursive helper to collapse multiple hyphens into a single one.

```

128 \cs_new_protected:Nn \__spintent_collapse_hyphens:N
129 {
130   \tl_if_in:NnT #1 { -- }
131   {
132     \tl_replace_all:Nnn #1 { -- } { - }
133     \__spintent_collapse_hyphens:N #1
134   }
135 }
```

(End of definition for `\__spintent_collapse_hyphens:N`.)

`\__spintent_sanitize_affix:Nn` The function `\__spintent_sanitize_affix:Nn` checks if the `{⟨argument⟩}` passed in ‘#2’ is *blank*. If so, it clears the target token list variable passed in ‘#1’. Otherwise, trimming bounds, converting spaces ‘ ’ to hyphens ‘-’, collapsing multiple hyphens, and safely stripping leading/trailing hyphens. For example, if you pass a target variable and raw text:

```
\__spintent_sanitize_affix:Nn \l__spintent_spsomesym_prefix_tl{ foo bar baz }
```

The function will “store” the normalized string `foo-bar-baz` inside `\l__spintent_spsomesym_prefix_tl`, ready to be conditionally appended.

```

136 \cs_new_protected:Nn \__spintent_sanitize_affix:Nn
137 {
138   \tl_if_blank:nTF {#2}
139   { \tl_clear:N #1 }
140   {
141     \tl_set:Ne #1 {#2}
142     \tl_trim_spaces:N #1
143     \tl_replace_all:Nnn #1 { ~ } { - }
144     \__spintent_collapse_hyphens:N #1
145     \tl_put_left:Nn #1 { \q_mark }
146     \tl_put_right:Nn #1 { \q_stop }
147     \tl_replace_all:Nnn #1 { \q_mark - } { \q_mark }
148     \tl_replace_all:Nnn #1 { - \q_stop } { \q_stop }
149     \tl_remove_all:Nn #1 { \q_mark }
150     \tl_remove_all:Nn #1 { \q_stop }
151   }
152 }
153 \cs_generate_variant:Nn \__spintent_sanitize_affix:Nn { cn }
```

(End of definition for `\__spintent_sanitize_affix:Nn`.)

## 11.10 The command `\setspintent`

The `\setspintent` command is used to configure `⟨keys⟩` “globally” for the utilities provided by `spintent`.

### 11.10.1 Internal variables and error messages for `\setspintent`

`\l__spintent_setspintent_input_arg_cmd_tl` The variable `\l__spintent_setspintent_input_arg_cmd_tl` will *store* the command or command name passed to the “first” `{⟨argument⟩}` ‘#1’ of the command `\setspintent`.

```
154 \tl_new:N \l__spintent_setspintent_input_arg_cmd_tl
```

Error message if the `{⟨argument⟩}` ‘#1’ passed to `\setspintent` is a “non-existent” command.

```

155 \msg_new:nnnn { spintent } { unknown-command-setup }
156 {
157   Cannot~configure~unknown~command~'\exp_not:c{#1}'.
158 }
159 {
160   You~attempted~to~use~\token_to_str:N \setspintent{#1}{...},~but~the~
161   command~\exp_not:c{#1}~is~not~defined.~Please~check~for~typos.
162 }
```

(End of definition for `\l__spintent_setspintent_input_arg_cmd_tl` and `unknown-command-setup`.)

```

\__spintent_setspintent_set_keys:nn
\__spintent_setspintent_set_keys:en

```

The function `\__spintent_setspintent_set_keys:nn` will check if the command passed as `{⟨argument⟩}` exists in ‘#1’ and set the `⟨keys⟩` passed as `{⟨argument⟩}` in ‘#2’ to the correct path; otherwise, it will return an error.

```

163 \cs_new_protected:Nn \__spintent_setspintent_set_keys:nn
164 {
165   \cs_if_exist:cTF { #1 }
166   {
167     \keys_set:nn { spintent / #1 } { #2 }
168   }
169   {
170     \msg_error:nnn { spintent } { unknown-command-setup } { #1 }
171   }
172 }
173 \cs_generate_variant:Nn \__spintent_setspintent_set_keys:nn { en }

```

(End of definition for `\__spintent_setspintent_set_keys:nn`.)

```

\__spintent_setspintent_parse_args:nn
\__spintent_setspintent_parse_args:Vn

```

The function `\__spintent_setspintent_parse_args:nn` will check if the argument passed in #1 begins with ‘\’ or is just a *command name*. If it begins with ‘\’, it will remove it using `\cs_to_str:N` and then execute `\__spintent_setspintent_set_keys:en`. Otherwise, i.e., if ‘\’ was not passed, it will directly execute `\__spintent_setspintent_set_keys:nn`.

```

174 \cs_new_protected:Nn \__spintent_setspintent_parse_args:nn
175 {
176   \tl_if_single:nTF { #1 }
177   {
178     \token_if_cs:NTF #1
179     {
180       \__spintent_setspintent_set_keys:en { \cs_to_str:N #1 } { #2 }
181     }
182     {
183       \__spintent_setspintent_set_keys:nn { #1 } { #2 }
184     }
185   }
186   {
187     \__spintent_setspintent_set_keys:nn { #1 } { #2 }
188   }
189 }
190 \cs_generate_variant:Nn \__spintent_setspintent_parse_args:nn { Vn }

```

(End of definition for `\__spintent_setspintent_parse_args:nn`.)

`\setspintent`

Now we define the user command `\setspintent`.

```

191 \NewDocumentCommand \setspintent { m m }
192 {
193   \tl_if_blank:nTF { #1 }
194   {
195     \msg_error:nnn { spintent } { unknown-command-setup } { <empty> }
196   }
197   {
198     \tl_set:Nn \l__spintent_setspintent_input_arg_cmd_tl { #1 }
199     \tl_trim_spaces:N \l__spintent_setspintent_input_arg_cmd_tl
200     \__spintent_setspintent_parse_args:Vn \l__spintent_setspintent_input_arg_cmd_tl { #2 }
201   }
202 }

```

(End of definition for `\setspintent`. This function is documented on page 3.)

## 11.11 Handling unknown keys

```

\l__spintent_command_name_str
unknown-choice
cmd-key-unknown
cmd-key-value-unknown

```

We define the variable `\l__spintent_command_name_str` which will *store* the “name” of each command within the implementation along with the messages `unknown-choice` used by `⟨keys⟩` that use `.choice:n` in their implementation and the messages `cmd-key-unknown`, `cmd-key-value-unknown` used by the `⟨key⟩` `unknown` in commands that use `[⟨key = val⟩]`.

```

203 \str_new:N \l__spintent_command_name_str
204 \msg_new:nnn { spintent } { unknown-choice }
205 {
206   The~value~'~#3'~for~'~#1'~key~is~invalid~use~('~#2').
207 }
208 \msg_new:nnnn { spintent } { cmd-key-unknown }
209 {
210   The~key~'~#1'~is~unknown~by~command~

```

```

211     \c_backslash_str \l__spintent_command_name_str \c_space_tl
212     and~is~being~ignored~\msg_line_context:.
213 }
214 {
215     The~command~\c_backslash_str \l__spintent_command_name_str \c_space_tl
216     does~not~have~a~key~called~'#1'.\\
217     Check~that~you~have~spelled~the~key~name~correctly.
218 }
219 \msg_new:nnnn { spintent } { cmd-key-value-unknown }
220 {
221     The~key~'#1=#2'~is~unknown~by~command~
222     \c_backslash_str \l__spintent_command_name_str \c_space_tl
223     and~is~being~ignored~\msg_line_context:.
224 }
225 {
226     The~command~\c_backslash_str \l__spintent_command_name_str \c_space_tl
227     does~not~have~a~key~called~'#1'.\\
228     Check~that~you~have~spelled~the~key~name~correctly.
229 }

```

(End of definition for `\l__spintent_command_name_str` and others.)

```

\__spintent_unknown_keys:nn
\__spintent_unknown_keys:Vn
\__spintent_unknown_keys:n

```

The internal functions `\__spintent_unknown_keys:nn` and `\__spintent_unknown_keys:n` used by the `<key>` unknown.

```

230 \cs_new_protected:Nn \__spintent_unknown_keys_cmd:nn
231 {
232     \tl_if_blank:nTF {#2}
233     { \msg_error:nnn { spintent } { cmd-key-unknown } {#1} }
234     { \msg_error:nnnn { spintent } { cmd-key-value-unknown } {#1} {#2} }
235 }
236 \cs_generate_variant:Nn \__spintent_unknown_keys_cmd:nn { Vn }
237 \cs_new_protected:Nn \__spintent_unknown_keys_cmd:n
238 {
239     \__spintent_unknown_keys_cmd:Vn \l_keys_key_str {#1}
240 }

```

(End of definition for `\__spintent_unknown_keys:nn` and `\__spintent_unknown_keys:n`.)

## 11.12 Definition of core base variables

```

\__spintent_luafun_clean_split_and_set:n
\__spintent_luafun_clean_split_and_set:V

```

The function `\__spintent_luafun_clean_split_and_set:n` defined in the Lua module `spintent.lua`, is common to several commands in this implementation. It analyzes, processes, cleans and splits the `{<argument>}` passed to commands using internal functions defined in the Lua module `spintent.lua`. It isolates signs, integer components, decimal separators, decimal parte, decimal periodic part, exponents and final physical “units”, assigning them directly to `expl3` variables.

This function assigns the states of the variables ‘\_str’ or ‘\_tl’ described below to `true`, `false`, `error` or `stores` information according to the role it fulfills within the implementation.

(End of definition for `\__spintent_luafun_clean_split_and_set:n`.)

```

\l__spintent_luaset_sign_tl
\l__spintent_luaset_part_int_tl
\l__spintent_luaset_dec_sep_tl
\l__spintent_luaset_part_dec_tl
\l__spintent_luaset_part_period_tl
\l__spintent_luaset_has_integer_str
\l__spintent_luaset_has_natural_str

```

The variable `\l__spintent_luaset_sign_tl` stores the explicit ‘+’ or ‘−’ sign if present. The variable `\l__spintent_luaset_part_int_tl` stores the raw digit sequence of the “integer part”. The variable `\l__spintent_luaset_dec_sep_tl` stores the matched ‘,’ or ‘.’ used as the decimal separator. The variable `\l__spintent_luaset_part_dec_tl` stores the raw digits of the “finite decimal part” that follow the separator, and the variable `\l__spintent_luaset_part_period_tl` stores the digits of the “infinite periodic decimal part” that follow a semicolon ‘;’.

The variable `\l__spintent_luaset_has_integer_str` store the state `true` when it is *only* an “integer”. The variable `\l__spintent_luaset_has_natural_str` store the state `true` when it is *only* an “natural number”.

```

241 \tl_new:N \l__spintent_luaset_sign_tl
242 \tl_new:N \l__spintent_luaset_part_int_tl
243 \tl_new:N \l__spintent_luaset_dec_sep_tl
244 \tl_new:N \l__spintent_luaset_part_dec_tl
245 \tl_new:N \l__spintent_luaset_part_period_tl
246 \str_new:N \l__spintent_luaset_has_integer_str
247 \str_new:N \l__spintent_luaset_has_natural_str

```

(End of definition for `\l__spintent_luaset_sign_tl` and others.)

```

\l__spinttent_luaset_only_part_int_str
\l__spinttent_luaset_only_part_dec_str
\l__spinttent_luaset_only_part_period_str
\l__spinttent_luaset_format_part_int_str
\l__spinttent_luaset_format_part_dec_str

```

The literal entries of the numbers that we will pass to the *intent function* :number of `\MathMLintent` are *stored* in `\l__spinttent_luaset_only_part_int_str` for the “*integer part*”, in `\l__spinttent_luaset_only_part_dec_str` for the “*finite decimal part*”, and in `\l__spinttent_luaset_only_part_period_str` for the “*infinite periodic decimal part*”.

Formatted versions (`cifras-sep=true`) are *stored* in `\l__spinttent_luaset_format_part_int_str` for the “*integer part*” and `\l__spinttent_luaset_format_part_dec_str` for the “*finite decimal part*”.

```

248 \str_new:N \l__spinttent_luaset_only_part_int_str
249 \str_new:N \l__spinttent_luaset_only_part_dec_str
250 \str_new:N \l__spinttent_luaset_only_part_period_str
251 \str_new:N \l__spinttent_luaset_format_part_int_str
252 \str_new:N \l__spinttent_luaset_format_part_dec_str

```

(End of definition for `\l__spinttent_luaset_only_part_int_str` and others.)

```

\l__spinttent_luaset_millions_str
\l__spinttent_luaset_decimal_period_str
\l__spinttent_luaset_not_decimal_period_str
\l__spinttent_luaset_not_decimal_not_period_str
\l__spinttent_luaset_dec_is_all_zeros_str
\l__spinttent_luaset_has_exponent_str
\l__spinttent_luaset_exponent_tl

```

The flag variable `\l__spinttent_luaset_millions_str` indicates whether the integer part translates exactly to a multiple of one million when passed to `\MathMLintent`. To properly structure the reading and writing of numbers in MathML, we use the flag variables `\l__spinttent_luaset_decimal_period_str`, `\l__spinttent_luaset_not_decimal_period_str`, and `\l__spinttent_luaset_not_decimal_not_period_str`. The flag variable `\l__spinttent_luaset_dec_is_all_zeros_str` monitors whether the “*finite decimal part*” is completely zero (useful for “*pure periodic decimals*”).

The flag variable `\l__spinttent_luaset_has_exponent_str` indicates whether active “*scientific notation*” was detected at the end of the entered number, passed with ‘e’ or ‘E’. If true, it will store the digits of the exponent in `\l__spinttent_luaset_exponent_tl`.

```

253 \str_new:N \l__spinttent_luaset_millions_str
254 \str_new:N \l__spinttent_luaset_decimal_period_str
255 \str_new:N \l__spinttent_luaset_not_decimal_period_str
256 \str_new:N \l__spinttent_luaset_not_decimal_not_period_str
257 \str_new:N \l__spinttent_luaset_dec_is_all_zeros_str
258 \str_new:N \l__spinttent_luaset_has_exponent_str
259 \tl_new:N \l__spinttent_luaset_exponent_tl

```

(End of definition for `\l__spinttent_luaset_millions_str` and others.)

```

\l__spinttent_luaset_has_units_str
\l__spinttent_luaset_status_str
\l__spinttent_luaset_arg_above_tl
\l__spinttent_luaset_arg_below_tl
\l__spinttent_luaset_denom_has_number_str

```

Since the function `\__spinttent_luafun_clean_split_and_set:n` is common to `\spnum` (§11.14) and `\spunit` (§11.15) (as well as others), it determines the following unit-related variables after processing the `{⟨arguments⟩}` via the Lua module `spintent.lua`.

First, it checks if there is any residual text *after* the processed number that is a candidate for physical “*units*”. It sets the flag variable `\l__spinttent_luaset_has_units_str` accordingly. Following this, it verifies if this text contains more than one “/”. It then sets the variable `\l__spinttent_luaset_status_str` to `valid` or `multislash`. If the structure is `valid`, it splits the unit into `\l__spinttent_luaset_arg_above_tl` and `\l__spinttent_luaset_arg_below_tl`. Finally, it analyzes `\l__spinttent_luaset_arg_below_tl`; if the denominator starts with a number, it sets `\l__spinttent_luaset_denom_has_number_str` to `true`, otherwise to `false`.

```

260 \str_new:N \l__spinttent_luaset_has_units_str
261 \str_new:N \l__spinttent_luaset_status_str
262 \tl_new:N \l__spinttent_luaset_arg_above_tl
263 \tl_new:N \l__spinttent_luaset_arg_below_tl
264 \str_new:N \l__spinttent_luaset_denom_has_number_str

```

(End of definition for `\l__spinttent_luaset_has_units_str` and others.)

## 11.13 Common messages for text mode and math mode

Messages shared by the different commands for text mode or math mode.

```

265 \msg_new:nnnn { spintent } { need-math-mode }
266 {
267   Math~mode~required~for~'#1'~\msg_line_context:.
268 }
269 {
270   Use~inside~$...$~or~in~math~environment.
271 }
272 \msg_new:nnnn { spintent } { need-text-mode }
273 {
274   Text~mode~required~for~'#1'~\msg_line_context:.
275 }
276 {
277   Use~outside~of~$...$~or~math~environment.
278 }

```

### 11.14 The command `\spnum`

The user command `\spnum` is *first* of “the big four” provided by the `spintent` package and is fundamental to the implementation of other commands. It handles the processing of numerical input, scientific notation, and units of measurement, using the Lua module `spintent.lua`, which is responsible for separating and assigning variables defined in `expl3`.

From the `expl3` side, we handle the printing logic and the *intent functions* passed to `\MathMLintent` to achieve a valid MathML structure that is accepted by MathCAT.

#### 11.14.1 Definition of keys and variables for `\spnum`

The variable `\l__spintenc_spnum_read_terse_bool` handled by the key `read-sign`, controls the *intent* passed to `\MathMLintent` to spoken the ‘+’ or ‘-’. If set to `clear`, it will say “*positivo*” or “*negativo*” after reading the number; if set to `terse`, it will say “*más*” or “*menos*” before reading the number.

The variable `\l__spintenc_spnum_read_period_sep_str` handled by the key `read-period-sep`, controls how the decimal separator ‘;’ (periodic part) is printed and spoken when the `{(argument)}` is a *pure decimal periodic* and is passed the *intent* to `\MathMLintent`. If set to `coma`, it will say “*coma*” and print ‘,’; if set to `punto` it will say “*punto*” and print ‘.’.

The variable `\l__spintenc_spnum__notation_sym_tl`, handled by the key `notation-sym`, determines whether ‘x’ will be printed when set to `times` or ‘.’ when set to `cdot`.

The variable `\l__spintenc_spnum_intent_read_sign_str` will *store* the spoken translation used in the *intent* function passed to `\MathMLintent` for the ‘+’ or ‘-’ according to the value passed to the key `read-sign`.

The variable `\l__spintenc_spnum_intent_read_dec_sep_str` *stores* the spoken translation that will be used in the *intent* function passed to `\MathMLintent` for reading the decimal separation as a “*coma*” or a “*punto*” according to the value found in the variable `\l__spintenc_luaset_dec_sep_tl`.

```
279 \bool_new:N \l__spintenc_spnum_read_terse_bool
280 \str_new:N \l__spintenc_spnum_read_period_sep_str
281 \tl_new:N \l__spintenc_spnum_notation_sym_tl
282 \str_new:N \l__spintenc_spnum_intent_read_sign_str
283 \str_new:N \l__spintenc_spnum_intent_read_dec_sep_str
```

(End of definition for `\l__spintenc_spnum_read_terse_bool` and others.)

```
cifras-sep      Now we add the keys cifras-sep, read-sign, read-period-sep and notation-sym.
read-sign
read-period-sep
notation-sym

284 \keys_define:nn { spintenc/spnum }
285 {
286   cifras-sep          .bool_set:N = \l__spintenc_spnum_cifras_sep_bool,
287   cifras-sep          .initial:n = true,
288   cifras-sep          .value_required:n = true,
289   read-sign           .choices:nn =
290     { terse , clear }
291     {
292       \int_case:nn { \l_keys_choice_int }
293       {
294         {1} { \bool_set_true:N \l__spintenc_spnum_read_terse_bool }
295         {2} { \bool_set_false:N \l__spintenc_spnum_read_terse_bool }
296       }
297     },
298   read-sign           .initial:n = terse,
299   read-sign           .value_required:n = true,
300   read-sign / unknown .code:n =
301     \msg_error:nneee { spintenc } { unknown-choice }
302     { read-sign } { terse , ~clear } { \exp_not:n {#1} },
303   read-period-sep     .choices:nn =
304     { coma , punto }
305     {
306       \int_case:nn { \l_keys_choice_int }
307       {
308         {1} { \str_set:Nn \l__spintenc_spnum_read_period_sep_str { coma } }
309         {2} { \str_set:Nn \l__spintenc_spnum_read_period_sep_str { punto } }
310       }
311     },
312   read-period-sep     .initial:n = coma,
313   read-period-sep     .value_required:n = true,
314   read-period-sep / unknown .code:n =
315     \msg_error:nneee { spintenc } { unknown-choice }
316     { read-period-sep } { coma , ~punto } { \exp_not:n {#1} },
317   notation-sym        .choices:nn =
318     { times , cdot }
```

```

319     {
320       \int_case:nn { \l_keys_choice_int }
321       {
322         {1} { \tl_set:Nn \__spintent_snum_notation_sym_tl { \times } }
323         {2} { \tl_set:Nn \__spintent_snum_notation_sym_tl { \cdot } }
324       }
325     },
326     notation-sym .initial:n = cdot,
327     notation-sym .value_required:n = true,
328     notation-sym / unknown .code:n =
329       \msg_error:nneee { spintent } { unknown-choice }
330       { notation-sym } { times , ~ cdot } { \exp_not:n {#1} },
331     unknown .code:n =
332       {
333         \str_set:Nn \__spintent_command_name_str { snum }
334         \__spintent_unknown_keys_cmd:n {#1}
335       },
336   }

```

(End of definition for *cifras-sep* and others.)

#### 11.14.2 Internal functions for \snum

```

\__spintent_snum_render_part:n
\__spintent_snum_render_int:
\__spintent_snum_render_dec:

```

The function `\__spintent_snum_render_part:n` processes the  $\langle argument \rangle$  passed in ‘#1’ used by the functions `\__spintent_snum_render_int:` and `\__spintent_snum_render_dec:` to define the *intent function* :number for the integer and decimal part passed to `\MathMLintenc`. It evaluates whether the key *cifras-sep* is active or not.

```

337 \cs_new_protected:Nn \__spintent_snum_render_part:n
338 {
339   \MathMLintenc { \str_use:c { l__spintenc_luaset_only_part_#1_str } :number }
340   {
341     \bool_if:NTF \__spintenc_snum_cifras_sep_bool
342     { \str_use:c { l__spintenc_luaset_format_part_#1_str } }
343     { \tl_use:c { l__spintenc_luaset_part_#1_tl } }
344   }
345 }
346 \cs_new_protected:Nn \__spintenc_snum_render_int:
347 {
348   \__spintenc_snum_render_part:n { int }
349 }
350 \cs_new_protected:Nn \__spintenc_snum_render_dec:
351 {
352   \__spintenc_snum_render_part:n { dec }
353 }

```

(End of definition for `\__spintenc_snum_render_part:n`, `\__spintenc_snum_render_int:`, and `\__spintenc_snum_render_dec:`.)

```

\__spintenc_snum_render_only_period:
\__spintenc_snum_print_period_bar:
\__spintenc_snum_render_period_sep:

```

The function `\__spintenc_snum_render_only_period:` isolates the “infinite periodic decimal part” from the “finite decimal part”, assigning the *intent function* :number passed to `\MathMLintenc` which will be used by `\MathMLarg` in the function `\__spintenc_snum_print_period_bar:`.

```

354 \cs_new_protected:Nn \__spintenc_snum_render_only_period:
355 {
356   \MathMLintenc { \l__spintenc_luaset_only_part_period_str :number }
357   {
358     \l__spintenc_luaset_part_period_tl
359   }
360 }

```

The function `\__spintenc_snum_print_period_bar:` inserts the function `\__spintenc_snum_render_only_period:` as an argument within the *intent function* `_periódico:postfix($dp)` passed to `\MathMLintenc`, generating the correct spoken for the “infinite periodic decimal part”.

```

361 \cs_new_protected:Nn \__spintenc_snum_print_period_bar:
362 {
363   \MathMLintenc { _periódico:postfix($dp) }
364   {
365     {
366       \overline
367       {
368         \MathMLarg { dp } { \__spintenc_snum_render_only_period: }
369       }
370     }
371   }

```

```
372 }
```

The function `\__spintent_snum_render_period_sep:` sets how the decimal separator handled by the key `read-period-sep` will be printed and spoken when a “*pure periodic decimal*” is passed.

```
373 \cs_new_protected:Nn \__spintent_snum_render_period_sep:
374 {
375   \str_if_eq:VnTF \l__spintent_snum_read_period_sep_str { coma }
376   {
377     \MathMLintent { _coma:prefix($pp) }
378     {
379       \mathord{,}
380       \MathMLarg { pp } { \__spintent_snum_print_period_bar: }
381     }
382   }
383   {
384     \MathMLintent { _punto:prefix($pp) }
385     {
386       \mathord{.}
387       \MathMLarg { pp } { \__spintent_snum_print_period_bar: }
388     }
389   }
390 }
```

(End of definition for `\__spintent_snum_render_only_period:`, `\__spintent_snum_print_period_bar:`, and `\__spintent_snum_render_period_sep:`.)

```
\__spintent_snum_render_decimal_sep:
```

The function `\__spintent_snum_render_decimal_sep:` parses the value of the variable `\l__spintent_luaset` and sets the variable `\l__spintent_snum_intent_read_dec_sep_str` with the function `_coma:prefix($dp)` for reading a “*coma*” or `_punto:prefix($dp)` for reading a “*punto*” to `\MathMLintent`. We print the decimal separator stored in `\l__spintent_luaset_dec_sep_tl` using `\mathord` to remove spaces around it and then use the command `\MathMLarg` where we insert `\__spintent_invisible_sep:` for compatibility with MathML and finally call the function `\__spintent_snum_render_dec:` to print the finite decimal part.

```
391 \cs_new_protected:Nn \__spintent_snum_render_decimal_sep:
392 {
393   \str_if_eq:VnTF \l__spintent_luaset_dec_sep_tl { , }
394   { \str_set:Nn \l__spintent_snum_intent_read_dec_sep_str { _coma } }
395   { \str_set:Nn \l__spintent_snum_intent_read_dec_sep_str { _punto } }
396   \MathMLintent { \l__spintent_snum_intent_read_dec_sep_str :prefix($dp) }
397   {
398     \mathord { \l__spintent_luaset_dec_sep_tl }
399     \MathMLarg { dp }
400     {
401       \__spintent_inner_sep:
402       \__spintent_snum_render_dec:
403     }
404   }
405   % period part
406   \str_if_eq:VnTF \l__spintent_luaset_decimal_period_str { true }
407   {
408     \str_if_eq:VnTF \l__spintent_luaset_dec_is_all_zeros_str { true }
409     {
410       \__spintent_inner_sep:
411       \__spintent_snum_print_period_bar:
412     }
413     {
414       \MathMLintent { _con:prefix($pnum) }
415       {
416         \__spintent_inner_sep:
417         \MathMLarg { pnum }
418         {
419           \__spintent_snum_print_period_bar:
420         }
421       }
422     }
423   }
424 }
```

(End of definition for `\__spintent_snum_render_decimal_sep:`.)

```
\__spintent_snum_print_part_dec:
\__spintent_snum_print_part_period:
```

The function `\__spintent_snum_print_part_dec:` checks if a decimal part exists. If it does, it determines whether the decimal separator is a comma or a period to set the correct speech intent (“*coma*” or “*punto*”).

It then prints the separator, renders the decimal digits, and if a periodic sequence follows, it decides how to attach the periodic bar.

The function `\__spintenc_snum_print_part_period:` handles cases where a periodic part exists. It evaluates if it should render the periodic separator directly, particularly when the number has a periodic part but lacks a standard decimal part.

```

425 \cs_new_protected:Nn \__spintenc_snum_print_part_dec:
426 {
427   \tl_if_empty:NF \l__spintenc_luaset_part_dec_tl
428   {
429     \__spintenc_snum_render_decimal_sep:
430   }
431 }
432 \cs_new_protected:Nn \__spintenc_snum_print_part_period:
433 {
434   \tl_if_empty:NF \l__spintenc_luaset_part_period_tl
435   {
436     \str_if_eq:VnTF \l__spintenc_luaset_not_decimal_period_str { true }
437     { \__spintenc_snum_render_period_sep: }
438     {
439       \tl_if_empty:NT \l__spintenc_luaset_part_dec_tl
440       {
441         \__spintenc_snum_render_period_sep:
442       }
443     }
444   }
445 }

```

(End of definition for `\__spintenc_snum_print_part_dec:` and `\__spintenc_snum_print_part_period:`.)

```

\__spintenc_snum_print_integer:
\__spintenc_snum_print_num_start:
\__spintenc_snum_print_number_core:

```

The function `\__spintenc_snum_print_integer:` checks if the number consists only of an integer. If so, it simply renders it. If a decimal or periodic part exists, it sets up a `MathML` intent to link the integer part (respecting digit separators if enabled) to the subsequent decimal or periodic functions.

The function `\__spintenc_snum_print_num_start:` checks if an explicit sign is present. If it finds one, it evaluates the terse reading flag to decide whether the sign should be read as a prefix (“*más*” or “*menos*”) or as a postfix (“*positivo*” or “*negativo*”), and wraps the intent around the rest of the number.

The function `\__spintenc_snum_print_number_core:` acts as the main wrapper. It triggers the sign and base number evaluation, and then checks if a scientific exponent is present. If it is, it appends the correct base-10 notation symbol and exponent to complete the sequence.

 Note for `\__spintenc_invisible_sep:`.

```

446 \cs_new_protected:Nn \__spintenc_snum_print_integer:
447 {
448   \str_if_eq:VnTF \l__spintenc_luaset_not_decimal_not_period_str { true }
449   { \__spintenc_snum_render_int: }
450   {
451     \MathMLintent { \l__spintenc_luaset_only_part_int_str :number:prefix($next) }
452     {
453       \bool_if:NTF \l__spintenc_snum_cifras_sep_bool
454       { \l__spintenc_luaset_format_part_int_str }
455       { \l__spintenc_luaset_part_int_tl }
456       \MathMLarg { next }
457       {
458         \tl_if_empty:NTF \l__spintenc_luaset_part_dec_tl
459         { \__spintenc_snum_print_part_period: }
460         { \__spintenc_snum_print_part_dec: }
461       }
462     }
463   }
464 }
465 \cs_new_protected:Nn \__spintenc_snum_print_num_start:
466 {
467   \tl_if_empty:NTF \l__spintenc_luaset_sign_tl
468   {
469     \__spintenc_invisible_sep: % need
470     \__spintenc_snum_print_integer:
471   }
472   {
473     \bool_if:NTF \l__spintenc_snum_read_terse_bool
474     {
475       \str_if_eq:VnTF \l__spintenc_luaset_sign_tl { + }

```

```

476         { \str_set:Nn \__spinttent_snum_intent_read_sign_str { _más:prefix($n)} }
477         { \str_set:Nn \__spinttent_snum_intent_read_sign_str { _menos:prefix($n)} }
478     }
479     {
480         \str_if_eq:VnTF \__spinttent_luaset_sign_tl { + }
481         { \str_set:Nn \__spinttent_snum_intent_read_sign_str { _positivo:postfix($n) } }
482         { \str_set:Nn \__spinttent_snum_intent_read_sign_str { _negativo:postfix($n) } }
483     }
484     \__spinttent_invisible_sep: % need
485     \MathMLintent { \__spinttent_snum_intent_read_sign_str }
486     {
487         \__spinttent_luaset_sign_tl
488         \MathMLarg { n } { \__spinttent_snum_print_integer: }
489     }
490 }
491 }
492 \cs_new_protected:Nn \__spinttent_snum_print_number_core:
493 {
494     \__spinttent_snum_print_num_start:
495     \str_if_eq:VnTF \__spinttent_luaset_has_exponent_str { true }
496     {
497         \MathMLintent { _por } { \__spinttent_snum_notation_sym_tl }
498         10^{\__spinttent_luaset_exponent_tl }
499     }
500 }

```

(End of definition for `\__spinttent_snum_print_integer:`, `\__spinttent_snum_print_num_start:`, and `\__spinttent_snum_print_number_core:`.)

### 11.14.3 Definition of error messages

spnum-has-units  
spnum-no-digits

We define two error messages for the `\spnum` command. The first “error” triggers if the user includes “units” in the  $\langle argument \rangle$  passed in ‘#1’, advising them to use the `\spunit` command instead. The second “error” is raised if the provided  $\langle argument \rangle$  contains no numerical digits.

```

501 \msg_new:nnnn { spinttent } { spnum-has-units }
502 {
503     The~\token_to_str:N \spnum \c_space_tl command~does~not~accept~units~(''#1'').
504 }
505 {
506     Use~\token_to_str:N \spunit \c_space_tl if~you~need~to~print~numbers~with~units.
507 }
508 \msg_new:nnnn { spinttent } { spnum-no-digits }
509 {
510     The~\token_to_str:N \spnum \c_space_tl command~requires~a~valid~numerical~value.
511 }
512 {
513     You~provided~''#1',~which~contains~no~digits.
514 }

```

(End of definition for `spnum-has-units` and `spnum-no-digits`.)

### 11.14.4 Implementation of the `\spum` command

`\spnum` The public user command `\spnum` accepts an optional  $[\langle key = val \rangle]$  argument in ‘#1’ and a mandatory  $\langle rational\ number \rangle$  in ‘#2’. It first checks if it is executed inside math mode, raising an error if it is not. If correct, it opens a local group to apply the options and passes the input to Lua for parsing via `\__spinttent_luafun_clean_split_and_set:n`.

Before printing, it validates the extracted data. It throws an error if it finds trailing units in the input or if the integer part is completely empty (no valid digits). If all checks pass, it calls `\__spinttent_snum_print_number_core:` to render the final number, and finally closes the group.

```

515 \NewDocumentCommand \spnum { O{} m }
516 {
517     \mode_if_math:TF
518     {
519         \group_begin:
520         \keys_set:nn { spinttent/spnum } { #1 }
521         \__spinttent_luafun_clean_split_and_set:n { \exp_not:n { #2 } }
522         \str_if_eq:VnTF \__spinttent_luaset_has_units_str { true }
523         { \msg_error:nnn { spinttent } { spnum-has-units } { #2 } }
524         {
525             \tl_if_empty:NTF \__spinttent_luaset_part_int_tl
526             { \msg_error:nnn { spinttent } { spnum-no-digits } { #2 } }
527             { \__spinttent_snum_print_number_core: }

```

```

528         }
529     \group_end:
530 }
531 { \msg_error:nnn { spintent } { need-math-mode } { \spnum } }
532 }

```

(End of definition for `\spnum`. This function is documented on page 3.)

## 11.15 The command `\spunit`

The user command `\spunit` is the *second* of “the big four” provided by the `spintent` package and is used to parse and print physical units and angular systems. It sends the  $\langle argument \rangle$  to Lua module `spintent.lua` via `__spintent_luafun_clean_split_and_set:n` to separate and clean the  $\langle argument \rangle$ .

This ensures that all “units” (such as SI fractions, products, or sexagesimal angles) are processed consistently by the `spintent.lua`.

### 11.15.1 Definition of variables

```

\__spintent_spunit_luaset_canonical_str
\__spintent_spunit_luaset_lookup_status_str
\__spintent_spunit_luaset_read_str
\__spintent_spunit_luaset_is_compact_str
\__spintent_spunit_luaset_compact_exp_str
\__spintent_spunit_luaset_is_sexagesimal_str
\__spintent_spunit_luaset_status_str

```

These string variables store the data returned by the Lua module `spintent.lua`. They hold the canonical “unit symbol”, the dictionary lookup status, the correct speech text for NVDA/MathCAT, and specific flags that indicate if the “unit” is compact or represents a sexagesimal angle.

```

533 \str_new:N \__spintent_spunit_luaset_canonical_str
534 \str_new:N \__spintent_spunit_luaset_lookup_status_str
535 \str_new:N \__spintent_spunit_luaset_read_str
536 \str_new:N \__spintent_spunit_luaset_is_compact_str
537 \str_new:N \__spintent_spunit_luaset_compact_exp_str
538 \str_new:N \__spintent_spunit_luaset_is_sexagesimal_str
539 \str_new:N \__spintent_spunit_luaset_status_str

```

(End of definition for `\__spintent_spunit_luaset_canonical_str` and others.)

```

\__spintent_spunit_exponent_tl
\__spintent_spunit_unit_name_tl
\__spintent_spunit_read_slash_str
\__spintent_spunit_exp_seq
\__spintent_spunit_mult_seq
\__spintent_spunit_is_mult_bool
\__spintent_spunit_read_cdot_bool

```

These variables are used to process and format the “unit”. The sequences `\__spintent_spunit_mult_seq` and `\__spintent_spunit_exp_seq` split compound “units” at multiplication ‘`*`’ or exponent ‘`^`’ boundaries.

The boolean variables control specific rendering options, such as whether the multiplication dot ‘`\cdot`’ is explicitly read aloud or silently skipped.

```

540 \tl_new:N \__spintent_spunit_exponent_tl
541 \tl_new:N \__spintent_spunit_unit_name_tl
542 \str_new:N \__spintent_spunit_read_slash_str
543 \seq_new:N \__spintent_spunit_exp_seq
544 \seq_new:N \__spintent_spunit_mult_seq
545 \bool_new:N \__spintent_spunit_is_mult_bool
546 \bool_new:N \__spintent_spunit_read_cdot_bool

```

(End of definition for `\__spintent_spunit_exponent_tl` and others.)

### 11.15.2 Definition of error messages

```

spunit-invalid-unit
spunit-no-units
inline-numeric-denominator

```

We define three error messages for the `\spunit` command. The first “error” triggers if the “unit expression” provided in ‘`#1`’ is unknown or malformed. The second “error” occurs if no “units” are found in the  $\langle argument \rangle$ , advising the user to use `\spnum` for pure numbers. The third “error” is raised if a numeric coefficient is illegally placed in the denominator of an inline fraction (after a slash).

```

547 \msg_new:nnnn { spintent } { spunit-invalid-unit }
548 {
549     The~expression~'#1'~contains~an~unknown~or~malformed~unit~structure.
550 }
551 {
552     Check~spelling~or~verify~the~unit~is~defined~in~the~dictionary.
553 }
554 \msg_new:nnnn { spintent } { spunit-no-units }
555 {
556     The~\token_to_str:N \spunit \c_space_tl command~requires~at~least~one~unit~('#1').
557 }
558 {
559     Use~\token_to_str:N \spnum \c_space_tl if~you~only~need~to~format~a~pure~number.
560 }
561 \msg_new:nnnn { spintent } { inline-numeric-denominator }
562 {
563     Illegal~numeric~coefficient~in~inline~denominator~'#1'.
564 }
565 {
566     The~inline~mode~of~\token_to_str:N \spunit \c_space_tl
567     does~not~allow~numbers~after~the~slash~(/).
568 }

```

(End of definition for `spunit-invalid-unit`, `spunit-no-units`, and `inline-numeric-denominator`.)

### 11.15.3 Definition of keys for \spunit

Now we add the keys `print-cdot`, `read-cdot`, `read-slash`, `cifras-sep`, `read-period-sep` and `notation-sym`.

```

read-cdot
read-slash
cifras-sep
read-period-sep
notation-sym
569 \keys_define:nn { spintent/spunit }
570 {
571   print-cdot      .bool_set:N = \l__spintent_spunit_print_cdot_bool,
572   print-cdot      .initial:n   = false,
573   print-cdot      .value_required:n = true,
574   read-cdot       .bool_set:N = \l__spintent_spunit_read_cdot_bool,
575   read-cdot       .initial:n   = false,
576   read-cdot       .value_required:n = true,
577   read-slash      .choices:nn =
578     { por , partido-por }
579   {
580     \int_case:nn { \l_keys_choice_int }
581     {
582       {1} { \str_set:Nn \l__spintent_spunit_read_slash_str { por } }
583       {2} { \str_set:Nn \l__spintent_spunit_read_slash_str { partido-por } }
584     }
585   },
586   read-slash      .initial:n   = por,
587   read-slash      .value_required:n = true,
588   read-slash / unknown .code:n =
589     \msg_error:nneee { spintent } { unknown-choice }
590     { read-slash } { por , ~partido-por } { \exp_not:n {#1} },
591   cifras-sep      .meta:nn = { spintent/spnum } { cifras-sep = {#1} },
592   cifras-sep      .value_required:n = true,
593   read-period-sep .meta:nn = { spintent/spnum } { read-period-sep = {#1} },
594   read-period-sep .value_required:n = true,
595   notation-sym    .meta:nn = { spintent/spnum } { notation-sym = {#1} },
596   notation-sym    .value_required:n = true,
597   unknown         .code:n =
598     {
599       \str_set:Nn \l__spintent_command_name_str { spunit }
600       \__spintent_unknown_keys_cmd:n {#1}
601     },
602 }

```

(End of definition for `read-cdot` and others.)

### 11.15.4 Internal functions for \spunit

`\__spintent_spunit_format_canonical:` The function `\__spintent_spunit_format_canonical:` handles the visual formatting of the “unit symbol”. It specifically checks for the *liter symbol* ‘*ℓ*’ to print it as `\ell`, and formats all other valid “units” in standard upright text using `\mathrm`.

```

603 \cs_new_protected:Nn \__spintent_spunit_format_canonical:
604 {
605   \str_case:VnF \l__spintent_spunit_luaset_canonical_str
606   {
607     { ℓ } { \ell }
608   }
609   { \mathrm { \l__spintent_spunit_luaset_canonical_str } }
610 }

```

(End of definition for `\__spintent_spunit_format_canonical:`.)

`\__spintent_spunit_process_base_exp:n` The function `\__spintent_spunit_process_base_exp:n` evaluates a “single unit” along with its exponent, ‘*#1*’. It first looks up the unit in the dictionary; if not found, flags it as invalid. Otherwise, unless this is the first unit in a product (`\__spintent_spunit_is_mult_bool`), it handles the spacing and multiplication dot: `\cdot` when `print-cdot` is `true` (read as `por` when `read-cdot` is also `true`, silent otherwise), or a thin space `\,` when `print-cdot` is `false` (still read as `por` under `read-cdot`).

```

611 \cs_new_protected:Nn \__spintent_spunit_process_base_exp:n
612 {
613   \tl_clear:N \l__spintent_spunit_exponent_tl
614   \__spintent_luafun_spunit_lookup_alias:n { #1 }
615   \str_if_eq:VnTF \l__spintent_spunit_luaset_lookup_status_str { notfound }
616   {
617     \str_set:Nn \l__spintent_luaset_status_str { invalid }
618   }
619   {
620     \bool_if:NF \l__spintent_spunit_is_mult_bool

```

```

621     {
622       \bool_if:NTF \l__spinttent_spunit_print_cdot_bool
623       {
624         \bool_if:NTF \l__spinttent_spunit_read_cdot_bool
625         { \MathMLintent { por } { \cdot } }
626         { \MathMLintent { :silent } { \cdot } }
627       }
628       {
629         \bool_if:NTF \l__spinttent_spunit_read_cdot_bool
630         { \MathMLintent { por } { \, } }
631         { \, }
632       }
633     }

```

Then renders the base unit and its exponent: either a pre-defined compact exponent from Lua (`\l__spinttent_spunit_lua`) or by manually splitting ‘#1’ at the ‘^’ symbol otherwise.

```

634     \str_if_eq:VnTF \l__spinttent_spunit_lua_set_is_compact_str { true }
635     {
636       {
637         \MathMLintent { \l__spinttent_spunit_lua_set_read_str }
638         {
639           \l__spinttent_spunit_format_canonical:
640         }
641       }
642       ^ { \l__spinttent_spunit_lua_set_compact_exp_str }
643     }
644     {
645       \seq_set_split:Nnn \l__spinttent_spunit_exp_seq { ^ } { #1 }
646       \seq_pop_left:NN \l__spinttent_spunit_exp_seq \l__spinttent_spunit_unit_name_tl
647       \seq_if_empty:NF \l__spinttent_spunit_exp_seq
648       {
649         \seq_pop_left:NN \l__spinttent_spunit_exp_seq \l__spinttent_spunit_exponent_tl
650       }
651       \MathMLintent { \l__spinttent_spunit_lua_set_read_str }
652       {
653         \l__spinttent_spunit_format_canonical:
654       }
655       \tl_if_empty:NF \l__spinttent_spunit_exponent_tl
656       {
657         ^ { \l__spinttent_spunit_exponent_tl }
658       }
659     }
660     \bool_set_false:N \l__spinttent_spunit_is_mult_bool
661   }
662 }

```

(End of definition for `\l__spinttent_spunit_process_base_exp:n`)

```

\l__spinttent_spunit_process_mult:n
\l__spinttent_spunit_process_mult:V

```

The function `\l__spinttent_spunit_process_mult:n` handles compound “units” joined by an explicit multiplication star ‘\*’. It splits the `{(argument)}` at each “star symbol” and applies `\l__spinttent_spunit_process_base_exp:n` to evaluate and render each resulting unit individually.

```

663 \cs_new_protected:Nn \l__spinttent_spunit_process_mult:n
664 {
665   \bool_set_true:N \l__spinttent_spunit_is_mult_bool
666   \seq_set_split:Nnn \l__spinttent_spunit_mult_seq { * } { #1 }
667   \seq_map_function:NN \l__spinttent_spunit_mult_seq \l__spinttent_spunit_process_base_exp:n
668 }
669 \cs_generate_variant:Nn \l__spinttent_spunit_process_mult:n { V }

```

(End of definition for `\l__spinttent_spunit_process_mult:n`)

```
\l__spinttent_print_spunit_hierarchy:
```

The function `\l__spinttent_print_spunit_hierarchy:` handles the layout of inline “unit” fractions. It first processes the “units” before the slash (the numerator). If a denominator exists, it prints the slash / ‘/’ with its corresponding speech intent, and then processes the remaining “units”.

```

670 \cs_new_protected:Nn \l__spinttent_print_spunit_hierarchy:
671 {
672   \tl_if_empty:NF \l__spinttent_lua_set_arg_above_tl
673   {
674     \l__spinttent_spunit_process_mult:V \l__spinttent_lua_set_arg_above_tl
675     \tl_if_empty:NF \l__spinttent_lua_set_arg_below_tl
676     {
677       \MathMLintent { \l__spinttent_spunit_read_slash_str } { / }

```

```

678         \__spinttent_spunit_process_mult:V \l__spinttent_luaset_arg_below_tl
679     }
680 }
681 }

```

(End of definition for \\_\_spinttent\_print\_spunit\_hierarchy:.)

\\_\_spinttent\_spunit\_safe\_exec:n

The function `\__spinttent_spunit_safe_exec:n` performs the main validation and layout for “units”. It first checks if the `{\langle argument \rangle}` has units, is a valid expression, and does not contain illegal numbers in the denominator, throwing the appropriate “errors” if any check fails.

If the unit is valid, it checks if it is a sexagesimal angle (degrees, minutes, or seconds). For sexagesimal “units”, it prints the number and the symbol “without any space in between”, applying a slight negative space ‘\!’ for arcseconds. For standard units, it prints the number followed by a thin space ‘\,’ and then processes the unit sequence.

```

682 \cs_new_protected:Nn \__spinttent_spunit_safe_exec:n
683 {
684     \str_if_eq:VnTF \l__spinttent_luaset_has_units_str { false }
685     { \msg_error:nnn { spinttent } { spunit-no-units } { #1 } }
686     {
687         \str_if_eq:VnTF \l__spinttent_luaset_status_str { valid }
688         {
689             \str_if_eq:VnTF
690             \l__spinttent_luaset_denom_has_number_str { true }
691             {
692                 \msg_error:nnn { spinttent }
693                 { inline-numeric-denominator } { #1 }
694             }
695             {
696                 \__spinttent_luafun_spunit_lookup_alias:V \l__spinttent_luaset_arg_above_tl
697                 \str_if_eq:VnTF \l__spinttent_spunit_luaset_is_sexagesimal_str { true }
698                 {
699                     \tl_if_empty:NF \l__spinttent_luaset_part_int_tl
700                     {
701                         \__spinttent_spnum_print_number_core:
702                     }
703                     \MathMLintent { \l__spinttent_spunit_luaset_read_str }
704                     {
705                         \mathord
706                         {
707                             \str_case:Vn \l__spinttent_spunit_luaset_canonical_str
708                             {
709                                 { " } { \prime \! \prime }
710                                 { ' } { \prime
711                                 { ° } { °
712                             }
713                         }
714                     }
715                 }
716                 {
717                     \tl_if_empty:NF \l__spinttent_luaset_part_int_tl
718                     {
719                         \__spinttent_spnum_print_number_core: \,
720                     }
721                     \__spinttent_print_spunit_hierarchy:
722                 }
723             }
724         }
725         { \msg_error:nnn { spinttent } { spunit-invalid-unit } { #1 } }
726     }
727 }

```

(End of definition for \\_\_spinttent\_spunit\_safe\_exec:n.)

### 11.15.5 Implementation of the \spunit command

\spunit

The public user command `\spunit` accepts an optional `[\langle key = val \rangle]` argument in ‘#1’ and a mandatory `{\langle unit expression \rangle}` in ‘#2’. It first checks if it is executed inside *math mode*, raising an “error” if it is not. If correct, it opens a local group to apply the options and passes the `{\langle argument \rangle}` to Lua module for parsing via `\__spinttent_luafun_clean_split_and_set:n` function.

Finally, it delegates the execution to the function `\__spinttent_spunit_safe_exec:n` to perform the safety checks and render the parsed data, before closing the group.

```

728 \NewDocumentCommand \spunit { 0{ } m }

```

```

729   {
730     \mode_if_math:TF
731     {
732       \group_begin:
733         \keys_set:nn { spintent/spunit } { #1 }
734         \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #2 } }
735         \__spintent_spunit_safe_exec:n { #2 }
736       \group_end:
737     }
738     { \msg_error:nnn { spintent } { need-math-mode } { \spunit } }
739   }

```

(End of definition for `\spunit`. This function is documented on page 4.)

## 11.16 The commands `\spnewunit` and `\spunitalias`

The user commands `\spnewunit` and `\spunitalias` allow users to add new custom “units” or “aliases” to the package’s dictionary. Similar to `\spunit`, these commands delegate the processing and validation to the `spintent.lua` via `\__spintent_luafun_define_custom_unit:nn` and `\__spintent_luafun_define_spunit`.

During the definition, the Lua module checks if the new “unit” or “alias” is valid and ensures it does not conflict with existing “units”. It returns the result of this operation to TeX using the `\l__spintent_spunit_luaset_status_str` variable. If a collision is detected or the  $\langle argument \rangle$  is invalid, the package throws the corresponding “error” message.

### 11.16.1 Definition of error messages

spnewunit-duplicate  
spunitalias-duplicate  
spunitalias-invalid-expr

We define three error messages for the creation of “units” and “aliases”. The first two “errors” trigger if the user attempts to define a “new unit” or alias passed in ‘#1’ that already exists in the dictionary. The third “error” is raised by `\spunitalias` if the target expression provided in ‘#2’ is invalid (for example, if it contains unknown “units” or incorrect syntax).

```

740 \msg_new:nnnn { spintent } { spnewunit-duplicate }
741 {
742   The~unit~symbol~or~alias~'#1'~is~already~defined!
743 }
744 {
745   You~cannot~redefine~with~\token_to_str:N \spnewunit.
746 }
747 \msg_new:nnnn { spintent } { spunitalias-duplicate }
748 {
749   The~alias~'#1'~is~already~defined~or~reserved!
750 }
751 {
752   You~cannot~redefine~an~existing~unit~or~another~alias.
753 }
754 \msg_new:nnnn { spintent } { spunitalias-invalid-expr }
755 {
756   The~expression~'#2'~for~alias~'#1'~is~invalid.
757 }
758 {
759   It~must~contain~only~defined~units,~exponents~and~one~'/'~.
760 }

```

(End of definition for `spnewunit-duplicate`, `spunitalias-duplicate`, and `spunitalias-invalid-expr`.)

`\spnewunit`

The public command `\spnewunit` defines a new custom unit. It passes the new unit symbol ‘#1’ and its speech text ‘#2’ to Lua via `\__spintent_luafun_define_custom_unit:nn`. If the module reports that the unit already exists, it throws the corresponding duplicate error.

```

761 \NewDocumentCommand \spnewunit { m m }
762 {
763   \mode_if_math:TF
764   {
765     \msg_error:nnn { spintent } { need-text-mode } { \spnewunit }
766   }
767   {
768     \__spintent_luafun_define_custom_unit:nn { #1 } { #2 }
769     \str_if_eq:VnT \l__spintent_spunit_luaset_status_str { duplicate }
770     {
771       \msg_error:nnn { spintent } { spnewunit-duplicate } { #1 }
772     }
773   }
774 }

```

(End of definition for `\spnewunit`. This function is documented on page 6.)

`\spunitalias` The public command `\spunitalias` creates a shortcut for a complex unit expression. It passes the new alias name ‘`#1`’ and the target expression ‘`#2`’ to Lua via `\__spintent_luafun_define_spunit_alias:nn`. It throws an error if the alias already exists, or if the provided target expression contains invalid syntax or unknown units.

```

775 \NewDocumentCommand \spunitalias { m m }
776 {
777   \mode_if_math:TF
778   {
779     \msg_error:nnn { spintent } { need-text-mode } { \spunitalias }
780   }
781   {
782     \__spintent_luafun_define_spunit_alias:nn { #1 } { #2 }
783     \str_if_eq:VnT \l__spintent_spunit_luaset_status_str { duplicate }
784     {
785       \msg_error:nnn { spintent } { spunitalias-duplicate } { #1 }
786     }
787     \str_if_eq:VnT \l__spintent_spunit_luaset_status_str { invalid-expr }
788     {
789       \msg_error:nnnn { spintent } { spunitalias-invalid-expr } { #1 } { #2 }
790     }
791   }
792 }

```

(End of definition for `\spunitalias`. This function is documented on page 6.)

### 11.17 The command `\spmoney`

The user command `\spmoney` is the *third* of “the big four” provided by the `spintent` package, designed to format monetary values and currencies. It delegates `{\argument}` parsing to the Lua module `spintent.lua` via `\__spintent_luafun_clean_split_and_set:n`. The module then retrieves the correct “currency symbol”, pluralization rules, and layout structures from the database.

The command accepts *optional arguments* to override the default “currency symbol” or enable “subunit” rendering. Finally, it constructs the MathML intent for MathCAT.

These string variables store the currency data returned by the Lua module `spintent.lua`. Upon a successful database lookup, the module populates these variables with the “currency symbol”, its layout position (pre or post), the appropriate singular and plural names for both the main unit and subunits, and the execution status flags. T<sub>E</sub>X then uses this information to render the visual layout and construct the correct MathML speech intents.

```

\l__spintent_spmoney_luaset_symbol_str
\l__spintent_spmoney_luaset_position_str
\l__spintent_spmoney_luaset_sing_str
\l__spintent_spmoney_luaset_plur_str
\l__spintent_spmoney_luaset_conde_str
\l__spintent_spmoney_luaset_subsing_str
\l__spintent_spmoney_luaset_subplur_str
\l__spintent_spmoney_luaset_print_iso_str
\l__spintent_spmoney_luaset_currency_arg_str
\l__spintent_spmoney_luaset_status_str

793 \str_new:N \l__spintent_spmoney_luaset_symbol_str
794 \str_new:N \l__spintent_spmoney_luaset_position_str
795 \str_new:N \l__spintent_spmoney_luaset_sing_str
796 \str_new:N \l__spintent_spmoney_luaset_plur_str
797 \str_new:N \l__spintent_spmoney_luaset_conde_str
798 \str_new:N \l__spintent_spmoney_luaset_subsing_str
799 \str_new:N \l__spintent_spmoney_luaset_subplur_str
800 \str_new:N \l__spintent_spmoney_luaset_print_iso_str
801 \str_new:N \l__spintent_spmoney_luaset_currency_arg_str
802 \str_new:N \l__spintent_spmoney_luaset_status_str

```

(End of definition for `\l__spintent_spmoney_luaset_symbol_str` and others.)

```

\l__spintent_spmoney_sub_units_bool
\l__spintent_spmoney_intent_money_str
\l__spintent_spmoney_intent_sign_str
\l__spintent_spmoney_intent_sub_unit_str
\l__spintent_spmoney_res_main_voice_str
\l__spintent_spmoney_subnum_int
\l__spintent_spmoney_extracted_curr_str
\l__spintent_spmoney_extracted_num_tl

803 \bool_new:N \l__spintent_spmoney_sub_units_bool
804 \str_new:N \l__spintent_spmoney_intent_money_str
805 \str_new:N \l__spintent_spmoney_intent_sign_str
806 \str_new:N \l__spintent_spmoney_intent_sub_unit_str
807 \str_new:N \l__spintent_spmoney_res_main_voice_str
808 \int_new:N \l__spintent_spmoney_subnum_int
809 \str_new:N \l__spintent_spmoney_extracted_curr_str
810 \tl_new:N \l__spintent_spmoney_extracted_num_tl

```

(End of definition for `\l__spintent_spmoney_sub_units_bool` and others.)

### 11.17.1 Definition of error messages

currency-no-subunits  
invalid-currency

We define two messages to handle invalid currency configurations. The first “error” warns the user if subunit rendering is requested for a currency that does not support fractional divisions (such as JPY or CLP), safely ignoring the flag. The second “error” is triggered if the provided currency identifier is not registered in the Lua database.

```

811 \msg_new:nnnn { spintent } { currency-no-subunits }
812 {
813     The~currency~'#1'~does~not~support~subunits.~
814     The~option~'sub-units=true'~is~ignored.
815 }
816 {
817     Monies~like~the~Japanese~Yen~(JPY)~or~Chilean~Peso~(CLP)~
818     do~not~have~fractional~subunits~registered~in~Lua~module.~
819 }
820 \msg_new:nnnn { spintent } { invalid-currency }
821 {
822     Invalid~currency~identifier~'#1'~in~\token_to_str:N \spmoney .
823 }
824 {
825     The~provided~symbol,~alias,~or~ISO~code~does~not~exist~in~
826     the~Lua~module.
827 }
```

(End of definition for currency-no-subunits and invalid-currency.)

### 11.17.2 Auxiliary functions for \spmoney

\\_\_spintenc\_spmoney\_set\_currency:n  
\\_\_spintenc\_spmoney\_normalize\_key:n  
\\_\_spintenc\_spmoney\_normalize\_key:V

The function `\__spintenc_spmoney_set_currency:n` validates the currency identifier provided in ‘#1’ against the database. It delegates the lookup to the helper function `\__spintenc_spmoney_normalize_key:n`, which interfaces with the Lua module `spintenc.lua`. If the module returns a not found status, an “error” is triggered. Otherwise, it stores the valid currency identifier in `\__spintenc_spmoney_luaset_currency_arg_str` for subsequent processing.

```

828 \cs_new_protected:Nn \__spintenc_spmoney_set_currency:n
829 {
830     \__spintenc_spmoney_normalize_key:n {#1}
831     \str_if_eq:VnTF \__spintenc_spmoney_luaset_status_str { notfound }
832     {
833         \msg_error:nnn { spintent } { invalid-currency } {#1}
834     }
835     { \str_set:Nn \__spintenc_spmoney_luaset_currency_arg_str {#1} }
836 }
837 \cs_new_protected:Nn \__spintenc_spmoney_normalize_key:n
838 {
839     \__spintenc_luafun_spmoney_normalize_key:n { #1 }
840 }
841 \cs_generate_variant:Nn \__spintenc_spmoney_normalize_key:n { V }
```

(End of definition for \\_\_spintenc\_spmoney\_set\_currency:n and \\_\_spintenc\_spmoney\_normalize\_key:n.)

### 11.17.3 Definition of keys for \spmoney

currency  
sub-units  
dolar-math  
cifras-sep

Now we add the keys `currency`, `sub-units`, `dolar-math` and `cifras-sep`.

```

842 \keys_define:nn { spintenc / spmoney }
843 {
844     currency      .code:n      = { \__spintenc_spmoney_set_currency:n {#1} },
845     currency      .initial:n    = { pesos },
846     currency      .value_required:n = true,
847     sub-units     .bool_set:N    = \__spintenc_spmoney_sub_units_bool,
848     sub-units     .value_required:n = true,
849     dolar-math    .bool_set:N    = \__spintenc_spmoney_dolar_math_bool,
850     dolar-math    .initial:n    = { true },
851     dolar-math    .value_required:n = true,
852     cifras-sep    .meta:nn      = { spintenc/spnum } { cifras-sep = {#1} },
853     cifras-sep    .value_required:n = true,
854     unknown       .code:n      =
855     {
856         \str_set:Nn \__spintenc_command_name_str { spmoney }
857         \__spintenc_unknown_keys_cmd:n {#1}
858     },
859 }
```

(End of definition for currency and others.)

#### 11.17.4 Internal functions for \spmoney

\\_\_spintrent\_spmoney\_validate\_money:

The function `\__spintrent_spmoney_validate_money`: verifies if the parsed input contains an embedded currency identifier by checking `\l__spintrent_luaset_has_units_str`. If a currency is detected, it extracts and normalizes the key from `\l__spintrent_luaset_arg_above_tl`. It throws an “error” if the currency is not found in the database. If valid, it updates `\l__spintrent_spmoney_luaset_currency_arg_str` and retrieves the full currency metadata from the `spintrent.lua` via `\__spintrent_luafun_spmoney_lookup_data:V`.

```

860 \cs_new_protected:Nn \__spintrent_spmoney_validate_money:
861 {
862   \str_if_eq:VnT \l__spintrent_luaset_has_units_str { true }
863   {
864     \__spintrent_spmoney_normalize_key:V \l__spintrent_luaset_arg_above_tl
865     \str_if_eq:VnTF \l__spintrent_spmoney_luaset_status_str { notfound }
866     { \msg_error:nne { spintrent } { invalid-currency } { \l__spintrent_luaset_arg_above_tl } }
867     {
868       \str_set:NV
869         \l__spintrent_spmoney_luaset_currency_arg_str
870         \l__spintrent_luaset_arg_above_tl
871     }
872   }
873   \str_set:Ne \l__spintrent_spmoney_luaset_status_str { found }
874   \__spintrent_luafun_spmoney_lookup_data:V \l__spintrent_spmoney_luaset_currency_arg_str
875 }

```

(End of definition for \\_\_spintrent\_spmoney\_validate\_money:.)

\\_\_spintrent\_spmoney\_print\_symbol:

The function `\__spintrent_spmoney_print_symbol`: renders the “*currency symbol*”. If `\l__spintrent_spmoney_luaset_print_iso_str` is `true`, it outputs the ISO code using `\mathrm`. Otherwise, it prints the “*standard symbol*”. For dollar signs ‘\$’, it checks `\l__spintrent_spmoney_dolar_math_bool` to determine whether to render it directly as a math character ‘\$’ or wrap it inside `\text`.

```

876 \cs_new_protected:Nn \__spintrent_spmoney_print_symbol:
877 {
878   \str_if_eq:VnTF \l__spintrent_spmoney_luaset_print_iso_str { true }
879   { \mathrm { \l__spintrent_spmoney_luaset_currency_arg_str } }
880   {
881     \str_if_eq:VnTF \l__spintrent_spmoney_luaset_symbol_str { $ }
882     {
883       \bool_if:NTF \l__spintrent_spmoney_dolar_math_bool
884       { \$ } { \text { \$ } }
885     }
886     { \l__spintrent_spmoney_luaset_symbol_str }
887   }
888 }

```

(End of definition for \\_\_spintrent\_spmoney\_print\_symbol:.)

\\_\_spintrent\_spmoney\_render\_layout:

The function `\__spintrent_spmoney_render_layout`: determines the layout order of the “*currency symbol*” relative to the numeric value. It evaluates `\l__spintrent_spmoney_luaset_position_str`; if set to `pre`, it renders the “*currency symbol*” *before* the number node. Otherwise, it places the number first, followed by the “*currency symbol*”.

```

889 \cs_new_protected:Nn \__spintrent_spmoney_render_layout:
890 {
891   \str_if_eq:VnTF \l__spintrent_spmoney_luaset_position_str { pre }
892   {
893     \__spintrent_spmoney_print_symbol:
894     \MathMLarg { n } { \__spintrent_spmoney_render_number_node: }
895   }
896   {
897     \MathMLarg { n } { \__spintrent_spmoney_render_number_node: }
898     \__spintrent_spmoney_print_symbol:
899   }
900 }

```

(End of definition for \\_\_spintrent\_spmoney\_render\_layout:.)

\\_\_spintrent\_spmoney\_render\_number\_node:  
\_\_spintrent\_spmoney\_render\_only\_integer\_node:

The function `\__spintrent_spmoney_render_number_node`: renders the full numeric value, encompassing both the integer and decimal components. Conversely, the function `\__spintrent_spmoney_render_only_integer_node` outputs exclusively the integer portion, omitting any decimal data.

```

901 \cs_new_protected:Nn \__spintrent_spmoney_render_number_node:
902 {
903   \__spintrent_spmoney_render_int:

```

```

904     \__spintent_snum_print_part_dec:
905   }
906 \cs_new_protected:Nn \__spintent_spmoney_render_only_integer_node:
907 {
908     \__spintent_snum_render_int:
909 }

```

(End of definition for `\__spintent_spmoney_render_number_node:` and `\__spintent_spmoney_render_only_integer_node:`.)

`\__spintent_spmoney_print_int:`

The function `\__spintent_spmoney_print_int:` evaluates the presence of an explicit sign in `\l__spintent_luaset_sign_tl`. If a sign is detected, it assigns a prefix speech intent ‘más’ or ‘menos’ and embeds the sign and subsequent layout inside a `\MathMLintent`. If no sign is present, it routes execution based on whether subunits are enabled.

```

910 \cs_new_protected:Nn \__spintent_spmoney_print_int:
911 {
912     \tl_if_empty:NTF \l__spintent_luaset_sign_tl
913     {
914         \bool_if:NTF \l__spintent_spmoney_sub_units_bool
915         { \__spintent_spmoney_set_and_print_sub_units: }
916         { \__spintent_spmoney_intent_int: }
917     }
918     {
919         \str_if_eq:VnTF \l__spintent_luaset_sign_tl { + }
920         {
921             \str_set:Nn \l__spintent_spmoney_intent_sign_str { _más:prefix ($n) }
922         }
923         {
924             \str_set:Nn \l__spintent_spmoney_intent_sign_str { _menos:prefix ($n) }
925         }
926         \__spintent_invisible_sep:
927         \MathMLintent { \l__spintent_spmoney_intent_sign_str }
928         {
929             \l__spintent_luaset_sign_tl
930             \MathMLarg { n }
931             {
932                 \bool_if:NTF \l__spintent_spmoney_sub_units_bool
933                 { \__spintent_spmoney_set_and_print_sub_units: }
934                 { \__spintent_spmoney_intent_int: }
935             }
936         }
937     }
938 }

```

(End of definition for `\__spintent_spmoney_print_int:`.)

`\__spintent_spmoney_intent_int:`

The function `\__spintent_spmoney_intent_int:` determines the correct grammatical inflection for the currency string (singular, plural, or the prepositional ‘de’ variant for exact millions). It evaluates the integer component, decimal boundaries, and zero-fills to select the appropriate string register, applying it as a postfix `MathML` intent over the layout rendered by `\__spintent_spmoney_render_layout:`.

```

939 \cs_new_protected:Nn \__spintent_spmoney_intent_int:
940 {
941     \tl_if_empty:NTF \l__spintent_luaset_dec_sep_tl
942     {
943         \str_if_eq:VnTF \l__spintent_luaset_only_part_int_str { 1 }
944         {
945             \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_sing_str
946         }
947         {
948             \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plur_str
949         }
950         \str_if_eq:VnT \l__spintent_luaset_millions_str { true }
951         {
952             \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_conde_s
953         }
954     }
955     {
956         \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plur_str
957         \str_if_eq:VnT \l__spintent_luaset_millions_str { true }
958         {
959             \str_if_eq:VnTF \l__spintent_luaset_dec_is_all_zeros_str { true }
960             {

```

```

961         \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_con
962     }
963     {
964         \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plu
965     }
966 }
967 }
968 \__spintent_invisible_sep:
969 \MathMLint { \l__spintent_spmoney_intent_money_str :postfix($n) }
970 {
971     \__spintent_spmoney_render_layout:
972 }
973 }

```

(End of definition for `\__spintent_spmoney_intent_int:`)

`\__spintent_spmoney_check_sub_units:`  
`spintent_spmoney_set_and_print_sub_units:`

The function `\__spintent_spmoney_check_sub_units:` verifies whether the selected currency supports “*subunits*”. If the database returns empty subunit data, it triggers a warning and automatically disables the “*subunit*” rendering flag.

The function `\__spintent_spmoney_set_and_print_sub_units:` processes and formats the speech intents when “*subunits*” are active. It normalizes the decimal component (for instance, padding a single-digit decimal like ‘5’ to ‘50’), evaluates the correct singular or plural inflections for both the main “*currency*” and the “*subunits*”, and delegates the final MathML tree assembly to `\__spintent_spmoney_build_tree:`.

```

974 \cs_new_protected:Nn \__spintent_spmoney_check_sub_units:
975 {
976     \str_if_empty:NT \l__spintent_spmoney_luaset_subsing_str
977     {
978         \msg_warning:nne { spintent } { currency-no-subunits }
979         { \l__spintent_spmoney_luaset_currency_arg_str }
980         \bool_set_false:N \l__spintent_spmoney_sub_units_bool
981     }
982 }
983 \cs_new_protected:Nn \__spintent_spmoney_set_and_print_sub_units:
984 {
985     \int_compare:nTF { \str_count:N \l__spintent_luaset_only_part_dec_str == 1 }
986     {
987         \int_set:Nn \l__spintent_spmoney_subnum_int
988         { \l__spintent_luaset_only_part_dec_str * 10 }
989         \str_set:Ne \l__spintent_luaset_only_part_dec_str
990         { \l__spintent_luaset_only_part_dec_str 0 }
991     }
992     {
993         \int_set:Nn \l__spintent_spmoney_subnum_int
994         { \l__spintent_luaset_only_part_dec_str }
995     }
996     \int_compare:nTF { \l__spintent_spmoney_subnum_int == 1 }
997     {
998         \str_set:Ne \l__spintent_spmoney_intent_sub_unit_str
999         { \l__spintent_spmoney_luaset_subsing_str :postfix($sub) }
1000     }
1001     {
1002         \str_set:Ne \l__spintent_spmoney_intent_sub_unit_str
1003         { \l__spintent_spmoney_luaset_subplur_str :postfix($sub) }
1004     }
1005     \str_if_eq:VnTF \l__spintent_luaset_only_part_int_str { 1 }
1006     {
1007         \str_set:NV \l__spintent_spmoney_res_main_voice_str
1008         \l__spintent_spmoney_luaset_sing_str
1009     }
1010     {
1011         \str_set:NV \l__spintent_spmoney_res_main_voice_str
1012         \l__spintent_spmoney_luaset_plur_str
1013     }
1014     \str_if_eq:VnT \l__spintent_luaset_millions_str { true }
1015     {
1016         \str_set:NV \l__spintent_spmoney_res_main_voice_str
1017         \l__spintent_spmoney_luaset_conde_str
1018     }
1019     \__spintent_spmoney_build_tree:
1020 }

```

(End of definition for `\__spintent_spmoney_check_sub_units:` and `\__spintent_spmoney_set_and_print_sub_units:`.)

```
\__spintent_spmoney_build_tree:
\__spintent_spmoney_print_sep_con:
\__spintent_spmoney_print_dec_sub_unit:
```

The function `\__spintent_spmoney_build_tree:` structures the final MathML tree assembly for fractional “*currencies*”. It arranges the integer component, the decimal separator, and the “*subunit*” component based on the layout position (pre or post) of the “*currency symbol*”.

Furthermore, `\__spintent_spmoney_print_sep_con:` applies the specific speech intent ‘con’ to the decimal separator, while `\__spintent_spmoney_print_dec_sub_unit:` binds the computed “*subunit*” intent to the decimal digits, ensuring accurate accessibility output.

```
1021 \cs_new_protected:Nn \__spintent_spmoney_build_tree:
1022 {
1023   \str_if_eq:VnTF \__spintent_spmoney_luaset_position_str { pre }
1024   {
1025     \MathMLintent { \__spintent_spmoney_res_main_voice_str :postfix($n) }
1026     {
1027       \__spintent_spmoney_print_symbol:
1028       \MathMLarg { n } { \__spintent_spmoney_render_only_integer_node: }
1029     }
1030     \__spintent_spmoney_print_sep_con:
1031     \__spintent_spmoney_print_dec_sub_unit:
1032   }
1033   {
1034     \MathMLintent { \__spintent_spmoney_res_main_voice_str :postfix($n) }
1035     {
1036       \__spintent_inner_sep:
1037       \MathMLarg { n } { \__spintent_spmoney_render_only_integer_node: }
1038     }
1039     \__spintent_spmoney_print_sep_con:
1040     \__spintent_spmoney_print_dec_sub_unit:
1041     \MathMLintent {:silent} { \__spintent_spmoney_print_symbol: }
1042   }
1043 }
1044 \cs_new_protected:Nn \__spintent_spmoney_print_sep_con:
1045 {
1046   \MathMLintent { _con }
1047   {
1048     \mathord { \__spintent_luaset_dec_sep_tl }
1049   }
1050 }
1051 \cs_new_protected:Nn \__spintent_spmoney_print_dec_sub_unit:
1052 {
1053   \MathMLintent { \__spintent_spmoney_intent_sub_unit_str }
1054   {
1055     \__spintent_inner_sep:
1056     \MathMLarg { sub }
1057     {
1058       \__spintent_spmoney_render_dec:
1059     }
1060   }
1061 }
```

(End of definition for `\__spintent_spmoney_build_tree:`, `\__spintent_spmoney_print_sep_con:`, and `\__spintent_spmoney_print_dec_sub_unit:`.)

```
\__spintent_spmoney_parse_and_route:n
```

The internal function `\__spintent_spmoney_parse_and_route:n` processes the `{\argument}` passed in ‘#1’ using Lua module to extract the numeric value and any “*currency*” identifier. If a “*currency*” is found (`\__spintent_spmoney_extracted_curr_str`), it sets the corresponding variables. It then calls `\__spintent_spmoney_validate_money:` to verify the “*currency*”. If “*subunit*” rendering is enabled (`\__spintent_spmoney_sub_units_bool`), it calls `\__spintent_spmoney_check_sub_units:`. Finally, it calls `\__spintent_spmoney_print_int:` to format the output.

```
1062 \cs_new_protected:Nn \__spintent_spmoney_parse_and_route:n
1063 {
1064   \__spintent_luafun_spmoney_prepare_input:n { #1 }
1065   \__spintent_luafun_clean_split_and_set:V \__spintent_spmoney_extracted_num_tl
1066   \str_if_empty:NF \__spintent_spmoney_extracted_curr_str
1067   {
1068     \str_set:Ne \__spintent_luaset_has_units_str { true }
1069     \str_set:NV \__spintent_luaset_arg_above_tl \__spintent_spmoney_extracted_curr_str
1070   }
1071   \__spintent_spmoney_validate_money:
1072   \bool_if:NT \__spintent_spmoney_sub_units_bool
```

```

1073     { \__spintent_spmoney_check_sub_units: }
1074     \__spintent_spmoney_print_int:
1075 }

```

(End of definition for \\_\_spintent\_spmoney\_parse\_and\_route:n.)

**\spmoney** The public document command **\spmoney** formats monetary values with accessible speech intents. It accepts an optional argument ‘#1’ for local key-value configurations and a mandatory argument ‘#2’ containing the numeric value and optional embedded currency identifier. It first evaluates the typesetting context, triggering a fatal error if executed outside of math mode. Upon validation, it establishes a local  $\TeX$  group to safely isolate the key assignments, delegates the raw input string to **\\_\_spintent\_spmoney\_parse\_and\_route:n** for parsing and rendering, and subsequently closes the group to prevent scope leakage.

```

1076 \NewDocumentCommand \spmoney { 0{} m }
1077 {
1078     \mode_if_math:TF
1079     {
1080         \group_begin:
1081         \keys_set:nn { spintent / spmoney } { #1 }
1082         \__spintent_spmoney_parse_and_route:n { #2 }
1083         \group_end:
1084     }
1085     { \msg_error:nnn { spintent } { need-math-mode } { \spmoney } }
1086 }

```

(End of definition for \spmoney. This function is documented on page 6.)

## 11.18 The command \spshort

The user command **\spshort** is the *fourth* and *final* of “the big four”. It manages the semantic injection and typographical formatting of *text mode* abbreviations, ordinals, and acronyms. Unlike *math mode* commands, which rely on **MathML** attributes, this command uses the PDF *tagging tools* provided by the **tagpdf**[29] package. Specifically, it uses the **E** (expanded) key to ensure that NVDA interpret the full form of the abbreviation, ordinal, or acronym in *text mode*.

These internal string variables serve as the primary data interface with the Lua module **spintent.lua**. During processing, the Lua module “stores” within these registers the execution status, the targeted typographical layout parameters, the phonetic replacement text for the **E** key intended for NVDA, the final sanitized output string, and the isolated structural components for ordinals (base and suffix).

```

1087 \str_new:N \__spintent_spshort_luaset_status_str
1088 \str_new:N \__spintent_spshort_luaset_layout_str
1089 \str_new:N \__spintent_spshort_luaset_expanded_str
1090 \str_new:N \__spintent_spshort_luaset_output_str
1091 \str_new:N \__spintent_spshort_luaset_base_str
1092 \str_new:N \__spintent_spshort_luaset_suffix_str

```

(End of definition for \\_\_spintent\_spshort\_luaset\_status\_str and others.)

### 11.18.1 Definition of error messages for \spshort

**unknown-abbreviation** The internal “error” message **unknown-abbreviation** is triggered when the command **\spshort** encounters an invalid lookup state from the Lua module. It acts as a strict fallback exception when the provided  $\langle argument \rangle$  cannot be mapped to any registered database entry in **spintent.lua** nor successfully resolved through the grammatical ordinal expansion rules.

```

1093 \msg_new:nnnn { spintent } { unknown-abbreviation }
1094 { The~abbreviation~or~ordinal~' #1 ' ~is~invalid~or~not~recognized. }
1095 { Verify~the~spelling~or~check~the~grammar~rules~for~ordinals. }

```

(End of definition for unknown-abbreviation.)

### 11.18.2 Accessibility interface

**\\_\_spintent\_spshort\_tag\_span:nn** The internal function **\\_\_spintent\_spshort\_tag\_span:nn** processes the  $\langle argument \rangle$  passed in ‘#2’ using the tools provided by **tagpdf**. By utilizing the **tag** and **E** keys, it creates a native PDF /E ( #1 ) structure and expands the  $\langle argument \rangle$  passed in ‘#1’ for proper reading with NVDA.

```

1096 \cs_new_protected:Nn \__spintent_spshort_tag_span:nn
1097 {
1098     \tag_mc_end_push:
1099     \tag_struct_begin:n { tag=Span, E={ #1 } }
1100     \tag_mc_begin:n { tag=Span }
1101     \tag_suspend:n { \spshort }
1102     #2
1103     \tag_resume:n { \spshort }
1104     \tag_mc_end:

```

```

1105     \tag_struct_end:
1106     \tag_mc_begin_pop:n {}
1107 }
1108 \cs_generate_variant:Nn \__spintent_spshort_tag_span:nn { en }

```

(End of definition for \\_\_spintent\_spshort\_tag\_span:nn.)

### 11.18.3 Definition of keys for \spshort

Now we add the keys `small-caps` and `read-arg`.

```

small-caps read-arg 1109 \keys_define:nn { spintent / spshort }
1110 {
1111     small-caps .bool_set:N = \__spintent_spshort_smallcaps_bool,
1112     small-caps .initial:n = false,
1113     small-caps .value_required:n = true,
1114     read-arg .str_set:N = \__spintent_spshort_read_arg_str,
1115     read-arg .initial:n = {},
1116     read-arg .value_required:n = true,
1117     unknown .code:n =
1118     {
1119         \str_set:Nn \__spintent_command_name_str { spshort }
1120         \__spintent_unknown_keys_cmd:n {#1}
1121     },
1122 }

```

(End of definition for `small-caps` and `read-arg`.)

### 11.18.4 Internal functions for \spshort

The internal function `\__spintent_spshort_print:nn` acts as the primary formatter for text abbreviations. It clears the string variable `\__spintent_spshort_read_arg_str` and evaluates the optional `<keys>` passed in ‘#1’. It then checks if a manual reading override was provided via those keys. If that string variable remains empty, it calls `\__spintent_spshort_process_and_render:n` to process the `{<argument>}` passed in ‘#2’. Otherwise, if an override is detected, it directly delegates the execution to `\__spintent_spshort_render_bypass:n`.

```

1123 \cs_new_protected:Nn \__spintent_spshort_print:nn
1124 {
1125     \str_clear:N \__spintent_spshort_read_arg_str
1126     \keys_set:nn { spintent / spshort } {#1}
1127     \str_if_empty:NTF \__spintent_spshort_read_arg_str
1128     {
1129         \__spintent_spshort_process_and_render:n {#2}
1130     }
1131     {
1132         \__spintent_spshort_render_bypass:n {#2}
1133     }
1134 }

```

(End of definition for `\__spintent_spshort_print:nn`.)

The internal function `\__spintent_spshort_process_and_render:n` interfaces with the Lua module `spintent.lua` function `\__spintent_luafun_spshort_process:n` to resolve the abbreviation or ordinal `{<argument>}` passed in ‘#1’.

It subsequently evaluates the execution state stored within the string variable `\__spintent_spshort_luaset_status` and dynamically routes execution to the corresponding rendering subroutine `success`, `fallback` or `error`. Any unrecognized status automatically defaults to the “error” handler.

```

1135 \cs_new_protected:Nn \__spintent_spshort_process_and_render:n
1136 {
1137     \__spintent_luafun_spshort_process:n {#1}
1138     \str_case:VnF \__spintent_spshort_luaset_status_str
1139     {
1140         { success } { \__spintent_spshort_render_success: }
1141         { fallback } { \__spintent_spshort_render_fallback: }
1142         { error } { \__spintent_spshort_render_error:n {#1} }
1143     }
1144     { \__spintent_spshort_render_error:n {#1} }
1145 }

```

(End of definition for `\__spintent_spshort_process_and_render:n`.)

```

\__spintent_spshort_render_success:
\__spintent_spshort_render_fallback:
\__spintent_spshort_render_bypass:n
\__spintent_spshort_render_error:n

```

The function `\__spintent_spshort_render_success:` injects the key `E` and executes the targeted typographical layout. The `\__spintent_spshort_render_fallback:` function handles default cases, applying optional formatting such as `SMALL CAPS` to unrecognized or partially resolved terms.

Conversely, the function `\__spintent_spshort_render_bypass:n` directly maps the user-defined “*speech*” to the  $\{\langle argument \rangle\}$ , bypassing the automated lookup Lua module. Finally, the function `\__spintent_spshort_rende` triggers a structural exception while yielding the unformatted  $\{\langle argument \rangle\}$  to prevent fatal compilation.

```

1146 \cs_new_protected:Nn \__spintent_spshort_render_success:
1147 {
1148   \__spintent_spshort_tag_span:en
1149   { \l__spintent_spshort_luaset_expanded_str }
1150   { \__spintent_spshort_execute_layout: }
1151 }
1152 \cs_new_protected:Nn \__spintent_spshort_render_fallback:
1153 {
1154   \__spintent_spshort_tag_span:en
1155   { \l__spintent_spshort_luaset_expanded_str }
1156   {
1157     \bool_if:NTF \l__spintent_spshort_smallcaps_bool
1158     { \textsc{ \l__spintent_spshort_luaset_output_str } }
1159     { \l__spintent_spshort_luaset_output_str }
1160   }
1161 }
1162 \cs_new_protected:Nn \__spintent_spshort_render_bypass:n
1163 {
1164   \__spintent_spshort_tag_span:en { \l__spintent_spshort_read_arg_str } { #1 }
1165 }
1166 \cs_new_protected:Nn \__spintent_spshort_render_error:n
1167 {
1168   \msg_error:nnn { spintent } { unknown-abbreviation } {#1}
1169 }

```

(End of definition for `\__spintent_spshort_render_success:` and others.)

```

\__spintent_spshort_execute_layout:

```

The internal function `\__spintent_spshort_execute_layout:` executes the typographical layout. It evaluates the descriptor “*stored*” within the string variable `\l__spintent_spshort_luaset_layout_str` (such as `linear_regular`, `superscript`, or `small_caps_pure`) and maps it to the corresponding  $\text{\TeX}$  formatting. Furthermore, it manages complex nested configurations, dynamically applying combinations such as superscripts integrated with conditional `SMALL CAPS` based on the active boolean variables.

```

1170 \cs_new_protected:Nn \__spintent_spshort_execute_layout:
1171 {
1172   \str_case:VnF \l__spintent_spshort_luaset_layout_str
1173   {
1174     { linear_regular }
1175     {
1176       \bool_if:NTF \l__spintent_spshort_smallcaps_bool
1177       { \textsc{ \l__spintent_spshort_luaset_output_str } }
1178       { \l__spintent_spshort_luaset_output_str }
1179     }
1180     { linear_caps }
1181     { \l__spintent_spshort_luaset_output_str }
1182     { linear_mixed }
1183     { \l__spintent_spshort_luaset_output_str }
1184     { superscript }
1185     {
1186       \l__spintent_spshort_luaset_base_str
1187       \bool_if:NTF \l__spintent_spshort_smallcaps_bool
1188       {
1189         \textsuperscript{
1190           \textsc{
1191             \str_lowercase:V \l__spintent_spshort_luaset_suffix_str
1192           }
1193         }
1194       }
1195       {
1196         \textsuperscript{ \l__spintent_spshort_luaset_suffix_str }
1197       }
1198     }
1199     { small_caps_pure }
1200     {
1201       \bool_if:NTF \l__spintent_spshort_smallcaps_bool

```

```

1202         {
1203             \textsc{
1204                 \str_lowercase:V \l__spintent_spsshort_luaset_output_str
1205             }
1206         }
1207         { \l__spintent_spsshort_luaset_output_str }
1208     }
1209     { acronym_long }
1210     { \l__spintent_spsshort_luaset_output_str }
1211 }
1212 { \l__spintent_spsshort_luaset_output_str }
1213 }

```

(End of definition for `\l__spintent_spsshort_execute_layout:`.)

`\spsshort` The public document command `\spsshort` manages the accessible formatting of *text mode* abbreviations, acronyms, and ordinals. It first triggers an error if invoked within a *math mode*, upon validation, puts `\mode_leave_vertical:` and establishes a local group to call the function `\l__spintent_spsshort_print:nn` for processing and printing.

```

1214 \NewDocumentCommand \spsshort { 0{} m }
1215 {
1216     \mode_if_math:TF
1217     {
1218         \msg_error:nnn { spintent } { need-text-mode } { \spsshort }
1219     }
1220     {
1221         \mode_leave_vertical:
1222         \group_begin:
1223         \l__spintent_spsshort_print:nn {#1} {#2}
1224         \group_end:
1225     }
1226 }

```

(End of definition for `\spsshort`. This function is documented on page 7.)

## 11.19 The command `\spsiglo`

The public document command `\spsiglo` provides a semantic interface for typesetting century designations (e.g., siglo XXI). It accepts both Arabic and Roman numerals as `{⟨argument⟩}`, interfacing with the Lua module `spintent.lua` to validate, parse, and normalize chronological values.

This command work in dual-mode context awareness, operating uniformly across both *text mode* and *math mode*. When executed within *math mode*, it constructs an accessible **MathML** intent expression tree. Conversely, when invoked in standard *text mode*, it encapsulates the rendered output within a native PDF tagging structure (such as `/Alt (⟨argument⟩)` containing an `alt` key.

`\l__spintent_spsiglo_luaset_error_str` These internal string variables handle data transfer and synchronization with the Lua module `spintent.lua`. They isolate the exception evaluation states, the Arabic numeral required for speech synthesis, the lowercase Roman numeral designated for small caps typesetting via `\textsc`, and the localized chronological era modifier.

```

\l__spintent_spsiglo_luaset_arabic_str
\l__spintent_spsiglo_luaset_roman_min_str
\l__spintent_spsiglo_print_era_str
1227 \str_new:N \l__spintent_spsiglo_luaset_error_str
1228 \str_new:N \l__spintent_spsiglo_luaset_arabic_str
1229 \str_new:N \l__spintent_spsiglo_luaset_roman_min_str
1230 \str_new:N \l__spintent_spsiglo_print_era_str

```

(End of definition for `\l__spintent_spsiglo_luaset_error_str` and others.)

### 11.19.1 Definition of error messages for `\spsiglo`

`invalid-century-format` This internal error message is issued when the provided century identifier fails both Arabic and Roman numeral validation. It is triggered when the Lua module `spintent.lua` cannot map the `{⟨argument⟩}` to an integer within the valid range of 1 to 4000, preventing incorrect speech synthesis.

```

1231 \msg_new:nnnn { spintent } { invalid-century-format }
1232 {
1233     Invalid~century~format~in~'~#1'~.
1234 }
1235 {
1236     The~argument~'~#2'~is~not~a~valid~Arabic~or~Roman~numeral.~
1237     Ensure~the~value~is~between~1~and~4000.
1238 }

```

(End of definition for `invalid-century-format`.)

### 11.19.2 Definition of keys for \spsiglo

`print-era` Now we add the key `print-era`. User configuration determining if an era designation is appended, available values are `none`, `ac` (antes de cristo) and `dc` (después de cristo).

```

1239 \keys_define:nn { spintent / spsiglo }
1240 {
1241   print-era           .choices:nn =
1242     { none , ac , dc }
1243     {
1244       \int_case:nn { \l_keys_choice_int }
1245       {
1246         {1} { \str_set:Nn \l__spintent_spsiglo_print_era_str { none } }
1247         {2} { \str_set:Nn \l__spintent_spsiglo_print_era_str { ac } }
1248         {3} { \str_set:Nn \l__spintent_spsiglo_print_era_str { dc } }
1249       }
1250     },
1251   print-era           .initial:n = none,
1252   print-era           .value_required:n = true,
1253   print-era / unknown .code:n =
1254     \msg_error:nneee { spintent } { unknown-choice }
1255     { print-era } { none, ~ac, ~dc } { \exp_not:n {#1} },
1256   unknown             .code:n =
1257     {
1258       \str_set:Nn \l__spintent_command_name_str { spsiglo }
1259       \__spintent_unknown_keys_cmd:n {#1}
1260     },
1261 }

```

(End of definition for `print-era`.)

### 11.19.3 Internal functions for \spsiglo

`\__spintent_spsiglo_render_simple:nn` The functions `\__spintent_spsiglo_render_simple:nn` and `\__spintent_spsiglo_render_with_era:nnn` generate MathML structures when century designations are evaluated within *math mode*. They construct MathML intent structures, mapping the Arabic numerals to their phonetic equivalents. Furthermore, `\__spintent_spsiglo_render_with_era:nnn` integrates the era modifiers `ac` or `dc`. It adds invisible spacing and postfix intents to ensure correct NVDA/MathCAT pronunciation.

```

1262 \cs_new_protected:Nn \__spintent_spsiglo_render_simple:nn
1263 {
1264   \MathMLintent { #1:number } { \text { \textsc {#2} } } }
1265 }
1266 \cs_generate_variant:Nn \__spintent_spsiglo_render_simple:nn { VV }
1267 \cs_new_protected:Nn \__spintent_spsiglo_render_with_era:nnn
1268 {
1269   \str_if_eq:nnTF { #1 } { ac }
1270   {
1271     \MathMLintent { antes-de-cristo:postfix($z) }
1272     {
1273       \__spintent_inner_sep:
1274       \MathMLarg { z }
1275       {
1276         \MathMLintent { #2:number }
1277         {
1278           \text { \textsc {#3}~a.~C. }
1279         }
1280       }
1281     }
1282   }
1283   {
1284     \MathMLintent { después-de-cristo:postfix($z) }
1285     {
1286       \__spintent_inner_sep:
1287       \MathMLarg { z }
1288       {
1289         \MathMLintent { #2:number }
1290         {
1291           \text { \textsc {#3}~d.~C. }
1292         }
1293       }
1294     }
1295   }
1296 }

```

```
1297 \cs_generate_variant:Nn \__spintent_spsiglo_render_with_era:nnn { VVV }
```

(End of definition for `\__spintent_spsiglo_render_simple:nn` and `\__spintent_spsiglo_render_with_era:nnn`.)

```
\__spintent_spsiglo_tag_span:nn
\__spintent_spsiglo_tag_span:en
```

The function `\__spintent_spsiglo_tag_span:nn` handles century formatting within *text mode*. It utilizes the `tagpdf` package to enclose the content inside a PDF /Alt (`#1`) structure. By passing the `{⟨argument⟩}` to the `alt` key, it embeds the chronological value directly into the PDF, ensuring correct NVDA/MathCAT pronunciation without relying on MathML intents.

```
1298 \cs_new_protected:Nn \__spintent_spsiglo_tag_span:nn
1299 {
1300   \tag_mc_end_push:
1301   \tag_struct_begin:n { tag=Span, alt={ #1 } }
1302   \tag_mc_begin:n { tag=Span }
1303   \tag_suspend:n { \spsiglo }
1304   #2
1305   \tag_resume:n { \spsiglo }
1306   \tag_mc_end:
1307   \tag_struct_end:
1308   \tag_mc_begin_pop:n {}
1309 }
1310 \cs_generate_variant:Nn \__spintent_spsiglo_tag_span:nn { en }
```

(End of definition for `\__spintent_spsiglo_tag_span:nn`.)

```
\__spintent_spsiglo_print_math:n
\__spintent_spsiglo_print_text:n
```

The functions `\__spintent_spsiglo_print_math:n` and `\__spintent_spsiglo_print_text:n` format century designations based on the active mode. Both first check the “error” status in the variable `\l__spintent_spsiglo_luaset_error_str` returned by the Lua module `spintent.lua` and issue an “error” if validation fails.

If the `{⟨argument⟩}` is valid, the function `\__spintent_spsiglo_print_math:n` operates in *math mode*, applying MathML structures depending on whether an era modifier is present. Conversely, the function `\__spintent_spsiglo_print_text:n` operates in *text mode*, utilizing the `tag` and `alt` keys to expand the value of the `print-era` option into full text (e.g., “antes de cristo”) for NVDA/MathCAT.

```
1311 \cs_new_protected:Nn \__spintent_spsiglo_print_math:n
1312 {
1313   \str_if_eq:VnTF \l__spintent_spsiglo_luaset_error_str { true }
1314   {
1315     \msg_error:nnnn { spintent } { invalid-century-format }
1316     { \spsiglo } { #1 }
1317   }
1318   {
1319     \str_if_eq:VnTF \l__spintent_spsiglo_print_era_str { none }
1320     {
1321       \__spintent_spsiglo_render_simple:VV
1322       \l__spintent_spsiglo_luaset_arabic_str
1323       \l__spintent_spsiglo_luaset_roman_min_str
1324     }
1325     {
1326       \__spintent_spsiglo_render_with_era:VVV
1327       \l__spintent_spsiglo_print_era_str
1328       \l__spintent_spsiglo_luaset_arabic_str
1329       \l__spintent_spsiglo_luaset_roman_min_str
1330     }
1331   }
1332 }
1333 \cs_new_protected:Nn \__spintent_spsiglo_print_text:n
1334 {
1335   \str_if_eq:VnTF \l__spintent_spsiglo_luaset_error_str { true }
1336   {
1337     \msg_error:nnnn { spintent } { invalid-century-format }
1338     { \spsiglo } { #1 }
1339   }
1340   {
1341     \str_case:Vn \l__spintent_spsiglo_print_era_str
1342     {
1343       { none }
1344       {
1345         \__spintent_spsiglo_tag_span:en
1346         { \l__spintent_spsiglo_luaset_arabic_str }
1347         { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } }
1348       }
1349       { ac }
```

```

1350         {
1351             \__spintent_spsiglo_tag_span:en
1352             { \l__spintent_spsiglo_luaset_arabic_str ~ antes ~ de ~ cristo }
1353             { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } ~ a.~C. }
1354         }
1355         { dc }
1356         {
1357             \__spintent_spsiglo_tag_span:en
1358             { \l__spintent_spsiglo_luaset_arabic_str ~ después ~ de ~ cristo }
1359             { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } ~ d.~C. }
1360         }
1361     }
1362 }
1363 }

```

(End of definition for \\_\_spintent\_spsiglo\_print\_math:n and \\_\_spintent\_spsiglo\_print\_text:n.)

**\spsiglo** The public command `\spsiglo` is the main interface for century typesetting. It opens a local group and evaluates any optional `<keys>` passed in ‘#1’, then executes the function `\__spintent_luafun_spsiglo_parse:n` defined in the Lua module `spintent.lua` passed in ‘#2’ and formats the output for *math mode* or *text mode*.

```

1364 \NewDocumentCommand \spsiglo { 0{} m }
1365 {
1366     \group_begin:
1367     \keys_set:nn { spintent / spsiglo } { #1 }
1368     \__spintent_luafun_spsiglo_parse:n { #2 }
1369     \mode_if_math:TF
1370     { \__spintent_spsiglo_print_math:n { #2 } }
1371     {
1372         \mode_leave_vertical:
1373         \__spintent_spsiglo_print_text:n { #2 }
1374     }
1375     \group_end:
1376 }

```

(End of definition for `\spsiglo`. This function is documented on page 11.)

## 11.20 The command \sptime

The command `\sptime` is the main public interface for typesetting and tagging time expressions. It parses time string formats (such as HH:MM or HH:MM:SS) using the Lua function `\__spintent_luafun_sptime_parse:n`. This separates the hours, minutes, seconds, and fractions to construct a MathML *intent* tree for NVDA/MathCAT.

```

\l__spintent_sptime_luaset_seconds_str
\l__spintent_sptime_luaset_has_seconds_str
\l__spintent_sptime_luaset_error_str
\l__spintent_sptime_luaset_is_fraction_str
\l__spintent_sptime_luaset_base_str

```

These internal string variables store the values returned from the Lua module `spintent.lua`. They hold the extracted time units, error flags, and boolean indicators (such as the presence of seconds or decimal fractions) used during formatting.

```

1377 \str_new:N \l__spintent_sptime_luaset_seconds_str
1378 \str_new:N \l__spintent_sptime_luaset_has_seconds_str
1379 \str_new:N \l__spintent_sptime_luaset_error_str
1380 \str_new:N \l__spintent_sptime_luaset_is_fraction_str
1381 \str_new:N \l__spintent_sptime_luaset_base_str

```

(End of definition for `\l__spintent_sptime_luaset_seconds_str` and others.)

### 11.20.1 Definition of error messages

`invalid-time-format`

The error message `invalid-time-format` defines the text displayed when an invalid time input is detected. It is triggered when an argument fails the format checks or contains values outside normal ranges, such as hours exceeding 23 or minutes and seconds exceeding 59.

```

1382 \msg_new:nnnn { spintent } { invalid-time-format }
1383 {
1384     Invalid~time~format~in~#1. \\\
1385     The~argument~'~#2'~does~not~match~any~supported~time~format.
1386 }
1387 {
1388     Allowed~formats:~HH:MM,~HH:MM:SS,~or~HH:MM:SS.ss. \\\
1389     Please~verify~that~hours~are~0-23~and~minutes/seconds~are~0-59.
1390 }

```

(End of definition for `invalid-time-format`.)

### 11.20.2 Internal functions for \sptime

\\_\_spintent\_sptime\_build\_seconds:nn

The internal function `\__spintent_sptime_build_seconds:nn` builds the MathML structure for the seconds component of a time expression. It wraps the numeric value ‘#2’ in speech intent tags, applying a connector prefix (passed in ‘#1’) and the “segundos” postfix so it is read correctly by NVDA/MathCAT.

```

1391 \cs_new_protected:Nn \__spintent_sptime_build_seconds:nn
1392 {
1393   \MathMLintent { #1:prefix($z) }
1394   {
1395     \text { : }
1396     \MathMLarg { z }
1397     {
1398       \MathMLintent { _segundos:postfix($m) }
1399       {
1400         \__spintent_inner_sep:
1401         \MathMLarg { m }
1402         {
1403           \text { #2 }
1404         }
1405       }
1406     }
1407   }
1408 }

```

(End of definition for \\_\_spintent\_sptime\_build\_seconds:nn.)

\\_\_spintent\_sptime\_build\_tree:

The internal function `\__spintent_sptime_build_tree:` builds the complete MathML structure for the time expression. It checks the string variables `\l__spintent_sptime_luaset_has_seconds_str` and `\l__spintent_sptime_luaset_is_fraction_str` to determine if the time includes seconds or decimal fractions. Based on these values, it inserts the base time string and either calls `\__spintent_sptime_build_seconds:` with the appropriate prefix (‘con’ or ‘minutos-con’), or simply appends the ‘minutos’ intent tag if no seconds are present.

```

1409 \cs_new_protected:Nn \__spintent_sptime_build_tree:
1410 {
1411   \str_if_eq:VnTF \l__spintent_sptime_luaset_has_seconds_str { true }
1412   {
1413     \MathMLintent { :time }
1414     {
1415       \text { \l__spintent_sptime_luaset_base_str }
1416     }
1417     \str_if_eq:VnTF \l__spintent_sptime_luaset_is_fraction_str { true }
1418     {
1419       \__spintent_sptime_build_seconds:nn { con }
1420       { \l__spintent_sptime_luaset_seconds_str }
1421     }
1422     {
1423       \__spintent_sptime_build_seconds:nn { _minutos-con }
1424       { \l__spintent_sptime_luaset_seconds_str }
1425     }
1426   }
1427   {
1428     \MathMLintent { :time }
1429     {
1430       \text { \l__spintent_sptime_luaset_base_str }
1431     }
1432     \str_if_eq:VnTF \l__spintent_sptime_luaset_is_fraction_str { false }
1433     {
1434       \MathMLintent { _minutos }
1435       {
1436         \c__spintent_invisible_sep_int
1437       }
1438     }
1439   }
1440 }

```

(End of definition for \\_\_spintent\_sptime\_build\_tree:.)

\sptime

The command `\sptime` is the public interface for formatting time expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it parses the input using `\__spintent_luafun_sptime_parse:n` and checks the error status in `\l__spintent_sptime_luaset_error_str`. If

an invalid format is detected, it issues the `invalid-time-format` error; otherwise, it calls `\__spintent_sptime_build_tree:` to assemble the MathML output.

```

1441 \NewDocumentCommand \sptime { m }
1442 {
1443   \mode_if_math:TF
1444   {
1445     \group_begin:
1446     \__spintent_luafun_sptime_parse:n { #1 }
1447     \str_if_eq:VnTF \__spintent_sptime_luaset_error_str { true }
1448     {
1449       \msg_error:nnnn { spintent } { invalid-time-format }
1450       { \sptime } { #1 }
1451     }
1452     {
1453       \__spintent_sptime_build_tree:
1454     }
1455   \group_end:
1456 }
1457 { \msg_error:nnn { spintent } { need-math-mode } { \sptime } }
1458 }

```

(End of definition for `\sptime`. This function is documented on page 11.)

### 11.21 The command `\spdate`

The command `\spdate` is the main public command for typesetting and tagging date expressions. It parses date string formats (such as DD/MM/YYYY or YYYY-MM-DD) using the Lua function `\__spintent_luafun_spdate_parse:n`. This verifies the input and constructs a MathML `intent` tree for NVDA/MathCAT.

```

\__spintent_spdate_luaset_output_str
\__spintent_spdate_luaset_error_str

```

These internal string variables store the values returned from the Lua module `spintent.lua`. They hold the formatted date string and the error status used during formatting.

```

1459 \str_new:N \__spintent_spdate_luaset_output_str
1460 \str_new:N \__spintent_spdate_luaset_error_str

```

(End of definition for `\__spintent_spdate_luaset_output_str` and `\__spintent_spdate_luaset_error_str`.)

#### 11.21.1 Definition of error messages

```
invalid-date-format
```

The error message `invalid-date-format` defines the text displayed when an invalid date input is detected. It is triggered when an argument fails the format checks or represents an invalid calendar date, such as `31/02/2024` or using an unsupported pattern like `YYYY/MM/DD`.

```

1461 \msg_new:nnnn { spintent } { invalid-date-format }
1462 {
1463   Invalid~date~or~unsupported~format~in~#1. \
1464   The~argument~'#2'~does~not~match~a~supported~format.
1465 }
1466 {
1467   Allowed~formats:~DD\slash MM\slash YYYY,~DD-MM-YYYY,~or~YYYY-MM-DD. \
1468   Ensure~the~day~and~month~are~valid~(e.g.,~avoid~31\slash 02\slash 2024).
1469 }

```

(End of definition for `invalid-date-format`.)

#### 11.21.2 Internal functions for `\spdate`

```
\__spintent_spdate_process:n
```

The internal function `\__spintent_spdate_process:n` parses the input using `\__spintent_luafun_spdate_parse:n` and checks the error status in `\__spintent_spdate_luaset_error_str`. If an invalid format is detected, it issues the `invalid-date-format` error. Otherwise, it builds the MathML `intent` structure using the formatted date stored in `\__spintent_spdate_luaset_output_str`.

```

1470 \cs_new_protected:Nn \__spintent_spdate_process:n
1471 {
1472   \__spintent_luafun_spdate_parse:n { #1 }
1473   \str_if_eq:VnTF \__spintent_spdate_luaset_error_str { true }
1474   {
1475     \msg_error:nnnn { spintent } { invalid-date-format }
1476     { \spdate } { #1 }
1477   }
1478   {
1479     \MathMLintent { :date }
1480     {
1481       \text { \__spintent_spdate_luaset_output_str }
1482     }
1483   }
1484 }

```

(End of definition for `\__spintent_sdate_process:n`.)

`\sdate` The command `\sdate` is the public interface for formatting date expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it calls `\__spintent_sdate_process:n` to process the input.

```

1485 \NewDocumentCommand \sdate { m }
1486 {
1487   \mode_if_math:TF
1488   {
1489     \group_begin:
1490       \__spintent_sdate_process:n { #1 }
1491     \group_end:
1492   }
1493   { \msg_error:nnn { spintent } { need-math-mode } { \sdate } }
1494 }

```

(End of definition for `\sdate`. This function is documented on page 11.)

## 11.22 Semantic Operators

This section provides purely semantic operators. Unlike their formatting counterparts (which parse and format numbers), these commands do not process mandatory arguments. Instead, they solely inject a mathematical glyph wrapped in its corresponding regional `MathCAT` intent. This grants teachers absolute control when building complex algebraic expressions, mixed numbers, or fraction divisions.

### 11.23 The command `\spusep`

For the use of parentheses, we will provide the command `\spusep` with only two `<keys>` `size` and `silent`. Most of the time we can use `\left` and `\right` or `\Uleft` and `\Uright`, but both forms create a group, and this hinders the code for *tagged* PDF when we want to interrupt something between them.

`spusep-empty-arg`  
`spusep-invalid-arg`

The first “error” message `spusep-empty-arg` is raised when `\spusep` receives an empty or blank `{<argument>}`, the second this is when an input is invalid.

```

1495 \msg_new:nnnn { spintent } { spusep-empty-arg }
1496 {
1497   Empty~argument~'#1'~for~\token_to_str:N \spusep.
1498 }
1499 {
1500   \token_to_str:N \spusep~requires~a~non-empty~argument.
1501 }
1502 \msg_new:nnn { spintent } { spusep-invalid-arg }
1503 {
1504   Invalid~argument~'#1'~for~\token_to_str:N \spusep.
1505 }

```

(End of definition for `spusep-empty-arg` and `spusep-invalid-arg`.)

`\__spintent_spusep_arg_tl`  
`\__spintent_spusep_left_tl`  
`\__spintent_spusep_right_tl`

The variable `\__spintent_spusep_arg_tl` stores the `{<argument>}`, the variable `\__spintent_spusep_left_tl` and `\__spintent_spusep_right_tl` stores the parentheses.

```

1506 \tl_new:N \__spintent_spusep_arg_tl
1507 \tl_new:N \__spintent_spusep_left_tl
1508 \tl_new:N \__spintent_spusep_right_tl

```

(End of definition for `\__spintent_spusep_arg_tl`, `\__spintent_spusep_left_tl`, and `\__spintent_spusep_right_tl`.)

`size`  
`silent`  
`unknown`

We define the `<keys>` `size` and `silent`.

```

1509 \keys_define:nn { spintent / spusep }
1510 {
1511   size .choices:nn =
1512     { none , big , Big , bigg , Bigg }
1513     {
1514       \int_case:nn { \l_keys_choice_int }
1515       {
1516         { 1 }
1517         {
1518           \tl_clear:N \__spintent_spusep_left_tl
1519           \tl_clear:N \__spintent_spusep_right_tl
1520         }
1521         { 2 }
1522         {
1523           \tl_set:Nn \__spintent_spusep_left_tl { \bigl }
1524           \tl_set:Nn \__spintent_spusep_right_tl { \bigr }
1525         }
1526       }
1527     }

```

```

1526         { 3 }
1527         {
1528             \tl_set:Nn \l__spintent_spusep_left_tl { \Bigl }
1529             \tl_set:Nn \l__spintent_spusep_right_tl { \Bigr }
1530         }
1531         { 4 }
1532         {
1533             \tl_set:Nn \l__spintent_spusep_left_tl { \biggl }
1534             \tl_set:Nn \l__spintent_spusep_right_tl { \biggr }
1535         }
1536         { 5 }
1537         {
1538             \tl_set:Nn \l__spintent_spusep_left_tl { \Biggl }
1539             \tl_set:Nn \l__spintent_spusep_right_tl { \Biggr }
1540         }
1541     }
1542 },
1543 size .initial:n = none,
1544 size .value_required:n = true,
1545 size / unknown .code:n =
1546     \msg_error:nneee { spintent } { unknown-choice }
1547     { size } { none,~big,~Big,~bigg,~Bigg } { \exp_not:n {#1} },
1548 silent .bool_set:N = \l__spintent_spusep_silent_bool,
1549 silent .initial:n = false,
1550 silent .value_required:n = true,
1551 unknown .code:n =
1552     {
1553         \str_set:Nn \l__spintent_command_name_str { spusep }
1554         \__spintent_unknown_keys_cmd:n {#1}
1555     },
1556 }

```

(End of definition for `size`, `silent`, and `unknown`.)

`\__spintent_spusep_set_parens:nn`

The function `\__spintent_spusep_set_parens:nn` will place the argument passed in ‘#2’ to the right of the variables `\l__spintent_spusep_left_tl` and `\l__spintent_spusep_right_tl` and then check the state of the variable `\l__spintent_spusep_silent_bool` handled by the key `silent` to print.

```

1557 \cs_new_protected:Nn \__spintent_spusep_set_parens:nn
1558 {
1559     \tl_put_right:cn { \l__spintent_spusep_ #1 _tl } { #2 }
1560     \bool_if:NTF \l__spintent_spusep_silent_bool
1561     {
1562         \MathMLint { :silent } { \tl_use:c { \l__spintent_spusep_ #1 _tl } }
1563     }
1564     {
1565         \tl_use:c { \l__spintent_spusep_ #1 _tl }
1566     }
1567 }

```

(End of definition for `\__spintent_spusep_set_parens:nn`.)

`\__spintent_spusep_convert:n`

The function `\__spintent_spusep_convert:n` captures and stores the  $\langle argument \rangle$  passed in ‘#1’ in the variable `\l__spintent_spusep_arg_tl`, validates the cases by converting them to the macros provided by `unicode-math`, and passes them to the function `\__spintent_spusep_set_parens:nn` for printing. If the input is not supported or is empty, we return an “error”.

```

1568 \cs_new_protected:Nn \__spintent_spusep_convert:n
1569 {
1570     \tl_set:Nn \l__spintent_spusep_arg_tl { \tl_trim_spaces:n {#1} }
1571     \tl_if_empty:NT \l__spintent_spusep_arg_tl
1572     {
1573         \msg_error:nnn { spintent } { spusep-empty-arg } {#1}
1574     }
1575     \str_case:VnF \l__spintent_spusep_arg_tl
1576     {
1577         { ( }
1578         {
1579             \__spintent_spusep_set_parens:nn { left } { \lparen }
1580         }
1581         { ) }
1582         {
1583             \__spintent_spusep_set_parens:nn { right } { \rparen }

```

```

1584     }
1585     { [ ] }
1586     {
1587         \__spintent_spusep_set_parens:nn { left } { \lbrack }
1588     }
1589     { ] }
1590     {
1591         \__spintent_spusep_set_parens:nn { right } { \rbrack }
1592     }
1593     { \{ }
1594     {
1595         \__spintent_spusep_set_parens:nn { left } { \lbrace }
1596     }
1597     { \} }
1598     {
1599         \__spintent_spusep_set_parens:nn { right } { \rbrace }
1600     }
1601 }
1602 {
1603     \msg_error:nnn { spintent } { spusep-invalid-arg } {#1}
1604 }
1605 }

```

(End of definition for \\_\_spintent\_spusep\_convert:n.)

**\spusep** The command `\spusep` checks that it is running in math mode, opens a local group, sets the keys passed in ‘#1’ and processes the  $\langle argument \rangle$  passed in ‘#2’ by means of the function `\__spintent_spusep_convert:n`.

```

1606 \NewDocumentCommand \spusep { O{} m }
1607 {
1608     \mode_if_math:TF
1609     {
1610         \group_begin:
1611         \keys_set:nn { spintent / spusep } {#1}
1612         \__spintent_spusep_convert:n {#2}
1613         \group_end:
1614     }
1615     { \msg_error:nnn { spintent } { need-math-mode } { \spusep } }
1616 }

```

(End of definition for \spusep. This function is documented on page 12.)

### 11.23.1 The command \spread

The command `\spread` provides a semantic read for MathCAT of mathematical symbols for situations in which it is necessary to specify gender or plurality in Spanish.

The “error” `spnewsym-empty-args` is raised when `\spnewsym` receives an empty or blank symbol or reading. The “error” `spnewsym-duplicate` is raised when it registers a symbol already in the dictionary. The “error” `spread-unknown-sym` is raised when `\spread` is given a symbol that is neither registered nor accompanied by `read-arg`.

```

1617 \msg_new:nnnn { spintent } { spnewsym-empty-args }
1618 {
1619     Empty~argument~in~\token_to_str:N \spnewsym.
1620 }
1621 {
1622     \token_to_str:N \spnewsym~requires~two~non-empty~arguments.
1623 }
1624 \msg_new:nnnn { spintent } { spnewsym-duplicate }
1625 {
1626     The~symbol~'~#1'~is~already~defined!
1627 }
1628 {
1629     You~cannot~redefine~with~\token_to_str:N \spnewsym.
1630 }
1631 \msg_new:nnnn { spintent } { spread-empty-sym }
1632 {
1633     Empty~symbol~argument~in~\token_to_str:N \spread.
1634 }
1635 {
1636     \token_to_str:N \spread~requires~a~non-empty~symbol.
1637 }
1638 \msg_new:nnnn { spintent } { spread-unknown-sym }

```

```

1639 {
1640   The~symbol~'#1'~is~not~registered~and~no~'read-arg'~was~given.
1641 }
1642 {
1643   Register~it~first~with~\token_to_str:N \spnewsym,~or~pass~'read-arg'.
1644 }

```

(End of definition for `spnewsym-empty-args`, `spnewsym-duplicate`, and `spread-unknown-sym`.)

`\l__spinttent_spread_luaset_status_str` and `\l__spinttent_spread_luaset_reading_str` are set by `\__spinttent_luafun_spread_new_sym:nn`/`\__spinttent_luafun_spread_lookup_sym:n`. `\l__spinttent_spread_read_arg_tl` stores `read-arg`.

```

1645 \str_new:N \l__spinttent_spread_luaset_status_str
1646 \str_new:N \l__spinttent_spread_luaset_reading_str
1647 \tl_new:N \l__spinttent_spread_read_arg_tl

```

(End of definition for `\l__spinttent_spread_luaset_status_str`, `\l__spinttent_spread_luaset_reading_str`, and `\l__spinttent_spread_read_arg_tl`.)

`read-arg` We define the *(keys)* `read-arg` and `silent-cmd`.  
`silent-cmd`  
`unknown`

```

1648 \keys_define:nn { spinttent / spread }
1649 {
1650   read-arg .code:n =
1651     { \__spinttent_sanitizize_affix:Nn \l__spinttent_spread_read_arg_tl {#1} },
1652   read-arg .initial:n = ,
1653   read-arg .value_required:n = true,
1654   silent-cmd .bool_set:N = \l__spinttent_spread_silent_cmd_bool,
1655   silent-cmd .initial:n = false,
1656   silent-cmd .value_required:n = true,
1657   unknown .code:n =
1658     {
1659       \str_set:Nn \l__spinttent_command_name_str { spread }
1660       \__spinttent_unknown_keys_cmd:n {#1}
1661     },
1662 }

```

(End of definition for `read-arg`, `silent-cmd`, and `unknown`.)

`\__spinttent_spread_exec:nn` The function `\__spinttent_spread_exec:nn` receives the optional *(keys)* in ‘#1’ and the *(symbol)* in ‘#2’. With `silent-cmd` true, the symbol is typeset with a `:silent` intent, nothing else evaluated. Otherwise, with `read-arg` given, it is used directly, overriding any registered reading for this call alone. With `read-arg` empty, calls `\__spinttent_luafun_spread_lookup_sym:n`, using the registered reading when found, reporting `spread-unknown-sym` otherwise.

```

1663 \cs_new_protected:Nn \__spinttent_spread_exec:nn
1664 {
1665   \keys_set:nn { spinttent / spread } { #1 }
1666   \tl_if_blank:nTF { #2 }
1667   { \msg_error:nn { spinttent } { spread-empty-sym } }
1668   {
1669     \bool_if:NTF \l__spinttent_spread_silent_cmd_bool
1670     { \MathMLintent { :silent } { #2 } }
1671     {
1672       \tl_if_empty:NTF \l__spinttent_spread_read_arg_tl
1673       {
1674         \__spinttent_luafun_spread_lookup_sym:n { \exp_not:n {#2} }
1675         \str_if_eq:VnTF \l__spinttent_spread_luaset_status_str { found }
1676         {
1677           \MathMLintent { \l__spinttent_spread_luaset_reading_str } { #2 }
1678         }
1679         {
1680           \msg_error:nne { spinttent } { spread-unknown-sym } { \exp_not:n {#2} }
1681         }
1682       }
1683       {
1684         \MathMLintent { \l__spinttent_spread_read_arg_tl } { #2 }
1685       }
1686     }
1687   }
1688 }

```

(End of definition for `\__spinttent_spread_exec:nn`.)

`\spread` The public command `\spread` enforces *math mode*, opens a local group, and delegates execution to `__spintent_spread_exec:nn`.

```

1689 \NewDocumentCommand \spread { O{} m }
1690 {
1691   \mode_if_math:TF
1692   {
1693     \group_begin:
1694     __spintent_spread_exec:nn { #1 } { #2 }
1695     \group_end:
1696   }
1697   { \msg_error:nnn { spintent } { need-math-mode } { \spread } }
1698 }

```

(End of definition for `\spread`. This function is documented on page 12.)

`\spnewsym` The public command `\spnewsym` registers a new symbol/reading pair. It passes the symbol ‘#1’ and its reading ‘#2’ to Lua via `__spintent_luafun_spread_new_sym:nn`, reporting `spnewsym-duplicate` when the symbol is already registered.

```

1699 \NewDocumentCommand \spnewsym { m m }
1700 {
1701   \mode_if_math:TF
1702   {
1703     \msg_error:nnn { spintent } { need-text-mode } { \spnewsym }
1704   }
1705   {
1706     \bool_lazy_or:nnTF { \tl_if_blank_p:n { #1 } } { \tl_if_blank_p:n { #2 } }
1707     { \msg_error:nn { spintent } { spnewsym-empty-args } }
1708     {
1709       __spintent_luafun_spread_new_sym:nn { \exp_not:n {#1} } { #2 }
1710       \str_if_eq:VnT \__spintent_spread_luaset_status_str { duplicate }
1711       {
1712         \msg_error:nnn { spintent } { spnewsym-duplicate } { #1 }
1713       }
1714     }
1715   }
1716 }

```

(End of definition for `\spnewsym`. This function is documented on page 11.)

### 11.23.2 The command `\spdsym`

`\__spintent_spdsym_read_sym_tl` Variables to store the regional reading string, the visual operator, and the optional semantic affixes specifically for `\spdsym`.

```

\__spintent_spdsym_symbol_tl
\__spintent_spdsym_prefix_tl
\__spintent_spdsym_suffix_tl
1717 \tl_new:N \__spintent_spdsym_read_sym_tl
1718 \tl_new:N \__spintent_spdsym_symbol_tl
1719 \tl_new:N \__spintent_spdsym_prefix_tl
1720 \tl_new:N \__spintent_spdsym_suffix_tl

```

(End of definition for `\__spintent_spdsym_read_sym_tl` and others.)

```

prefix We define the (keys) prefix, suffix, set-sym and read-sym for \spdsym.
suffix
set-sym
read-sym
unknown
1721 \keys_define:nn { spintent / spdsym }
1722 {
1723   prefix .code:n =
1724   { __spintent_sanitize_affix:Nn \__spintent_spdsym_prefix_tl {#1} },
1725   prefix .initial:n = ,
1726   prefix .value_required:n = true,
1727   suffix .code:n =
1728   { __spintent_sanitize_affix:Nn \__spintent_spdsym_suffix_tl {#1} },
1729   suffix .initial:n = ,
1730   suffix .value_required:n = true,
1731   set-sym .choices:nn =
1732   { old-style , two-dots , slash }
1733   {
1734     \int_case:nn { \l_keys_choice_int }
1735     {
1736       {1} { \tl_set:Nn \__spintent_spdsym_symbol_tl { \div } }
1737       {2} { \tl_set:Nn \__spintent_spdsym_symbol_tl { : } }
1738       {3} { \tl_set:Nn \__spintent_spdsym_symbol_tl { / } }
1739     }
1740   },
1741   set-sym .initial:n = two-dots,

```

```

1742     set-sym                .value_required:n = true,
1743     set-sym / unknown     .code:n =
1744         \msg_error:nneee { spintent } { unknown-choice }
1745         { set-sym } { old-style, ~two-dots, ~slash } { \exp_not:n {#1} },
1746     read-sym              .choices:nn =
1747         { dividido , dividido-por , dividido-entre }
1748         {
1749             \int_case:nn { \l_keys_choice_int }
1750             {
1751                 {1} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido } }
1752                 {2} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido-por } }
1753                 {3} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido-entre } }
1754             }
1755         },
1756     read-sym              .initial:n = dividido,
1757     read-sym              .value_required:n = true,
1758     read-sym / unknown     .code:n =
1759         \msg_error:nneee { spintent } { unknown-choice }
1760         { read-sym } { dividido, ~dividido-por, ~dividido-entre }
1761         { \exp_not:n {#1} },
1762     unknown               .code:n =
1763         {
1764             \str_set:Nn \l__spintent_command_name_str { spdsym }
1765             \l__spintent_unknown_keys_cmd:n {#1}
1766         },
1767 }

```

(End of definition for *prefix* and others.)

**\spdsym** Defines the **\spdsym** operator. It takes a single optional argument for local setup and outputs the configured intent structure, safely attaching prefixes and suffixes if present.

```

1768 \NewDocumentCommand \spdsym { 0{ } }
1769 {
1770     \mode_if_math:TF
1771     {
1772         \group_begin:
1773         \keys_set:nn { spintent / spdsym } { #1 }
1774         \MathMLintent
1775         {
1776             \tl_if_empty:NF \l__spintent_spdsym_prefix_tl
1777             { \l__spintent_spdsym_prefix_tl - }
1778             \l__spintent_spdsym_read_sym_tl
1779             \tl_if_empty:NF \l__spintent_spdsym_suffix_tl
1780             { - \l__spintent_spdsym_suffix_tl }
1781         }
1782         { \l__spintent_spdsym_symbol_tl }
1783         \group_end:
1784     }
1785     { \msg_error:nnn { spintent } { need-math-mode } { \spdsym } }
1786 }

```

(End of definition for **\spdsym**. This function is documented on page 12.)

### 11.23.3 The command **\spmsym**

**\l\_\_spintent\_spmsym\_read\_sym\_tl** Variables to store the regional reading string, the visual operator, and the optional semantic affixes specifically for **\spmsym**.

```

\l__spintent_spmsym_symbol_tl
\l__spintent_spmsym_prefix_tl
\l__spintent_spmsym_suffix_tl
1787 \tl_new:N \l__spintent_spmsym_read_sym_tl
1788 \tl_new:N \l__spintent_spmsym_symbol_tl
1789 \tl_new:N \l__spintent_spmsym_prefix_tl
1790 \tl_new:N \l__spintent_spmsym_suffix_tl

```

(End of definition for **\l\_\_spintent\_spmsym\_read\_sym\_tl** and others.)

prefix We define the *<keys>* prefix, suffix, set-sym, read-sym and not-print for **\spmsym**.

```

suffix 1791 \keys_define:nn { spintent / spmsym }
set-sym 1792 {
read-sym 1793     prefix                .code:n =
not-print 1794     { \l__spintent_sanitize_affix:Nn \l__spintent_spmsym_prefix_tl {#1} },
unknown 1795     prefix                .initial:n = ,
1796     prefix                .value_required:n = true,
1797     suffix                .code:n =
1798     { \l__spintent_sanitize_affix:Nn \l__spintent_spmsym_suffix_tl {#1} },

```

```

1799     suffix                .initial:n = ,
1800     suffix                .value_required:n = true,
1801     set-sym               .choices:nn =
1802     { cross , dot , asterisk }
1803     {
1804       \int_case:nn { \l_keys_choice_int }
1805       {
1806         {1} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \times } }
1807         {2} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \cdot } }
1808         {3} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \ast } }
1809       }
1810     },
1811     set-sym               .initial:n = dot,
1812     set-sym               .value_required:n = true,
1813     set-sym / unknown     .code:n =
1814     \msg_error:nnee { spintent } { unknown-choice }
1815     { set-sym } { cross, ~dot, ~asterisk } { \exp_not:n {#1} },
1816     read-sym              .choices:nn =
1817     { por , multiplicado-por , veces }
1818     {
1819       \int_case:nn { \l_keys_choice_int }
1820       {
1821         {1} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { por } }
1822         {2} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { multiplicado-por } }
1823         {3} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { veces } }
1824       }
1825     },
1826     read-sym              .initial:n = por,
1827     read-sym              .value_required:n = true,
1828     read-sym / unknown     .code:n =
1829     \msg_error:nnee { spintent } { unknown-choice }
1830     { read-sym } { por, ~multiplicado-por, ~veces } { \exp_not:n {#1} },
1831     not-print             .bool_set:N = \l__spintent_spmsym_not_print_bool,
1832     not-print             .initial:n = false,
1833     not-print             .value_required:n = true,
1834     unknown               .code:n =
1835     {
1836       \str_set:Nn \l__spintent_command_name_str { spmsym }
1837       \__spintent_unknown_keys_cmd:n {#1}
1838     },
1839   }

```

(End of definition for prefix and others.)

**\spmsym** Defines the `\spmsym` operator. Similar to `\spdsym`, it outputs the correct mathematical glyph based on the local *(keys)*, safely attaching prefixes and suffixes if present.

```

1840 \NewDocumentCommand \spmsym { 0{} }
1841 {
1842   \mode_if_math:TF
1843   {
1844     \group_begin:
1845     \keys_set:nn { spintent / spmsym } { #1 }
1846     \MathMLintent
1847     {
1848       \tl_if_empty:NF \l__spintent_spmsym_prefix_tl
1849       { \l__spintent_spmsym_prefix_tl - }
1850       \l__spintent_spmsym_read_sym_tl
1851       \tl_if_empty:NF \l__spintent_spmsym_suffix_tl
1852       { - \l__spintent_spmsym_suffix_tl }
1853     }
1854     {
1855       \bool_if:NTF \l__spintent_spmsym_not_print_bool
1856       {
1857         \invisibletimes
1858       }
1859       { \l__spintent_spmsym_symbol_tl }
1860     }
1861     \group_end:
1862   }
1863   { \msg_error:nnn { spintent } { need-math-mode } { \spmsym } }
1864 }

```

(End of definition for `\spmsym`. This function is documented on page 13.)

## 11.24 The command `\spdiv`

The user-level command `\spdiv` formats division representations. It takes two mandatory arguments: the  $\langle \textit{dividend} \rangle$  and the  $\langle \textit{divisor} \rangle$ . Both  $\langle \textit{arguments} \rangle$  are internally processed by the `\spnum` command to ensure proper digit separation. This command enforces strict validation, ensuring that neither the  $\langle \textit{dividend} \rangle$  nor the  $\langle \textit{divisor} \rangle$  is *blank*.

### 11.24.1 Definition of keys for `\spdiv`

We define the  $\langle \textit{keys} \rangle$  `wrap-parens` and `read-parens` for `\spdiv` command. Number-related  $\langle \textit{keys} \rangle$  `cifras-sep`, `read-sign`, `read-period-sep` and `notation-sym` are forwarded to `\spnum` command using `.meta:nn`, while semantic operator  $\langle \textit{keys} \rangle$  `set-sym` and `read-sym` are forwarded to the `\spdsym` command using `.meta:nn`.

```

1865 \keys_define:nn { spintenc / spdiv }
1866 {
1867   cifras-sep      .meta:nn = { spintenc / spnum } { cifras-sep      = {#1} },
1868   read-sign       .meta:nn = { spintenc / spnum } { read-sign       = {#1} },
1869   read-period-sep .meta:nn = { spintenc / spnum } { read-period-sep = {#1} },
1870   notation-sym    .meta:nn = { spintenc / spnum } { notation-sym    = {#1} },
1871   set-sym         .meta:nn = { spintenc / spdsym } { set-sym         = {#1} },
1872   read-sym        .meta:nn = { spintenc / spdsym } { read-sym        = {#1} },
1873   wrap-parens     .bool_set:N = \l__spintenc_spdiv_wrap_parens_bool,
1874   wrap-parens     .initial:n  = false,
1875   wrap-parens     .value_required:n = true,
1876   read-parens     .bool_set:N = \l__spintenc_spdiv_read_parens_bool,
1877   read-parens     .initial:n  = true,
1878   read-parens     .value_required:n = true,
1879   unknown         .code:n =
1880   {
1881     \str_set:Nn \l__spintenc_command_name_str { spdiv }
1882     \__spintenc_unknown_keys_cmd:n {#1}
1883   },
1884 }

```

(End of definition for `cifras-sep` and others.)

### 11.24.2 Definition of error messages

`spdiv-empty-argument` Raised when either the  $\langle \textit{dividend} \rangle$  or the  $\langle \textit{divisor} \rangle$  is missing or *blank*.

```

1885 \msg_new:nnnn { spintenc } { spdiv-empty-argument }
1886 { The~arguments~for~\token_to_str:N \spdiv~cannot~be~empty. }
1887 {
1888   You~must~provide~both~a~dividend~and~a~divisor.\\
1889 }

```

(End of definition for `spdiv-empty-argument`.)

### 11.24.3 Internal functions for `\spdiv`

`\__spintenc_spdiv_render_op:nn` The function `\__spintenc_spdiv_render_op:nn` parses both  $\langle \textit{arguments} \rangle$  through the `\spnum` command and embeds the semantic operator from `\spdsym`.

```

1890 \cs_new_protected:Nn \__spintenc_spdiv_render_op:nn
1891 {
1892   \spnum {#1}
1893   \MathMLintent { \l__spintenc_spdsym_read_sym_tl }
1894   {
1895     \l__spintenc_spdsym_symbol_tl
1896   }
1897   \spnum {#2}
1898 }

```

(End of definition for `\__spintenc_spdiv_render_op:nn`.)

`\__spintenc_spdiv_render_parens:nn` The function `\__spintenc_spdiv_render_parens:nn` handles the structural parentheses around the division.

```

1899 \cs_new_protected:Nn \__spintenc_spdiv_render_parens:nn
1900 {
1901   \bool_if:NTF \l__spintenc_spdiv_read_parens_bool
1902   {
1903     ( \__spintenc_spdiv_render_op:nn {#1} {#2} )
1904   }
1905   {
1906     \MathMLintent { :silent } { ( ) }

```

```

1907     \__spintent_spdiv_render_op:nn {#1} {#2}
1908     \MathMLintent { :silent } { ) }
1909   }
1910 }

```

(End of definition for \\_\_spintent\_spdiv\_render\_parens:nn.)

\\_\_spintent\_spdiv\_process:nnn The function \\_\_spintent\_spdiv\_process:nnn sets  $\langle keys \rangle$ , validates  $\{ \langle arguments \rangle \}$ , and decides on parenthesis wrapping.

```

1911 \cs_new_protected:Nn \__spintent_spdiv_process:nnn
1912 {
1913   \keys_set:nn { spintent / spdiv } { #1 }
1914   \tl_if_blank:nTF { #2 }
1915   { \msg_error:nnn { spintent } { spdiv-empty-argument } }
1916   {
1917     \tl_if_blank:nTF { #3 }
1918     { \msg_error:nnn { spintent } { spdiv-empty-argument } }
1919     {
1920       \bool_if:NTF \__spintent_spdiv_wrap_parens_bool
1921       { \__spintent_spdiv_render_parens:nn {#2} {#3} }
1922       { \__spintent_spdiv_render_op:nn {#2} {#3} }
1923     }
1924   }
1925 }

```

(End of definition for \\_\_spintent\_spdiv\_process:nnn.)

\spdiv Defines the user command \spdiv. It strictly relies on *math mode*.

```

1926 \NewDocumentCommand \spdiv { O{} m m }
1927 {
1928   \mode_if_math:TF
1929   {
1930     \group_begin:
1931     \__spintent_spdiv_process:nnn { #1 } { #2 } { #3 }
1932     \group_end:
1933   }
1934   { \msg_error:nnn { spintent } { need-math-mode } { \spdiv } }
1935 }

```

(End of definition for \spdiv. This function is documented on page 13.)

## 11.25 The command \spmul

The user-level command \spmul formats multiplication representations. It takes two mandatory arguments: the  $\{ \langle first\ factor \rangle \}$  and the  $\{ \langle second\ factor \rangle \}$ . Both  $\{ \langle arguments \rangle \}$  are internally processed by the \spnum command to ensure proper digit separation. This command enforces strict validation, ensuring that neither the  $\{ \langle first\ factor \rangle \}$  nor the  $\{ \langle second\ factor \rangle \}$  is *blank*.

### 11.25.1 Definition of keys for \spmul

We define the  $\langle keys \rangle$  wrap-parens and read-parens for \spmul command. Number-related  $\langle keys \rangle$  cifras-sep, read-sign, read-period-sep and notation-sym are forwarded to \spnum command using .meta:nn, while semantic operator  $\langle keys \rangle$  set-sym and read-sym are forwarded to the \spmsym command using .meta:nn.

```

1936 \keys_define:nn { spintent / spmul }
1937 {
1938   cifras-sep      .meta:nn = { spintent / spnum } { cifras-sep      = {#1} },
1939   read-sign       .meta:nn = { spintent / spnum } { read-sign       = {#1} },
1940   read-period-sep .meta:nn = { spintent / spnum } { read-period-sep = {#1} },
1941   notation-sym    .meta:nn = { spintent / spnum } { notation-sym    = {#1} },
1942   set-sym         .meta:nn = { spintent / spmsym } { set-sym         = {#1} },
1943   read-sym        .meta:nn = { spintent / spmsym } { read-sym        = {#1} },
1944   wrap-parens     .bool_set:N = \__spintent_spmul_wrap_parens_bool,
1945   wrap-parens     .initial:n = false,
1946   wrap-parens     .value_required:n = true,
1947   read-parens     .bool_set:N = \__spintent_spmul_read_parens_bool,
1948   read-parens     .initial:n = true,
1949   read-parens     .value_required:n = true,
1950   unknown         .code:n =
1951   {
1952     \str_set:Nn \__spintent_command_name_str { spmul }
1953     \__spintent_unknown_keys_cmd:n {#1}
1954   },
1955 }

```

(End of definition for `cifras-sep` and others.)

### 11.25.2 Definition of error messages

`spmul-empty-argument` Raised when either the  $\langle \textit{first factor} \rangle$  or the  $\langle \textit{second factor} \rangle$  is missing or *blank*.

```

1956 \msg_new:nnnn { spintent } { spmul-empty-argument }
1957 { The~arguments~for~\token_to_str:N \spmul~cannot~be~empty. }
1958 {
1959   You~must~provide~both~first~factor~and~second~factor.\\
1960 }
```

(End of definition for `spmul-empty-argument`.)

### 11.25.3 Internal functions for `\spmul`

`\__spinttent_spmul_render_op:nn` The function `\__spinttent_spmul_render_op:nn` parses both  $\langle \textit{arguments} \rangle$  through the `\spnum` command and embeds the semantic operator from `\spmsym`.

```

1961 \cs_new_protected:Nn \__spinttent_spmul_render_op:nn
1962 {
1963   \spnum {#1}
1964   \MathMLintent { \__spinttent_spmsym_read_sym_tl }
1965   {
1966     \__spinttent_spmsym_symbol_tl
1967   }
1968   \spnum {#2}
1969 }
```

(End of definition for `\__spinttent_spmul_render_op:nn`.)

`\__spinttent_spmul_render_parens:nn` The function `\__spinttent_spmul_render_parens:nn` handles the structural parentheses around the multiplication.

```

1970 \cs_new_protected:Nn \__spinttent_spmul_render_parens:nn
1971 {
1972   \bool_if:NTF \__spinttent_spmul_read_parens_bool
1973   {
1974     ( \__spinttent_spmul_render_op:nn {#1} {#2} )
1975   }
1976   {
1977     \MathMLintent { :silent } { ( }
1978     \__spinttent_spmul_render_op:nn {#1} {#2}
1979     \MathMLintent { :silent } { ) }
1980   }
1981 }
```

(End of definition for `\__spinttent_spmul_render_parens:nn`.)

`\__spinttent_spmul_process:nnn` The function `\__spinttent_spmul_process:nnn` sets  $\langle \textit{keys} \rangle$ , validates  $\langle \textit{arguments} \rangle$ , and decides on parenthesis wrapping.

```

1982 \cs_new_protected:Nn \__spinttent_spmul_process:nnn
1983 {
1984   \keys_set:nn { spintent / spmul } { #1 }
1985   \tl_if_blank:nTF { #2 }
1986   { \msg_error:nnn { spintent } { spmul-empty-argument } }
1987   {
1988     \tl_if_blank:nTF { #3 }
1989     { \msg_error:nnn { spintent } { spmul-empty-argument } }
1990     {
1991       \bool_if:NTF \__spinttent_spmul_wrap_parens_bool
1992       { \__spinttent_spmul_render_parens:nn {#2} {#3} }
1993       { \__spinttent_spmul_render_op:nn {#2} {#3} }
1994     }
1995   }
1996 }
```

(End of definition for `\__spinttent_spmul_process:nnn`.)

`\spmul` Defines the user command `\spmul`. It strictly relies on *math mode*.

```

1997 \NewDocumentCommand \spmul { 0{} m m }
1998 {
1999   \mode_if_math:TF
2000   {
2001     \group_begin:
2002     \__spinttent_spmul_process:nnn { #1 } { #2 } { #3 }
2003     \group_end:

```

```

2004     }
2005     { \msg_error:nnn { spintent } { need-math-mode } { \spmul } }
2006 }

```

(End of definition for `\spmul`. This function is documented on page 14.)

## 11.26 The commands `\spgcd` and `\spmcn`

The user-level commands `\spgcd` and `\spmcn` provide a semantic interface for the Greatest Common Divisor (MCD in Spanish) and the Least Common Multiple (MCM in Spanish). They delegate argument validation and calculation to the Lua module `spintent.lua`, constructing a valid `MathCAT` intent.

### 11.26.1 Variables for `\spgcd` and `\spmcn`

Internal variables generated dynamically via a comma-separated list to avoid code duplication for both symmetric commands.

```

\__spintenc_spmcn_spmgcd_luaset_error_str
\__spintenc_mc_luaset_types_str
\__spintenc_mc_luaset_all_numeric_str
\__spintenc_mc_args_seq
\__spintenc_mc_raw_args_seq
\__spintenc_spmgcd_print_result_bool
\__spintenc_spmcn_print_result_bool
\__spintenc_spmgcd_read_terse_bool
\__spintenc_spmcn_read_terse_bool
\__spintenc_spmgcd_print_upper_bool
\__spintenc_spmcn_print_upper_bool
\__spintenc_spmgcd_prefix_tl
\__spintenc_spmcn_prefix_tl
\__spintenc_spmgcd_suffix_tl
\__spintenc_spmcn_suffix_tl

```

```

2007 \str_new:N \__spintenc_spmcn_spmgcd_luaset_error_str
2008 \clist_new:N \__spintenc_mc_luaset_types_clist
2009 \str_new:N \__spintenc_mc_luaset_all_numeric_str
2010 \seq_new:N \__spintenc_mc_args_seq
2011 \seq_new:N \__spintenc_mc_raw_args_seq
2012 \seq_new:N \__spintenc_mc_types_seq
2013 \int_new:N \__spintenc_mc_idx_int
2014 \clist_map_inline:nn { mcd , mcm }
2015 {
2016   \tl_new:c { \__spintenc_sp #1 _luaset_ #1 _value_tl }
2017   \bool_new:c { \__spintenc_sp #1 _read_terse_bool }
2018   \tl_new:c { \__spintenc_sp #1 _prefix_tl }
2019 }

```

(End of definition for `\__spintenc_spmcn_spmgcd_luaset_error_str` and others.)

### 11.26.2 Definition of error messages

`mc-invalid-arguments`

Exception triggered when the comma-separated sequence contains negative integers, floats, or alphabetical characters.

```

2020 \msg_new:nnnn { spintent } { mc-invalid-arguments }
2021 {
2022   Invalid~arguments~provided~to~#1.
2023 }
2024 {
2025   The~arguments~must~be~a~comma-separated~list. \\\
2026 }

```

(End of definition for `mc-invalid-arguments`.)

### 11.26.3 Keys for `\spgcd` and `\spmcn`

`result`  
`prefix`  
`read-cmd`  
`upper-case`  
`concept`  
`silent-cmd`  
`unknown`

We define the shared *(keys)* `result`, `prefix`, `read-cmd`, `upper-case`, `concept` and `silent-cmd`.

```

2027 \clist_map_inline:nn { mcd , mcm }
2028 {
2029   \keys_define:nn { spintent / sp #1 }
2030   {
2031     result .bool_set:c = { \__spintenc_sp #1 _print_result_bool },
2032     result .initial:n = false,
2033     result .value_required:n = true,
2034     prefix .code:n =
2035       { \__spintenc_sanitizc_affix:cn { \__spintenc_sp #1 _prefix_tl } {##1} },
2036     prefix .initial:n = ,
2037     prefix .value_required:n = true,
2038     read-cmd .choices:nn =
2039       { terse , clear }
2040     {
2041       \int_case:nn { \l_keys_choice_int }
2042       {
2043         {1} { \bool_set_true:c { \__spintenc_sp #1 _read_terse_bool } }
2044         {2} { \bool_set_false:c { \__spintenc_sp #1 _read_terse_bool } }
2045       }
2046     },
2047     read-cmd .initial:n = clear,
2048     read-cmd .value_required:n = true,
2049     read-cmd / unknown .code:n =
2050       \msg_error:nneee { spintent } { unknown-choice }
2051       { read-cmd } { terse , ~clear } { \exp_not:n {##1} },
2052     upper-case .bool_set:c = { \__spintenc_sp #1 _print_upper_bool },

```

```

2053     upper-case      .initial:n = false,
2054     upper-case      .value_required:n = true,
2055     concept         .bool_set:c = { \__spintent_sp #1 _concept_bool },
2056     concept         .initial:n = true,
2057     concept         .value_required:n = true,
2058     silent-cmd      .bool_set:c = { \__spintent_sp #1 _silent_cmd_bool },
2059     silent-cmd      .initial:n = false,
2060     silent-cmd      .value_required:n = true,
2061     unknown         .code:n =
2062     {
2063         \str_set:Nn \__spintent_command_name_str { sp #1 }
2064         \__spintent_unknown_keys_cmd:n {##1}
2065     },
2066 }
2067 }

```

(End of definition for *result* and others.)

#### 11.26.4 Internal functions for \spmcd and \spmcm

\\_\_spintent\_mc\_process\_args:nn

The function `\__spintent_mc_process_args:nn` parses the operation ‘#1’ and its raw argument list ‘#2’. Calls `\__spintent_luafun_mc_classify_args:n`, splits ‘#2’ into `\__spintent_mc_raw_args_seq` and `\__spintent_mc_luaset_types_clist` into `\__spintent_mc_types_seq`, then walks the former by position, fetching the matching type from the latter through `\seq_item:Nn`, to build `\__spintent_mc_args_seq`: each argument wrapped through `\spnum` when its type is `number`, verbatim otherwise. Injects MathML intents around the already-formatted `\__spintent_mc_args_seq`.

```

2068 \cs_new_protected:Nn \__spintent_mc_process_args:nn
2069 {
2070     \__spintent_luafun_mc_classify_args:n {#2}
2071     \seq_set_from_clist:Nn \__spintent_mc_raw_args_seq {#2}
2072     \seq_set_from_clist:NN \__spintent_mc_types_seq \__spintent_mc_luaset_types_clist
2073     \seq_clear:N \__spintent_mc_args_seq
2074     \int_zero:N \__spintent_mc_idx_int
2075     \seq_map_inline:Nn \__spintent_mc_raw_args_seq
2076     {
2077         \int_incr:N \__spintent_mc_idx_int
2078         \str_if_eq:eeTF { \seq_item:Nn \__spintent_mc_types_seq { \__spintent_mc_idx_int } } {
2079             { \seq_put_right:Nn \__spintent_mc_args_seq { \spnum {##1} } }
2080             { \seq_put_right:Nn \__spintent_mc_args_seq {##1} }
2081         }
2082     }

```

Injects MathML intents around the already-formatted `\__spintent_mc_args_seq`, exactly as before.

```

2082     \bool_if:cTF { \__spintent_sp #1 _silent_cmd_bool }
2083     {
2084         \seq_set_map:Nnn \__spintent_mc_args_seq \__spintent_mc_args_seq
2085         { \MathMLintent { :silent } { ##1 } }
2086         \MathMLintent { _entre } { \c__spintent_invisible_sep_int }
2087         \MathMLintent { :silent } { ( }
2088         \seq_use:Nnnn \__spintent_mc_args_seq
2089         { \MathMLintent { _y:silent } { , } }
2090         { \MathMLintent { :silent } { , } }
2091         { \MathMLintent { _y:silent } { , } }
2092     }
2093     {
2094         \MathMLintent { _entre } { \c__spintent_invisible_sep_int }
2095         \MathMLintent { :silent } { ( }
2096         \seq_use:Nnnn \__spintent_mc_args_seq
2097         { \MathMLintent { _y } { , } }
2098         { \MathMLintent { :silent } { , } }
2099         { \MathMLintent { _y } { , } }
2100     }
2101     \MathMLintent { :silent } { ) }
2102 }

```

(End of definition for `\__spintent_mc_process_args:nn`.)

\\_\_spintent\_spmc\_process:nnnn

The internal function `\__spintent_spmc_process:nnnn` is the shared handler for both `\spmcd` and `\spmcm`. Takes the short identifier in ‘#1’, its full reading string in ‘#2’, the local *(keys)* in ‘#3’, and the optional arguments in ‘#4’.

The operator name (through `\operatorname`, per `upper-case`) is silenced with `silent-cmd=true` — `\__spintent_mc_process_args:nn` silences the arguments too, in that case — or with `concept=false`

alone, leaving the arguments untouched. With `concept=true`, it carries the literal `intent` instead — `#1` or `#2` per `read-cmd`, prefixed with `prefix` when given.

With `#4` given, classifies it through `\__spintent_luafun_mc_classify_args:n`. Only when every argument is a number does it call the Lua calculation (`calculate_#1`), reporting `mc-invalid-arguments` on failure, and, when `result` is `true`, printing the computed value. `\__spintent_mc_process_args:n` prints the argument list itself either way.

```

2103 \cs_new_protected:Nn \__spintent_spmc_process:nnnn
2104 {
2105   \keys_set:nn { spintent / sp #1 } { #3 }
2106   \bool_if:cTF { \__spintent_sp #1 _silent_cmd_bool }
2107   {
2108     \MathMLintent { :silent }
2109     {
2110       \operatorname
2111       {
2112         \bool_if:cTF { \__spintent_sp #1 _print_upper_bool }
2113         { \text_uppercase:n {#1} } { #1 }
2114       }
2115     }
2116   }
2117   {
2118     \bool_if:cTF { \__spintent_sp #1 _concept_bool }
2119     {
2120       \MathMLintent
2121       {
2122         \tl_if_empty:cF { \__spintent_sp #1 _prefix_tl }
2123         {
2124           \tl_use:c { \__spintent_sp #1 _prefix_tl } -
2125         }
2126         \bool_if:cTF { \__spintent_sp #1 _read_terse_bool } { #1 :nofix } { #2 :nofix }
2127       }
2128       {
2129         \operatorname
2130         {
2131           \bool_if:cTF { \__spintent_sp #1 _print_upper_bool }
2132           { \text_uppercase:n {#1} } { #1 }
2133         }
2134       }
2135     }
2136   }
2137   \MathMLintent { :silent }
2138   {
2139     \operatorname
2140     {
2141       \bool_if:cTF { \__spintent_sp #1 _print_upper_bool }
2142       { \text_uppercase:n {#1} } { #1 }
2143     }
2144   }
2145 }
2146

```

With `#4` given, classifies it through `\__spintent_luafun_mc_classify_args:n`. Only when every argument is a number does it call the Lua calculation (`calculate_#1`), reporting `mc-invalid-arguments` on failure — which should not happen, since every argument already passed classification — and, when `result` is `true`, printing the computed value. With any non-numeric argument, no calculation is attempted at all. Either way, `\__spintent_mc_process_args:n` prints the argument list itself.

```

2147   \tl_if_novalue:nF { #4 }
2148   {
2149     \__spintent_luafun_mc_classify_args:n {#4}
2150     \str_if_eq:VnT \__spintent_mc_luaset_all_numeric_str { true }
2151     {
2152       \use:c { __spintent_luafun_calculate_ #1 :n } { #4 }
2153       \str_if_eq:VnT \__spintent_spmcm_spmcd_luaset_error_str { true }
2154       {
2155         \msg_error:nne { spintent } { mc-invalid-arguments } { \exp_not:c { sp #1 } }
2156       }
2157     }
2158     \__spintent_mc_process_args:nn {#1} {#4}
2159     \str_if_eq:VnT \__spintent_mc_luaset_all_numeric_str { true }
2160     {

```

```

2161         \bool_if:cT { l__spintent_sp #1 _print_result_bool }
2162         {
2163             \MathMLintent { es-igual-a } { = }
2164             \tl_use:c { l__spintent_sp #1 _luaset_ #1 _value_tl }
2165         }
2166     }
2167 }
2168 }

```

(End of definition for `\__spintent_spmc_process:nnnn`.)

`\spmc` The commands `\spmc` and `\spmc` act as the public interface wrappers for typesetting the greatest common divisor and least common multiple. They enforce math mode, open a local group, and defer execution to the main processing macro `\__spintent_spmc_process:nnnn`, passing along the appropriate reading strings and arguments.

```

2169 \NewDocumentCommand \spmc { 0{} d() }
2170 {
2171     \mode_if_math:TF
2172     {
2173         \group_begin:
2174         \__spintent_spmc_process:nnnn { mcd } { máximo-común-divisor } {#1} {#2}
2175         \group_end:
2176     }
2177     { \msg_error:nnn { spintent } { need-math-mode } { \spmc } }
2178 }
2179 \NewDocumentCommand \spmc { 0{} d() }
2180 {
2181     \mode_if_math:TF
2182     {
2183         \group_begin:
2184         \__spintent_spmc_process:nnnn { mcm } { mínimo-común-múltiplo } {#1} {#2}
2185         \group_end:
2186     }
2187     { \msg_error:nnn { spintent } { need-math-mode } { \spmc } }
2188 }

```

(End of definition for `\spmc` and `\spmc`. These functions are documented on page 15.)

### 11.27 The commands `\spnM` and `\spnD`

The user-level commands `\spnM` and `\spnD` provide a semantic interface for the Multiples of a number (“*múltiplos*” in Spanish) and the Divisors of a number (“*divisores*” in Spanish). They delegate argument validation and calculation to the Lua module `spintent.lua`, constructing a valid `MathCAT` intent.

#### 11.27.1 Definition of variables for `\spnM` and `\spnD`

Internal variables generated dynamically for multiples and divisors.

```

\l__spintent_md_out_seq
\l__spintent_spmD_luaset_is_number_str
\l__spintent_spmM_spmD_luaset_error_str
\l__spintent_spmM_luaset_result_tl
\l__spintent_spmD_luaset_result_tl
\l__spintent_spmM_luaset_is_truncated_str
\l__spintent_spmD_luaset_is_truncated_str
\l__spintent_spmM_prefix_tl
\l__spintent_spmD_prefix_tl
\l__spintent_spmM_print_result_num_tl
\l__spintent_spmD_print_result_num_tl
2189 \seq_new:N \l__spintent_md_out_seq
2190 \str_new:N \l__spintent_spmM_spmD_luaset_error_str
2191 \str_new:N \l__spintent_spmM_luaset_result_tl
2192 \clist_map_inline:nn { nM , nD }
2193 {
2194     \tl_new:c { l__spintent_sp #1 _luaset_result_tl }
2195     \str_new:c { l__spintent_sp #1 _luaset_is_truncated_str }
2196     \tl_new:c { l__spintent_sp #1 _prefix_tl }
2197 }

```

(End of definition for `\l__spintent_md_out_seq` and others.)

#### 11.27.2 Keys for `\spnM` and `\spnD`

We define the shared `<keys>` `result`, `silent-cmd`, `result-num`, `prefix` and `concept`.

```

2198 \clist_map_inline:nn { nM , nD }
2199 {
2200     \keys_define:nn { spintent / sp #1 }
2201     {
2202         result .bool_set:c = { l__spintent_sp #1 _print_result_bool },
2203         result .initial:n = false,
2204         result .value_required:n = true,
2205         silent-cmd .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
2206         silent-cmd .initial:n = false,
2207         silent-cmd .value_required:n = true,
2208         result-num .tl_set:c = { l__spintent_sp #1 _print_result_num_tl },
2209         result-num .initial:n = ,

```

```

2210     prefix      .code:n =
2211     { \__spintent_sanitizetize_affix:cn { \__spintent_sp #1 _prefix_tl } {##1} },
2212     prefix      .initial:n = ,
2213     prefix      .value_required:n = true,
2214     concept     .bool_set:c = { \__spintent_sp #1 _concept_bool },
2215     concept     .initial:n = true,
2216     concept     .value_required:n = true,
2217     unknown     .code:n =
2218     {
2219         \str_set:Nn \__spintent_command_name_str { sp #1 }
2220         \__spintent_unknown_keys_cmd:n {##1}
2221     },
2222 }
2223 }

```

(End of definition for *result* and others.)

### 11.27.3 Definition of error messages for \spnM and \spnD

md-invalid-argument

Exception triggered when the argument for multiples or divisors is invalid.

```

2224 \msg_new:nnnn { spintent } { md-invalid-argument }
2225 {
2226     Invalid~argument~provided~to~#1~\msg_line_context:.
2227 }
2228 {
2229     The~argument~must~be~a~single~positive~integer. \
2230 }

```

(End of definition for *md-invalid-argument*.)

### 11.27.4 Internal functions for \spnM and \spnD

\\_\_spintent\_spnM\_spnD\_process:nnnnn

The internal function `\__spintent_spnM_spnD_process:nnnnn` acts as the primary entry point for the multiples and divisors commands. It takes the command identifier ‘#1’, its printed mathematical symbol ‘#2’, its base reading string ‘#3’, the local key-value options ‘#4’, and the optional argument ‘#5’. Its sole purpose is to evaluate and set the provided keys within the current scope before delegating the core branching and typesetting logic to `\__spintent_spn_M_and_D_exec:nnnn`.

```

2231 \cs_new_protected:Nn \__spintent_spnM_spnD_process:nnnnn
2232 {
2233     \keys_set:nn { spintent / sp #1 } { #4 }
2234     \__spintent_spnM_spnD_exec:nnnn { #1 } { #2 } { #3 } { #5 }
2235 }

```

(End of definition for `\__spintent_spnM_spnD_process:nnnnn`.)

\\_\_spintent\_spnM\_spnD\_intent:nn

The internal function `\__spintent_spnM_spnD_intent:nn` resolves the main reading intent string. It takes the command identifier ‘#1’ (such as “nM” or “nD”) and the base reading string ‘#2’. It automatically prepends the localized prefix if defined, normalizes accentuation for “*múltiplos*”, and appends ‘-de’.

```

2236 \cs_new:Nn \__spintent_spnM_spnD_intent:nn
2237 {
2238     \tl_if_empty:cF { \__spintent_sp #1 _prefix_tl }
2239     { \tl_use:c { \__spintent_sp #1 _prefix_tl } - }
2240     \str_case:nnF { #2 }
2241     { { múltiplos } { múltiplos } }
2242     { { #2 }
2243     -de
2244 }

```

(End of definition for `\__spintent_spnM_spnD_intent:nn`.)

\\_\_spintent\_spnM\_spnD\_silent:nn

The internal function `\__spintent_spn_M_and_D_silent:nn` prints the atomized notation structure in full silent mode. It takes the mathematical symbol ‘#1’ (such as “D” or “M”) and the variable or explicit argument ‘#2’. Every component is wrapped in a silent intent, and an invisible separator is injected between the parentheses to prevent assistive technologies from applying default contextual reading rules.

```

2245 \cs_new_protected:Nn \__spintent_spnM_spnD_silent:nn
2246 {
2247     \MathMLintent { :silent } { \operatorname { #1 } }
2248     \MathMLintent { :silent } { ( }
2249     \MathMLintent { :silent } { \c__spintent_invisible_sep_int }
2250     \MathMLintent { :silent } { #2 }
2251     \MathMLintent { :silent } { ) }
2252 }

```

(End of definition for `\__spintent_spnM_spnD_silent:nn`.)

`\__spintent_spnM_spnD_result:n`

The internal function `\__spintent_spnM_and_D_result:n` handles the calculation and accessible formatting of the computed mathematical set. It takes the command identifier ‘#1’. It invokes the corresponding Lua backend calculation, reporting `md-invalid-argument` if that fails — a safety net beyond `\__spintent_luafun_spnmd_classify:n`’s own check, since the Lua calculation never trusted that check blindly. On success, it maps the resulting clist into a sequence, and formats the final printed output utilizing custom accessibility intents for both separators and conjunctions. It also safely manages truncation indicators.

```

2253 \cs_new_protected:Nn \__spintent_spnM_spnD_result:n
2254 {
2255   \use:c { __spintent_luafun_calcular_ #1 :nn }
2256   { \__spintent_luaset_part_int_tl }
2257   { \tl_use:c { l__spintent_sp #1 _print_result_num_tl } }
2258   \str_if_eq:VnTF \__spintent_spnM_spnD_luaset_error_str { true }
2259   {
2260     \msg_error:nne { spintent } { md-invalid-argument } { \exp_not:c { sp #1 } }
2261   }
2262   {
2263     \MathMLintent { _son } { = \{ }
2264     \seq_set_from_clist:Ne
2265       \__spintent_md_out_seq { \tl_use:c { l__spintent_sp #1 _luaset_result_tl } }
2266     \seq_use:Nnnn \__spintent_md_out_seq
2267       { \MathMLintent { _y } { , } }
2268       { \MathMLintent { :silent } { , } }
2269       { \MathMLintent { _y } { , } }
2270     \str_if_eq:vnT
2271       { l__spintent_sp #1 _luaset_is_truncated_str } { true }
2272     {
2273       \MathMLintent { :silent } { { , } ~ \dots
2274     }
2275     \MathMLintent { :silent } { \} }
2276   }
2277 }

```

(End of definition for `\__spintent_spnM_spnD_result:n`.)

`\__spintent_spnM_spnD_exec:nnnn`

The function `\__spintent_spnM_spnD_exec:nnnn` takes the command name in ‘#1’, the symbol in ‘#2’, the base reading string in ‘#3’, and the optional argument in ‘#4’. The parentheses are always silenced.

With ‘#4’ absent, the placeholder `n` stands in for the argument. With ‘#4’ given, `\__spintent_luafun_spnmd_classify:n` decides whether it is wrapped through `\spnum` — and, when it is, whether `\__spintent_spnM_spnD_result:n` also prints the computed set, per `result` — or left verbatim, with no result attempted.

Either way, `silent-cmd` silences the whole thing through `\__spintent_spnM_spnD_silent:nn`; otherwise `concept true` carries the literal intent from `\__spintent_spnM_spnD_intent:nn`, `false` silences only the operator symbol, leaving the argument to read normally.

```

2278 \cs_new_protected:Nn \__spintent_spnM_spnD_exec:nnnn
2279 {
2280   \tl_if_novalue:nTF { #4 }
2281   {
2282     \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
2283     { \__spintent_spnM_spnD_silent:nn { #2 } { n } }
2284     {
2285       \bool_if:cTF { l__spintent_sp #1 _concept_bool }
2286       {
2287         \MathMLintent
2288           { \__spintent_spnM_spnD_intent:nn { #1 } { #3 } }
2289           { \operatorname { #2 } }
2290       }
2291       { \MathMLintent { :silent } { \operatorname { #2 } } }
2292       \MathMLintent { :silent } { ( }
2293       n
2294       \MathMLintent { :silent } { ) }
2295     }
2296   }
2297   {
2298     \__spintent_luafun_spnmd_classify:n { \exp_not:n { #4 } }
2299     \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
2300     {
2301       \str_if_eq:VnTF \__spintent_spnmd_luaset_is_number_str { true }
2302       { \__spintent_spnM_spnD_silent:nn { #2 } { \spnum { #4 } } }
2303       { \__spintent_spnM_spnD_silent:nn { #2 } { #4 } }
2304     }
2305   }

```

```

2306         \bool_if:cTF { \l__spinttent_sp #1 _concept_bool }
2307         {
2308             \MathMLIntent
2309             { \l__spinttent_spnM_spnD_intent:nn { #1 } { #3 } }
2310             { \operatorname { #2 } }
2311         }
2312         { \MathMLIntent { :silent } { \operatorname { #2 } } }
2313         \MathMLIntent { :silent } { ( }
2314         \str_if_eq:VnTF \l__spinttent_spnmd_luaset_is_number_str { true }
2315         { \spnum { #4 } }
2316         { #4 }
2317         \MathMLIntent { :silent } { ) }
2318     }
2319     \str_if_eq:VnT \l__spinttent_spnmd_luaset_is_number_str { true }
2320     {
2321         \bool_if:cT { \l__spinttent_sp #1 _print_result_bool }
2322         { \l__spinttent_spnM_spnD_result:n { #1 } }
2323     }
2324 }
2325 }

```

(End of definition for `\l__spinttent_spnM_spnD_exec:nnnn`.)

`\spnM` The commands `\spnM` and `\spnD` act as the public interface wrappers for typesetting multiples and divisors, respectively. They enforce math mode, open a local group, and defer execution to the function `\l__spinttent_spnM_spnD_process:nnnnn`, passing along the optional *(keys)* ‘#1’ and the numeric *{(argument)}* ‘#2’.

```

2326 \NewDocumentCommand \spnM { 0{} d() }
2327 {
2328     \mode_if_math:TF
2329     {
2330         \group_begin:
2331         \l__spinttent_spnM_spnD_process:nnnnn { nM } { M } { multiplos } { #1 } { #2 }
2332         \group_end:
2333     }
2334     { \msg_error:nnn { spinttent } { need-math-mode } { \spnM } }
2335 }
2336 \NewDocumentCommand \spnD { 0{} d() }
2337 {
2338     \mode_if_math:TF
2339     {
2340         \group_begin:
2341         \l__spinttent_spnM_spnD_process:nnnnn { nD } { D } { divisores } { #1 } { #2 }
2342         \group_end:
2343     }
2344     { \msg_error:nnn { spinttent } { need-math-mode } { \spnD } }
2345 }

```

(End of definition for `\spnM` and `\spnD`. These functions are documented on page 15.)

## 11.28 The command `\sprazon`

The user command `\sprazon` typesets a ratio  $a:b$  as a fraction (`\dfrac` or `\frac`, per *use-dfrac*), read natively — no Lua parsing involved.

### 11.28.1 Definition of keys for `\sprazon`

`silent-cmd` We define the *(keys)* `silent-cmd`, `use-dfrac` and `inline`.

```

2346 \keys_define:nn { spinttent / sprazon }
2347 {
2348     silent-cmd .bool_set:N = \l__spinttent_sprazon_silent_cmd_bool,
2349     silent-cmd .initial:n = false,
2350     silent-cmd .value_required:n = true,
2351     use-dfrac .bool_set:N = \l__spinttent_sprazon_use_dfrac_bool,
2352     use-dfrac .initial:n = false,
2353     use-dfrac .value_required:n = true,
2354     inline .bool_set:N = \l__spinttent_sprazon_inline_bool,
2355     inline .initial:n = false,
2356     inline .value_required:n = true,
2357     unknown .code:n =
2358     {
2359         \str_set:Nn \l__spinttent_command_name_str { sprazon }
2360         \l__spinttent_unknown_keys_cmd:n { #1 }

```

```

2361     },
2362 }

```

(End of definition for `silent-cmd` and others.)

`\__spintent_sprazon_exec:nnn`

The function `\__spintent_sprazon_exec:nnn` receives the optional  $\langle keys \rangle$  in ‘#1’, the numerator in ‘#2’ and the denominator in ‘#3’. Under `silent-cmd`, the whole thing is silenced as one unit — `\frac/\dfrac` (per `use-dfrac`) normally, or `#2 : #3` under `inline`.

Otherwise, with `inline=false` (the default), numerator and denominator are each tagged through a native `\MathMLarg` (num/denom) inside `\frac/\dfrac`, and the outer `\MathMLintent` reads `_es-a:infix($num,$denom)`. With `inline=true`, the same two references are tagged side by side instead — no `\frac/\dfrac` involved — with the `:` between them carrying the `_es-a:infix($num,$denom)` intent on its own.

```

2363 \cs_new_protected:Nn \__spintent_sprazon_exec:nnn
2364 {
2365   \keys_set:nn { spintent / sprazon } { #1 }
2366   \__spintent_invisible_sep:
2367   \bool_if:NTF \__spintent_sprazon_silent_cmd_bool
2368   {
2369     \MathMLintent { _- }
2370     {
2371       \__spintent_inner_sep:
2372       \bool_if:NTF \__spintent_sprazon_inline_bool
2373       { #2 : #3 }
2374       {
2375         \bool_if:NTF \__spintent_sprazon_use_dfrac_bool
2376         { \dfrac { #2 } { #3 } }
2377         { \frac { #2 } { #3 } }
2378       }
2379     }
2380   }
2381   {
2382     \bool_if:NTF \__spintent_sprazon_inline_bool
2383     {
2384       \MathMLintent { _es-a:infix($num,$denom) }
2385       {
2386         \MathMLarg { num } { #2 }
2387         :
2388         \MathMLarg { denom } { #3 }
2389       }
2390     }
2391     {
2392       \MathMLintent { _es-a:infix($num,$denom) }
2393       {
2394         \__spintent_inner_sep:
2395         \bool_if:NTF \__spintent_sprazon_use_dfrac_bool
2396         {
2397           \dfrac { \MathMLarg { num } { #2 } } { \MathMLarg { denom } { #3 } }
2398         }
2399         { \frac { \MathMLarg { num } { #2 } } { \MathMLarg { denom } { #3 } } }
2400       }
2401     }
2402   }
2403 }

```

(End of definition for `\__spintent_sprazon_exec:nnn`.)

`\sprazon`

The public command `\sprazon` enforces *math mode* and delegates execution to `\__spintent_sprazon_exec:nnn`.

```

2404 \NewDocumentCommand \sprazon { O{} m m }
2405 {
2406   \mode_if_math:TF
2407   {
2408     \group_begin:
2409     \__spintent_sprazon_exec:nnn { #1 } { #2 } { #3 }
2410     \group_end:
2411   }
2412   { \msg_error:nnn { spintent } { need-math-mode } { \sprazon } }
2413 }

```

(End of definition for `\sprazon`. This function is documented on page ??.)

### 11.29 The command `\spProp`

The user command `\spProp` typesets a proportion  $a:b = c:d$  as two fractions joined by  $=$ , reusing `\__spintent_sprazon_exec:nnn` directly (not the public `\sprazon` command) for each ratio.

`\spProp` The public command `\spProp` enforces *math mode*, and forwards the same  $\langle keys \rangle$  in ‘#1’ to both ratios:  $a/b$  (‘#2’/‘#3’) and  $c/d$  (‘#4’/‘#5’), joined by a native `\MathMLintent` reading `_como` over the  $=$  sign.

```

2414 \NewDocumentCommand \spProp { 0{} m m m m }
2415 {
2416   \mode_if_math:TF
2417   {
2418     \group_begin:
2419     \__spintent_sprazon_exec:nnn { #1 } { #2 } { #3 }
2420     \bool_if:NTF \__spintent_sprazon_silent_cmd_bool
2421     {
2422       \MathMLintent { :silent } { = }
2423     }
2424     { \MathMLintent { _como } { = } }
2425     \__spintent_sprazon_exec:nnn { #1 } { #4 } { #5 }
2426   \group_end:
2427 }
2428 { \msg_error:nnn { spintent } { need-math-mode } { \spProp } }
2429 }
```

(End of definition for `\spProp`. This function is documented on page ??.)

### 11.30 The command `\spang`

The command `\spang` is the main public command for typesetting and tagging sexagesimal angle expressions (degrees, minutes, and seconds). It parses the input using the Lua function `\__spintent_luafun_spang_parse:n`. This separates the degrees, minutes, and seconds to construct a `MathML intent` tree with the correct postfixes for NVDA/MathCAT.

```

\__spintent_spang_luaset_grado_str
\__spintent_spang_luaset_minuto_str
\__spintent_spang_luaset_segundo_str
\__spintent_spang_luaset_error_str
```

These internal string variables store the values returned from the Lua module `spintent.lua`. They hold the extracted numerical values for degrees, minutes, and seconds, as well as the error status used during formatting.

```

2430 \str_new:N \__spintent_spang_luaset_grado_str
2431 \str_new:N \__spintent_spang_luaset_minuto_str
2432 \str_new:N \__spintent_spang_luaset_segundo_str
2433 \str_new:N \__spintent_spang_luaset_error_str
```

(End of definition for `\__spintent_spang_luaset_grado_str` and others.)

#### 11.30.1 Definition of error messages for `\spang`

`invalid-angle-format`

The error message `invalid-angle-format` defines the text displayed when an invalid angle input is detected. It is triggered when an argument fails the format checks, does not match the expected sexagesimal pattern, or contains invalid measurement symbols.

```

2434 \msg_new:nnnn { spintent } { invalid-angle-format }
2435 { Invalid~angle~format~in~#1. }
2436 {
2437   The~argument~'~#2'~could~not~be~parsed. \\\
2438 }
```

(End of definition for `invalid-angle-format`.)

#### 11.30.2 Internal functions for `\spang`

`\__spintent_spang_process:n`

The internal function `\__spintent_spang_process:n` parses the angle input using `\__spintent_luafun_spang_parse:n` after resetting the error flag in `\__spintent_spang_luaset_error_str`. If an invalid format is detected, it issues the `invalid-angle-format` error. Otherwise, it checks the extracted string variables `\__spintent_spang_luaset_grado_str`, `\__spintent_spang_luaset_minuto_str`, and `\__spintent_spang_luaset_segundo_str`. For each non-empty variable, it constructs the corresponding `MathML intent` structure with the correct measurement symbols and postfixes for NVDA/MathCAT.

```

2439 \cs_new_protected:Nn \__spintent_spang_process:n
2440 {
2441   \str_set:Nn \__spintent_spang_luaset_error_str { false }
2442   \__spintent_luafun_spang_parse:n { #1 }
2443   \str_if_eq:NnTF \__spintent_spang_luaset_error_str { true }
2444   {
2445     \msg_error:nnnn { spintent } { invalid-angle-format } { \spang } { #1 }
2446   }
2447 }
```

```

2448 \str_if_empty:NF \l__spintent_spang_luaset_grado_str
2449 {
2450   \spunit{ \l__spintent_spang_luaset_grado_str °}
2451 }
2452 \str_if_empty:NF \l__spintent_spang_luaset_minuto_str
2453 {
2454   \MathMLintent { minutos : postfix($m) }
2455   {
2456     \MathMLarg { m }
2457     {
2458       \spunit{ \l__spintent_spang_luaset_minuto_str ' }
2459     }
2460   }
2461 }
2462 \str_if_empty:NF \l__spintent_spang_luaset_segundo_str
2463 {
2464   \MathMLintent { segundos : postfix($s) }
2465   {
2466     \MathMLarg { s }
2467     {
2468       \spunit { \l__spintent_spang_luaset_segundo_str " }
2469     }
2470   }
2471 }
2472 }
2473 }

```

(End of definition for `\l__spintent_spang_process:n`.)

**\spang** The command `\spang` is the public interface for formatting sexagesimal angle expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it calls `\l__spintent_spang_process:n` to process the  $\langle argument \rangle$ .

```

2474 \NewDocumentCommand \spang { m }
2475 {
2476   \mode_if_math:TF
2477   {
2478     \group_begin:
2479     \l__spintent_spang_process:n { #1 }
2480     \group_end:
2481   }
2482   { \msg_error:nnn { spintent } { need-math-mode } { \spang } }
2483 }

```

(End of definition for `\spang`. This function is documented on page 23.)

### 11.31 The command `\spnZ`

The user-level command `\spnZ` formats integers (the set  $\mathbb{Z}$ ). It includes specific options to automatically enclose numbers (particularly negative ones) in visual parentheses and allows you to control whether these parentheses are read by MathCAT.

This command verifies that the  $\langle argument \rangle$  is indeed an *integer*, checking for the complete absence of decimal separators or attached units of measurement.

#### 11.31.1 Definition of keys for `\spnZ`

Now we define the  $\langle keys \rangle$  `cifras-sep` and `read-sign`. These are `.meta:nn`  $\langle keys \rangle$  passed directly to the `\spnum` command. We also introduce `wrap-parens`, which determines whether the number is enclosed in parentheses, and `read-parens`, which controls if these parentheses are read by MathCAT.

```

2484 \keys_define:nn { spintent / spnZ }
2485 {
2486   cifras-sep .meta:nn = { spintent / spnum } { cifras-sep = {#1} },
2487   read-sign .meta:nn = { spintent / spnum } { read-sign = {#1} },
2488   wrap-parens .bool_set:N = \l__spintent_spnz_wrap_parens_bool,
2489   wrap-parens .initial:n = false,
2490   wrap-parens .value_required:n = true,
2491   read-parens .bool_set:N = \l__spintent_spnz_read_parens_bool,
2492   read-parens .initial:n = true,
2493   read-parens .value_required:n = true,
2494   unknown .code:n =
2495   {
2496     \str_set:Nn \l__spintent_command_name_str { spnZ }
2497     \__spintent_unknown_keys_cmd:n {#1}

```

```

2498     },
2499 }

```

(End of definition for *cifras-sep* and others.)

### 11.31.2 Definition of error messages

spnz-not-integer

Raised when the  $\langle \textit{argument} \rangle$  contains decimal separators or attached units, which conflict with the constraints of a pure integer representation.

```

2500 \msg_new:nnnn { spintent } { spnz-not-integer }
2501 { The~value~'#1'~is~not~a~valid~integer. }
2502 {
2503   Command~\token_to_str:N \spnZ~only~accepts~integers.\\
2504 }

```

(End of definition for *spnz-not-integer*.)

### 11.31.3 Internal functions for $\backslash\textit{spnZ}$

$\backslash\textit{__spintent\_spnz\_render\_parens}$ :

The function  $\backslash\textit{__spintent\_spnz\_render\_parens}$ : first checks the state of the boolean variable  $\backslash\textit{__spintent\_spnz\_read\_parens\_bool}$  (controlled by the *read-parens* key). If it is *true*, it prints the number surrounded by parentheses. Otherwise, it applies the *:silent function intent* to the parentheses while printing the number.

```

2505 \cs_new_protected:Nn \__spintent_spnz_render_parens:
2506 {
2507   \bool_if:NTF \__spintent_spnz_read_parens_bool
2508   {
2509     ( \__spintent_spnum_print_num_start: )
2510   }
2511   {
2512     \MathMLintent { :silent } { ( }
2513     \__spintent_spnum_print_num_start:
2514     \MathMLintent { :silent } { ) }
2515   }
2516 }

```

(End of definition for  $\backslash\textit{__spintent\_spnz\_render\_parens}$ .)

$\backslash\textit{__spintent\_spnz\_process:nn}$

The function  $\backslash\textit{__spintent\_spnz\_process:nn}$  first processes the  $\langle \textit{keys} \rangle$  passed in ‘#1’ and then calls  $\backslash\textit{__spintent\_luafun\_clean\_split\_and\_set:n}$  on ‘#2’. It verifies that the input is a valid integer by checking if the variable  $\backslash\textit{__spintent\_luaset\_has\_integer\_str}$  is *true*. If not *true*, an “error” is raised.

If it is *true*, it checks whether  $\backslash\textit{__spintent\_luaset\_part\_int\_tl}$  is *empty* and raises an “error” if so. Otherwise, it checks  $\backslash\textit{__spintent\_spnz\_wrap\_parens\_bool}$  set by *wrap-parens* and calls either the function  $\backslash\textit{__spintent\_spnz\_render\_parens}$ : or  $\backslash\textit{__spintent\_spnum\_print\_num\_start:}$ .

```

2517 \cs_new_protected:Nn \__spintent_spnz_process:nn
2518 {
2519   \keys_set:nn { spintent / spnZ } {#1}
2520   \__spintent_luafun_clean_split_and_set:n { \exp_not:n {#2} }
2521   \str_if_eq:VnTF \__spintent_luaset_has_integer_str { true }
2522   {
2523     \tl_if_empty:NTF \__spintent_luaset_part_int_tl
2524     { \msg_error:nnn { spintent } { spnum-no-digits } {#2} }
2525     {
2526       \bool_if:NTF \__spintent_spnz_wrap_parens_bool
2527       { \__spintent_spnz_render_parens: }
2528       { \__spintent_spnum_print_num_start: }
2529     }
2530   }
2531   { \msg_error:nnn { spintent } { spnz-not-integer } {#2} }
2532 }

```

(End of definition for  $\backslash\textit{__spintent\_spnz\_process:nn}$ .)

$\backslash\textit{spnZ}$

Defines the user command  $\backslash\textit{spnZ}$ , which is executed within a *group* and strictly enforces *math mode*.

```

2533 \NewDocumentCommand \spnZ { O{} m }
2534 {
2535   \mode_if_math:TF
2536   {
2537     \group_begin:
2538     \__spintent_spnz_process:nn { #1 } { #2 }
2539     \group_end:
2540   }
2541   { \msg_error:nnn { spintent } { need-math-mode } { \spnZ } }
2542 }

```

(End of definition for `\spnZ`. This function is documented on page 16.)

### 11.32 The command `\spmQ`

The command `\spmQ` typesets “mixed numbers” within Spanish mathematical contexts (e.g.,  $3\frac{1}{2}$ ). It delegates the scaling of the integer part to the Lua module `spintent.lua` via `\__spintent_luafun_spmQ_scale_int:Vn`, which measures both the integer glyph and the fraction box at the node level and computes a scaling factor to align their heights visually. Finally, it constructs the correct MathML intent for MathCAT.

#### 11.32.1 Definition of variables for `\spmQ`

```
\__spintent_spmQ_join_word_str
\__spintent_spmQ_intent_read_sign_str
\__spintent_spmQ_wrap_parens_l_tl
\__spintent_spmQ_wrap_parens_r_tl
\__spintent_saved_luamml_flag_int
```

The variable `\__spintent_spmQ_join_word_str` stores the value set by the key `join-word` which will be verbalized by MathCAT as a grammatical connector between the integer and fractional parts.

The variable `\__spintent_spmQ_intent_read_sign_str` stores the `:intent` passed to `\MathMLintent` when the signs ‘+’ or ‘−’ are present in the integer part, used by the function `\__spintent_spmQ_print_scale_int:n`.

The variables `\__spintent_spmQ_wrap_parens_l_tl` and `\__spintent_spmQ_wrap_parens_r_tl` store the sized parentheses set by the function `\__spintent_spmQ_set_wrap_parens_size:` and used by the key `wrap-parens`.

The integer variable `\__spintent_saved_luamml_flag_int` temporarily stores the current value of `\__luamml_flag_int` before disabling it during the injection of the scaled integer node, and restores it afterwards. This prevents `luamml` from generating a spurious `<math>` element around the sub\_box processed in isolation, before `\frac` has expanded.

```
2543 \str_new:N \__spintent_spmQ_join_word_str
2544 \str_new:N \__spintent_spmQ_intent_read_sign_str
2545 \tl_new:N \__spintent_spmQ_wrap_parens_l_tl
2546 \tl_new:N \__spintent_spmQ_wrap_parens_r_tl
2547 \int_new:N \__spintent_saved_luamml_flag_int
```

(End of definition for `\__spintent_spmQ_join_word_str` and others.)

#### 11.32.2 Definition of error messages for `\spmQ`

```
spmQ-zero-denominator
spmQ-not-integer
```

The error message `spmQ-zero-denominator` is raised when a zero denominator is passed to `\spmQ`. The second error `spmQ-not-integer` is returned when the integer part passed in  $\{\langle argument \rangle\}$  is not actually an integer.

```
2548 \msg_new:nnnn { spintent } { spmQ-zero-denominator }
2549 {
2550   Zero~denominator~in~\token_to_str:N \spmQ~\msg_line_context:.
2551 }
2552 {
2553   The~denominator~of~a~fraction~cannot~be~zero.\\
2554 }
2555 \msg_new:nnnn { spintent } { spmQ-not-integer }
2556 {
2557   The~value~'#1'~is~not~a~valid~integer~\msg_line_context:.
2558 }
2559 {
2560   Command~\token_to_str:N \spmQ \c_space_tl only~accepts~integers.\\
2561 }
```

(End of definition for `spmQ-zero-denominator` and `spmQ-not-integer`.)

#### 11.32.3 Definition of keys for `\spmQ`

```
join-word
use-spnZ
wrap-parens
read-parens
print-dfrac
unknown
```

Now we add the keys `join-word`, `use-spnZ`, `wrap-parens`, `read-parens` and `print-dfrac`.

```
2562 \keys_define:nn { spintent / spmQ }
2563 {
2564   join-word .choices:nn =
2565     { entero , enteros , y }
2566     {
2567       \int_case:nn { \l_keys_choice_int }
2568       {
2569         {1} { \str_set:Nn \__spintent_spmQ_join_word_str { entero } }
2570         {2} { \str_set:Nn \__spintent_spmQ_join_word_str { enteros } }
2571         {3} { \str_set:Nn \__spintent_spmQ_join_word_str { y } }
2572       }
2573     },
2574   join-word .initial:n = y,
2575   join-word .value_required:n = true,
2576   join-word/unknown .code:n =
2577     \msg_error:nnee { spintent } { unknown-choice }
2578     { join-word } { entero , ~enteros , ~y } { \exp_not:n {#1} },
2579   use-spnZ .bool_set:N = \__spintent_spmQ_use_spnZ_bool,
```

```

2580     use-spnZ             .initial:n = true,
2581     use-spnZ             .value_required:n = true,
2582     print-dfrac          .bool_set:N = \__spintent_spmQ_print_dfrac_bool,
2583     print-dfrac          .initial:n = false,
2584     print-dfrac          .value_required:n = true,
2585     wrap-parens          .bool_set:N = \__spintent_spmQ_wrap_parens_bool,
2586     wrap-parens          .initial:n = false,
2587     wrap-parens          .value_required:n = true,
2588     read-parens          .bool_set:N = \__spintent_spmQ_read_parens_bool,
2589     read-parens          .initial:n = true,
2590     read-parens          .value_required:n = true,
2591     unknown              .code:n =
2592     {
2593         \str_set:Nn \__spintent_command_name_str { spmQ }
2594         \__spintent_unknown_keys_cmd:n {#1}
2595     },
2596 }

```

(End of definition for join-word and others.)

#### 11.32.4 Internal functions for \spmQ

\\_\_spintent\_spmQ\_set\_wrap\_parens\_size:

The function `\__spintent_spmQ_set_wrap_parens_size:` captures the current mathematical style using the primitive `\mathstyle` from LuaTeX and sets the variables `\__spintent_spmQ_wrap_parens_l_tl` and `\__spintent_spmQ_wrap_parens_r_tl` with the appropriate size for the parentheses. Display styles (0 and 1) receive `\Bigl\lparen` and `\Bigr\rparen`; text styles (2 and 3) receive `\bigr\lparen` and `\bigr\rparen`; any other style falls back to unsized `\lparen` and `\rparen`.

If the key `print-dfrac` is active, the size is always forced to `\Bigl\lparen`/`\Bigr\rparen` regardless of the current mathematical style, since `\dfrac` always produces a display-height fraction.

```

2597 \cs_new_protected:Nn \__spintent_spmQ_set_wrap_parens_size:
2598 {
2599     \int_case:nnF { \mathstyle }
2600     {
2601         { 0 }
2602         {
2603             \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \Bigl \lparen }
2604             \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \Bigr \rparen }
2605         }
2606         { 1 }
2607         {
2608             \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \Bigl \lparen }
2609             \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \Bigr \rparen }
2610         }
2611         { 2 }
2612         {
2613             \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \bigr \lparen }
2614             \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \bigr \rparen }
2615         }
2616         { 3 }
2617         {
2618             \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \bigr \lparen }
2619             \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \bigr \rparen }
2620         }
2621     }
2622     {
2623         \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \lparen }
2624         \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \rparen }
2625     }

```

If the key `print-dfrac` is active, we must reset the variables `\__spintent_spmQ_wrap_parens_l_tl` and `\__spintent_spmQ_wrap_parens_r_tl` to `\Bigl\lparen`/`\Bigr\rparen` unconditionally.

```

2626     \bool_if:NT \__spintent_spmQ_print_dfrac_bool
2627     {
2628         \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \Bigl \lparen }
2629         \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \Bigr \rparen }
2630     }
2631 }

```

(End of definition for `\__spintent_spmQ_set_wrap_parens_size:`.)

\\_\_spintent\_spmQ\_start\_wrap\_parens:  
\\_\_spintent\_spmQ\_stop\_wrap\_parens:

The functions `\__spintent_spmQ_start_wrap_parens:` and `\__spintent_spmQ_stop_wrap_parens:` open and close the parentheses around the “mixed number” when the key `wrap-parens` is active. They first call

`\__spintent_spmQ_set_wrap_parens_size:` to determine the correct size for the current mathematical style, then print the appropriate parenthesis stored in `\__spintent_spmQ_wrap_parens_l_tl` or `\__spintent_spmQ_wrap_parens_r_tl`.

If the variable `\__spintent_spmQ_read_parens_bool`, controlled by the key `read-parens`, is *false*, each parenthesis is wrapped inside `\MathMLintent{ :silent }` so that NVDA/MathCAT does not verbalize them.

```

2632 \cs_new_protected:Nn \__spintent_spmQ_start_wrap_parens:
2633 {
2634   \bool_if:NT \__spintent_spmQ_wrap_parens_bool
2635   {
2636     \__spintent_spmQ_set_wrap_parens_size:
2637     \bool_if:NTF \__spintent_spmQ_read_parens_bool
2638     { \__spintent_spmQ_wrap_parens_l_tl }
2639     { \MathMLintent { :silent } { \__spintent_spmQ_wrap_parens_l_tl } }
2640   }
2641 }
2642 \cs_new_protected:Nn \__spintent_spmQ_stop_wrap_parens:
2643 {
2644   \bool_if:NT \__spintent_spmQ_wrap_parens_bool
2645   {
2646     \__spintent_spmQ_set_wrap_parens_size:
2647     \bool_if:NTF \__spintent_spmQ_read_parens_bool
2648     { \__spintent_spmQ_wrap_parens_r_tl }
2649     { \MathMLintent { :silent } { \__spintent_spmQ_wrap_parens_r_tl } }
2650   }
2651 }

```

(End of definition for `\__spintent_spmQ_start_wrap_parens:` and `\__spintent_spmQ_stop_wrap_parens:`.)

`\__spintent_spmQ_set_join_word:`

The function `\__spintent_spmQ_set_join_word:` injects the grammatical connector between the integer part and the fraction by dispatching on the value stored in `\__spintent_spmQ_join_word_str`, set by the key `join-word`. In all three cases the visual output is the Unicode invisible separator U+2063 via `\c__spintent_invisible_sep_tl`, which produces a `<mi>` node in the MathML structure; what differs is the `:intent` text attached to it via `\MathMLintent`, which NVDA/MathCAT will verbalize as «entero», «enteros» or «y» respectively.

```

2652 \cs_new_protected:Nn \__spintent_spmQ_set_join_word:
2653 {
2654   \str_case:Nn \__spintent_spmQ_join_word_str
2655   {
2656     { entero } { \MathMLintent { _entero } { \c__spintent_invisible_sep_int } }
2657     { enteros } { \MathMLintent { _enteros } { \c__spintent_invisible_sep_int } }
2658     { y } { \MathMLintent { _y- } { \c__spintent_invisible_sep_int } }
2659   }
2660 }

```

(End of definition for `\__spintent_spmQ_set_join_word:`.)

`\__spintent_spmQ_print_scale_int:`

The function `\__spintent_spmQ_print_scale_int:` prints the integer part of the “mixed number” scaled to match the visual height of the accompanying fraction. Before calling the Lua function `\__spintent_luafun_spmQ_scale_int` it temporarily disables `\luamml` by saving the current value of `\__luamml_flag_int` into `\__spintent_saved_luamml_flag_int` and setting it to 0 (skip completely). This prevents `\luamml` from generating a spurious `<math>` wrapper around the `sub_box` node that holds the scaled integer, which is processed in isolation before `\frac` has expanded. Once the Lua function returns, the original flag value is restored via `\__spintent_saved_luamml_flag_int`.

The Lua function receives the integer digits from `\__spintent_luaset_part_int_tl` and the fraction command name (`\frac` or `\dfrac`, determined by the key `print-dfrac`) as its two arguments, and inserts the appropriately scaled `sub_box` node directly into the current horizontal list.

```

2661 \cs_new_protected:Nn \__spintent_spmQ_print_scale_int:
2662 {
2663   \int_set_eq:Nc \__spintent_saved_luamml_flag_int { \__luamml_flag_int }
2664   \int_set:Nn \__luamml_flag_int { 0 }
2665   \bool_if:NTF \__spintent_spmQ_print_dfrac_bool
2666   { \__spintent_luafun_spmQ_scale_int:Nn \__spintent_luaset_part_int_tl { dfrac } }
2667   { \__spintent_luafun_spmQ_scale_int:Nn \__spintent_luaset_part_int_tl { frac } }
2668   \int_set_eq:cN { \__luamml_flag_int } \__spintent_saved_luamml_flag_int
2669 }

```

(End of definition for `\__spintent_spmQ_print_scale_int:`.)

`\__spintrent_spmQ_print_scale_int:n`

The function `\__spintrent_spmQ_print_scale_int:n` takes the integer part of the “mixed number” passed in ‘#1’. It first calls `\__spintrent_luafun_clean_split_and_set:n` on ‘#1’ to parse and populate the shared Lua variables, then checks the status of `\__spintrent_luaset_has_integer_str`.

If the status is `true`, it inspects `\__spintrent_luaset_sign_tl`. When the sign is *empty*, it delegates directly to `\__spintrent_spmQ_print_scale_int:` to render the scaled integer. When a sign is present, it sets `\__spintrent_spmQ_intent_read_sign_str` to either `_más:prefix($e)` or `_menos:prefix($e)` and wraps the sign glyph together with a `MathML` argument (`$e`) containing the scaled integer inside a `\MathMLintent`, so that NVDA/MathCAT verbalizes the sign as a prefix before the integer value. An `\__spintrent_invisible_sep:` call between the sign and the `MathMLarg` forces the opening of a new `<mrow>` in the `MathML` structure, which is required for the intent tree to be well-formed.

If the status is `false`, the `spmQ-not-integer` “error” is raised immediately.

```

2670 \cs_new_protected:Nn \__spintrent_spmQ_print_scale_int:n
2671 {
2672   \__spintrent_luafun_clean_split_and_set:n { \exp_not:n {#1} }
2673   \str_if_eq:VnTF \__spintrent_luaset_has_integer_str { true }
2674   {
2675     \tl_if_empty:NTF \__spintrent_luaset_sign_tl
2676     {
2677       \__spintrent_spmQ_print_scale_int:
2678     }
2679     {
2680       \str_if_eq:VnTF \__spintrent_luaset_sign_tl { + }
2681       {
2682         \str_set:Nn
2683           \__spintrent_spmQ_intent_read_sign_str { _más:prefix($e) }
2684       }
2685       {
2686         \str_set:Nn
2687           \__spintrent_spmQ_intent_read_sign_str { _menos:prefix($e) }
2688       }
2689       \MathMLintent { \__spintrent_spmQ_intent_read_sign_str }
2690       {
2691         \__spintrent_luaset_sign_tl \c__spintrent_invisible_sep_int % \__spintrent_invisible_s
2692         \MathMLarg { e } { \__spintrent_spmQ_print_scale_int: }
2693       }
2694     }
2695   }
2696   { \msg_error:nnn { spintrent } { spmQ-not-integer } { #1 } }
2697 }

```

(End of definition for `\__spintrent_spmQ_print_scale_int:n`)

`\__spintrent_spmQ_print_spmZ_dfrac:nn`

The function `\__spintrent_spmQ_print_spmZ_dfrac:nn` typesets the fractional part of the “mixed number”. It takes the numerator in ‘#1’ and the denominator in ‘#2’. It first calls `\__spintrent_luafun_clean_split_and_set:` on ‘#1’ to validate that the numerator is an integer; if `\__spintrent_luaset_has_integer_str` is `false`, the `spmQ-not-integer` “error” is raised immediately.

If the validation passes, the function dispatches on the two boolean variables `\__spintrent_spmQ_use_spmZ_bool` and `\__spintrent_spmQ_print_dfrac_bool`, controlled respectively by the keys `use-spmZ` and `print-dfrac`. When `use-spmZ` is `true`, both the numerator and denominator are passed through `\spnZ` to gain `MathML` integer semantics; otherwise, the raw token lists ‘#1’ and ‘#2’ are used directly. In both branches, the fraction command is selected dynamically via `\use:c:` if `print-dfrac` is `true`, `dfrac` is assembled; otherwise, `frac` is used.

```

2698 \cs_new_protected:Nn \__spintrent_spmQ_print_spmZ_dfrac:nn
2699 {
2700   \__spintrent_luafun_clean_split_and_set:n { \exp_not:n {#1} }
2701   \str_if_eq:VnTF \__spintrent_luaset_has_integer_str { false }
2702   {
2703     \msg_error:nnn { spintrent } { spmQ-not-integer } { #1 }
2704   }
2705   \bool_if:NTF \__spintrent_spmQ_use_spmZ_bool
2706   {
2707     \use:c { \bool_if:NT \__spintrent_spmQ_print_dfrac_bool { d } frac }
2708     { \spnZ { #1 } } { \spnZ { #2 } }
2709   }
2710   {
2711     \use:c { \bool_if:NT \__spintrent_spmQ_print_dfrac_bool { d } frac }
2712     { #1 } { #2 }
2713   }
2714 }

```

(End of definition for `\__spintent_spmQ_print_spnZ_dfrac:nn`.)

`\__spintent_spmQ_print:nnn`

The function `\__spintent_spmQ_print:nnn` is the main rendering orchestrator for `\spmQ`. It takes three arguments: the integer part ‘#1’, the numerator ‘#2’, and the denominator ‘#3’. It opens the optional parentheses via `\__spintent_spmQ_start_wrap_parens:`, then calls `\__spintent_spmQ_print_scale_int:n` to scale and render the integer part. Next, it injects the grammatical connector between the integer and the fraction via `\__spintent_spmQ_set_join_word:`. Then it calls `\__spintent_spmQ_print_spnZ_dfrac:nn` to typeset the fraction using the keys `print-dfrac` and `use-spnZ`. Finally, it closes the optional parentheses via `\__spintent_spmQ_stop_wrap_parens:`.

```
2715 \cs_new_protected:Nn \__spintent_spmQ_print:nnn
2716 {
2717   \__spintent_spmQ_start_wrap_parens:
2718   \__spintent_spmQ_print_scale_int:n { #1 }
2719   \__spintent_spmQ_set_join_word:
2720   \__spintent_spmQ_print_spnZ_dfrac:nn { #2 } { #3 }
2721   \__spintent_spmQ_stop_wrap_parens:
2722 }
```

(End of definition for `\__spintent_spmQ_print:nnn`.)

`\__spintent_spmQ_exec:nnnn`

The internal function `\__spintent_spmQ_exec:nnnn` takes four arguments: the optional *(keys)* ‘#1’, the integer ‘#2’, the numerator ‘#3’, and the denominator ‘#4’. First sets the optional *(keys)* from ‘#1’, then processes the denominator input in ‘#4’ using `\__spintent_luafun_clean_split_and_set:n` to validate it is a non-zero integer.

It checks the status of `\__spintent_luaaset_has_integer_str`. If `true` and the integer part is exactly zero, it issues the `spmQ-zero-denominator` “error”. Otherwise, whether the status is `false` or the integer part is *non-zero*, it calls `\__spintent_spmQ_print:nnn` to format the “mixed number”.

```
2723 \cs_new_protected:Nn \__spintent_spmQ_exec:nnnn
2724 {
2725   \keys_set:nn { spintent / spmQ } { #1 }
2726   \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #4 } }
2727   \str_if_eq:VnTF \__spintent_luaaset_has_integer_str { true }
2728   {
2729     \int_compare:nNnTF { \__spintent_luaaset_part_int_tl } = { 0 }
2730     { \msg_error:nn { spintent } { spmQ-zero-denominator } }
2731     { \__spintent_spmQ_print:nnn { #2 } { #3 } { #4 } }
2732   }
2733   { \__spintent_spmQ_print:nnn { #2 } { #3 } { #4 } }
2734 }
```

(End of definition for `\__spintent_spmQ_exec:nnnn`.)

`\spmQ`

The command `\spmQ` is the public interface for formatting “mixed numbers”. It first checks if it is being used in *math mode*, issuing an “error” if not. Inside a local group, it calls `\__spintent_spmQ_exec:nnnn` to process the optional *(keys)* and the *{(arguments)}*.

```
2735 \NewDocumentCommand \spmQ { O{} m m m }
2736 {
2737   \mode_if_math:TF
2738   {
2739     \group_begin:
2740     \__spintent_spmQ_exec:nnnn { #1 } { #2 } { #3 } { #4 }
2741     \group_end:
2742   }
2743   { \msg_error:nnn { spintent } { need-math-mode } { \spmQ } }
2744 }
```

(End of definition for `\spmQ`. This function is documented on page 17.)

### 11.33 The commands of the geo family

The commands described in this section provide a semantic interface for geometric objects in Spanish mathematical contexts. They all delegate parsing to `spintent.lua`, sharing the variables `\__spintent_geo_luaaset_error_str`, `\__spintent_geo_luaaset_intent_str`, `\__spintent_geo_luaaset_print_tl`, `\__spintent_geo_luaaset_sub_str` and `\__spintent_geo_luaaset_tipo_str` — constructing a valid MathCAT intent — and a single point is always annotated through `\__spintent_mathmlarg_point:nnn`. Every command also activates `\__spintent_scope_attr_int` within its own group, so the shared queue never claims content typeset outside a geo command in the same formula.

### 11.33.1 Definition of shared variables

Internal variables shared across all commands of the family to avoid repeated calls into Lua.

```

\l__spintent_geo_luaset_error_str
\l__spintent_geo_luaset_intent_str
\l__spintent_geo_luaset_print_tl
\l__spintent_geo_luaset_is_name_str
\l__spintent_geo_luaset_poly_sym_str
\l__spintent_geo_luaset_angulo_case_str
\l__spintent_geo_luaset_concept_str
\l__spintent_geo_luaset_letra_word_str
\l__spintent_geo_luaset_sub_spoken_str
\l__spintent_geo_luaset_nombre_es_num_str
\l__spintent_geo_luaset_letra_str
\l__spintent_geo_luaset_sub_str
\l__spintent_geo_luaset_tipo_str
2745 \str_new:N \l__spintent_geo_luaset_error_str
2746 \str_new:N \l__spintent_geo_luaset_intent_str
2747 \tl_new:N \l__spintent_geo_luaset_print_tl
2748 \clist_new:N \l__spintent_geo_luaset_points_clist
2749 \str_new:N \l__spintent_geo_luaset_is_name_str
2750 \str_new:N \l__spintent_geo_luaset_poly_sym_str
2751 \str_new:N \l__spintent_geo_luaset_angulo_case_str
2752 \str_new:N \l__spintent_geo_luaset_concept_str
2753 \str_new:N \l__spintent_geo_luaset_letra_word_str
2754 \str_new:N \l__spintent_geo_luaset_sub_spoken_str
2755 \str_new:N \l__spintent_geo_luaset_nombre_es_num_str
2756 \str_new:N \l__spintent_geo_luaset_letra_str
2757 \str_new:N \l__spintent_geo_luaset_sub_str
2758 \str_new:N \l__spintent_geo_luaset_tipo_str

```

(End of definition for `\l__spintent_geo_luaset_error_str` and others.)

## 11.34 The command `\spcoord`

The user command `\spcoord` typesets an ordered pair  $(x,y)$ , with the comma (or semicolon) between them carrying its own `:infix` intent, and each side wrapped with `:prefix($vp)/:prefix($vn)` when it starts with  $+/-$ , so the sign is always read as más/menos — never positivo/negativo, which is not how MathCAT reads a bare sign in this position.

### 11.34.1 Definition of variables for `\spcoord`

Variables set by the function `\__spintent_luafun_spcoord_parse_and_set:n`. The string variable `\l__spintent_spcoord_luaset_sep_str` is `,` or `;`, whichever the argument actually used. For the abscissa and ordinate values, the variables `\l__spintent_spcoord_luaset_#1_sign_str` is `+`, `-`, or empty; the variables `\l__spintent_spcoord_luaset_#1_type_str` is `number` or `raw`; the variables `\l__spintent_spcoord_luaset_#1_value_str` is the value with the sign already removed; and the variables `\l__spintent_spcoord_luaset_#1_has_frac_str` is `none`, `frac`, or `dfrac`.

```

\l__spintent_spcoord_luaset_error_str
\l__spintent_spcoord_luaset_sep_str
\l__spintent_spcoord_luaset_x_sign_str
\l__spintent_spcoord_luaset_x_type_str
\l__spintent_spcoord_luaset_x_value_str
\l__spintent_spcoord_luaset_x_has_frac_str
\l__spintent_spcoord_luaset_y_sign_str
\l__spintent_spcoord_luaset_y_type_str
\l__spintent_spcoord_luaset_y_value_str
\l__spintent_spcoord_luaset_y_has_frac_str
2759 \str_new:N \l__spintent_spcoord_luaset_error_str
2760 \str_new:N \l__spintent_spcoord_luaset_sep_str
2761 \clist_map_inline:nn { x , y }
2762 {
2763   \str_new:c { \l__spintent_spcoord_luaset_#1_sign_str }
2764   \str_new:c { \l__spintent_spcoord_luaset_#1_type_str }
2765   \str_new:c { \l__spintent_spcoord_luaset_#1_value_str }
2766   \str_new:c { \l__spintent_spcoord_luaset_#1_has_frac_str }
2767 }

```

(End of definition for `\l__spintent_spcoord_luaset_error_str` and others.)

```

\l__spintent_spcoord_raw_arg_tl
\l__spintent_spcoord_intent_str
\l__spintent_spcoord_wrap_parens_l_tl
\l__spintent_spcoord_wrap_parens_r_tl

```

The variable `\l__spintent_spcoord_raw_arg_tl` holds the `{(argument)}` exactly as given. The variable `\l__spintent_spcoord_intent_str` holds the `:infix` word for the separator `_coma:infix` or `_punto-y-coma:infix`. The variables `\l__spintent_spcoord_wrap_parens_l_tl`/`\l__spintent_spcoord_wrap_parens_r_tl` hold the opening/closing delimiter, sized by the function `\__spintent_spcoord_set_wrap_parens_size:.`

```

2768 \tl_new:N \l__spintent_spcoord_raw_arg_tl
2769 \str_new:N \l__spintent_spcoord_intent_str
2770 \tl_new:N \l__spintent_spcoord_wrap_parens_l_tl
2771 \tl_new:N \l__spintent_spcoord_wrap_parens_r_tl

```

(End of definition for `\l__spintent_spcoord_raw_arg_tl` and others.)

### 11.34.2 Definition of error messages for `\spcoord`

`spcoord-invalid-arg`

```

2772 \msg_new:nnn { spintent } { spcoord-invalid-arg }
2773 {
2774   Invalid~argument~'#1'~for~\token_to_str:N \spcoord.
2775 }

```

(End of definition for `spcoord-invalid-arg`.)

### 11.34.3 Definition of keys for \spcoord

silent-cmd We define the key `silent-cmd`.

```

2776 \keys_define:nn { spintrent / spcoord }
2777 {
2778   silent-cmd .bool_set:N = \l__spintrent_spcoord_silent_cmd_bool,
2779   silent-cmd .initial:n = false,
2780   silent-cmd .value_required:n = true,
2781   unknown .code:n =
2782     {
2783       \str_set:Nn \l__spintrent_command_name_str { spcoord }
2784       \__spintrent_unknown_keys_cmd:n {#1}
2785     },
2786 }
```

(End of definition for `silent-cmd`.)

### 11.34.4 Internal functions for \spcoord

\\_\_spintrent\_spcoord\_set\_wrap\_parens\_size: With neither side carrying a fraction, the parentheses stay unsized. Otherwise, `\mathstyle` picks `\Bigl/\Bigr` for display, `\bigl/\bigr` for text. When either side is `dfrac`, the parentheses are forced to `\Bigl/\Bigr` regardless of `\mathstyle`, since `\dfrac` always renders at display height.

```

2787 \cs_new_protected:Nn \__spintrent_spcoord_set_wrap_parens_size:
2788 {
2789   \tl_set:Nn \l__spintrent_spcoord_wrap_parens_l_tl { \lparen }
2790   \tl_set:Nn \l__spintrent_spcoord_wrap_parens_r_tl { \rparen }
2791   \bool_if:nT
2792     {
2793       \bool_lazy_or_p:nn
2794         { \bool_not_p:n { \str_if_eq_p:Vn \l__spintrent_spcoord_luaset_x_has_frac_str { none } } }
2795         { \bool_not_p:n { \str_if_eq_p:Vn \l__spintrent_spcoord_luaset_y_has_frac_str { none } } }
2796     }
2797     {
2798       \int_case:nnF { \mathstyle }
2799       {
2800         { 0 }
2801         {
2802           \tl_set:Nn \l__spintrent_spcoord_wrap_parens_l_tl { \Bigl \lparen }
2803           \tl_set:Nn \l__spintrent_spcoord_wrap_parens_r_tl { \Bigr \rparen }
2804         }
2805         { 1 }
2806         {
2807           \tl_set:Nn \l__spintrent_spcoord_wrap_parens_l_tl { \Bigl \lparen }
2808           \tl_set:Nn \l__spintrent_spcoord_wrap_parens_r_tl { \Bigr \rparen }
2809         }
2810         { 2 }
2811         {
2812           \tl_set:Nn \l__spintrent_spcoord_wrap_parens_l_tl { \bigl \lparen }
2813           \tl_set:Nn \l__spintrent_spcoord_wrap_parens_r_tl { \bigr \rparen }
2814         }
2815         { 3 }
2816         {
2817           \tl_set:Nn \l__spintrent_spcoord_wrap_parens_l_tl { \bigl \lparen }
2818           \tl_set:Nn \l__spintrent_spcoord_wrap_parens_r_tl { \bigr \rparen }
2819         }
2820       }
2821     }
2822   \bool_if:nT
2823     {
2824       \bool_lazy_or_p:nn
2825         { \str_if_eq_p:Vn \l__spintrent_spcoord_luaset_x_has_frac_str { dfraction } }
2826         { \str_if_eq_p:Vn \l__spintrent_spcoord_luaset_y_has_frac_str { dfraction } }
2827     }
2828     {
2829       \tl_set:Nn \l__spintrent_spcoord_wrap_parens_l_tl { \Bigl \lparen }
2830       \tl_set:Nn \l__spintrent_spcoord_wrap_parens_r_tl { \Bigr \rparen }
2831     }
2832 }
2833 }
```

(End of definition for `\__spintrent_spcoord_set_wrap_parens_size:.`)

`\__spintent_spcoord_value:n` With `\__spintent_spcoord_luaset_ #1 _type_str` equal to `number`, calls `\spnum` on `\__spintent_spcoord_luaset_ #1 _value_str`. Otherwise, typesets it as given — a `\frac\dfraction` or a letter expression, untouched. Receives `x` or `y` in ‘`#1`’.

```

2834 \cs_new_protected:Nn \__spintent_spcoord_value:n
2835 {
2836   \str_if_eq:eeTF
2837     { \str_use:c { \__spintent_spcoord_luaset_ #1 _type_str } } { number }
2838   {
2839     \exp_args:Ne \spnum { \str_use:c { \__spintent_spcoord_luaset_ #1 _value_str } }
2840   }
2841   { \str_use:c { \__spintent_spcoord_luaset_ #1 _value_str } }
2842 }

```

(End of definition for `\__spintent_spcoord_value:n`.)

`\__spintent_spcoord_print:n` With `\__spintent_spcoord_luaset_ #1 _sign_str` equal to `+` or `-`, wraps the sign and `\__spintent_spcoord_luaset_ #1 _value_str` together with `_más:prefix($vp)/_menos:prefix($vn)`, referencing the value through `read` as `más/menos`. With no sign, delegates directly to `\__spintent_spcoord_value:n`. Receives `x` or `y` in ‘`#1`’.

```

2843 \cs_new_protected:Nn \__spintent_spcoord_print:n
2844 {
2845   \str_case:enF { \str_use:c { \__spintent_spcoord_luaset_ #1 _sign_str } }
2846   {
2847     { + }
2848     {
2849       \MathMLintent { _más:prefix ($vp) }
2850       {
2851         \str_use:c { \__spintent_spcoord_luaset_ #1 _sign_str }
2852         \MathMLarg{ vp } { \__spintent_spcoord_value:n { #1 } }
2853       }
2854     }
2855     { - }
2856     {
2857       \MathMLintent { _menos :prefix($vn) }
2858       {
2859         \str_use:c { \__spintent_spcoord_luaset_ #1 _sign_str }
2860         \MathMLarg{ vn } { \__spintent_spcoord_value:n { #1 } }
2861       }
2862     }
2863   }
2864   { \__spintent_spcoord_value:n { #1 } }
2865 }

```

(End of definition for `\__spintent_spcoord_print:n`.)

`\__spintent_spcoord_intent_infix:` Sets `\__spintent_spcoord_intent_str` to `_coma:infix` or `_punto-y-coma:infix`, per `\__spintent_spcoord_intent_str`. Then wraps both sides — each referenced through with it, `$x/$y` standing for `\__spintent_spcoord_print:n`’s own output.

```

2866 \cs_new_protected:Nn \__spintent_spcoord_intent_infix:
2867 {
2868   \str_if_eq:VnTF \__spintent_spcoord_luaset_sep_str { ; }
2869   { \str_set:Nn \__spintent_spcoord_intent_str { _punto-y-coma:infix } }
2870   { \str_set:Nn \__spintent_spcoord_intent_str { _coma:infix } }
2871   \MathMLintent{ \__spintent_spcoord_intent_str ($x, $y)}
2872   {
2873     \MathMLarg{x}{ \__spintent_spcoord_print:n { x } }
2874     \MathMLintent { :silent } { \__spintent_spcoord_luaset_sep_str }
2875     \MathMLarg{y}{ \__spintent_spcoord_print:n { y } }
2876   }
2877 }

```

(End of definition for `\__spintent_spcoord_intent_infix:.`)

`\__spintent_spcoord_mathml:` With `silent-cmd=true`, everything — parentheses and argument alike — is wrapped in a single `_-`, reading nothing at all: `\__spintent_spcoord_raw_arg_tl` is typeset exactly as given, untouched by `\__spintent_spcoord_print:n/\__spintent_spcoord_intent_infix:.` Otherwise, the parentheses are silenced individually and `\__spintent_spcoord_intent_infix:` provides the reading in between.

```

2878 \cs_new_protected:Nn \__spintent_spcoord_mathml:
2879 {
2880   \bool_if:NTF \__spintent_spcoord_silent_cmd_bool
2881   {
2882     \MathMLintent { _- }
2883     {

```

```

2884         \l__spintenc_spcoord_wrap_parens_l_tl
2885         \l__spintenc_spcoord_raw_arg_tl
2886         \l__spintenc_spcoord_wrap_parens_r_tl
2887     }
2888 }
2889 {
2890     \MathMLintent { :silent } { \l__spintenc_spcoord_wrap_parens_l_tl }
2891     \l__spintenc_spcoord_intent_infix:
2892     \MathMLintent { :silent } { \l__spintenc_spcoord_wrap_parens_r_tl }
2893 }
2894 }

```

(End of definition for `\l__spintenc_spcoord_mathml:`)

`\l__spintenc_spcoord_exec:nn`

Sets the *(keys)* in ‘#1’ and `\l__spintenc_spcoord_raw_arg_tl` to ‘#2’. Calls `\l__spintenc_luafun_spcoord_parse_and_set:n` either way — even with `silent-cmd=true`, since `\l__spintenc_spcoord_set_wrap_parens_size:` still needs `\l__spintenc_spcoord_luaaset_ #1 _has_frac_str` — but with `silent-cmd=true`, forces `\l__spintenc_spcoord_luaaset_error_str` back to `false` rather than reporting it: the raw *{(argument)}* is typeset regardless of whether it parsed. Otherwise, reports `spcoord-invalid-arg` on error. Either way, sizes the parentheses, then delegates to `\l__spintenc_spcoord_mathml:` unless there was an error.

```

2895 \cs_new_protected:Nn \l__spintenc_spcoord_exec:nn
2896 {
2897     \keys_set:nn { spintenc / spcoord } {#1}
2898     \tl_set:Nn \l__spintenc_spcoord_raw_arg_tl {#2}
2899     \bool_if:NTF \l__spintenc_spcoord_silent_cmd_bool
2900     {
2901         \l__spintenc_luafun_spcoord_parse_and_set:n { \exp_not:n {#2} }
2902         \str_set:Nn \l__spintenc_spcoord_luaaset_error_str { false }
2903     }
2904     {
2905         \l__spintenc_luafun_spcoord_parse_and_set:n { \exp_not:n {#2} }
2906         \str_if_eq:VnT \l__spintenc_spcoord_luaaset_error_str { true }
2907         { \msg_error:nnn { spintenc } { spcoord-invalid-arg } { \exp_not:n {#2} } }
2908     }
2909     \l__spintenc_spcoord_set_wrap_parens_size:
2910     \str_if_eq:VnF \l__spintenc_spcoord_luaaset_error_str { true }
2911     { \l__spintenc_spcoord_mathml: }
2912 }

```

(End of definition for `\l__spintenc_spcoord_exec:nn`.)

`\spcoord`

The public command enforces *math mode*, opens a local group, and delegates to `\l__spintenc_spcoord_exec:nn`.

```

2913 \NewDocumentCommand \spcoord { 0{} m }
2914 {
2915     \mode_if_math:TF
2916     {
2917         \group_begin:
2918         \attribute \l__spintenc_scope_attr_int = 1
2919         \l__spintenc_spcoord_exec:nn {#1} {#2}
2920         \group_end:
2921     }
2922     { \msg_error:nnn { spintenc } { need-math-mode } { \spcoord } }
2923 }

```

(End of definition for `\spcoord`. This function is documented on page 18.)

### 11.35 The function `\l__spintenc_mathmlarg_point:nnn`

`\l__spintenc_mathmlarg_point_ref_tl`  
`\l__spintenc_mathmlarg_point_mode_str`  
`\l__spintenc_mathmlarg_point_kv_tl`  
`\l__spintenc_mathmlarg_point_silent_bool`

Scratch variables for `\l__spintenc_mathmlarg_point:nnn`: the reference name (#1), and the calling command’s own `read-cmd/silent-cmd` state, looked up by name (#2).

```

2924 \tl_new:N \l__spintenc_mathmlarg_point_ref_tl
2925 \str_new:N \l__spintenc_mathmlarg_point_mode_str
2926 \tl_new:N \l__spintenc_mathmlarg_point_kv_tl
2927 \bool_new:N \l__spintenc_mathmlarg_point_silent_bool

```

(End of definition for `\l__spintenc_mathmlarg_point_ref_tl` and others.)

`\l__spintenc_mathmlarg_point:nnn`  
`\l__spintenc_mathmlarg_point:nnV`

Annotates one point (#3) with `arg` and, when applicable, a literal *intent*, without wrapping it in its own `<mrow>`. Concept *intent* is passed empty on purpose: the caller applies it externally, via a native `\MathMLintent`.

With `\l__spintenc_geo_luaaset_tipo_str` equal to `sub`, for `read-cmd=school` the letter alone is tagged through `\l__spintenc_luafun_mrow_intent_arg:nn`, and the subscript is appended separately, silenced.

For `read-cmd=mathcat`, letter and subscript are typeset together, tagged as a whole through a native `\MathMLarg`.

```

2928 \cs_new_protected:Nn \__spintent_mathmlarg_point:nnn
2929 {
2930   \tl_set:Nn \__spintent_mathmlarg_point_ref_tl {#1}
2931   \__spintent_luafun_geo_point_parse_and_set:n { \exp_not:n {#3} }
2932   \bool_set_eq:Nc \__spintent_mathmlarg_point_silent_bool { \__spintent_ #2 _silent_cmd_bool }
2933   \str_set_eq:Nc \__spintent_mathmlarg_point_mode_str { \__spintent_ #2 _mode_str }
2934   \__spintent_invisible_sep:
2935   \str_case:VnF \__spintent_geo_luaset_tipo_str
2936   {
2937     { sub }
2938     {
2939       \str_if_eq:VnTF \__spintent_mathmlarg_point_mode_str { school }
2940       {
2941         \tl_set:Nc \__spintent_mathmlarg_point_kv_tl
2942           { arg = "\__spintent_mathmlarg_point_ref_tl", intent = "_ \__spintent_geo_lua
2943         \__spintent_luafun_mrow_intent_arg:nn
2944           { } { \__spintent_mathmlarg_point_kv_tl }
2945         {
2946           \__spintent_geo_luaset_letra_str
2947           \c_math_subscript_token
2948             { \MathMLint { :silent } { \__spintent_geo_luaset_sub_str } }
2949         }
2950       }
2951     }
2952     {
2953       \__spintent_inner_sep:
2954       \MathMLarg { \__spintent_mathmlarg_point_ref_tl }
2955       {
2956         {
2957           \__spintent_geo_luaset_letra_str
2958           \c_math_subscript_token { \__spintent_geo_luaset_sub_str }
2959         }
2960       }
2961     }
2962   }
2963 }

```

With `\__spintent_geo_luaset_tipo_str` equal to `prima`, the same distinction applies: `read-cmd=school` tags the letter with a literal `intent` and appends the mark unsilenced (a prime is not spoken on the base, so nothing to repeat); `read-cmd=mathcat` tags letter and mark together through a native `\MathMLarg`.

```

2964 { prima }
2965 {
2966   \str_if_eq:VnTF \__spintent_mathmlarg_point_mode_str { school }
2967   {
2968     \tl_set:Nc \__spintent_mathmlarg_point_kv_tl
2969       { arg = "\__spintent_mathmlarg_point_ref_tl", intent = "_ \__spintent_geo_lua
2970     \__spintent_luafun_mrow_intent_arg:nn
2971       { } { \__spintent_mathmlarg_point_kv_tl }
2972     { \__spintent_geo_luaset_letra_str \__spintent_geo_luaset_sub_str }
2973   }
2974   {
2975     {
2976       \__spintent_inner_sep:
2977       \MathMLarg { \__spintent_mathmlarg_point_ref_tl }
2978       {
2979         { \__spintent_geo_luaset_letra_str \__spintent_geo_luaset_sub_str }
2980       }
2981     }
2982   }
2983 }
2984 }

```

Otherwise (no subscript, no prime), the same distinction applies to the combined `\__spintent_geo_luaset_print_t` as a whole — the shared queue for `school` (or when `silent-cmd` is `true`), a native `\MathMLarg` for `mathcat`, never mixed within the same formula, since the queue does not recognize a native `\MathMLarg` claim as already taken.

```

2985 {
2986   \bool_if:NTF \__spintent_mathmlarg_point_silent_bool
2987   {

```

```

2988     \tl_set:Nc \l__spintent_mathmlarg_point_kv_tl
2989     { arg = "\l__spintent_mathmlarg_point_ref_tl", intent = "-" }
2990     \__spintent_luafun_mrow_intent_arg:nn
2991     { } % debe ir vacío
2992     { \l__spintent_mathmlarg_point_kv_tl }
2993     { \l__spintent_geo_luaset_print_tl }
2994   }
2995   {
2996     \str_if_eq:VnTF \l__spintent_mathmlarg_point_mode_str { school }
2997     {
2998       \tl_set:Nc \l__spintent_mathmlarg_point_kv_tl
2999       { arg = "\l__spintent_mathmlarg_point_ref_tl", intent = "- \l__spintent_geo_lua
3000       \__spintent_luafun_mrow_intent_arg:nn
3001       { } % debe ir vacío
3002       { \l__spintent_mathmlarg_point_kv_tl }
3003       { \l__spintent_geo_luaset_print_tl }
3004     }
3005     {
3006       \MathMLarg { \l__spintent_mathmlarg_point_ref_tl }
3007       { \l__spintent_geo_luaset_print_tl }
3008     }
3009   }
3010 }
3011 }
3012 \cs_generate_variant:Nn \__spintent_mathmlarg_point:nnn { nnV }

```

(End of definition for `\__spintent_mathmlarg_point:nnn`.)

### 11.36 The command `\spPoint`

The user-level command `\spPoint` provides a semantic interface for a single geometric point. It delegates argument validation to the Lua module `spintent.lua`, constructing a valid MathCAT intent.

#### 11.36.1 Definition of variables for `\spPoint`

Internal variables holding the values set by `read-cmd` and `prefix`.

```

\l__spintent_spPoint_mode_str
\l__spintent_spPoint_prefix_tl
\l__spintent_spPoint_intent_arg_tl
3013 \str_new:N \l__spintent_spPoint_mode_str
3014 \tl_new:N \l__spintent_spPoint_prefix_tl
3015 \tl_new:N \l__spintent_spPoint_intent_arg_tl

```

(End of definition for `\l__spintent_spPoint_mode_str`, `\l__spintent_spPoint_prefix_tl`, and `\l__spintent_spPoint_intent_arg_tl`.)

#### 11.36.2 Definition of error messages for `\spPoint`

`spPoint-invalid-point`

The “error” `spPoint-invalid-point` is raised when the argument passed in ‘#1’ is not a single uppercase letter with an optional subscript or prime modifier.

```

3016 \msg_new:nnnn { spintent } { spPoint-invalid-point }
3017 {
3018   Invalid~point~format~in~\token_to_str:N \spPoint:~'#1'.
3019 }
3020 {
3021   The~argument~must~be~a~single~uppercase~letter~with~an~optional~
3022   subscript~(\_{...})~or~prime~(').\
3023 }

```

(End of definition for `spPoint-invalid-point`.)

#### 11.36.3 Definition of keys for `\spPoint`

`read-cmd`  
`prefix`  
`silent-cmd`  
`unknown`

We define the *(keys)* `read-cmd`, `prefix`, `silent-cmd` and `concept`.

```

3024 \keys_define:nn { spintent / spPoint }
3025 {
3026   read-cmd .choices:nn = { school , mathcat }
3027   {
3028     \int_case:nn { \l_keys_choice_int }
3029     {
3030       {1} { \str_set:Nn \l__spintent_spPoint_mode_str { school } }
3031       {2} { \str_set:Nn \l__spintent_spPoint_mode_str { mathcat } }
3032     }
3033   },
3034   read-cmd .initial:n = school,
3035   read-cmd .value_required:n = true,
3036   read-cmd / unknown .code:n =
3037     \msg_error:nneee { spintent } { unknown-choice }

```

```

3038     { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
3039     prefix .code:n =
3040     { \__spintent_sanitize_affix:Nn \__spintent_spPoint_prefix_tl {#1} },
3041     prefix .initial:n = ,
3042     prefix .value_required:n = true,
3043     silent-cmd .bool_set:N = \__spintent_spPoint_silent_cmd_bool,
3044     silent-cmd .initial:n = false,
3045     silent-cmd .value_required:n = true,
3046     concept .bool_set:N = \__spintent_spPoint_concept_bool,
3047     concept .initial:n = true,
3048     concept .value_required:n = true,
3049     unknown .code:n =
3050     {
3051         \str_set:Nn \__spintent_command_name_str { spPoint }
3052         \__spintent_unknown_keys_cmd:n {#1}
3053     },
3054 }

```

(End of definition for read-cmd and others.)

### 11.36.4 Internal functions for \spPoint

The internal function `\__spintent_spPoint_print:n` processes the `{\argument}` passed in ‘#1’. Calls `\__spintent_luafun_geo_point_parse_and_set:n`, reporting spPoint-invalid-point on error. If `silent-cmd` is true or `concept` is false, sets `\__spintent_spPoint_intent_arg_tl` to `_{\pt}` — forced further to `_` — when `silent-cmd` is true and `read-cmd` is not `school`, since `\__spintent_spPoint_mathml:n` then tags through a native `\MathMLarg` that needs the concept silenced too. Otherwise sets it to `punto:prefix(\pt)`, prefixed with `prefix` when given. Finally calls `\__spintent_spPoint_mathml:n`.

```

3055 \cs_new_protected:Nn \__spintent_spPoint_print:n
3056 {
3057     \__spintent_luafun_geo_point_parse_and_set:n { \exp_not:n {#1} }
3058     \str_if_eq:NnT \__spintent_geo_luaset_error_str { true }
3059     {
3060         \msg_error:nne { spintent } { spPoint-invalid-point } { \exp_not:n {#1} }
3061     }
3062     \bool_if:nTF
3063     {
3064         \bool_lazy_or:nn
3065         { \__spintent_spPoint_silent_cmd_bool }
3066         { ! \__spintent_spPoint_concept_bool }
3067     }
3068     {
3069         \tl_set:Nn \__spintent_spPoint_intent_arg_tl { _(\pt) }
3070         \bool_if:NnT \__spintent_spPoint_silent_cmd_bool
3071         {
3072             \str_if_eq:NnF \__spintent_spPoint_mode_str { school }
3073             {
3074                 \tl_set:Nn \__spintent_spPoint_intent_arg_tl { _- }
3075             }
3076         }
3077     }
3078     {
3079         \tl_set:Nn \__spintent_spPoint_intent_arg_tl
3080         {
3081             \tl_if_empty:NF \__spintent_spPoint_prefix_tl
3082             {
3083                 \tl_use:N \__spintent_spPoint_prefix_tl -
3084             }
3085             punto:prefix(\pt)
3086         }
3087     }
3088     \__spintent_spPoint_mathml:n { #1 }
3089 }

```

(End of definition for \\_\_spintent\_spPoint\_print:n.)

The function `\__spintent_spPoint_mathml:n` calls `\__spintent_invisible_sep:` first, then wraps `\__spintent_mathmlarg_point:nnn` — which parses ‘#1’, tags it `arg="pt"`, and silences a real subscript or prime on its own — inside a native `\MathMLintent` carrying `\__spintent_spPoint_intent_arg_tl` as the concept `intent`, with `\__spintent_inner_sep:` first inside that group.

```

3090 \cs_new_protected:Nn \__spintent_spPoint_mathml:n
3091 {

```

```

3092     \__spintent_invisible_sep:
3093     \MathMLintend { \l__spintent_spPoint_intent_arg_tl }
3094     {
3095         \__spintent_inner_sep:
3096         \__spintent_mathmlarg_point:nnn { pt } { spPoint } { #1 }
3097     }
3098 }

```

(End of definition for \\_\_spintent\_spPoint\_mathml:n.)

\\_\_spintent\_spPoint\_exec:nnn

The function `\__spintent_spPoint_exec:nnn` receives the optional  $\langle keys \rangle$  in ‘#1’ and the point  $\{\langle argument \rangle\}$  in ‘#2’, delegating to `\__spintent_spPoint_print:n`. With ‘#3’ given (the optional coordinate), silences the gap between the point and the coordinate first (otherwise read as an implicit product), then temporarily aliases `\l__spintent_spcoord_silent_cmd_bool` to `\l__spintent_spPoint_silent_cmd_bool` and calls `\__spintent_luafun_spcoord_parse_and_set:n` directly, reporting `spcoord-invalid-arg` on error unless silenced, then `\__spintent_spcoord_set_wrap_parens_size:` and `\__spintent_spcoord_math`

```

3099 \cs_new_protected:Nn \__spintent_spPoint_exec:nnn
3100 {
3101     \keys_set:nn { spintent / spPoint } { #1 }
3102     \__spintent_spPoint_print:n { #2 }
3103     \tl_if_novalue:nF {#3}
3104     {
3105         \MathMLintend { :silent } { \c__spintent_invisible_sep_int } % silencia el "por"
3106         \bool_set_eq:NN \l__spintent_spcoord_silent_cmd_bool \l__spintent_spPoint_silent_cmd_bool
3107         \tl_set:Nn \l__spintent_spcoord_raw_arg_tl {#3}
3108         \bool_if:NTF \l__spintent_spcoord_silent_cmd_bool
3109         {
3110             \__spintent_luafun_spcoord_parse_and_set:n { \exp_not:n {#3} }
3111             \str_set:Nn \l__spintent_spcoord_luaerror_str { false }
3112         }
3113         {
3114             \__spintent_luafun_spcoord_parse_and_set:n { \exp_not:n {#3} }
3115             \str_if_eq:VnT \l__spintent_spcoord_luaerror_str { true }
3116             { \msg_error:nnn { spintent } { spcoord-invalid-arg } { \exp_not:n {#3} } }
3117         }
3118         \__spintent_spcoord_set_wrap_parens_size:
3119         \str_if_eq:VnF \l__spintent_spcoord_luaerror_str { true }
3120         { \__spintent_spcoord_mathml: }
3121     }
3122 }

```

(End of definition for \\_\_spintent\_spPoint\_exec:nnn.)

\spPoint

The public command `\spPoint` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spPoint_exec:nnn`.

```

3123 \NewDocumentCommand \spPoint { 0{} m d() }
3124 {
3125     \mode_if_math:TF
3126     {
3127         \group_begin:
3128         \attribute \l__spintent_scope_attr_int = 1
3129         \__spintent_spPoint_exec:nnn { #1 } { #2 } { #3 }
3130         \group_end:
3131     }
3132     { \msg_error:nnn { spintent } { need-math-mode } { \spPoint } }
3133 }

```

(End of definition for \spPoint. This function is documented on page 18.)

## 11.37 The command \spvec

The user command `\spvec` supports two types of notation for the input  $\{\langle argument \rangle\}$ : a pair of points (with `\overrightarrow` applied), or a one-letter name (with `\vec` applied instead). Both notations are automatically detected by the Lua module. An optional coordinate, in parentheses, may follow the  $\{\langle argument \rangle\}$ .

### 11.37.1 Definition of variables for \spvec

The variable `\l__spintent_spvec_mode_str` stores the value set by the key `read-cmd` and the variable `\l__spintent_spvec_prefix_tl` stores the value set by the key `prefix`.

```

3134 \str_new:N \l__spintent_spvec_mode_str
3135 \tl_new:N \l__spintent_spvec_prefix_tl
3136 \tl_new:N \l__spintent_spvec_intent_arg_tl
3137 \tl_new:N \l__spintent_spvec_kv_tl

```

```

3138 \seq_new:N \__spintent_sspec_pt_seq
3139 \tl_new:N \__spintent_sspec_refs_tl
3140 \int_new:N \__spintent_sspec_pt_idx_int

```

(End of definition for `\__spintent_sspec_mode_str` and others.)

### 11.37.2 Definition of error messages for `\sspec`

`sspec-invalid-arg` The “error” `sspec-invalid-arg` is raised when the argument fails to parse, when a one-letter name carries a prime, or when the point list is not exactly two points.

```

3141 \msg_new:nnn { spintent } { sspec-invalid-arg }
3142 {
3143   Invalid~argument~'#1'~for~\token_to_str:N \sspec.
3144 }

```

(End of definition for `sspec-invalid-arg`.)

### 11.37.3 Definition of keys for `\sspec`

`silent-cmd` We define the *(keys)* `silent-cmd`, `concept`, `autofit`, `read-cmd` and `prefix`.

```

3145 \keys_define:nn { spintent / sspec }
3146 {
3147   silent-cmd      .bool_set:N = \__spintent_sspec_silent_cmd_bool,
3148   silent-cmd      .initial:n = false,
3149   silent-cmd      .value_required:n = true,
3150   concept         .bool_set:N = \__spintent_sspec_concept_bool,
3151   concept         .initial:n = true,
3152   concept         .value_required:n = true,
3153   autofit         .bool_set:N = \__spintent_sspec_autofit_bool,
3154   autofit         .initial:n = false,
3155   autofit         .value_required:n = true,
3156   read-cmd        .choices:nn = { school , mathcat }
3157   {
3158     \int_case:nn { \_keys_choice_int }
3159     {
3160       {1} { \str_set:Nn \__spintent_sspec_mode_str { school } }
3161       {2} { \str_set:Nn \__spintent_sspec_mode_str { mathcat } }
3162     }
3163   },
3164   read-cmd        .initial:n = school,
3165   read-cmd        .value_required:n = true,
3166   read-cmd / unknown .code:n =
3167     \msg_error:nneee { spintent } { unknown-choice }
3168     { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
3169   prefix          .code:n =
3170     { \__spintent_sanitiz_abbrev:Nn \__spintent_sspec_prefix_tl {#1} },
3171   prefix          .initial:n = { },
3172   prefix          .value_required:n = true,
3173   unknown         .code:n =
3174     {
3175       \str_set:Nn \__spintent_command_name_str { sspec }
3176       \__spintent_unknown_keys_cmd:n {#1}
3177     },
3178 }

```

(End of definition for `silent-cmd` and others.)

### 11.37.4 Internal functions for `\sspec`

`\__spintent_sspec_print:n` The internal function `\__spintent_sspec_print:n` processes the *{(argument)}* passed in ‘#1’. Calls `\__spintent_luafun_geo_parse_and_set:n`, reporting `sspec-invalid-arg` on error, on a proper name with `\__spintent_geo_luasets_tipo_str` equal to `prima` (a one-letter vector cannot carry a prime), or on a point list that is not exactly two points. Otherwise calls `\__spintent_sspec_mathml:n`.

```

3179 \cs_new_protected:Nn \__spintent_sspec_print:n
3180 {
3181   \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#1} }
3182   \str_if_eq:VnT \__spintent_geo_luasets_is_name_str { true }
3183   {
3184     \str_if_eq:VnT \__spintent_geo_luasets_tipo_str { prima }
3185     { \str_set:Nn \__spintent_geo_luasets_error_str { true } }
3186   }
3187   \str_if_eq:VnT \__spintent_geo_luasets_is_name_str { false }
3188   {
3189     \int_compare:nT { \clist_count:N \__spintent_geo_luasets_points_clist != 2 }

```

```

3190         { \str_set:Nn \l__spintent_geo_luaset_error_str { true } }
3191     }
3192     \str_if_eq:NnTF \l__spintent_geo_luaset_error_str { true }
3193     { \msg_error:nne { spintent } { spvec-invalid-arg } { \exp_not:n {#1} } }
3194     { \l__spintent_spvec_mathml:n { #1 } }
3195 }

```

(End of definition for `\l__spintent_spvec_print:n`)

`\l__spintent_spvec_mathml:n`

The function `\l__spintent_spvec_mathml:n` branches on `\l__spintent_geo_luaset_is_name_str`. For a proper name (a single letter, no prime), the concept `intent` is `_( $\$pt$ )` when `silent-cmd` is `true` or `concept` is `false` — forced further to `_-` when `silent-cmd` is `true` and `read-cmd` is not `school`; otherwise `vector:prefix( $\$pt$ )`. With `\l__spintent_geo_luaset_tipo_str` equal to `sub`, `read-cmd=school` tags `\vec{letra}` through `\l__spintent_luafun_mrow_intent_arg:nn` with a literal `kv`, the subscript appended and silenced; `read-cmd=mathcat` tags `\vec{...}` through a native `\MathMLarg` instead, the subscript silenced only under `silent-cmd`. `autofit` inserts a `\vphantom{b}` inside `\vec{...}` to normalize its height. Otherwise (no subscript), the same `kv` logic wraps `\vec{print\_tl}` as a whole.

```

3196 \cs_new_protected:Nn \l__spintent_spvec_mathml:n
3197 {
3198     \str_if_eq:NnTF \l__spintent_geo_luaset_is_name_str { true }
3199     {
3200         \bool_if:NTF
3201         {
3202             \bool_lazy_or_p:nn
3203             { \l__spintent_spvec_silent_cmd_bool }
3204             { ! \l__spintent_spvec_concept_bool }
3205         }
3206         {
3207             \tl_set:Nn \l__spintent_spvec_intent_arg_tl { _( $\$pt$ ) }
3208             \bool_if:NT \l__spintent_spvec_silent_cmd_bool
3209             {
3210                 \str_if_eq:NnF \l__spintent_spvec_mode_str { school }
3211                 { \tl_set:Nn \l__spintent_spvec_intent_arg_tl { _- } }
3212             }
3213         }
3214         {
3215             \tl_set:Nn \l__spintent_spvec_intent_arg_tl { vector:prefix( $\$pt$ ) }
3216         }
3217     }
3218     \str_case:NnF \l__spintent_geo_luaset_tipo_str
3219     {
3220         { sub }
3221         {
3222             \str_if_eq:NnTF \l__spintent_spvec_mode_str { school }
3223             {
3224                 \bool_if:NTF \l__spintent_spvec_silent_cmd_bool
3225                 { \tl_set:Nn \l__spintent_spvec_kv_tl { arg = "pt", intent = "_-" } }
3226                 {
3227                     \tl_set:Nn \l__spintent_spvec_kv_tl
3228                     { arg = "pt", intent = "_ \l__spintent_geo_luaset_intent_str -" }
3229                 }
3230                 \l__spintent_invisible_sep:
3231                 \MathMLintent { \l__spintent_spvec_intent_arg_tl }
3232                 {
3233                     \l__spintent_inner_sep:
3234                     \l__spintent_luafun_mrow_intent_arg:nn
3235                     { } { \l__spintent_spvec_kv_tl }
3236                 }
3237                 {
3238                     \bool_if:NT \l__spintent_spvec_autofit_bool
3239                     { \vphantom{ b } }
3240                     \l__spintent_geo_luaset_letra_str
3241                 }
3242                 \c_math_subscript_token
3243                 { \MathMLintent { :silent } { \l__spintent_geo_luaset_sub_str } }
3244             }
3245         }
3246     }
3247     {
3248         \l__spintent_invisible_sep:
3249         \MathMLintent { \l__spintent_spvec_intent_arg_tl }

```

```

3250         {
3251             \__spintent_inner_sep:
3252             \vec
3253             {
3254                 \bool_if:NT \__spintent_sptvec_autofit_bool
3255                 { \vphantom{ b } }
3256                 \MathMLarg{ pt } { \__spintent_geo_luaset_letra_str }
3257             }
3258             \bool_if:NT \__spintent_sptvec_silent_cmd_bool
3259             { \c_math_subscript_token { \__spintent_geo_luaset_sub_str } }
3260         }
3261         \bool_if:NF \__spintent_sptvec_silent_cmd_bool
3262         { \c_math_subscript_token { \__spintent_geo_luaset_sub_str } }
3263     }
3264 }
3265 {
3266 {
3267     \bool_if:NTF \__spintent_sptvec_silent_cmd_bool
3268     { \tl_set:Nn \__spintent_sptvec_kv_tl { arg = "pt", intent = "-" } }
3269     {
3270         \tl_set:Nn \__spintent_sptvec_kv_tl
3271         { arg = "pt", intent = "_ \__spintent_geo_luaset_intent_str -" }
3272     }
3273     \__spintent_invisible_sep:
3274     \MathMLintent { \__spintent_sptvec_intent_arg_tl }
3275     {
3276         \__spintent_inner_sep:
3277         \__spintent_luafun_mrow_intent_arg:nn
3278         { } { \__spintent_sptvec_kv_tl }
3279     }
3280     {
3281         \vec
3282         {
3283             \bool_if:NT \__spintent_sptvec_autofit_bool
3284             { \vphantom{ b } }
3285             \__spintent_geo_luaset_print_tl
3286         }
3287     }
3288 }
3289 }

```

For a pair of points, `\__spintent_geo_luaset_points_clist` is split into `\__spintent_sptvec_pt_seq` and walked to build `\__spintent_sptvec_refs_tl`, the `$pts1`, `$pts2` reference list. The concept `intent` is `_(refs)` or `vector:prefix(refs)`, same conditions as above, applied with a native `\MathMLintent` wrapping `\overrightarrow`. The sequence is walked again, calling `\__spintent_mathmlarg_point:nnn` once per point.

```

3290     {
3291         \seq_set_from_clist:NN \__spintent_sptvec_pt_seq \__spintent_geo_luaset_points_clist
3292         \tl_clear:N \__spintent_sptvec_refs_tl
3293         \int_zero:N \__spintent_sptvec_pt_idx_int
3294         \seq_map_inline:Nn \__spintent_sptvec_pt_seq
3295         {
3296             \int_incr:N \__spintent_sptvec_pt_idx_int
3297             \tl_if_empty:NF \__spintent_sptvec_refs_tl
3298             { \tl_put_right:Nn \__spintent_sptvec_refs_tl { , } }
3299             \tl_put_right:Ne \__spintent_sptvec_refs_tl { $ pts \int_use:N \__spintent_sptvec_pt_idx_int }
3300         }
3301         \bool_if:nTF
3302         {
3303             \bool_lazy_or_p:nn
3304             { \__spintent_sptvec_silent_cmd_bool }
3305             { ! \__spintent_sptvec_concept_bool }
3306         }
3307         {
3308             \tl_set:Ne \__spintent_sptvec_intent_arg_tl { _ ( \__spintent_sptvec_refs_tl ) }
3309         }
3310         {
3311             \tl_set:Ne \__spintent_sptvec_intent_arg_tl
3312             {
3313                 \tl_if_empty:NF \__spintent_sptvec_prefix_tl
3314                 { \tl_use:N \__spintent_sptvec_prefix_tl - }

```

```

3315         vector:prefix( \l__spintent_svec_refs_tl )
3316     }
3317 }
3318 \__spintent_invisible_sep:
3319 \MathMLintent { \l__spintent_svec_intent_arg_tl }
3320 {
3321     \__spintent_inner_sep:
3322     \overrightarrow
3323     {
3324         \int_zero:N \l__spintent_svec_pt_idx_int
3325         \seq_map_inline:Nn \l__spintent_svec_pt_seq
3326         {
3327             \int_incr:N \l__spintent_svec_pt_idx_int
3328             \__spintent_mathmlarg_point:nnn
3329             { pts \int_use:N \l__spintent_svec_pt_idx_int }
3330             { svec }
3331             { ##1 }
3332         }
3333     }
3334 }
3335 }
3336 }

```

(End of definition for \\_\_spintent\_svec\_mathml:n.)

\\_\_spintent\_svec\_exec:nnn The function \\_\_spintent\_svec\_exec:nnn receives the optional  $\langle keys \rangle$  in ‘#1’ and the  $\{\langle argument \rangle\}$  in ‘#2’, delegating to \\_\_spintent\_svec\_print:n. With ‘#3’ given (the optional coordinate), aliases \l\_\_spintent\_spcord\_silent\_cmd\_bool to \l\_\_spintent\_svec\_silent\_cmd\_bool — reverted when the calling group closes — and calls \\_\_spintent\_luafun\_spcord\_parse\_and\_set:n directly, reporting spcoord-invalid-arg on error unless silenced, then \\_\_spintent\_spcord\_set\_wrap\_parens\_size: and \\_\_spintent\_spcord\_mathml: — the same rendering functions \spcoord itself uses.

```

3337 \cs_new_protected:Nn \__spintent_svec_exec:nnn
3338 {
3339     \keys_set:nn { spintent / svec } { #1 }
3340     \__spintent_svec_print:n { #2 }
3341     \tl_if_novalue:nF {#3}
3342     {
3343         \equal % notación de vector =( , )
3344         \MathMLintent { :silent } { \c__spintent_invisible_sep_int } % previene lectura de intervalo
3345         \bool_set_eq:NN \l__spintent_spcord_silent_cmd_bool \l__spintent_svec_silent_cmd_bool
3346         \tl_set:Nn \l__spintent_spcord_raw_arg_tl {#3}
3347         \bool_if:NTF \l__spintent_svec_silent_cmd_bool
3348         {
3349             \__spintent_luafun_spcord_parse_and_set:n { \exp_not:n {#3} }
3350             \str_set:Nn \l__spintent_spcord_luaset_error_str { false }
3351         }
3352         {
3353             \__spintent_luafun_spcord_parse_and_set:n { \exp_not:n {#3} }
3354             \str_if_eq:VnT \l__spintent_spcord_luaset_error_str { true }
3355             { \msg_error:nnn { spintent } { spcoord-invalid-arg } { \exp_not:n {#3} } }
3356         }
3357         \__spintent_spcord_set_wrap_parens_size:
3358         \str_if_eq:VnF \l__spintent_spcord_luaset_error_str { true }
3359         { \__spintent_spcord_mathml: }
3360     }
3361 }

```

(End of definition for \\_\_spintent\_svec\_exec:nnn.)

\spvec The public command \spvec enforces *math mode*, opens a local group, and delegates execution to \\_\_spintent\_svec\_exec:nnn.

```

3362 \NewDocumentCommand \spvec { 0{} m d() }
3363 {
3364     \mode_if_math:TF
3365     {
3366         \group_begin:
3367         \attribute \l__spintent_scope_attr_int = 1
3368         \__spintent_svec_exec:nnn { #1 } { #2 } { #3 }
3369         \group_end:
3370     }
3371     { \msg_error:nnn { spintent } { need-math-mode } { \spvec } }
3372 }

```

(End of definition for `\spvec`. This function is documented on page 19.)

### 11.38 The command `\sprecta`

The user command `\sprecta` supports two types of notation for the input  $\langle argument \rangle$ : when the argument contains two points (separated by an optional comma), `\overleftarrow{\phantom{x}}` is applied; when the argument is a single point (without comma) or a lowercase letter, it is formatted without any visual decorator. Both notations are automatically detected by the Lua module.

#### 11.38.1 Definition of variables for `\sprecta`

The variable `\l__spintent_sprecta_mode_str` store the values set by the key `read-cmd` and the variable `\l__spintent_sprecta_prefix_tl` store the value set by the key `prefix`.

```

3373 \str_new:N \l__spintent_sprecta_mode_str
3374 \tl_new:N \l__spintent_sprecta_prefix_tl
3375 \seq_new:N \l__spintent_sprecta_pt_seq
3376 \tl_new:N \l__spintent_sprecta_refs_tl
3377 \tl_new:N \l__spintent_sprecta_intent_arg_tl
3378 \tl_new:N \l__spintent_sprecta_kv_tl
3379 \int_new:N \l__spintent_sprecta_pt_idx_int

```

(End of definition for `\l__spintent_sprecta_mode_str` and others.)

#### 11.38.2 Definition of error messages for `\sprecta`

The “error” `sprecta-invalid-points` is raised when the argument matches neither the two-point notation nor the name notation.

```

3380 \msg_new:nnnn { spintent } { sprecta-invalid-points }
3381 {
3382   Invalid~format~in~\token_to_str:N \sprecta:~'#1'~\msg_line_context:.
3383 }
3384 {
3385   See~the~documentation~of~\token_to_str:N \sprecta~for~the~accepted~argument~forms.
3386 }

```

(End of definition for `sprecta-invalid-points`.)

#### 11.38.3 Definition of keys for `\sprecta`

We define the  $\langle keys \rangle$  `read-cmd`, `prefix` and `silent-cmd`.

```

read-cmd  \keys_define:nn { spintent / sprecta }
prefix    {
silent-cmd {
unknown   {
3387   read-cmd      .choices:nn = { school , mathcat }
3388   {
3389     read-cmd      .initial:n = school,
3390     {
3391       \int_case:nn { \l_keys_choice_int }
3392       {
3393         {1} { \str_set:Nn \l__spintent_sprecta_mode_str { school } }
3394         {2} { \str_set:Nn \l__spintent_sprecta_mode_str { mathcat } }
3395       }
3396     },
3397     read-cmd      .initial:n = school,
3398     read-cmd      .value_required:n = true,
3399     read-cmd / unknown .code:n =
3400     \msg_error:nneee { spintent } { unknown-choice }
3401     { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
3402     prefix        .code:n =
3403     { \__spintent_sanitize_affix:Nn \l__spintent_sprecta_prefix_tl {#1} },
3404     prefix        .initial:n = ,
3405     prefix        .value_required:n = true,
3406     silent-cmd    .bool_set:N = \l__spintent_sprecta_silent_cmd_bool,
3407     silent-cmd    .initial:n = false,
3408     silent-cmd    .value_required:n = true,
3409     concept       .bool_set:N = \l__spintent_sprecta_concept_bool,
3410     concept       .initial:n = true,
3411     concept       .value_required:n = true,
3412     unknown       .code:n =
3413     {
3414       \str_set:Nn \l__spintent_command_name_str { sprecta }
3415       \__spintent_unknown_keys_cmd:n {#1}
3416     },
3417   }
}
}

```

(End of definition for `read-cmd` and others.)

### 11.38.4 Internal functions for \sprecta

\\_spintent\_sprecta\_print:n

The internal function \\_spintent\_sprecta\_print:n processes the  $\{\langle argument \rangle\}$  passed in ‘#1’. Calls \\_spintent\_luafun\_geo\_parse\_and\_set:n, reporting sprecta-invalid-points on error, then \\_spintent\_

```

3418 \cs_new_protected:Nn \_spintent_sprecta_print:n
3419 {
3420   \_spintent_luafun_geo_parse_and_set:n { \exp_not:n {#1} }
3421   \str_if_eq:VnT \_spintent_geo_luaset_error_str { true }
3422   {
3423     \msg_error:nne { spintent } { sprecta-invalid-points } { \exp_not:n {#1} }
3424   }
3425   \_spintent_sprecta_mathml:n { #1 }
3426 }

```

(End of definition for \\_spintent\_sprecta\_print:n.)

\\_spintent\_sprecta\_mathml:n

The function \\_spintent\_sprecta\_mathml:n branches on \\_spintent\_geo\_luaset\_is\_name\_str. For a proper name (m, l, ...), the concept intent is :silent(\$pt) when silent-cmd is true or concept is false; otherwise recta:prefix(\$pt), prefixed with prefix when given.

With \\_spintent\_geo\_luaset\_tipo\_str equal to sub, read-cmd=school tags the base through \\_spintent\_luafun\_mrow\_intent\_arg:nn with a literal kv, the subscript appended and silenced; read-cmd=mathcat tags base and subscript together through a native \MathMLarg instead.

```

3427 \cs_new_protected:Nn \_spintent_sprecta_mathml:n
3428 {
3429   \str_if_eq:VnTF \_spintent_geo_luaset_is_name_str { true }
3430   {
3431     \_spintent_invisible_sep:
3432     \bool_if:nTF
3433     {
3434       \bool_lazy_or_p:nn
3435       { \_spintent_sprecta_silent_cmd_bool }
3436       { ! \_spintent_sprecta_concept_bool }
3437     }
3438     {
3439       \tl_set:Nn \_spintent_sprecta_intent_arg_tl { :silent($pt) }
3440     }
3441     {
3442       \tl_set:Ne \_spintent_sprecta_intent_arg_tl
3443       {
3444         \tl_if_empty:NF \_spintent_sprecta_prefix_tl
3445         { \tl_use:N \_spintent_sprecta_prefix_tl - }
3446         recta:prefix($pt)
3447       }
3448     }
3449     \str_case:VnF \_spintent_geo_luaset_tipo_str
3450     {
3451       { sub }
3452       {
3453         \str_if_eq:VnTF \_spintent_sprecta_mode_str { school }
3454         {
3455           \tl_set:Ne \_spintent_sprecta_kv_tl
3456           { arg = "pt", intent = "_ \_spintent_geo_luaset_intent_str -" }
3457           \MathMLintent { \_spintent_sprecta_intent_arg_tl }
3458           {
3459             \_spintent_inner_sep:
3460             \_spintent_luafun_mrow_intent_arg:nn
3461             { } { \_spintent_sprecta_kv_tl }
3462             {
3463               \_spintent_geo_luaset_letra_str
3464               \c_math_subscript_token
3465               { \MathMLintent { :silent } { \_spintent_geo_luaset_sub_str } }
3466             }
3467           }
3468         }
3469       }
3470     }
3471   }

```

For read-cmd=mathcat, the same sub case tags base and subscript together through a native \MathMLarg instead — never mixed with the shared queue, since the queue does not recognize a native \MathMLarg claim as already taken.

```

3469   {
3470     \MathMLintent { \_spintent_sprecta_intent_arg_tl }
3471     {

```

```

3472         {
3473             \__spintent_inner_sep:
3474             \MathMLarg { pt }
3475         {
3476             {
3477                 \l__spintent_geo_luaset_letra_str
3478                 \c_math_subscript_token { \l__spintent_geo_luaset_sub_str }
3479             }
3480         }
3481     }
3482 }
3483 }
3484 }

```

With `\l__spintent_geo_luaset_tipo_str` equal to `prima`, the same distinction applies: `read-cmd=school` tags the base with the literal `kv` and appends the mark, unsilenced (a prime is not spelled out on the base, so nothing to repeat); `read-cmd=mathcat` tags base and mark together through a native `\MathMLarg`.

```

3485 { prima }
3486 {
3487     \str_if_eq:VnTF \l__spintent_spectra_mode_str { school }
3488     {
3489         \tl_set:Nx \l__spintent_spectra_kv_tl
3490         { arg = "pt", intent = "_ \l__spintent_geo_luaset_intent_str -" }
3491         \MathMLintent { \l__spintent_spectra_intent_arg_tl }
3492         {
3493             \__spintent_inner_sep:
3494             \__spintent_luafun_mrow_intent_arg:nn
3495             { } { \l__spintent_spectra_kv_tl }
3496             { \l__spintent_geo_luaset_letra_str \l__spintent_geo_luaset_sub_str }
3497         }
3498     }
3499     {
3500         \MathMLintent { \l__spintent_spectra_intent_arg_tl }
3501         {
3502             {
3503                 \__spintent_inner_sep:
3504                 \MathMLarg { pt }
3505             {
3506                 { \l__spintent_geo_luaset_letra_str \l__spintent_geo_luaset_sub_str }
3507             }
3508         }
3509     }
3510 }
3511 }
3512 }

```

Otherwise (no subscript, no prime), the same `kv` logic wraps the combined `\l__spintent_geo_luaset_print_tl` as a whole — the shared queue for `school`, its own `arg` alone for `mathcat`.

```

3513 {
3514     \bool_if:NTF \l__spintent_spectra_silent_cmd_bool
3515     {
3516         \tl_set:Nx \l__spintent_spectra_kv_tl { arg = "pt", intent = "- " }
3517     }
3518     {
3519         \str_if_eq:VnTF \l__spintent_spectra_mode_str { school }
3520         {
3521             \tl_set:Nx \l__spintent_spectra_kv_tl
3522             { arg = "pt", intent = "_ \l__spintent_geo_luaset_intent_str -" }
3523         }
3524         {
3525             \tl_set:Nx \l__spintent_spectra_kv_tl { arg = "pt" }
3526         }
3527     }
3528     \__spintent_luafun_mrow_intent_arg:nn
3529     { \l__spintent_spectra_intent_arg_tl }
3530     { \l__spintent_spectra_kv_tl }
3531     { \l__spintent_geo_luaset_print_tl }
3532 }
3533 }

```

For a list of points, `\l__spintent_geo_luaset_points_clist` is split into `\l__spintent_spectra_pt_seq` and walked to build `\l__spintent_spectra_refs_tl`, the `$pt1, $pt2, ...` reference list. The concept `intent` is `_(refs)` or `recta:prefix(refs)`, same conditions as above, applied with a native

`\MathMLintent` wrapping `\overleftrightharrow`. The sequence is walked again, calling `\__spintent_mathmlarg` once per point.

```

3534     {
3535       \seq_set_from_clist:NN \l__spintent_spectra_pt_seq \l__spintent_geo_luaset_points_clist
3536       \tl_clear:N \l__spintent_spectra_refs_tl
3537       \int_zero:N \l__spintent_spectra_pt_idx_int
3538       \seq_map_inline:Nn \l__spintent_spectra_pt_seq
3539         {
3540           \int_incr:N \l__spintent_spectra_pt_idx_int
3541           \tl_if_empty:NF \l__spintent_spectra_refs_tl
3542             { \tl_put_right:Nn \l__spintent_spectra_refs_tl { , } }
3543           \tl_put_right:Ne \l__spintent_spectra_refs_tl { $ pt \int_use:N \l__spintent_spectra_
3544         }
3545       \bool_if:nTF
3546         {
3547           \bool_lazy_or_p:nn
3548             { \l__spintent_spectra_silent_cmd_bool }
3549             { ! \l__spintent_spectra_concept_bool }
3550         }
3551         {
3552           \tl_set:Ne \l__spintent_spectra_intent_arg_tl { _ ( \l__spintent_spectra_refs_tl ) }
3553         }
3554         {
3555           \tl_set:Ne \l__spintent_spectra_intent_arg_tl
3556             {
3557               \tl_if_empty:NF \l__spintent_spectra_prefix_tl
3558                 { \tl_use:N \l__spintent_spectra_prefix_tl - }
3559               recta:prefix( \l__spintent_spectra_refs_tl )
3560             }
3561         }
3562       \__spintent_invisible_sep:
3563       \MathMLintent { \l__spintent_spectra_intent_arg_tl }
3564       {
3565         \__spintent_inner_sep:
3566         \overleftrightharrow
3567         {
3568           \int_zero:N \l__spintent_spectra_pt_idx_int
3569           \seq_map_inline:Nn \l__spintent_spectra_pt_seq
3570             {
3571               \int_incr:N \l__spintent_spectra_pt_idx_int
3572               \__spintent_mathmlarg_point:nnn
3573                 { pt \int_use:N \l__spintent_spectra_pt_idx_int }
3574                 { spectra }
3575                 { ##1 }
3576             }
3577         }
3578       }
3579     }
3580   }

```

(End of definition for `\__spintent_spectra_mathml:n`.)

`\__spintent_spectra_exec:nn`

The function `\__spintent_spectra_exec:nn` receives the optional *(keys)* passed in ‘#1’ and the *(argument)* passed in ‘#2’ delegating the rendering of the latter to `\__spintent_spectra_print:n`.

```

3581 \cs_new_protected:Nn \__spintent_spectra_exec:nn
3582 {
3583   \keys_set:nn { spintent / spectra } { #1 }
3584   \__spintent_spectra_print:n { #2 }
3585 }

```

(End of definition for `\__spintent_spectra_exec:nn`.)

`\spectra`

The public command `\spectra` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spectra_exec:nn`.

```

3586 \NewDocumentCommand \spectra { 0{} m }
3587 {
3588   \mode_if_math:TF
3589     {
3590       \group_begin:
3591       \attribute \l__spintent_scope_attr_int = 1
3592       \__spintent_spectra_exec:nn { #1 } { #2 }

```

```

3593     \group_end:
3594   }
3595   { \msg_error:nnn { spintent } { need-math-mode } { \sprecta } }
3596 }

```

(End of definition for `\sprecta`. This function is documented on page 19.)

### 11.39 The command `\sprayo`

The user command `\sprayo` provides a semantic interface for a ray. The key `read-sty` selects the verbalized term, `rayo` or `semirrecta`, both rendered identically with `\overrightarrow`.

#### 11.39.1 Definition of variables for `\sprayo`

The variable `\l__spintent_sprayo_mode_str` store the values set by the key `read-cmd`; the variable `\l__spintent_sprayo_prefix_tl` store the value set by the key `prefix` and the variable `\l__spintent_sprayo_read_sty_str` stores the value set by the key `read-sty`.

The variables defined here are assigned using `.choices:nn` or `.code:n` within `\keys_define:nn`.

```

3597 \str_new:N \l__spintent_sprayo_mode_str
3598 \tl_new:N \l__spintent_sprayo_prefix_tl
3599 \str_new:N \l__spintent_sprayo_read_sty_str

```

(End of definition for `\l__spintent_sprayo_mode_str`, `\l__spintent_sprayo_prefix_tl`, and `\l__spintent_sprayo_read_sty_str`.)

#### 11.39.2 Definition of error messages for `\sprayo`

`sprayo-invalid-points` Raised when the argument is not a comma-separated pair of uppercase letters.

```

3600 \msg_new:nnnn { spintent } { sprayo-invalid-points }
3601 {
3602   Invalid~point~format~in~\token_to_str:N \sprayo:~'#1'~\msg_line_context:.
3603 }
3604 {
3605   See~the~documentation~of~\token_to_str:N \sprayo~for~the~accepted~point~formats.
3606 }

```

(End of definition for `sprayo-invalid-points`.)

#### 11.39.3 Definition of keys for `\sprayo`

We define the `(keys)` `read-cmd`, `prefix`, `silent-cmd`, `concept` and `read-sty`.

```

3607 \keys_define:nn { spintent / sprayo }
3608 {
3609   read-cmd      .choices:nn = { school , mathcat }
3610   {
3611     \int_case:nn { \l_keys_choice_int }
3612     {
3613       {1} { \str_set:Nn \l__spintent_sprayo_mode_str { school } }
3614       {2} { \str_set:Nn \l__spintent_sprayo_mode_str { mathcat } }
3615     }
3616   },
3617   read-cmd      .initial:n = school,
3618   read-cmd      .value_required:n = true,
3619   read-cmd / unknown .code:n =
3620     \msg_error:nnee { spintent } { unknown-choice }
3621     { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
3622   prefix        .code:n =
3623     { \__spintent_sanitize_affix:Nn \l__spintent_sprayo_prefix_tl {#1} },
3624   prefix        .initial:n = ,
3625   prefix        .value_required:n = true,
3626   silent-cmd    .bool_set:N = \l__spintent_sprayo_silent_cmd_bool,
3627   silent-cmd    .initial:n = false,
3628   silent-cmd    .value_required:n = true,
3629   concept       .bool_set:N = \l__spintent_sprayo_concept_bool,
3630   concept       .initial:n = true,
3631   concept       .value_required:n = true,
3632   read-sty      .choices:nn = { rayo , semirrecta }
3633   {
3634     \int_case:nn { \l_keys_choice_int }
3635     {
3636       {1} { \str_set:Nn \l__spintent_sprayo_read_sty_str { rayo } }
3637       {2} { \str_set:Nn \l__spintent_sprayo_read_sty_str { semirrecta } }
3638     }
3639   },
3640   read-sty      .initial:n = rayo,

```

```

3641     read-sty             .value_required:n = true,
3642     read-sty / unknown  .code:n =
3643       \msg_error:nneee { spintent } { unknown-choice }
3644       { read-sty } { rayo,~semirrecta } { \exp_not:n {#1} },
3645     unknown             .code:n =
3646       {
3647         \str_set:Nn \l__spintent_command_name_str { sprayo }
3648         \__spintent_unknown_keys_cmd:n {#1}
3649       },
3650   }

```

(End of definition for read-cmd and others.)

#### 11.39.4 Internal functions for \spray

`\l__spintent_sprayo_pt_seq` Auxiliary variables used by `\__spintent_sprayo_mathml:` to split `\l__spintent_geo_luaset_points_clist` and build the `$pts1, $pts2, ...` reference list.

```

\l__spintent_sprayo_refs_tl
\l__spintent_sprayo_intent_arg_tl
\l__spintent_sprayo_pt_idx_int
3651 \seq_new:N \l__spintent_sprayo_pt_seq
3652 \tl_new:N \l__spintent_sprayo_refs_tl
3653 \tl_new:N \l__spintent_sprayo_intent_arg_tl
3654 \int_new:N \l__spintent_sprayo_pt_idx_int

```

(End of definition for `\l__spintent_sprayo_pt_seq` and others.)

`\__spintent_sprayo_print:n` The internal function `\__spintent_sprayo_print:n` process the `{(argument)}` passed in ‘#1’. It calls `\__spintent_luafun_geo_parse_and_set:n` to process the `{(argument)}`, returning an “error” if the input is invalid, then calls `\__spintent_sprayo_mathml:`. Unlike `\sprecta`, `\spray` has no proper-name argument: it always takes a list of points, so no `intent` is built here.

```

3655 \cs_new_protected:Nn \__spintent_sprayo_print:n
3656 {
3657   \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#1} }
3658   \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
3659   {
3660     \msg_error:nne { spintent } { sprayo-invalid-points } { \exp_not:n {#1} }
3661   }
3662   \__spintent_sprayo_mathml:
3663 }

```

(End of definition for `\__spintent_sprayo_print:n`)

`\__spintent_sprayo_mathml:` The function `\__spintent_sprayo_mathml:` splits `\l__spintent_geo_luaset_points_clist` (set by Lua, one raw point per element) into `\l__spintent_sprayo_pt_seq` and walks it once to build `\l__spintent_sprayo_refs_tl`, the comma-separated `$pts1, $pts2, ...` reference list, naming each point by position.

The concept `intent` is `_(refs)` when `silent-cmd` is `true` or `concept` is `false`; otherwise it is `\l__spintent_sprayo_read_sty_str:prefix(refs)` — the concept word itself (`rayo` or `semirrecta`) comes from the `read-sty` key, not a literal, since `\spray` verbalizes under two names — prefixed with `prefix` when given. It is applied with a native `\MathMLintent` wrapping `\overrightarrow`, preceded by `\__spintent_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`). The sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `ptsn` reference name and `silent-cmd/read-cmd` looked up under `sprayo`: it alone decides `arg/literal intent` per point, so concept plays no role at that level.

```

3664 \cs_new_protected:Nn \__spintent_sprayo_mathml:
3665 {
3666   \seq_set_from_clist:NN \l__spintent_sprayo_pt_seq \l__spintent_geo_luaset_points_clist
3667   \tl_clear:N \l__spintent_sprayo_refs_tl
3668   \int_zero:N \l__spintent_sprayo_pt_idx_int
3669   \seq_map_inline:Nn \l__spintent_sprayo_pt_seq
3670   {
3671     \int_incr:N \l__spintent_sprayo_pt_idx_int
3672     \tl_if_empty:NF \l__spintent_sprayo_refs_tl
3673     { \tl_put_right:Nn \l__spintent_sprayo_refs_tl { , } }
3674     \tl_put_right:Ne \l__spintent_sprayo_refs_tl { $ pts \int_use:N \l__spintent_sprayo_pt_idx_int }
3675   }
3676   \bool_if:nTF
3677   {
3678     \bool_lazy_or_p:nn
3679     { \l__spintent_sprayo_silent_cmd_bool }
3680     { ! \l__spintent_sprayo_concept_bool }
3681   }
3682   {

```

```

3683     \tl_set:Nx \l__spintent_sprayo_intent_arg_tl { _( \l__spintent_sprayo_refs_tl ) }
3684   }
3685   {
3686     \tl_set:Nx \l__spintent_sprayo_intent_arg_tl
3687     {
3688       \tl_if_empty:NF \l__spintent_sprayo_prefix_tl
3689       { \tl_use:N \l__spintent_sprayo_prefix_tl - }
3690       \l__spintent_sprayo_read_sty_str :prefix( \l__spintent_sprayo_refs_tl )
3691     }
3692   }
3693   \__spintent_invisible_sep:
3694   \MathMLintent { \l__spintent_sprayo_intent_arg_tl }
3695   {
3696     \__spintent_inner_sep:
3697     \overrightarrow
3698     {
3699       \int_zero:N \l__spintent_sprayo_pt_idx_int
3700       \seq_map_inline:Nn \l__spintent_sprayo_pt_seq
3701       {
3702         \int_incr:N \l__spintent_sprayo_pt_idx_int
3703         \__spintent_mathmlarg_point:nnn
3704         { pts \int_use:N \l__spintent_sprayo_pt_idx_int }
3705         { sprayo }
3706         { ##1 }
3707       }
3708     }
3709   }
3710 }

```

(End of definition for `\__spintent_sprayo_mathml:`)

`\__spintent_sprayo_exec:nn` The function `\__spintent_sprayo_exec:nn` receives the optional  $\langle keys \rangle$  passed in ‘#1’ and the  $\{ \langle argument \rangle \}$  passed in ‘#2’ delegating the rendering of the latter to `\__spintent_sprayo_print:n`.

```

3711 \cs_new_protected:Nn \__spintent_sprayo_exec:nn
3712 {
3713   \keys_set:nn { spintent / sprayo } { #1 }
3714   \__spintent_sprayo_print:n { #2 }
3715 }

```

(End of definition for `\__spintent_sprayo_exec:nn`)

`\sprayo` The public command `\sprayo` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_sprayo_exec:nn`.

```

3716 \NewDocumentCommand \sprayo { 0{} m }
3717 {
3718   \mode_if_math:TF
3719   {
3720     \group_begin:
3721     \attribute \l__spintent_scope_attr_int = 1
3722     \__spintent_sprayo_exec:nn { #1 } { #2 }
3723     \group_end:
3724   }
3725   { \msg_error:nnn { spintent } { need-math-mode } { \sprayo } }
3726 }

```

(End of definition for `\sprayo`. This function is documented on page 20.)

## 11.40 The commands `\spside` and `\spmside`

The user commands `\spside` and `\spmside` provide a semantic interface for a side, segment or trace determined by two points. The key `read-sty` selects the verbalized term — *lado*, *segmento*, or *trazo*, the latter used in straightedge-and-compass constructions — and the key `read-arg` overrides that term entirely with a free-form string. Additionally, the command `\spmside` provides the key `print-m`, controlling whether `\operatorname{m}` is typeset before the argument.

### 11.40.1 Definition of variables for `\spside` and `\spmside`

`\l__spintent_spside_mode_str` Each pair of variables is declared once for both `\spside` and `\spmside` via `\clist_map_inline:nn`. The variables `\l__spintent_spside_mode_str` and `\l__spintent_spside_mode_str` store the values set by the key `read-cmd`; the variables `\l__spintent_spside_prefix_tl` and `\l__spintent_spmside_prefix_tl` store the values set by the key `prefix`.

`\l__spintent_spside_read_sty_str` The variables `\l__spintent_spside_read_sty_str` and `\l__spintent_spmside_read_sty_str` stores the value set by the key `read-sty`.

The variables `\l__spintent_spside_read_arg_tl` and `\l__spintent_spmside_read_arg_tl` stores the value set by the key `read-arg`.

- The variables are defined here as they are assigned using `.choices:nn` or `.code:n` rather than a direct property within `\keys_define:nn`.

```

3727 \clist_map_inline:nn { side , mside }
3728 {
3729   \str_new:c { l__spintent_sp #1 _mode_str      }
3730   \tl_new:c { l__spintent_sp #1 _prefix_tl      }
3731   \str_new:c { l__spintent_sp #1 _read_sty_str  }
3732   \tl_new:c { l__spintent_sp #1 _read_arg_tl   }
3733 }

```

(End of definition for `\l__spintent_spside_mode_str` and others.)

#### 11.40.2 Definition of error messages for `\spside` and `\spmside`

spside-invalid-points  
spmside-invalid-points

We define an error message for the commands `\spside` and `\spmside`. The “error” is triggered if the user includes a invalid “*point*” in the `{\argument}` passed in ‘#1’ to the command.

```

3734 \clist_map_inline:nn { side , mside }
3735 {
3736   \msg_new:nnnn { spintent } { sp #1 -invalid-points }
3737   {
3738     Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
3739   }
3740   {
3741     See~the~documentation~of~##1~for~the~accepted~point~formats.
3742   }
3743 }

```

(End of definition for `spside-invalid-points` and `spmside-invalid-points`.)

#### 11.40.3 Definition of keys for `\spside` and `\spmside`

read-cmd  
prefix  
silent-cmd  
concept  
read-sty  
read-arg  
print-m  
unknown

We define the shared `\keys` `read-cmd`, `prefix`, `silent-cmd`, `concept`, `read-sty`, `read-arg` and `print-m`.

```

3744 \clist_map_inline:nn { side , mside }
3745 {
3746   \keys_define:nn { spintent / sp #1 }
3747   {
3748     read-cmd      .choices:nn = { school , mathcat }
3749     {
3750       \int_case:nn { \l_keys_choice_int }
3751       {
3752         {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school } }
3753         {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
3754       }
3755     },
3756     read-cmd      .initial:n = school,
3757     read-cmd      .value_required:n = true,
3758     read-cmd / unknown .code:n =
3759     \msg_error:nnee { spintent } { unknown-choice }
3760     { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
3761     prefix        .code:n =
3762     { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
3763     prefix        .initial:n = ,
3764     prefix        .value_required:n = true,
3765     silent-cmd    .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
3766     silent-cmd    .initial:n = false,
3767     silent-cmd    .value_required:n = true,
3768     concept       .bool_set:c = { l__spintent_sp #1 _concept_bool },
3769     concept       .initial:n = true,
3770     concept       .value_required:n = true,
3771     read-sty      .choices:nn = { lado , segmento , trazo }
3772     {
3773       \int_case:nn { \l_keys_choice_int }
3774       {
3775         {1} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { lado } }
3776         {2} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { segmento } }
3777         {3} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { trazo } }
3778       }
3779     },
3780     read-sty      .initial:n = lado,
3781     read-sty      .value_required:n = true,
3782     read-sty / unknown .code:n =

```

```

3783         \msg_error:nnee { spintent } { unknown-choice }
3784         { read-sty } { lado,~segmento,~trazo } { \exp_not:n {##1} },
3785 read-arg          .code:n =
3786     {
3787         \__spintent_sanitize_affix:cn { \__spintent_sp #1 _read_arg_tl } {##1}
3788     },
3789 read-arg          .initial:n = ,
3790 read-arg          .value_required:n = true,
3791 print-m           .bool_set:c = { \__spintent_sp #1 _print_m_bool },
3792 print-m           .initial:n = false,
3793 print-m           .value_required:n = true,
3794 unknown          .code:n =
3795     {
3796         \str_set:Nn \__spintent_command_name_str { sp #1 }
3797         \__spintent_unknown_keys_cmd:n {##1}
3798     },
3799 }
3800 }

```

🔗 The keys `print-m` and `concept` are defined for both commands for simplicity, but has no effect on `\spside`.

(End of definition for `read-cmd` and others.)

```

\__spintent_spside_intent_tl \__spintent_spside_pt_seq \__spintent_spside_refs_tl
\__spintent_spside_intent_arg_tl \__spintent_spside_pt_idx_int

```

`\__spintent_spside_intent_tl` holds the concept word (`read-arg` or `read-sty`). The rest are scratch variables shared by both commands (the function itself is shared) to split `\__spintent_geo_luaset_points_clist` and build the `$pts1, $pts2, ...` reference list.

```

3801 \tl_new:N \__spintent_spside_intent_tl
3802 \seq_new:N \__spintent_spside_pt_seq
3803 \tl_new:N \__spintent_spside_refs_tl
3804 \tl_new:N \__spintent_spside_intent_arg_tl
3805 \int_new:N \__spintent_spside_pt_idx_int

```

(End of definition for `\__spintent_spside_intent_tl` and others.)

```
\__spintent_spside_process:nn
```

The internal function `\__spintent_spside_process:nn` receives the short name of the command (`side` or `mside`) in ‘#1’ and the `{⟨argument⟩}` in ‘#2’. It calls `\__spintent_luafun_geo_parse_and_set:n` to process ‘#2’, returning an “error” if invalid, then builds `\__spintent_spside_intent_tl`. When `read-arg` is empty, it is set from `read-sty`, prefixed with ‘medida-del-’ for `\spmside`. When `read-arg` is given, it is used as is, and ‘medida-del-’ is prepended only if `concept` is `true`: with `concept=false`, `read-arg` is the verbatim escape hatch for a full user-written phrase (e.g. `read-arg={medida de la hipotenusa}`), gender and article left to the user. No `intent` is built here: it needs the reference list, which `\__spintent_spside_math` builds after splitting the points.

```

3806 \cs_new_protected:Nn \__spintent_spside_process:nn
3807 {
3808     \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#2} }
3809     \str_if_eq:VnT \__spintent_geo_luaset_error_str { true }
3810     {
3811         \msg_error:nnee { spintent } { sp #1 -invalid-points }
3812         { \c_backslash_str sp #1 } { \exp_not:n {#2} }
3813     }
3814     \tl_if_empty:cTF { \__spintent_sp #1 _read_arg_tl }
3815     {
3816         \tl_set:Ne \__spintent_spside_intent_tl { \str_use:c { \__spintent_sp #1 _read_sty_str } }
3817         \str_if_eq:eeT {#1} { mside }
3818         {
3819             \tl_put_left:Nn \__spintent_spside_intent_tl { medida-del- }
3820         }
3821     }
3822     {
3823         \tl_set:Ne \__spintent_spside_intent_tl { \tl_use:c { \__spintent_sp #1 _read_arg_tl } }
3824         \bool_if:cT { \__spintent_sp #1 _concept_bool }
3825         {
3826             \str_if_eq:eeT {#1} { mside }
3827             {
3828                 \tl_put_left:Nn \__spintent_spside_intent_tl { medida-del- }
3829             }
3830         }
3831     }
3832     \__spintent_spside_mathml:n { #1 }
3833 }

```

(End of definition for `\__spintent_spside_process:nn`.)

`\__spintent_spside_mathml:n` The function `\__spintent_spside_mathml:n` receives the short command name in ‘#1’. It splits `\l__spintent_geo_luaset_points_clist` into `\l__spintent_spside_pt_seq` and walks it once to build `\l__spintent_spside_refs_tl`, the comma-separated `$pts1,$pts2,...` reference list, naming each point by position.

```

3834 \cs_new_protected:Nn \__spintent_spside_mathml:n
3835 {
3836   \seq_set_from_clist:NN \l__spintent_spside_pt_seq \l__spintent_geo_luaset_points_clist
3837   \tl_clear:N \l__spintent_spside_refs_tl
3838   \int_zero:N \l__spintent_spside_pt_idx_int
3839   \seq_map_inline:Nn \l__spintent_spside_pt_seq
3840   {
3841     \int_incr:N \l__spintent_spside_pt_idx_int
3842     \tl_if_empty:NF \l__spintent_spside_refs_tl
3843     { \tl_put_right:Nn \l__spintent_spside_refs_tl { , } }
3844     \tl_put_right:Ne \l__spintent_spside_refs_tl { $ pts \int_use:N \l__spintent_spside_pt_idx_int }
3845   }

```

The concept `intent` is `_(refs)` when `silent-cmd` is `true`; otherwise, if `read-arg` is empty, it is `_(refs)` when `concept` is `false` or `\l__spintent_spside_intent_tl:prefix(refs)` when `true`; if `read-arg` is given, it is always `\l__spintent_spside_intent_tl:prefix(refs)` — `concept` never silences a user-given `read-arg` (see `\__spintent_spside_process:nn`). The `:prefix(...)` form is prefixed with `prefix` when given.

```

3846   \bool_if:cTF { \l__spintent_sp #1 _silent_cmd_bool }
3847   {
3848     \tl_set:Ne \l__spintent_spside_intent_arg_tl { _ ( \l__spintent_spside_refs_tl ) }
3849   }
3850   {
3851     \tl_if_empty:cTF { \l__spintent_sp #1 _read_arg_tl }
3852     {
3853       \bool_if:cTF { \l__spintent_sp #1 _concept_bool }
3854       {
3855         \tl_set:Ne \l__spintent_spside_intent_arg_tl
3856         {
3857           \tl_if_empty:cF { \l__spintent_sp #1 _prefix_tl }
3858           { \tl_use:c { \l__spintent_sp #1 _prefix_tl } - }
3859           \l__spintent_spside_intent_tl :prefix( \l__spintent_spside_refs_tl )
3860         }
3861       }
3862       {
3863         \tl_set:Ne \l__spintent_spside_intent_arg_tl { _ ( \l__spintent_spside_refs_tl ) }
3864       }
3865     }
3866     {
3867       \tl_set:Ne \l__spintent_spside_intent_arg_tl
3868       {
3869         \tl_if_empty:cF { \l__spintent_sp #1 _prefix_tl }
3870         { \tl_use:c { \l__spintent_sp #1 _prefix_tl } - }
3871         \l__spintent_spside_intent_tl :prefix( \l__spintent_spside_refs_tl )
3872       }
3873     }
3874   }

```

Applied with a native `\MathMLintent`, preceded by `\__spintent_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`). Inside, for `#1=side` the points are always wrapped in `\overline`; for `#1=mside`, when `print-m` is `true` a silent `\operatorname{m}` precedes the `\overline`-wrapped points; when `false`, the points are typeset bare, with no `\overline` at all — the recursive lookup in the shared `mathml_filter` callback finds each point either way, whether nested inside `\overline` or sitting directly in the `\MathMLintent` group. In every case the sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `ptsn` reference name and `silent-cmd/read-cmd` looked up under `sp#1`: it alone decides `arg/literal intent` per point, so `concept` plays no role at that level.

```

3875   \__spintent_invisible_sep: % (luamml issues #26)
3876   \MathMLintent { \l__spintent_spside_intent_arg_tl }
3877   {
3878     \__spintent_inner_sep:
3879     \str_if_eq:eeTF {#1} { side }
3880     {
3881       \overline { \l__spintent_spside_points_loop:n {#1} }
3882     }
3883     {

```

```

3884         \bool_if:cTF { \__spintent_sp #1 _print_m_bool }
3885         {
3886             \__spintent_luafun_mrow_intent_arg:nn
3887             { } { intent = ":silent" }
3888             { \operatorname { m } }
3889             \overline { \__spintent_spside_points_loop:n {#1} }
3890         }
3891         {
3892             \__spintent_spside_points_loop:n {#1}
3893         }
3894     }
3895 }
3896 }

```

(End of definition for `\__spintent_spside_mathml:n`.)

`\__spintent_spside_points_loop:n` Walks `\__spintent_spside_pt_seq` a second time, calling `\__spintent_mathmlarg_point:nnn` once per point. Factored out of `\__spintent_spside_mathml:n` since it is needed identically in three places there (`side`, `mside` with and without `print-m`); it takes the short command name in ‘`#1`’ again since a separate macro does not inherit its caller’s own `#1`.

```

3897 \cs_new_protected:Nn \__spintent_spside_points_loop:n
3898 {
3899     \int_zero:N \__spintent_spside_pt_idx_int
3900     \seq_map_inline:Nn \__spintent_spside_pt_seq
3901     {
3902         \int_incr:N \__spintent_spside_pt_idx_int
3903         \__spintent_mathmlarg_point:nnn
3904         { pts \int_use:N \__spintent_spside_pt_idx_int }
3905         { sp #1 }
3906         { ##1 }
3907     }
3908 }

```

(End of definition for `\__spintent_spside_points_loop:n`.)

`\__spintent_spside_exec:nnn` The function `\__spintent_spside_exec:nnn` receives the *short name* of the command passed in ‘`#1`’, the optional *keys* passed in ‘`#2`’ and the *argument* passed in ‘`#3`’, delegating the rendering to `\__spintent_spside_process:nn`.

```

3909 \cs_new_protected:Nn \__spintent_spside_exec:nnn
3910 {
3911     \keys_set:nn { spintent / sp #1 } { #2 }
3912     \__spintent_spside_process:nn { #1 } { #3 }
3913 }

```

(End of definition for `\__spintent_spside_exec:nnn`.)

`\spside` Both public commands `\spside` and `\spmside` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spside_exec:nnn`.

```

3914 \NewDocumentCommand \spside { O{} m }
3915 {
3916     \mode_if_math:TF
3917     {
3918         \group_begin:
3919         \attribute \__spintent_scope_attr_int = 1
3920         \__spintent_spside_exec:nnn { side } { #1 } { #2 }
3921         \group_end:
3922     }
3923     { \msg_error:nnn { spintent } { need-math-mode } { \spside } }
3924 }
3925 \NewDocumentCommand \spmside { O{} m }
3926 {
3927     \mode_if_math:TF
3928     {
3929         \group_begin:
3930         \attribute \__spintent_scope_attr_int = 1
3931         \__spintent_spside_exec:nnn { mside } { #1 } { #2 }
3932         \group_end:
3933     }
3934     { \msg_error:nnn { spintent } { need-math-mode } { \spmside } }
3935 }

```

(End of definition for `\spside` and `\spmside`. These functions are documented on page 20.)

### 11.41 The commands `\spangle` and `\spmangle`

The user commands `\spangle` and `\spmangle`, provide a semantic interface for angles, following the same generic pattern as `\spside/\spmside`. The key `print-m` chooses between the `\angle/\rightanglesqr` pair, typeset together with an explicit `\operatorname{m}`, and the `\measuredangle/\measuredrightangle` pair, which already conveys the notion of measure on its own; the key `recto` independently chooses the right-angle variant of whichever pair is active.

#### 11.41.1 Definition of variables for `\spangle` and `\spmangle`

```
\l__spintent_spangle_mode_str
\l__spintent_spmangle_mode_str
\l__spintent_spangle_prefix_tl
\l__spintent_spmangle_prefix_tl
```

Each pair of variables is declared once for both `\spangle` and `\spmangle` via `\clist_map_inline:nn`. The variables `\l__spintent_spangle_mode_str` and `\l__spintent_spmangle_mode_str` store the values set by the key `read-cmd`; the variables `\l__spintent_spangle_prefix_tl` and `\l__spintent_spmangle_prefix_tl` store the values set by the key `prefix`.

🌱 The variables defined here are assigned using `.choices:nn` or `.code:n` within `\keys_define:nn`.

```
3936 \clist_map_inline:nn { angle , mangle }
3937 {
3938   \str_new:c { l__spintent_sp #1 _mode_str }
3939   \tl_new:c { l__spintent_sp #1 _prefix_tl }
3940 }
```

(End of definition for `\l__spintent_spangle_mode_str` and others.)

#### 11.41.2 Definition of error messages for `\spangle` and `\spmangle`

```
spangle-invalid-arg
spmangle-invalid-arg
```

We define an “error” message for the commands `\spangle` and `\spmangle`. The “error” is triggered if the user includes a invalid `{\langle argument \rangle}` passed in ‘`#1`’ to the command.

```
3941 \clist_map_inline:nn { angle , mangle }
3942 {
3943   \msg_new:nnnn { spintent } { sp #1 -invalid-arg }
3944   {
3945     Invalid~format~in~##1!~'##2'~\msg_line_context:.
3946   }
3947   {
3948     See~the~documentation~of~##1~for~the~accepted~argument~forms.
3949   }
3950 }
```

(End of definition for `spangle-invalid-arg` and `spmangle-invalid-arg`.)

#### 11.41.3 Definition of keys for `\spangle` and `\spmangle`

```
read-cmd
prefix
recto
print-m
silent-cmd
concept
unknown
```

We define the shared `\keys` `read-cmd`, `prefix`, `recto`, `print-m`, `silent-cmd` and `concept`,

```
3951 \clist_map_inline:nn { angle , mangle }
3952 {
3953   \keys_define:nn { spintent / sp #1 }
3954   {
3955     read-cmd .choices:nn = { school , mathcat }
3956     {
3957       \int_case:nn { \l_keys_choice_int }
3958       {
3959         {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school } }
3960         {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
3961       }
3962     },
3963     read-cmd .initial:n = school,
3964     read-cmd .value_required:n = true,
3965     read-cmd / unknown .code:n =
3966     \msg_error:nneee { spintent } { unknown-choice }
3967     { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
3968     prefix .code:n =
3969     { \l__spintent_sanitise_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
3970     prefix .initial:n = ,
3971     prefix .value_required:n = true,
3972     recto .bool_set:c = { l__spintent_sp #1 _recto_bool },
3973     recto .initial:n = false,
3974     recto .value_required:n = true,
3975     print-m .bool_set:c = { l__spintent_sp #1 _print_m_bool },
3976     print-m .initial:n = false,
3977     print-m .value_required:n = true,
3978     silent-cmd .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
3979     silent-cmd .initial:n = false,
3980     silent-cmd .value_required:n = true,
3981     concept .bool_set:c = { l__spintent_sp #1 _concept_bool },
```

```

3982         concept                .initial:n = true,
3983         concept                .value_required:n = true,
3984         unknown                .code:n =
3985         {
3986             \str_set:Nn \__spintent_command_name_str { sp #1 }
3987             \__spintent_unknown_keys_cmd:n {##1}
3988         },
3989     }
3990 }

```

• The keys `print-m` and `concept` are defined for both commands for simplicity, but has no effect on `\spangle`.

(End of definition for `read-cmd` and others.)

#### 11.41.4 Internal functions for `\spangle` and `\spmangle`

`\__spintent_spangle_sym:n` The internal function `\__spintent_spangle_sym:n` dispatches on `#1`, `print-m` and `recto` to select one of four symbols: `\angle/\rightanglesqr` when `print-m` is true; `\measuredangle/\measuredrightangle` otherwise. When `#1` is `angle`, only `recto` applies, choosing between `\angle` and `\rightangle`.

```

3991 \cs_new_protected:Nn \__spintent_spangle_sym:n
3992 {
3993     \str_if_eq:eeTF {#1} { mangle }
3994     {
3995         \bool_if:cTF { \__spintent_sp #1 _print_m_bool }
3996         {
3997             \bool_if:cTF { \__spintent_sp #1 _recto_bool }
3998             { \__spintent_right_angle_sqr: } { \angle }
3999         }
4000         {
4001             \bool_if:cTF { \__spintent_sp #1 _recto_bool }
4002             { \measuredrightangle } { \measuredangle }
4003         }
4004     }
4005     {
4006         \bool_if:cTF { \__spintent_sp #1 _recto_bool }
4007         { \__spintent_right_angle_sqr: } { \angle }
4008     }
4009 }

```

(End of definition for `\__spintent_spangle_sym:n`.)

`\l__spintent_spangle_intent_tl` `\l__spintent_spangle_pt_seq` `\l__spintent_spangle_refs_tl` `\l__spintent_spangle_intent_arg_tl` `\l__spintent_spangle_pt_idx_int` `\l__spintent_spangle_print_m_now_bool` `\l__spintent_spangle_intent_tl` holds the concept word (ángulo or medida-del-ángulo, ‘-recto’-suffixed when `recto` is true). The rest are scratch variables shared by both commands to split `\l__spintent_geo_lua` and build the reference list — used only in the tres-puntos and un-punto cases (see `\__spintent_spangle_mathml:n`):

```

4010 \tl_new:N \l__spintent_spangle_intent_tl
4011 \seq_new:N \l__spintent_spangle_pt_seq
4012 \tl_new:N \l__spintent_spangle_refs_tl
4013 \tl_new:N \l__spintent_spangle_intent_arg_tl
4014 \int_new:N \l__spintent_spangle_pt_idx_int
4015 \str_new:N \l__spintent_spangle_kv_tl
4016 \bool_new:N \l__spintent_spangle_print_m_now_bool

```

(End of definition for `\l__spintent_spangle_intent_tl` and others.)

`\__spintent_spangle_process:nn` The internal function `\__spintent_spangle_process:nn` receives the short name of the command (`angle` or `mangle`) in ‘`#1`’ and the `{\argument}` in ‘`#2`’. Calls `\__spintent_luafun_geo_angulo_parse_and_set:n`, reporting `sp#1-invalid-arg` on error. For a proper name that is a single letter (not numeric), also reports the same error when that letter is lowercase — angle names take an uppercase letter, number, or Greek letter, never a lowercase Latin one; the length check keeps a Greek letter command such as `\alpha` out of this comparison. Then sets `\l__spintent_spangle_intent_tl` to `ángulo` (or `medida-del-ángulo` for `\spmangle`), suffixed with `-recto` when `recto` is true, and calls `\__spintent_spangle_mathml:n`.

```

4017 \cs_new_protected:Nn \__spintent_spangle_process:nn
4018 {
4019     \__spintent_luafun_geo_angulo_parse_and_set:n { \exp_not:n {#2} }
4020     \str_if_eq:VnT \l__spintent_geo_lua_set_error_str { true }
4021     {
4022         \msg_error:nnee { spintent } { sp #1 -invalid-arg }
4023         { \c_backslash_str sp #1 } { \exp_not:n {#2} }
4024     }
4025     \str_if_eq:VnT \l__spintent_geo_lua_set_angulo_case_str { nombre }
4026     {
4027         \str_if_eq:VnF \l__spintent_geo_lua_set_nombre_es_num_str { true }
4028         {

```

```

4029         \int_compare:nT { \str_count:N \l__spintent_geo_luaset_letra_str = 1 }
4030         {
4031             \str_if_eq:eeF { \l__spintent_geo_luaset_letra_str } { \str_uppercase:n { \l__spintent_geo_luaset_letra_str } }
4032             {
4033                 \msg_error:nnee { spintent } { sp #1 -invalid-arg }
4034                 { \c_backslash_str sp #1 } { \exp_not:n {#2} }
4035             }
4036         }
4037     }
4038 }
4039 \str_if_eq:eeTF {#1} { mangle }
4040 {
4041     \tl_set:Nn \l__spintent_spangle_intent_tl { medida-del-ángulo }
4042 }
4043 {
4044     \tl_set:Nn \l__spintent_spangle_intent_tl { ángulo }
4045 }
4046 \bool_if:cT { \l__spintent_sp #1 _recto_bool }
4047 {
4048     \tl_put_right:Nn \l__spintent_spangle_intent_tl { -recto }
4049 }
4050 \l__spintent_spangle_mathml:n {#1}
4051 }

```

(End of definition for `\l__spintent_spangle_process:nn`.)

`\l__spintent_spangle_mathml:n` The function `\l__spintent_spangle_mathml:n` branches on `\l__spintent_geo_luaset_angulo_case_str`. For `nombre`, the symbol (`\l__spintent_spangle_sym:n`) is tagged through `\l__spintent_luafun_mrow_intent_arg:n` on its own, with `\l__spintent_spangle_intent_tl:prefix` (no `$`-reference: `nombre` never refers to more than one thing) — `_-` when `silent-cmd` is `true` or `concept` is `false`. With `silent-cmd` `true`, the name that follows is silenced as a whole.

```

4052 \cs_new_protected:Nn \l__spintent_spangle_mathml:n
4053 {
4054     \l__spintent_invisible_sep: % (luamml issues #26)
4055     \str_if_eq:VnTF \l__spintent_geo_luaset_angulo_case_str { nombre }
4056     {
4057         \bool_if:cTF { \l__spintent_sp #1 _silent_cmd_bool }
4058         {
4059             \tl_set:Nn \l__spintent_spangle_intent_arg_tl { _- }
4060         }
4061         {
4062             \bool_if:cTF { \l__spintent_sp #1 _concept_bool }
4063             {
4064                 \tl_set:Nn \l__spintent_spangle_intent_arg_tl
4065                 {
4066                     \tl_if_empty:cF { \l__spintent_sp #1 _prefix_tl }
4067                     { \tl_use:c { \l__spintent_sp #1 _prefix_tl } - }
4068                     \l__spintent_spangle_intent_tl :prefix
4069                 }
4070             }
4071             {
4072                 \tl_set:Nn \l__spintent_spangle_intent_arg_tl { _- }
4073             }
4074         }
4075         \tl_set:Nn \l__spintent_spangle_kv_tl
4076         { intent = "\l__spintent_spangle_intent_arg_tl" }
4077         \l__spintent_luafun_mrow_intent_arg:nn
4078         { } { \l__spintent_spangle_kv_tl }
4079         { \l__spintent_spangle_sym:n {#1} }
4080         \bool_if:cTF { \l__spintent_sp #1 _silent_cmd_bool }
4081         {
4082             \l__spintent_luafun_mrow_intent_arg:nn { } { intent = "_-" }
4083             { \l__spintent_geo_luaset_print_tl }
4084         }

```

Otherwise, a numeric name is tagged as a whole, through `\l__spintent_luafun_mrow_intent_arg:nn`, with a literal `intent` built from `\l__spintent_geo_luaset_intent_str`.

```

4085     {
4086         \str_if_eq:VnTF \l__spintent_geo_luaset_nombre_es_num_str { true }
4087         {
4088             \tl_set:Nn \l__spintent_spangle_kv_tl

```

```

4089         { intent = "_ \l__spintent_geo_luaset_intent_str -" }
4090         \l__spintent_luafun_mrow_intent_arg:nn
4091         { } { \l__spintent_spangle_kv_tl }
4092     { \l__spintent_geo_luaset_print_tl }
4093 }

```

With `\l__spintent_geo_luaset_tipo_str` equal to `sub`, `read-cmd=school` tags the letter through the same function, with a literal `intent`, and the subscript is appended and silenced; `read-cmd=mathcat` leaves both letter and subscript untouched, for MathCAT to read the whole `<msub>` natively (no `arg` is set in either case: there is no `$`-reference to resolve).

```

4094     {
4095         \str_case:VnF \l__spintent_geo_luaset_tipo_str
4096         {
4097             { sub }
4098             {
4099                 \str_if_eq:eeTF { \use:c { \l__spintent_sp #1 _mode_str } } { school }
4100                 {
4101                     \tl_set:Ne \l__spintent_spangle_kv_tl
4102                     { intent = "_ \l__spintent_geo_luaset_intent_str -" }
4103                     \l__spintent_luafun_mrow_intent_arg:nn
4104                     { } { \l__spintent_spangle_kv_tl }
4105                     {
4106                         \l__spintent_geo_luaset_letra_str
4107                         \c_math_subscript_token
4108                         { \MathMLintent { :silent } { \l__spintent_geo_luaset_sub_str } }
4109                     }
4110                 }
4111             }
4112             \l__spintent_geo_luaset_letra_str
4113             \c_math_subscript_token { \l__spintent_geo_luaset_sub_str }
4114         }
4115     }

```

With `\l__spintent_geo_luaset_tipo_str` equal to `prima`, the prime is always read — spoken as part of the literal for `read-cmd=school`, native for `read-cmd=mathcat` — since a prime is not a repeated word to silence.

```

4116         { prima }
4117         {
4118             \str_if_eq:eeTF { \use:c { \l__spintent_sp #1 _mode_str } } { school }
4119             {
4120                 \tl_set:Ne \l__spintent_spangle_kv_tl
4121                 { intent = "_ \l__spintent_geo_luaset_intent_str -" }
4122                 \l__spintent_luafun_mrow_intent_arg:nn
4123                 { } { \l__spintent_spangle_kv_tl }
4124                 { \l__spintent_geo_luaset_letra_str \l__spintent_geo_luaset_sub_str }
4125             }
4126             {
4127                 \l__spintent_geo_luaset_letra_str \l__spintent_geo_luaset_sub_str
4128             }
4129         }
4130     }
4131     { \l__spintent_geo_luaset_print_tl }
4132 }
4133 }
4134 }

```

For `tres-puntos/un-punto`, `\l__spintent_geo_luaset_points_clist` is split into `\l__spintent_spangle_pt_seq` and walked to build `\l__spintent_spangle_refs_tl`, the `$pt1, $pt2, ...` reference list. A single native `\MathMLintent` wraps the symbol and the points together, with `\l__spintent_spangle_intent_tl:prefix(refs` — `_` — when `silent-cmd` is `true`, `_(refs)` when `concept` is `false` alone. For `\spmangle` with `print-m` `true`, a silent `\operatorname{m}` precedes `\l__spintent_spangle_sym:n` inside the same `\MathMLintent`. The sequence is walked again, calling `\l__spintent_mathmlarg_point:nnn` once per point.

```

4135     {
4136         \seq_set_from_clist:NN \l__spintent_spangle_pt_seq \l__spintent_geo_luaset_points_clist
4137         \tl_clear:N \l__spintent_spangle_refs_tl
4138         \int_zero:N \l__spintent_spangle_pt_idx_int
4139         \seq_map_inline:Nn \l__spintent_spangle_pt_seq
4140         {
4141             \int_incr:N \l__spintent_spangle_pt_idx_int
4142             \tl_if_empty:NF \l__spintent_spangle_refs_tl
4143             { \tl_put_right:Nn \l__spintent_spangle_refs_tl { , } }
4144             \tl_put_right:Ne \l__spintent_spangle_refs_tl { $ pt \int_use:N \l__spintent_spangle_

```

```

4145     }
4146     \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
4147     {
4148         \tl_set:Nn \l__spintent_spangle_intent_arg_tl { _- }
4149     }
4150     {
4151         \bool_if:cTF { l__spintent_sp #1 _concept_bool }
4152         {
4153             \tl_set:Ne \l__spintent_spangle_intent_arg_tl
4154             {
4155                 \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4156                 { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4157                 \l__spintent_spangle_intent_tl :prefix( \l__spintent_spangle_refs_tl )
4158             }
4159         }
4160         {
4161             \tl_set:Ne \l__spintent_spangle_intent_arg_tl { _( \l__spintent_spangle_refs_tl )
4162         }
4163     }
4164     \bool_set_false:N \l__spintent_spangle_print_m_now_bool
4165     \str_if_eq:eeT {#1} { mangle }
4166     {
4167         \bool_if:cT { l__spintent_sp #1 _print_m_bool }
4168         { \bool_set_true:N \l__spintent_spangle_print_m_now_bool }
4169     }
4170     \MathMLintent { \l__spintent_spangle_intent_arg_tl }
4171     {
4172         \__spintent_inner_sep:
4173         \bool_if:NT \l__spintent_spangle_print_m_now_bool
4174         {
4175             \__spintent_luafun_mrow_intent_arg:nn
4176             { } { intent = ":silent" }
4177             { \operatorname { m } } }
4178         }
4179         \__spintent_luafun_mrow_intent_arg:nn
4180         { } { intent = ":silent" }
4181         { \__spintent_spangle_sym:n {#1} }
4182         \int_zero:N \l__spintent_spangle_pt_idx_int
4183         \seq_map_inline:Nn \l__spintent_spangle_pt_seq
4184         {
4185             \int_incr:N \l__spintent_spangle_pt_idx_int
4186             \__spintent_mathmlarg_point:nnn
4187             { pt \int_use:N \l__spintent_spangle_pt_idx_int }
4188             { sp#1 }
4189             { ##1 }
4190         }
4191     }
4192 }
4193 }

```

(End of definition for `\__spintent_spangle_mathml:n`.)

`\__spintent_spangle_exec:nnn`

The function `\__spintent_spangle_exec:nnn` receives the *short name* of the command passed in ‘#1’, the optional *⟨keys⟩* passed in ‘#2’ and the *⟨argument⟩* passed in ‘#3’, delegating the rendering to `\__spintent_spangle_process:nn`.

```

4194 \cs_new_protected:Nn \__spintent_spangle_exec:nnn
4195 {
4196     \keys_set:nn { spintent / sp #1 } { #2 }
4197     \__spintent_spangle_process:nn { #1 } { #3 }
4198 }

```

(End of definition for `\__spintent_spangle_exec:nnn`.)

`\spangle`  
`\spmangle`

Both public commands `\spangle` and `\spmangle` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spangle_exec:nnn`.

```

4199 \NewDocumentCommand \spangle { 0{} m }
4200 {
4201     \mode_if_math:TF
4202     {
4203         \group_begin:
4204         \attribute \l__spintent_scope_attr_int = 1

```

```

4205     \__spintent_spangle_exec:nnn { angle } { #1 } { #2 }
4206   \group_end:
4207 }
4208 { \msg_error:nnn { spintent } { need-math-mode } { \spangle } }
4209 }
4210 \NewDocumentCommand \spmangle { 0{} m }
4211 {
4212   \mode_if_math:TF
4213   {
4214     \group_begin:
4215     \attribute \__spintent_scope_attr_int = 1
4216     \__spintent_spangle_exec:nnn { mangle } { #1 } { #2 }
4217     \group_end:
4218   }
4219   { \msg_error:nnn { spintent } { need-math-mode } { \spmangle } }
4220 }

```

(End of definition for `\spangle` and `\spmangle`. These functions are documented on page 22.)

## 11.42 The commands `\sparc` and `\spmarc`

The user commands `\sparc` and `\spmarc` provide a semantic interface for arcs. Unlike `\spmside` and `\spmangle`, `\spmarc` has no `print-m` key: `\operatorname{m}` is always present, since  $m\widehat{AB}$  is the only standard notation for the measure of an arc.

### 11.42.1 Definition of variables for `\sparc` and `\spmarc`

Each pair of variables is declared once for both `\sparc` and `\spmarc` via `\clist_map_inline:nn`. The variables `\__spintent_sparc_mode_str` and `\__spintent_spmarc_mode_str` store the values set by the key `read-cmd`; the variables `\__spintent_sparc_prefix_tl` and `\__spintent_spmarc_prefix_tl` store the values set by the key `prefix`.

```

4221 \clist_map_inline:nn { arc , marc }
4222 {
4223   \str_new:c { \__spintent_sp #1 _mode_str }
4224   \tl_new:c { \__spintent_sp #1 _prefix_tl }
4225 }

```

(End of definition for `\__spintent_sparc_mode_str` and others.)

### 11.42.2 Definition of error messages for `\sparc` and `\spmarc`

We define an “error” message for the commands `\sparc` and `\spmarc`. The “error” is triggered if the user includes a invalid `{⟨argument⟩}` passed in ‘#1’ to the command.

```

4226 \clist_map_inline:nn { arc , marc }
4227 {
4228   \msg_new:nnnn { spintent } { sp #1 -invalid-points }
4229   {
4230     Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
4231   }
4232   {
4233     See~the~documentation~of~##1~for~the~accepted~point~formats.
4234   }
4235 }

```

(End of definition for `sparc-invalid-points` and `spmarc-invalid-points`.)

### 11.42.3 Definition of keys for `\sparc` and `\spmarc`

We define the shared `⟨keys⟩` `read-cmd`, `prefix`, `silent-cmd` and `concept`.

```

read-cmd  \clist_map_inline:nn { arc , marc }
prefix    {
silent-cmd {
concept   \keys_define:nn { spintent / sp #1 }
unknown   {
4236         read-cmd .choices:nn = { school , mathcat }
4237         {
4238           \int_case:nn { \l_keys_choice_int }
4239           {
4240             {1} { \str_set:cn { \__spintent_sp #1 _mode_str } { school } }
4241             {2} { \str_set:cn { \__spintent_sp #1 _mode_str } { mathcat } }
4242           }
4243         },
4244         read-cmd .initial:n = school,
4245         read-cmd .value_required:n = true,
4246         read-cmd / unknown .code:n =
4247         \msg_error:nnee { spintent } { unknown-choice }
4248       }
4249     }
4250   }
4251 }

```

```

4252         { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
4253     prefix                .code:n =
4254     { \__spintent_sanitize_affix:cn { \__spintent_sp #1 _prefix_tl } {##1} },
4255     prefix                .initial:n = ,
4256     prefix                .value_required:n = true,
4257     silent-cmd            .bool_set:c = { \__spintent_sp #1 _silent_cmd_bool },
4258     silent-cmd            .initial:n = false,
4259     silent-cmd            .value_required:n = true,
4260     concept               .bool_set:c = { \__spintent_sp #1 _concept_bool },
4261     concept               .initial:n = true,
4262     concept               .value_required:n = true,
4263     unknown               .code:n =
4264     {
4265         \str_set:Nn \__spintent_command_name_str { sp #1 }
4266         \__spintent_unknown_keys_cmd:n {##1}
4267     },
4268 }
4269 }

```

(End of definition for read-cmd and others.)

#### 11.42.4 Internal functions for \sparc and \spmarc

\\_\_spintent\_sparc\_wide\_arc:n

The function `\__spintent_sparc_wide_arc:n` acts as a fallback when the mathematical font used in the document does not provide the command `\widearc`. The definition of this function is a direct adaptation of a response given by Clea F. Rees (@cfr) to a query in the chat room of TeX-SX.

```

4270 \cs_new_protected_nopar:Nn \__spintent_sparc_wide_arc:n
4271 {
4272     \Umathaccent 7 \symoperators "023DC \scan_stop: {#1}
4273 }

```

(End of definition for \\_\_spintent\_sparc\_wide\_arc:n.)

\\_\_spintent\_sparc\_print\_arc:n

The function `\__spintent_sparc_print_arc:n` applies the command `\widearc` to the argument passed to `\sparc` or `\spmarc`. If the mathematical font set in the document does not provide the command `\widearc`, we will use the built-in function `\__spintent_sparc_wide_arc:n`.

```

4274 \cs_new_protected:Nn \__spintent_sparc_print_arc:n
4275 {
4276     \cs_if_exist:NTF \widearc
4277     { \widearc {#1} }
4278     { \__spintent_sparc_wide_arc:n {#1} }
4279 }

```

(End of definition for \\_\_spintent\_sparc\_print\_arc:n.)

\l\_\_spintent\_sparc\_intent\_tl

`\l__spintent_sparc_intent_tl` holds the concept word (arco or medida-del-arco). The rest are scratch variables to split `\l__spintent_geo_luaset_points_clist` and build the `$pt1, $pt2, ...` reference list used in the concept intent.

```

4280 \tl_new:N \l__spintent_sparc_intent_tl
4281 \seq_new:N \l__spintent_sparc_pt_seq
4282 \tl_new:N \l__spintent_sparc_refs_tl
4283 \tl_new:N \l__spintent_sparc_intent_arg_tl
4284 \int_new:N \l__spintent_sparc_pt_idx_int

```

(End of definition for \l\_\_spintent\_sparc\_intent\_tl and others.)

\\_\_spintent\_sparc\_process:nn

The internal function `\__spintent_sparc_process:nn` receives the short name of the command (arc or marc) in ‘#1’ and the `{\argument}` in ‘#2’. It calls `\__spintent_luafun_geo_parse_and_set:n` to process ‘#2’, returning an “error” if invalid, then sets `\l__spintent_sparc_intent_tl` to arco (or medida-del-arco for `\spmarc`). No intent is built here: it needs the reference list, which `\__spintent_sparc_mathl:n` builds after splitting the points.

```

4285 \cs_new_protected:Nn \__spintent_sparc_process:nn
4286 {
4287     \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#2} }
4288     \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
4289     {
4290         \msg_error:nnee { spintent } { sp #1 -invalid-points }
4291         { \c_backslash_str sp #1 } { \exp_not:n {#2} }
4292     }
4293     \str_if_eq:eeTF {#1} { marc }
4294     { \tl_set:Nn \l__spintent_sparc_intent_tl { medida-del-arco } }
4295     { \tl_set:Nn \l__spintent_sparc_intent_tl { arco } }
4296     \__spintent_sparc_mathl:n {#1}
4297 }

```

(End of definition for `\__spintent_sparc_process:nn`.)

`\__spintent_sparc_mathml:n`

The function `\__spintent_sparc_mathml:n` receives the short command name in ‘`#1`’. It splits `\__spintent_geo_` into `\__spintent_sparc_pt_seq` and walks it once to build `\__spintent_sparc_refs_tl`, the comma-separated `$pt1,$pt2,...` reference list, naming each point by position.

```

4298 \cs_new_protected:Nn \__spintent_sparc_mathml:n
4299 {
4300   \seq_set_from_clist:NN \__spintent_sparc_pt_seq \__spintent_geo_luaset_points_clist
4301
4302   \tl_clear:N \__spintent_sparc_refs_tl
4303   \int_zero:N \__spintent_sparc_pt_idx_int
4304   \seq_map_inline:Nn \__spintent_sparc_pt_seq
4305   {
4306     \int_incr:N \__spintent_sparc_pt_idx_int
4307     \tl_if_empty:NF \__spintent_sparc_refs_tl
4308     { \tl_put_right:Nn \__spintent_sparc_refs_tl { , } }
4309     \tl_put_right:Ne \__spintent_sparc_refs_tl { $ pt \int_use:N \__spintent_sparc_pt_idx_int }
4310   }

```

The concept `intent` is `_` - when `silent-cmd` is `true` or `concept` is `false`; otherwise `\__spintent_sparc_intent_` prefixed with `prefix` when given.

```

4311 \bool_if:cTF { \__spintent_sp #1 _silent_cmd_bool }
4312 {
4313   \tl_set:Nn \__spintent_sparc_intent_arg_tl { _- }
4314 }
4315 {
4316   \bool_if:cTF { \__spintent_sp #1 _concept_bool }
4317   {
4318     \tl_set:Ne \__spintent_sparc_intent_arg_tl
4319     {
4320       \tl_if_empty:cF { \__spintent_sp #1 _prefix_tl }
4321       { \tl_use:c { \__spintent_sp #1 _prefix_tl } - }
4322       \__spintent_sparc_intent_tl :prefix( \__spintent_sparc_refs_tl )
4323     }
4324   }
4325   {
4326     \tl_set:Nn \__spintent_sparc_intent_arg_tl { _- }
4327   }
4328 }

```

It is applied with a single native `\MathMLintent` wrapping `\__spintent_sparc_print_arc:n` directly (same shape as `\__spintent_spside_mathml:n`), preceded by `\__spintent_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`, `luamml` issue #26). A silent `\operatorname{m}` precedes it for `\spmarc` (always, no `print-m` key here). The sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `ptn` reference name, looked up under `sp#1`.

```

4329 \__spintent_invisible_sep: % need by MathCAT (bug in luamml issues #26)
4330 \MathMLintent { \__spintent_sparc_intent_arg_tl }
4331 {
4332   \__spintent_inner_sep:
4333   \str_if_eq:eeT {#1} { marc }
4334   {
4335     \__spintent_luafun_mrow_intent_arg:nn
4336     { } { intent = ":silent" }
4337     { \operatorname { m } } }
4338   }
4339   \__spintent_sparc_print_arc:n
4340   {
4341     \int_zero:N \__spintent_sparc_pt_idx_int
4342     \seq_map_inline:Nn \__spintent_sparc_pt_seq
4343     {
4344       \int_incr:N \__spintent_sparc_pt_idx_int
4345       \__spintent_mathmlarg_point:nnn
4346       { pt \int_use:N \__spintent_sparc_pt_idx_int }
4347       { sp #1 }
4348       { ##1 }
4349     }
4350   }
4351 }
4352 }

```

(End of definition for `\__spintent_sparc_mathml:n`.)

`\__spintent_sparc_exec:nnn` The function `\__spintent_sparc_exec:nnn` receives the *short name* of the command passed in ‘#1’, the optional *keys* passed in ‘#2’ and the *{argument}* passed in ‘#3’, delegating the rendering to `\__spintent_sparc_process:nn`.

```

4353 \cs_new_protected:Nn \__spintent_sparc_exec:nnn
4354 {
4355     \keys_set:nn { spintent / sp #1 } { #2 }
4356     \__spintent_sparc_process:nn { #1 } { #3 }
4357 }

```

(End of definition for `\__spintent_sparc_exec:nnn`.)

`\sparc` Both public commands `\sparc` and `\spmarc` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_sparc_exec:nnn`.

```

4358 \NewDocumentCommand \sparc { O{} m }
4359 {
4360     \mode_if_math:TF
4361     {
4362         \group_begin:
4363         \attribute \__spintent_scope_attr_int = 1
4364         \__spintent_sparc_exec:nnn { arc } { #1 } { #2 }
4365         \group_end:
4366     }
4367     { \msg_error:nnn { spintent } { need-math-mode } { \sparc } }
4368 }
4369 \NewDocumentCommand \spmarc { O{} m }
4370 {
4371     \mode_if_math:TF
4372     {
4373         \group_begin:
4374         \attribute \__spintent_scope_attr_int = 1
4375         \__spintent_sparc_exec:nnn { marc } { #1 } { #2 }
4376         \group_end:
4377     }
4378     { \msg_error:nnn { spintent } { need-math-mode } { \spmarc } }
4379 }

```

(End of definition for `\sparc` and `\spmarc`. These functions are documented on page 24.)

### 11.43 The command `\spPoly`

The user command `\spPoly` typesets a polygon defined by its vertices. The figure type and its visual symbol are inferred from the vertex count: three points produce a triangle, four a square, five or more a generic polygon with no symbol. It never sends a boolean to Lua: `\__spintent_geo_luaset_intent_str` (the full *intent*, with the figure name).

#### 11.43.1 Definition of variables for `\spPoly`

`\__spintent_spPoly_mode_str` The variable `\__spintent_spPoly_mode_str` stores the value set by the key `read-cmd`; the variable `\__spintent_spPoly_prefix_tl` stores the value set by the key `prefix`.

```

4380 \str_new:N \__spintent_spPoly_mode_str
4381 \tl_new:N \__spintent_spPoly_prefix_tl

```

(End of definition for `\__spintent_spPoly_mode_str` and `\__spintent_spPoly_prefix_tl`.)

#### 11.43.2 Definition of error messages for `\spPoly`

`spPoly-invalid-points` Raised when the argument is not a comma-separated list of at least three uppercase letters.

```

4382 \msg_new:nnnn { spintent } { spPoly-invalid-points }
4383 {
4384     Invalid~point~format~in~\token_to_str:N \spPoly:~'#1'~\msg_line_context:.
4385 }
4386 {
4387     See~the~documentation~of~\token_to_str:N \spPoly~for~the~accepted~point~formats.
4388 }

```

(End of definition for `spPoly-invalid-points`.)

### 11.43.3 Definition of keys for \spPoly

We define the *(keys)* read-cmd, prefix, print-sym, silent-cmd and concept.

```

read-cmd  We define the (keys) read-cmd, prefix, print-sym, silent-cmd and concept.
prefix    4389 \keys_define:nn { spintent / spPoly }
print-sym 4390 {
silent-cmd 4391   read-cmd           .choices:nn = { school , mathcat }
concept    4392   {
unknown    4393     \int_case:nn { \l_keys_choice_int }
           4394     {
           4395       {1} { \str_set:Nn \l__spintent_spPoly_mode_str { school } }
           4396       {2} { \str_set:Nn \l__spintent_spPoly_mode_str { mathcat } }
           4397     }
           4398   },
           4399   read-cmd           .initial:n = school,
           4400   read-cmd           .value_required:n = true,
           4401   read-cmd / unknown .code:n =
           4402     \msg_error:nneee { spintent } { unknown-choice }
           4403     { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
           4404   prefix           .code:n =
           4405     { \__spintent_sanitize_affix:Nn \l__spintent_spPoly_prefix_tl {#1} },
           4406   prefix           .initial:n = ,
           4407   prefix           .value_required:n = true,
           4408   print-sym         .bool_set:N = \l__spintent_spPoly_print_sym_bool,
           4409   print-sym         .initial:n = true,
           4410   print-sym         .value_required:n = true,
           4411   silent-cmd        .bool_set:N = \l__spintent_spPoly_silent_cmd_bool,
           4412   silent-cmd        .initial:n = false,
           4413   silent-cmd        .value_required:n = true,
           4414   concept          .bool_set:N = \l__spintent_spPoly_concept_bool,
           4415   concept          .initial:n = true,
           4416   concept          .value_required:n = true,
           4417   unknown         .code:n =
           4418     {
           4419       \str_set:Nn \l__spintent_command_name_str { spPoly }
           4420       \__spintent_unknown_keys_cmd:n {#1}
           4421     },
           4422   }

```

(End of definition for read-cmd and others.)

### 11.43.4 Internal functions for \spPoly

Scratch variables to split \l\_\_spintent\_geo\_luaset\_points\_clist and build the \$pt1, \$pt2, ... reference list, and to hold the final concept intent once built.

```

\l__spintent_spPoly_pt_seq 4423 \seq_new:N \l__spintent_spPoly_pt_seq
\l__spintent_spPoly_refs_tl 4424 \tl_new:N \l__spintent_spPoly_refs_tl
\l__spintent_spPoly_intent_arg_tl 4425 \tl_new:N \l__spintent_spPoly_intent_arg_tl
\l__spintent_spPoly_pt_idx_int 4426 \int_new:N \l__spintent_spPoly_pt_idx_int

```

(End of definition for \l\_\_spintent\_spPoly\_pt\_seq and others.)

\\_\_spintent\_spPoly\_sym: Unchanged: reads \l\_\_spintent\_geo\_luaset\_poly\_sym\_str and typesets the corresponding symbol.

```

4427 \cs_new_protected:Nn \__spintent_spPoly_sym:
4428 {
4429   \str_case:Vn \l__spintent_geo_luaset_poly_sym_str
4430   {
4431     { triangle } { \triangle }
4432     { square } { \mdlgwhtsquare }
4433   }
4434 }

```

(End of definition for \\_\_spintent\_spPoly\_sym:.)

\\_\_spintent\_spPoly\_build\_refs: Splits \l\_\_spintent\_geo\_luaset\_points\_clist into \l\_\_spintent\_spPoly\_pt\_seq and walks it once to build \l\_\_spintent\_spPoly\_refs\_tl, the comma-separated \$pt1, \$pt2, ... reference list, naming each vertex by position.

```

4435 \cs_new_protected:Nn \__spintent_spPoly_build_refs:
4436 {
4437   \seq_set_from_clist:NN \l__spintent_spPoly_pt_seq \l__spintent_geo_luaset_points_clist
4438   \tl_clear:N \l__spintent_spPoly_refs_tl
4439   \int_zero:N \l__spintent_spPoly_pt_idx_int
4440   \seq_map_inline:Nn \l__spintent_spPoly_pt_seq
4441   {

```

```

4442         \int_incr:N \l__spintent_spPoly_pt_idx_int
4443         \tl_if_empty:NF \l__spintent_spPoly_refs_tl
4444         { \tl_put_right:Nn \l__spintent_spPoly_refs_tl { , } }
4445         \tl_put_right:Ne \l__spintent_spPoly_refs_tl { $ pt \int_use:N \l__spintent_spPoly_pt_idx_int }
4446     }
4447 }

```

(End of definition for `\__spintent_spPoly_build_refs:`)

`\__spintent_spPoly_build_concept:`

Builds `\l__spintent_spPoly_intent_arg_tl`, independently of `print-sym:_(refs)` when `silent-cmd` is `true` or `concept` is `false`; otherwise `\l__spintent_geo_luaset_concept_str:prefix(refs)`, prefixed with `prefix` when given. Must be called after `\__spintent_spPoly_build_refs:`.

```

4448 \cs_new_protected:Nn \__spintent_spPoly_build_concept:
4449 {
4450     \bool_if:NTF \l__spintent_spPoly_silent_cmd_bool
4451     {
4452         \tl_set:Ne \l__spintent_spPoly_intent_arg_tl { _ ( \l__spintent_spPoly_refs_tl ) }
4453     }
4454     {
4455         \bool_if:NTF \l__spintent_spPoly_concept_bool
4456         {
4457             \tl_set:Ne \l__spintent_spPoly_intent_arg_tl
4458             {
4459                 \tl_if_empty:NF \l__spintent_spPoly_prefix_tl
4460                 { \tl_use:N \l__spintent_spPoly_prefix_tl - }
4461                 \l__spintent_geo_luaset_concept_str :prefix( \l__spintent_spPoly_refs_tl )
4462             }
4463         }
4464         {
4465             \tl_set:Ne \l__spintent_spPoly_intent_arg_tl { _ ( \l__spintent_spPoly_refs_tl ) }
4466         }
4467     }
4468 }

```

(End of definition for `\__spintent_spPoly_build_concept:`)

`\__spintent_spPoly_points_loop:`

Walks `\l__spintent_spPoly_pt_seq` a second time, calling `\__spintent_mathmlarg_point:nnn` once per vertex with its `ptn` reference name, looked up under `spPoly:` it alone decides `arg/literal intent` per vertex — replacing the single combined `\MathMLarg{pts}{...}` that lost `arg` on any vertex with a subscript or prime (the same `annotate_attribute/sub-sup` limitation already found and worked around for `\spPoint`).

```

4469 \cs_new_protected:Nn \__spintent_spPoly_points_loop:
4470 {
4471     \int_zero:N \l__spintent_spPoly_pt_idx_int
4472     \seq_map_inline:Nn \l__spintent_spPoly_pt_seq
4473     {
4474         \int_incr:N \l__spintent_spPoly_pt_idx_int
4475         \__spintent_mathmlarg_point:nnn
4476         { pt \int_use:N \l__spintent_spPoly_pt_idx_int }
4477         { spPoly }
4478         { ##1 }
4479     }
4480 }

```

(End of definition for `\__spintent_spPoly_points_loop:`)

`\__spintent_spPoly_process:n`

Calls `\__spintent_luafun_geo_poly_parse_and_set:n`, reports `spPoly-invalid-points` on failure, then delegates to `\__spintent_spPoly_mathml:`. No `intent` is built here: it needs the reference list, which `\__spintent_spPoly_mathml:` builds after splitting the vertices.

```

4481 \cs_new_protected:Nn \__spintent_spPoly_process:n
4482 {
4483     \__spintent_luafun_geo_poly_parse_and_set:n { \exp_not:n {#1} }
4484     \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
4485     {
4486         \msg_error:nne { spintent } { spPoly-invalid-points }
4487         { \exp_not:n {#1} }
4488     }
4489     \__spintent_spPoly_mathml:
4490 }

```

(End of definition for `\__spintent_spPoly_process:n`)

`\__spintent_spPoly_mathml:` Builds the reference list and the concept `\__spintent_spPoly_build_refs:`, `\__spintent_spPoly_build_concept:` then wraps everything in a single native `\MathMLintent`, preceded by `\__spintent_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`, [luamml issue #26](#)).  
`print-sym` and `concept/silent-cmd` are independent: the former only decides whether the glyph (`\__spintent_spPoly_sym:`, silenced with `\__spintent_luafun_mrow_intent_arg:nn`) is attempted — it is skipped anyway when `\l__spintent_geo_luaset_poly_sym_str` is empty (no symbol defined for that vertex count, e.g. pentagon and above) — and never affects whether the concept word itself is spoken.

```

4491 \cs_new_protected:Nn \__spintent_spPoly_mathml:
4492 {
4493   \__spintent_invisible_sep: % (luamml issues #26)
4494   \__spintent_spPoly_build_refs:
4495   \__spintent_spPoly_build_concept:
4496   \MathMLintent { \l__spintent_spPoly_intent_arg_tl }
4497   {
4498     \__spintent_inner_sep:
4499     \bool_if:NT \l__spintent_spPoly_print_sym_bool
4500     {
4501       \str_if_empty:NF \l__spintent_geo_luaset_poly_sym_str
4502       {
4503         \__spintent_luafun_mrow_intent_arg:nn
4504         { } { intent = ":silent" }
4505         { \__spintent_spPoly_sym: }
4506       }
4507     }
4508     \__spintent_spPoly_points_loop:
4509   }
4510 }

```

(End of definition for `\__spintent_spPoly_mathml:`.)

`\__spintent_spPoly_exec:nn` The function `\__spintent_spPoly_exec:nn` receives the optional keys in `#1` and the argument in `#2`, delegating the rendering to `\__spintent_spPoly_process:n`.

```

4511 \cs_new_protected:Nn \__spintent_spPoly_exec:nn
4512 {
4513   \keys_set:nn { spintent / spPoly } { #1 }
4514   \__spintent_spPoly_process:n { #2 }
4515 }

```

(End of definition for `\__spintent_spPoly_exec:nn`.)

`\spPoly` The public command `\spPoly` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spPoly_exec:nn`.

```

4516 \NewDocumentCommand \spPoly { O{} m }
4517 {
4518   \mode_if_math:TF
4519   {
4520     \group_begin:
4521     \attribute \l__spintent_scope_attr_int = 1
4522     \__spintent_spPoly_exec:nn { #1 } { #2 }
4523     \group_end:
4524   }
4525   { \msg_error:nnn { spintent } { need-math-mode } { \spPoly } }
4526 }

```

(End of definition for `\spPoly`. This function is documented on [page 25](#).)

## 11.44 The commands `\spAfig` and `\spPfig`

The user commands `\spAfig` and `\spPfig` typeset the area and the perimeter of a polygonal figure, using the delimited-optional argument syntax `d()` already established by `\spcmd/\spmcm`. The key `print-sym` is never sent to Lua: the symbol type is returned unconditionally in `\l__spintent_geo_luaset_poly_sym_str`, and `expl3` decides whether to wrap it in `\MathMLintent{<:silent>}` — the same mechanism already used by `\spPoly`.

### 11.44.1 Definition of variables for `\spAfig` and `\spPfig`

`\l__spintent_geo_final_print_tl` Scratch variables used by `\__spintent_spfig_process:nn` to assemble the visual and the `intent`. `\l__spintent_geo_final_print_tl` holds the final visual placed under `\mathcal`'s subscript. The final literal `intent` and the concept forwarded when `read-cmd` is `mathcat`.

```

4527 \tl_new:N \l__spintent_geo_final_print_tl

```

(End of definition for `\l__spintent_geo_final_print_tl`.)

```

\__spintent_spAfig_mode_str
\__spintent_spPfig_mode_str
  \__spintent_spAfig_print_sym_str
  \__spintent_spPfig_print_sym_str
    \__spintent_spAfig_prefix_tl
    \__spintent_spPfig_prefix_tl
  \__spintent_spAfig_read_arg_tl
  \__spintent_spPfig_read_arg_tl

```

These variables store the value set by each corresponding key: `read-cmd`, `print-sym`, `prefix` and `read-arg`, respectively. The value set by `read-arg` overrides the figure name inferred from the vertex count, independently of `print-sym`.

```

4528 \clist_map_inline:nn { Afig , Pfig }
4529 {
4530   \str_new:c { \__spintent_sp #1 _mode_str      }
4531   \str_new:c { \__spintent_sp #1 _print_sym_str }
4532   \tl_new:c { \__spintent_sp #1 _prefix_tl      }
4533   \tl_new:c { \__spintent_sp #1 _read_arg_tl    }
4534 }

```

(End of definition for `\__spintent_spAfig_mode_str` and others.)

#### 11.44.2 Definition of error messages for `\spAfig` and `\spPfig`

All eight messages are generated by the same map, using the same `##1/##2` doubling convention as `\spside/\spmside`. `sp #1-invalid-points` is currently unused by `\__spintent_spfig_process:nn`, which raises only `sp #1-invalid-arg` for any parse failure, regardless of whether a comma was present. `sp #1-sym-letra-conflict` and `sp #1-sym-cardinality-conflict` are raised when `print-sym` is forced to a value incompatible with the actual argument.

```

4535 \clist_map_inline:nn { Afig , Pfig }
4536 {
4537   \msg_new:nnnn { spintent } { sp #1 -invalid-points }
4538   {
4539     Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
4540   }
4541   {
4542     See~the~documentation~of~##1~for~the~accepted~point~formats.
4543   }
4544   \msg_new:nnnn { spintent } { sp #1 -invalid-arg }
4545   {
4546     Invalid~argument~format~in~##1:~'##2'~\msg_line_context:.
4547   }
4548   {
4549     See~the~documentation~of~##1~for~the~accepted~argument~forms.
4550   }
4551   \msg_new:nnnn { spintent } { sp #1 -sym-letra-conflict }
4552   {
4553     print-sym~cannot~be~combined~with~a~lettered~argument~in~'##1'~\msg_line_context:.
4554   }
4555   {
4556     See~the~documentation~of~##1~for~details~on~print-sym.
4557   }
4558   \msg_new:nnnn { spintent } { sp #1 -sym-cardinality-conflict }
4559   {
4560     print-sym~does~not~match~the~vertex~count~in~'##1'~\msg_line_context:.
4561   }
4562   {
4563     See~the~documentation~of~##1~for~details~on~print-sym.
4564   }
4565   \msg_new:nnnn { spintent } { sp #1 -read-arg-key }
4566   {
4567     read-arg~has~no~effect~with~a~single~argument~in~'##1'~\msg_line_context:.
4568   }
4569   {
4570     See~the~documentation~of~##1~for~details~on~read-arg.
4571   }
4572 }

```

(End of definition for `spAfig-invalid-points` and others.)

#### 11.44.3 Definition of keys for `\spAfig` and `\spPfig`

```

read-cmd
prefix
print-sym
read-arg
read-sub
silent-cmd
concept
concept*
unknown

```

The key `print-sym` chooses the visual symbol: `auto` (default) infers it from the vertex count (three points give `\triangle`, four give `\mdlgwhtsquare`, five or more give none); `triangle/square` force one, requiring the vertex count to match; `none` omits it, even with vertices. With a bare number/letter argument, `auto` and `none` behave the same (no symbol); `triangle/square` nest the argument under the forced symbol instead, and cannot be combined with a table letter (F/P/B/L). The key `read-arg` overrides the spoken figure name for real vertices, independently of `print-sym`. The key `read-sub` controls whether the bare number/letter argument is read with the word «sub» or as a plain number.

```

4573 \clist_map_inline:nn { Afig , Pfig }
4574 {

```

```

4575 \keys_define:nn { spintent / sp #1 }
4576 {
4577   read-cmd          .choices:nn = { school , mathcat }
4578   {
4579     \int_case:nn { \l_keys_choice_int }
4580     {
4581       {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school } }
4582       {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
4583     }
4584   },
4585   read-cmd          .initial:n = school,
4586   read-cmd          .value_required:n = true,
4587   read-cmd / unknown .code:n =
4588     \msg_error:nneee { spintent } { unknown-choice }
4589     { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
4590   prefix            .code:n =
4591     { \__spintent_sanitiz_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
4592   prefix            .initial:n = ,
4593   prefix            .value_required:n = true,
4594   print-sym         .choices:nn = { auto , triangle , square , none }
4595   {
4596     \int_case:nn { \l_keys_choice_int }
4597     {
4598       {1} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { auto } }
4599       {2} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { triangle } }
4600       {3} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { square } }
4601       {4} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { none } }
4602     }
4603   },
4604   print-sym         .initial:n = auto,
4605   print-sym         .value_required:n = true,
4606   print-sym / unknown .code:n =
4607     \msg_error:nneee { spintent } { unknown-choice }
4608     { print-sym } { auto,~triangle,~square,~none } { \exp_not:n {##1} },
4609   read-arg          .code:n =
4610     { \__spintent_sanitiz_affix:cn { l__spintent_sp #1 _read_arg_tl } {##1} },
4611   read-arg          .initial:n = ,
4612   read-arg          .value_required:n = true,
4613   read-sub          .bool_set:c = { l__spintent_sp #1 _read_sub_bool },
4614   read-sub          .initial:n = false,
4615   read-sub          .value_required:n = true,
4616   silent-cmd        .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
4617   silent-cmd        .initial:n = false,
4618   silent-cmd        .value_required:n = true,
4619   concept           .bool_set:c = { l__spintent_sp #1 _concept_bool },
4620   concept           .initial:n = true,
4621   concept           .value_required:n = true,
4622   concept*          .meta:n =
4623   {
4624     concept = false,
4625     read-arg = { \str_if_eq:eeTF {#1} { Afig } { _área } { _perímetro } },
4626   },
4627   concept*          .value_forbidden:n = true,
4628   unknown           .code:n =
4629   {
4630     \str_set:Nn \l__spintent_command_name_str { sp #1 }
4631     \__spintent_unknown_keys_cmd:n {##1}
4632   },
4633 }
4634 }

```

(End of definition for read-cmd and others.)

#### 11.44.4 Internal functions for \spAfig and \spPfig

`\__spintent_spfig_sym:` The function `\__spintent_spfig_sym:` reads `\l__spintent_geo_print_sym_str` and typesets the corresponding symbol: `\triangle` for the value `triangle`, `\mdlgwhtsquare` for the value `square`.

```

4635 \cs_new_protected:Nn \__spintent_spfig_sym:
4636 {
4637   \str_case:Vn \l__spintent_geo_print_sym_str
4638   {
4639     { triangle } { \triangle }

```

```

4640         { square } { \mdlgwhtsquare }
4641     }
4642 }

```

(End of definition for `\__spintent_spfig_sym:`)

`\__spintent_spfig_op_tl` holds área or perímetro. `\__spintent_spfig_intent_tl` holds the concept word for the vertices case (`read-arg` or the auto-detected figure name). The rest are scratch variables to split `\__spintent_geo_luaset_points_clist` and build the `$pt1, $pt2, ...` reference list.

```

4643 \tl_new:N \__spintent_spfig_op_tl
4644 \tl_new:N \__spintent_spfig_intent_tl
4645 \tl_new:N \__spintent_spfig_desc_tl
4646 \tl_new:N \__spintent_spfig_symbol_tl
4647 \seq_new:N \__spintent_spfig_pt_seq
4648 \tl_new:N \__spintent_spfig_refs_tl
4649 \tl_new:N \__spintent_spfig_intent_arg_tl
4650 \int_new:N \__spintent_spfig_pt_idx_int
4651 \str_new:N \__spintent_geo_print_sym_str

```

(End of definition for `\__spintent_spfig_op_tl` and others.)

Receives the short command name (Afig or Pfig) in ‘#1’ and the  $\langle argument \rangle$  in ‘#2’. Calls `\__spintent_luafun_geo` or `\__spintent_luafun_geo_pfig_parse_and_set:n`, reports `sp#1-invalid-arg` on failure, and `sp#1-read-arg` when `read-arg` is given for a bare number/letter argument (no vertices), where it has no effect. Sets `\__spintent_spfig_op_tl` to área/perímetro, then delegates to `\__spintent_spfig_mathml:n`.

```

4652 \cs_new_protected:Nn \__spintent_spfig_process:nn
4653 {
4654     \str_if_eq:eeTF {#1} { Afig }
4655     { \__spintent_luafun_geo_afig_parse_and_set:n { \exp_not:n {#2} } }
4656     { \__spintent_luafun_geo_pfig_parse_and_set:n { \exp_not:n {#2} } }
4657     \str_if_eq:VnT \__spintent_geo_luaset_error_str { true }
4658     {
4659         \msg_error:nnee { spintent } { sp #1 -invalid-arg }
4660         { \c_backslash_str sp #1 } { \exp_not:n {#2} }
4661     }
4662     \clist_if_empty:NT \__spintent_geo_luaset_points_clist
4663     {
4664         \tl_if_empty:cF { \__spintent_sp #1 _read_arg_tl }
4665         {
4666             \msg_error:nne { spintent } { sp #1 -read-arg-key }
4667             { \c_backslash_str sp #1 }
4668         }
4669     }
4670     \str_if_eq:eeTF {#1} { Afig }
4671     { \tl_set:Nn \__spintent_spfig_op_tl { área } }
4672     { \tl_set:Nn \__spintent_spfig_op_tl { perímetro } }
4673     \str_set:Ne \__spintent_geo_print_sym_str { \str_use:c { \__spintent_sp #1 _print_sym_str } }
4674     \__spintent_spfig_mathml:n {#1}
4675 }

```

(End of definition for `\__spintent_spfig_process:nn`.)

Handles the argument-less, table-letter and bare number/letter cases (no vertices). Builds `\__spintent_geo_final_print` (the visual placed under `\mathcal`’s subscript) and `\__spintent_spfig_desc_tl` (the descriptive part, without área/perímetro). Receives the short command name in ‘#1’.

With an empty argument, both are cleared for `print-sym auto/ none`; a forced `print-sym` instead places the symbol alone, with its own name as the description.

```

4676 \cs_new_protected:Nn \__spintent_spfig_build_simple:n
4677 {
4678     \tl_if_empty:nTF { \__spintent_geo_luaset_print_tl }
4679     {
4680         \str_case:VnF \__spintent_geo_print_sym_str
4681         {
4682             { auto }
4683             {
4684                 \tl_clear:N \__spintent_geo_final_print_tl
4685                 \tl_clear:N \__spintent_spfig_desc_tl
4686             }
4687             { none }
4688             {
4689                 \tl_clear:N \__spintent_geo_final_print_tl

```

```

4690         \tl_clear:N \l__spintent_spfig_desc_tl
4691     }
4692 }
4693 {
4694     \tl_set:Nn \l__spintent_geo_final_print_tl { \__spintent_spfig_sym: }
4695     \tl_set:Ne \l__spintent_spfig_desc_tl
4696     {
4697         \str_if_eq:VnTF \l__spintent_geo_print_sym_str { triangle }
4698         { triángulo } { cuadrado }
4699     }
4700 }
4701 }

```

With a bare number/letter (empty `\l__spintent_geo_luaset_letra_word_str`), the value itself becomes the visual for `print-sym auto/none`, or its own subscript under a forced symbol otherwise; the description is the spoken value, prefixed with «sub-» when `read-sub` is `true` (further prefixed with the symbol's own name for a forced `print-sym`).

```

4702     {
4703         \str_if_empty:NTF \l__spintent_geo_luaset_letra_word_str
4704         {
4705             \str_case:VnF \l__spintent_geo_print_sym_str
4706             {
4707                 { auto } { \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_p
4708                 { none } { \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_p
4709             }
4710             {
4711                 \tl_set:Nn \l__spintent_geo_final_print_tl
4712                 {
4713                     \__spintent_spfig_sym:
4714                     \c_math_subscript_token { \l__spintent_geo_luaset_print_tl }
4715                 }
4716             }
4717             \str_case:VnF \l__spintent_geo_print_sym_str
4718             {
4719                 { auto }
4720                 {
4721                     \tl_set:Ne \l__spintent_spfig_desc_tl
4722                     {
4723                         \bool_if:cT { \l__spintent_sp #1 _read_sub_bool } { sub - }
4724                         \l__spintent_geo_luaset_sub_spoken_str
4725                     }
4726                 }
4727                 { none }
4728                 {
4729                     \tl_set:Ne \l__spintent_spfig_desc_tl
4730                     {
4731                         \bool_if:cT { \l__spintent_sp #1 _read_sub_bool } { sub - }
4732                         \l__spintent_geo_luaset_sub_spoken_str
4733                     }
4734                 }
4735             }
4736         }
4737         \tl_set:Ne \l__spintent_spfig_desc_tl
4738         {
4739             del-
4740             \str_if_eq:VnTF \l__spintent_geo_print_sym_str { triangle }
4741             { triángulo } { cuadrado }
4742             -
4743             \bool_if:cT { \l__spintent_sp #1 _read_sub_bool } { sub - }
4744             \l__spintent_geo_luaset_sub_spoken_str
4745         }
4746     }
4747 }

```

With a table letter (F/P/B/L/T, non-empty `\l__spintent_geo_luaset_letra_word_str`), a forced `print-sym` raises `sp#1-sym-letra-conflict` — a table letter is never nested under a symbol. The visual is the value as-is; the description is its own word, connected as `\l__spintent_geo_luaset_letra_conector_str` requires, with the same spoken subscript as above when present.

```

4748     {
4749         \str_case:VnF \l__spintent_geo_print_sym_str
4750         {
4751             { auto } { }

```

```

4752         { none } { }
4753     }
4754     {
4755         \msg_error:nne { spintent } { sp #1 -sym-letra-conflict }
4756         { \c_backslash_str sp #1 }
4757     }
4758     \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_print_tl }
4759     \tl_set:Ne \l__spintent_spfig_desc_tl
4760     {
4761         \str_if_empty:NF \l__spintent_geo_luaset_letra_conector_str
4762         { \l__spintent_geo_luaset_letra_conector_str - }
4763         \l__spintent_geo_luaset_letra_word_str
4764         \str_if_empty:NF \l__spintent_geo_luaset_sub_spoken_str
4765         {
4766             -
4767             \bool_if:cT { \l__spintent_sp #1 _read_sub_bool } { sub - }
4768             \l__spintent_geo_luaset_sub_spoken_str
4769         }
4770     }
4771 }
4772 }

```

The variable `\l__spintent_spfig_intent_tl` is always the `_word-` shape: `\l__spintent_spfig_op_tl` alone when `\l__spintent_spfig_desc_tl` is empty, both joined by a dash otherwise. `silent-cmd` overrides `\l__spintent_spfig_intent_arg_tl` with `_`; with `silent-cmd=false` and `concept false`, it becomes `_l__spintent_spfig_desc_tl-` (the descriptive part alone, without *área/perímetro*); with `concept true` it is `\l__spintent_spfig_intent_tl`.

```

4773     \tl_set:Ne \l__spintent_spfig_intent_tl
4774     {
4775         \tl_if_empty:NTF \l__spintent_spfig_desc_tl
4776         { _ \l__spintent_spfig_op_tl - }
4777         { _ \l__spintent_spfig_op_tl - \l__spintent_spfig_desc_tl - }
4778     }
4779     \bool_if:cTF { \l__spintent_sp #1 _silent_cmd_bool }
4780     { \tl_set:Nn \l__spintent_spfig_intent_arg_tl { _- } }
4781     {
4782         \bool_if:cTF { \l__spintent_sp #1 _concept_bool }
4783         { \tl_set:Ne \l__spintent_spfig_intent_arg_tl { \l__spintent_spfig_intent_tl } }
4784         { \tl_set:Ne \l__spintent_spfig_intent_arg_tl { _ \l__spintent_spfig_desc_tl - } }
4785     }
4786 }

```

(End of definition for `\__spintent_spfig_build_simple:n`)

`\__spintent_spfig_build_refs:`

Splits `\l__spintent_geo_luaset_points_clist` into `\l__spintent_spfig_pt_seq` and walks it once to build `\l__spintent_spfig_refs_tl`, the comma-separated `$pt1,$pt2,...` reference list, naming each vertex by position.

```

4787 \cs_new_protected:Nn \__spintent_spfig_build_refs:
4788 {
4789     \seq_set_from_clist:NN \l__spintent_spfig_pt_seq \l__spintent_geo_luaset_points_clist
4790     \tl_clear:N \l__spintent_spfig_refs_tl
4791     \int_zero:N \l__spintent_spfig_pt_idx_int
4792     \seq_map_inline:Nn \l__spintent_spfig_pt_seq
4793     {
4794         \int_incr:N \l__spintent_spfig_pt_idx_int
4795         \tl_if_empty:NF \l__spintent_spfig_refs_tl
4796         { \tl_put_right:Nn \l__spintent_spfig_refs_tl { , } }
4797         \tl_put_right:Ne \l__spintent_spfig_refs_tl { $ pt \int_use:N \l__spintent_spfig_pt_idx_int
4798     }
4799 }

```

(End of definition for `\__spintent_spfig_build_refs:`)

`\__spintent_spfig_build_concept_vertices:n`

Handles the vertices case. Receives the short command name in ‘#1’. Must be called after `\__spintent_spfig_build_refs:n`. `read-arg`, when given, overrides the auto-detected figure name in `\l__spintent_spfig_intent_tl`: with `concept true` it is still prefixed with *área-del-/perímetro-del-*, with `concept false` it is used verbatim. With `read-arg` empty, `\l__spintent_spfig_intent_tl` is simply `\l__spintent_geo_luaset_concept_str`.

```

4800 \cs_new_protected:Nn \__spintent_spfig_build_concept_vertices:n
4801 {
4802     \tl_if_empty:cTF { \l__spintent_sp #1 _read_arg_tl }
4803     { \tl_set:Ne \l__spintent_spfig_intent_tl { \l__spintent_geo_luaset_concept_str } }

```

```

4804     {
4805         \bool_if:cTF { l__spintent_sp #1 _concept_bool }
4806     {
4807         \tl_set:Ne \l__spintent_spfig_intent_tl
4808         {
4809             \l__spintent_spfig_op_tl -del-
4810             \tl_use:c { l__spintent_sp #1 _read_arg_tl }
4811         }
4812     }
4813     {
4814         \tl_set:Ne \l__spintent_spfig_intent_tl { \tl_use:c { l__spintent_sp #1 _read_arg_tl }
4815     }
4816 }

```

Determines `\l__spintent_spfig_symbol_tl` (empty when none applies): for `print-sym=auto`, the symbol matching `\l__spintent_geo_luaset_poly_sym_str(\triangle/\mdlgwhtsquare)`, or none when that is empty; `print-sym=none` always clears it; a forced `print-sym` requires it to match the actual vertex count, reporting `sp#1-sym-cardinality-conflict` otherwise.

```

4817 \str_case:VnF \l__spintent_geo_print_sym_str
4818 {
4819     { auto }
4820     {
4821         \str_case:Vn \l__spintent_geo_luaset_poly_sym_str
4822         {
4823             { triangle } { \tl_set:Nn \l__spintent_spfig_symbol_tl { \triangle } }
4824             { square } { \tl_set:Nn \l__spintent_spfig_symbol_tl { \mdlgwhtsquare } }
4825         }
4826         \str_if_empty:NT \l__spintent_geo_luaset_poly_sym_str
4827         { \tl_clear:N \l__spintent_spfig_symbol_tl }
4828     }
4829     { none }
4830     { \tl_clear:N \l__spintent_spfig_symbol_tl }
4831 }
4832 {
4833     \str_if_eq:VVTF \l__spintent_geo_print_sym_str \l__spintent_geo_luaset_poly_sym_str
4834     { \tl_set:Nn \l__spintent_spfig_symbol_tl { \l__spintent_spfig_sym: } }
4835     {
4836         \msg_error:nne { spintent } { sp #1 -sym-cardinality-conflict }
4837         { \c_backslash_str sp #1 }
4838     }
4839 }

```

`silent-cmd` sets `\l__spintent_spfig_intent_arg_tl` to `_(refs)`. Otherwise, with `read-arg` empty, `concept` chooses between `\l__spintent_spfig_intent_tl:prefix(refs)` (prefixed with `prefix` when given) and `_(refs)` alone; with `read-arg` given, it is always `\l__spintent_spfig_intent_tl:prefix(refs)` — `read-arg` is never silenced by `concept` alone, only by `silent-cmd`.

```

4840 \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
4841 {
4842     \tl_set:Ne \l__spintent_spfig_intent_arg_tl { _ ( \l__spintent_spfig_refs_tl ) }
4843 }
4844 {
4845     \tl_if_empty:cTF { l__spintent_sp #1 _read_arg_tl }
4846     {
4847         \bool_if:cTF { l__spintent_sp #1 _concept_bool }
4848         {
4849             \tl_set:Ne \l__spintent_spfig_intent_arg_tl
4850             {
4851                 \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4852                 { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4853                 \l__spintent_spfig_intent_tl :prefix( \l__spintent_spfig_refs_tl )
4854             }
4855         }
4856         {
4857             \tl_set:Ne \l__spintent_spfig_intent_arg_tl { _ ( \l__spintent_spfig_refs_tl ) }
4858         }
4859     }
4860     {
4861         \tl_set:Ne \l__spintent_spfig_intent_arg_tl
4862         {
4863             \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4864             { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4865             \l__spintent_spfig_intent_tl :prefix( \l__spintent_spfig_refs_tl )

```

```

4866         }
4867     }
4868 }
4869 }

```

(End of definition for `\__spintent_sfig_build_concept_vertices:n`.)

`\__spintent_sfig_points_loop:n`

Walks `\__spintent_sfig_pt_seq` a second time, calling `\__spintent_mathmlarg_point:nnn` once per vertex with its `ptn` reference name, looked up under `sp#1`. Receives the short command name in ‘`#1`’.

```

4870 \cs_new_protected:Nn \__spintent_sfig_points_loop:n
4871 {
4872     \int_zero:N \__spintent_sfig_pt_idx_int
4873     \seq_map_inline:Nn \__spintent_sfig_pt_seq
4874     {
4875         \int_incr:N \__spintent_sfig_pt_idx_int
4876         \__spintent_mathmlarg_point:nnn
4877         { pt \int_use:N \__spintent_sfig_pt_idx_int }
4878         { sp #1 }
4879         { ##1 }
4880     }
4881 }

```

(End of definition for `\__spintent_sfig_points_loop:n`.)

`\__spintent_sfig_mathml:n`

Receives the short command name in ‘`#1`’. Branches on whether `\__spintent_geo_luasets_points_clist` is empty.

Without vertices, `\__spintent_sfig_build_simple:n` builds the content; a `\MathMLintent` wraps it only when `concept` is `true` — with `concept=false` the content is typeset directly, with no `intent` at all. With vertices, `\__spintent_sfig_build_refs:` then `\__spintent_sfig_build_concept_vertices:n` build the reference list and the concept `intent`, applied with a single native `\MathMLintent`. Inside, `\__spintent_mathcal:n{⟨A | P⟩}` is silenced with `\__spintent_luafun_mrow_intent_arg:nn` (it would otherwise be claimed by the queue meant for the symbol/vertices, the same ordering issue already found and fixed for `\spangle`’s own symbol); `\__spintent_sfig_symbol_tl` is silenced the same way when non-empty; the sequence is then walked by `\__spintent_sfig_points_loop:n`. All of this sits nested inside a subscript (`\c_math_subscript_token`), reached by the shared `mathml_filter` callback’s recursive lookup through a noad’s sub field, not just its nucleus.

`\__spintent_invisible_sep:` precedes both branches (a second sibling so the sole `<mrow>` does not collapse into `<math>`).

```

4882 \cs_new_protected:Nn \__spintent_sfig_mathml:n
4883 {
4884     \__spintent_invisible_sep: % (luamml issues #26)
4885     \clist_if_empty:NTF \__spintent_geo_luasets_points_clist
4886     {
4887         \__spintent_sfig_build_simple:n {#1}
4888         \MathMLintent { \__spintent_sfig_intent_arg_tl }
4889         {
4890             \__spintent_inner_sep:
4891             \str_if_eq:eeTF {#1} { Afig } { \__spintent_mathcal:n { A } } { \__spintent_mathcal:n { A } }
4892             \tl_if_empty:NF \__spintent_geo_final_print_tl
4893             {
4894                 \c_math_subscript_token { \__spintent_geo_final_print_tl }
4895             }
4896         }
4897     }
4898     {
4899         \__spintent_sfig_build_refs:
4900         \__spintent_sfig_build_concept_vertices:n {#1}
4901         \MathMLintent { \__spintent_sfig_intent_arg_tl }
4902         {
4903             \__spintent_inner_sep:
4904             \__spintent_luafun_mrow_intent_arg:nn
4905             { } { intent = "silent" }
4906             {
4907                 \str_if_eq:eeTF {#1} { Afig } { \__spintent_mathcal:n { A } } { \__spintent_mathcal:n { A } }
4908             }
4909             \c_math_subscript_token
4910             {
4911                 \tl_if_empty:NF \__spintent_sfig_symbol_tl
4912                 {
4913                     \__spintent_luafun_mrow_intent_arg:nn

```

```

4914         { } { intent = ":silent" }
4915         { \tl_use:N \l__spintent_spfig_symbol_tl }
4916     }
4917     \__spintent_spfig_points_loop:n {#1}
4918 }
4919 }
4920 }
4921 }

```

(End of definition for `\__spintent_spfig_mathml:n`.)

`\__spintent_spfig_exec:nnn`

The function `\__spintent_spfig_exec:nnn` receives the short name of the command in `#1`, the optional keys in `#2`, and the argument in `#3`, delegating the rendering to `\__spintent_spfig_process:nn`.

```

4922 \cs_new_protected:Nn \__spintent_spfig_exec:nnn
4923 {
4924     \keys_set:nn { spintent / sp #1 } { #2 }
4925     \__spintent_spfig_process:nn { #1 } { #3 }
4926 }

```

(End of definition for `\__spintent_spfig_exec:nnn`.)

`\spAfig`

`\spPfig`

Both public commands `\spAfig` and `\spPfig` enforce math mode, open a local group, and delegate execution to `\__spintent_spfig_exec:nnn`.

```

4927 \NewDocumentCommand \spAfig { 0{} d() }
4928 {
4929     \mode_if_math:TF
4930     {
4931         \group_begin:
4932         \attribute \l__spintent_scope_attr_int = 1
4933         \tl_if_novalue:nTF {#2}
4934         { \__spintent_spfig_exec:nnn { Afig } { #1 } { } }
4935         { \__spintent_spfig_exec:nnn { Afig } { #1 } { #2 } }
4936         \group_end:
4937     }
4938     { \msg_error:nnn { spintent } { need-math-mode } { \spAfig } }
4939 }
4940 \NewDocumentCommand \spPfig { 0{} d() }
4941 {
4942     \mode_if_math:TF
4943     {
4944         \group_begin:
4945         \attribute \l__spintent_scope_attr_int = 1
4946         \tl_if_novalue:nTF {#2}
4947         { \__spintent_spfig_exec:nnn { Pfig } { #1 } { } }
4948         { \__spintent_spfig_exec:nnn { Pfig } { #1 } { #2 } }
4949         \group_end:
4950     }
4951     { \msg_error:nnn { spintent } { need-math-mode } { \spPfig } }
4952 }

```

(End of definition for `\spAfig` and `\spPfig`. These functions are documented on page 25.)

## 11.45 The commands `\sprad` and `\spdiam`

The user-level commands `\sprad` and `\spdiam` provide a semantic interface for the radius and diameter of a circle. They delegate argument validation to the Lua module `spintent.lua`, thereby constructing a valid intent for `MathCAT`.

### 11.45.1 Definition of variables for `\sprad` and `\spdiam`

Each pair of variables is declared once for both `\sprad` and `\spdiam` via `\clist_map_inline:nn`.

```

4953 \clist_map_inline:nn { rad , diam }
4954 {
4955     \str_new:c { \l__spintent_sp #1 _mode_str }
4956     \str_new:c { \l__spintent_sp #1 _sub_index_str }
4957     \tl_new:c { \l__spintent_sp #1 _read_arg_tl }
4958 }

```

(End of definition for `\l__spintent_sprad_mode_str` and others.)

Scratch variables used by the function `\__spintent_sp_rad_diam_print:n`.

```

4959 \tl_new:N \l__spintent_sp_rad_diam_letra_tl
4960 \tl_new:N \l__spintent_sp_rad_diam_word_tl

```

(End of definition for `\l__spintent_sp_rad_diam_letra_tl` and `\l__spintent_sp_rad_diam_word_tl`.)

### 11.45.2 Definition of keys for \sprad and \spdiam

We define the shared *keys* `read-cmd`, `upper-case`, `sub-index`, `read-sub`, `read-arg`, `concept`, `silent-cmd` and `print-sym`.

```

4961 \clist_map_inline:nn { rad , diam }
4962 {
4963   \keys_define:nn { spintent / sp #1 }
4964   {
4965     read-cmd      .choices:nn = { school , mathcat }
4966     {
4967       \int_case:nn { \l_keys_choice_int }
4968       {
4969         {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school } }
4970         {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
4971       }
4972     },
4973     read-cmd      .initial:n = school,
4974     read-cmd      .value_required:n = true,
4975     read-cmd / unknown .code:n =
4976       \msg_error:nneee { spintent } { unknown-choice }
4977       { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
4978     upper-case    .bool_set:c = { l__spintent_sp #1 _upper_case_bool },
4979     upper-case    .initial:n = false,
4980     upper-case    .value_required:n = true,
4981     sub-index     .code:n =
4982       { \str_set:cn { l__spintent_sp #1 _sub_index_str } {##1} },
4983     sub-index     .initial:n = ,
4984     sub-index     .value_required:n = true,
4985     read-sub      .bool_set:c = { l__spintent_sp #1 _read_sub_bool },
4986     read-sub      .initial:n = false,
4987     read-sub      .value_required:n = true,
4988     read-arg      .code:n =
4989       { l__spintent_sanitize_affix:cn { l__spintent_sp #1 _read_arg_tl } {##1} },
4990     read-arg      .initial:n = ,
4991     read-arg      .value_required:n = true,
4992     concept       .bool_set:c = { l__spintent_sp #1 _concept_bool },
4993     concept       .initial:n = true,
4994     concept       .value_required:n = true,
4995     silent-cmd    .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
4996     silent-cmd    .initial:n = false,
4997     silent-cmd    .value_required:n = true,
4998     print-sym     .bool_set:c = { l__spintent_sp #1 _print_sym_bool },
4999     print-sym     .initial:n = false,
5000     print-sym     .value_required:n = true,
5001     unknown       .code:n =
5002       {
5003         \str_set:Nn \l__spintent_command_name_str { sp #1 }
5004         \__spintent_unknown_keys_cmd:n {##1}
5005       },
5006   }
5007 }

```

🔗 The keys `print-sym` are defined for both commands for simplicity, but has no effect on `\sprad`.

(End of definition for `read-cmd` and others.)

### 11.45.3 Internal functions for \sprad and \spdiam

`\__spintent_sp_rad_diam_print:n` The function `\__spintent_sp_rad_diam_print:n` sets `\l__spintent_sp_rad_diam_letra_tl` to the bare letter — `r/R` for `\sprad`, `d/D` for `\spdiam` — per `upper-case`, and, for `\spdiam` with `print-sym` true, silently typesets `\diameter` first. Receives the short command name in ‘`#1`’ (rad/diam).

```

5008 \cs_new_protected:Nn \__spintent_sp_rad_diam_print:n
5009 {
5010   \bool_if:cTF { l__spintent_sp #1 _upper_case_bool }
5011   {
5012     \str_if_eq:eeTF {#1} { rad }
5013     { \tl_set:Nn \l__spintent_sp_rad_diam_letra_tl { R } }
5014     { \tl_set:Nn \l__spintent_sp_rad_diam_letra_tl { D } }
5015   }
5016   {
5017     \str_if_eq:eeTF {#1} { rad }
5018     { \tl_set:Nn \l__spintent_sp_rad_diam_letra_tl { r } }
5019     { \tl_set:Nn \l__spintent_sp_rad_diam_letra_tl { d } }

```

```

5020     }
5021     \__spintent_invisible_sep:
5022     \str_if_eq:eeT {#1} { diam }
5023     {
5024         \bool_if:NT \__spintent_spdiam_print_sym_bool
5025         { \MathMLintent { :silent } { \diameter } }
5026     }

```

With `silent-cmd true`, the intent is `_` — the letter alone through the shared queue when there is no `sub-index`, or letter and subscript together in one native `\MathMLintent` (subscript silenced) when there is.

```

5027     \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
5028     {
5029         \str_if_empty:cTF { l__spintent_sp #1 _sub_index_str }
5030         {
5031             \__spintent_luafun_mrow_intent_arg:nn
5032             { } { intent = "_" }
5033             { \l__spintent_sp_rad_diam_letra_tl }
5034         }
5035         {
5036             \MathMLintent { _- }
5037             {
5038                 \__spintent_inner_sep:
5039                 \l__spintent_sp_rad_diam_letra_tl
5040                 \c_math_subscript_token
5041                 {
5042                     \MathMLintent { :silent }
5043                     { \str_use:c { l__spintent_sp #1 _sub_index_str } }
5044                 }
5045             }
5046         }
5047     }

```

Otherwise, with `concept true` (the default), it verbalizes `radio/diámetro` (`read-arg` replaces this word when given), with `sub-index`, if present, appended right after (`read-sub` choosing `sub-⟨value⟩` or `⟨value⟩` alone) — the letter itself is never read here. With no `sub-index`, the shared queue tags the letter alone; with one, letter and subscript are wrapped together in one native `\MathMLintent` for `read-cmd=school` (subscript silenced, already spoken), or the shared queue tags the letter alone with the subscript left bare for `read-cmd=mathcat`.

```

5048     {
5049         \bool_if:cTF { l__spintent_sp #1 _concept_bool }
5050         {
5051             \tl_if_empty:cTF { l__spintent_sp #1 _read_arg_tl }
5052             {
5053                 \str_if_eq:eeTF {#1} { rad }
5054                 { \tl_set:Nn \l__spintent_sp_rad_diam_word_tl { radio } }
5055                 { \tl_set:Nn \l__spintent_sp_rad_diam_word_tl { diámetro } }
5056             }
5057             { \tl_set:Nn \l__spintent_sp_rad_diam_word_tl { \tl_use:c { l__spintent_sp #1 _read_arg_tl } } }
5058             \str_if_empty:cTF { l__spintent_sp #1 _sub_index_str }
5059             {
5060                 \__spintent_luafun_mrow_intent_arg:nn
5061                 { } { intent = "_" \l__spintent_sp_rad_diam_word_tl - :prefix" }
5062                 { \l__spintent_sp_rad_diam_letra_tl }
5063             }
5064             {
5065                 \str_if_eq:eeTF
5066                 { \str_use:c { l__spintent_sp #1 _mode_str } } { school }
5067                 {
5068                     \MathMLintent
5069                     {
5070                         _ \l__spintent_sp_rad_diam_word_tl -
5071                         \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub- }
5072                         \str_use:c { l__spintent_sp #1 _sub_index_str }
5073                         :prefix
5074                     }
5075                     {
5076                         \__spintent_inner_sep:
5077                         \l__spintent_sp_rad_diam_letra_tl
5078                         \c_math_subscript_token
5079                         {
5080                             \MathMLintent { :silent }
5081                             { \str_use:c { l__spintent_sp #1 _sub_index_str } }

```

```

5082         }
5083     }
5084 }
5085 {
5086     \__spintent_luafun_mrow_intent_arg:nn
5087     { } { intent = "_" \__spintent_sp_rad_diam_word_tl - :prefix" }
5088     { \__spintent_sp_rad_diam_letra_tl }
5089     \c_math_subscript_token
5090     { \str_use:c { \__spintent_sp #1 _sub_index_str } }
5091 }
5092 }
5093 }

```

With `concept false`, the letter itself is read instead — but only when `read-cmd` is `school`, built the same way from the letter (with `sub-index`, letter and subscript wrapped together, subscript silenced). With `read-cmd mathcat`, the intent is left empty: MathCAT applies its own rules to the letter and its subscript, unmarked by `us`.

```

5094 {
5095     \str_if_empty:cTF { \__spintent_sp #1 _sub_index_str }
5096     {
5097         \str_if_eq:eeTF
5098         { \str_use:c { \__spintent_sp #1 _mode_str } } { school }
5099         {
5100             \__spintent_luafun_mrow_intent_arg:nn
5101             { } { intent = "_" \__spintent_sp_rad_diam_letra_tl - :prefix" }
5102             { \__spintent_sp_rad_diam_letra_tl }
5103         }
5104         { \__spintent_sp_rad_diam_letra_tl }
5105     }
5106     {
5107         \str_if_eq:eeTF
5108         { \str_use:c { \__spintent_sp #1 _mode_str } } { school }
5109         {
5110             \MathMLintent
5111             {
5112                 _ \__spintent_sp_rad_diam_letra_tl -
5113                 \bool_if:cT { \__spintent_sp #1 _read_sub_bool } { sub- }
5114                 \str_use:c { \__spintent_sp #1 _sub_index_str }
5115                 :prefix
5116             }
5117             {
5118                 \__spintent_inner_sep:
5119                 \__spintent_sp_rad_diam_letra_tl
5120                 \c_math_subscript_token
5121                 {
5122                     \MathMLintent { :silent }
5123                     { \str_use:c { \__spintent_sp #1 _sub_index_str } }
5124                 }
5125             }
5126         }
5127     }
5128     \__spintent_sp_rad_diam_letra_tl
5129     \c_math_subscript_token
5130     { \str_use:c { \__spintent_sp #1 _sub_index_str } }
5131 }
5132 }
5133 }
5134 }
5135 }

```

(End of definition for `\__spintent_sp_rad_diam_print:n`.)

`\__spintent_sp_rad_diam_exec:nn`

The function `\__spintent_sp_rad_diam_exec:nn` receives the *short name* of the command passed in ‘#1’ and the optional (*keys*) passed in ‘#2’ then delegating the rendering to `\__spintent_sp_rad_diam_print:n`.

```

5136 \cs_new_protected:Nn \__spintent_sp_rad_diam_exec:nn
5137 {
5138     \keys_set:nn { spintent / sp #1 } { #2 }
5139     \__spintent_sp_rad_diam_print:n { #1 }
5140 }

```

(End of definition for `\__spintent_sp_rad_diam_exec:nn`.)

`\sprad` Both public commands `\sprad` and `\spdiam` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_sp_rad_diam_exec:nn`.

```

5141 \NewDocumentCommand \sprad { 0{} }
5142 {
5143   \mode_if_math:TF
5144   {
5145     \group_begin:
5146     \__spintent_sp_rad_diam_exec:nn { rad } {#1}
5147     \group_end:
5148   }
5149   { \msg_error:nnn { spintent } { need-math-mode } { \sprad } }
5150 }
5151 \NewDocumentCommand \spdiam { 0{} }
5152 {
5153   \mode_if_math:TF
5154   {
5155     \group_begin:
5156     \__spintent_sp_rad_diam_exec:nn { diam } {#1}
5157     \group_end:
5158   }
5159   { \msg_error:nnn { spintent } { need-math-mode } { \spdiam } }
5160 }

```

(End of definition for `\sprad` and `\spdiam`. These functions are documented on page 28.)

## 11.46 The command `\spcirc`

The user command `\spcirc` provides a semantic interface for a circle or a circumference, using the delimited-optional argument syntax `d()` already established by `\spcmd/\spmcm` and `\spAfig/\spPfig`. The argument holds a required center point and an optional radius, separated by a comma. The radius is never required to declare its own kind via a key: its shape alone decides whether it is a pure number, a measure (number + unit), a bare letter, or a two-letter segment — reusing, respectively, `\spnum`, `\spunit`, and `\spside/\spmside` directly as sub-commands, rather than reimplementing any of their parsing.

### 11.46.1 Definition of variables for `\spcirc`

Shared outputs of `\__spintent_luafun_geo_circ_parse_and_set:n`. `\__spintent_geo_luaset_circ_radio_type_str` holds one of *number*, *measure*, *label*, *segment*, or an empty string when no radius was given. `\__spintent_geo_luaset_circ_radio_raw_str` holds the corresponding raw text, ready to be forwarded to the sub-command that renders it.

```

5161 \str_new:N \__spintent_geo_luaset_circ_radio_type_str
5162 \str_new:N \__spintent_geo_luaset_circ_radio_raw_str

```

(End of definition for `\__spintent_geo_luaset_circ_radio_type_str` and `\__spintent_geo_luaset_circ_radio_raw_str`.)

`\__spintent_spcirc_mode_str` The variable `\__spintent_spcirc_mode_str` stores the value set by the key `read-cmd`; the variable `\__spintent_spcirc_prefix_tl` stores the value set by the key `prefix`. The variable `\__spintent_spcirc_print_sym_str` stores the value set by the key `print-sym` (*usual* or *odot*). The variable `\__spintent_spcirc_read_sty_str` stores the value set by the key `read-sty` (*circun* or *circle*). The variable `\__spintent_spcirc_radio_sty_str` stores the value set by the key `radio-sty`, only meaningful when the radius is a segment. The variable `\__spintent_spcirc_sub_index_str` stores the value set by the key `sub-index`. The variable `\__spintent_spcirc_articulo_str` is set internally by `\__spintent_spcirc_word_and_article:` to *la* or *el*, matching the grammatical gender of the chosen word.

```

5163 \str_new:N \__spintent_spcirc_mode_str
5164 \tl_new:N \__spintent_spcirc_prefix_tl
5165 \str_new:N \__spintent_spcirc_print_sym_str
5166 \str_new:N \__spintent_spcirc_read_sty_str
5167 \str_new:N \__spintent_spcirc_radio_sty_str
5168 \str_new:N \__spintent_spcirc_sub_index_str
5169 \str_new:N \__spintent_spcirc_articulo_str
5170 \tl_new:N \__spintent_spcirc_read_arg_tl

```

(End of definition for `\__spintent_spcirc_mode_str` and others.)

`\__spintent_spcirc_word_tl` Scratch variables used by `\__spintent_spcirc_process:n` to assemble the chosen word (*circunferencia/círculo* or the `read-arg` override) and the final literal *intent* string.

```

5171 \tl_new:N \__spintent_spcirc_word_tl

```

(End of definition for `\__spintent_spcirc_word_tl`.)

### 11.46.2 Definition of error messages for \spcirc

spcirc-invalid-arg Raised when the argument does not match a single center point, optionally followed by a comma and a valid radius shape.

```

5172 \msg_new:nnnn { spintent } { spcirc-invalid-arg }
5173 {
5174   Invalid~argument~format~in~\token_to_str:N \spcirc:~'#1'~\msg_line_context:.
5175 }
5176 {
5177   See~the~documentation~of~\token_to_str:N \spcirc~for~the~accepted~argument~forms.
5178 }

```

(End of definition for spcirc-invalid-arg.)

### 11.46.3 Definition of keys for \spcirc

The key `print-sym` chooses the visual symbol: `usual` for a plain C, `odot` (default) for `\odot`. The key `read-sty` chooses the verbalized word and its article: `circun` (default) for «la *circunferencia*», `circle` for «el *círculo*». The key `radio-sty` chooses, only when the radius is a two-letter segment, whether it is rendered as the *object* (`side`, via `\spside`) or its *measure* (`mside`, default, via `\spmside`). The key `sub-index` sets a subscript on the symbol; the key `read-sub` controls whether it is read with the word «sub» or as a plain number, matching the same convention as `\spAfig`.

```

5179 \keys_define:nn { spintent / spcirc }
5180 {
5181   read-cmd      .choices:nn = { school , mathcat }
5182   {
5183     \int_case:nn { \l_keys_choice_int }
5184     {
5185       {1} { \str_set:Nn \l__spintent_spcirc_mode_str { school } }
5186       {2} { \str_set:Nn \l__spintent_spcirc_mode_str { mathcat } }
5187     }
5188   },
5189   read-cmd      .initial:n = school,
5190   read-cmd      .value_required:n = true,
5191   prefix        .code:n =
5192     { \__spintent_sanitize_affix:Nn \l__spintent_spcirc_prefix_tl {#1} },
5193   prefix        .initial:n = ,
5194   prefix        .value_required:n = true,
5195   print-sym     .choices:nn = { usual , odot }
5196   {
5197     \int_case:nn { \l_keys_choice_int }
5198     {
5199       {1} { \str_set:Nn \l__spintent_spcirc_print_sym_str { usual } }
5200       {2} { \str_set:Nn \l__spintent_spcirc_print_sym_str { odot } }
5201     }
5202   },
5203   print-sym     .initial:n = odot,
5204   print-sym     .value_required:n = true,
5205   read-sty      .choices:nn = { circun , circle }
5206   {
5207     \int_case:nn { \l_keys_choice_int }
5208     {
5209       {1} { \str_set:Nn \l__spintent_spcirc_read_sty_str { circun } }
5210       {2} { \str_set:Nn \l__spintent_spcirc_read_sty_str { circle } }
5211     }
5212   },
5213   read-sty      .initial:n = circun,
5214   read-sty      .value_required:n = true,
5215   diameter      .bool_set:N = \l__spintent_spcirc_diameter_bool,
5216   diameter      .initial:n = false,
5217   diameter      .value_required:n = true,
5218   radio-sty     .choices:nn = { side , mside }
5219   {
5220     \int_case:nn { \l_keys_choice_int }
5221     {
5222       {1} { \str_set:Nn \l__spintent_spcirc_radio_sty_str { side } }
5223       {2} { \str_set:Nn \l__spintent_spcirc_radio_sty_str { mside } }
5224     }
5225   },
5226   radio-sty     .initial:n = mside,
5227   radio-sty     .value_required:n = true,
5228   sub-index     .code:n =

```

```

5229     { \str_set:Nn \l__spintent_spcirc_sub_index_str {#1} },
5230     sub-index      .initial:n = ,
5231     sub-index      .value_required:n = true,
5232     read-sub       .bool_set:N = \l__spintent_spcirc_read_sub_bool,
5233     read-sub       .initial:n = false,
5234     read-sub       .value_required:n = true,
5235     read-arg       .code:n =
5236     { \__spintent_sanitizé_affix:Nn \l__spintent_spcirc_read_arg_tl {#1} },
5237     read-arg       .initial:n = ,
5238     read-arg       .value_required:n = true,
5239     silent-cmd     .bool_set:N = \l__spintent_spcirc_silent_cmd_bool,
5240     silent-cmd     .initial:n = false,
5241     silent-cmd     .value_required:n = true,
5242     concept        .bool_set:N = \l__spintent_spcirc_concept_bool,
5243     concept        .initial:n = true,
5244     concept        .value_required:n = true,
5245     spside         .meta:nn = { spintent / spside } { #1 },
5246     spside         .value_required:n = true,
5247     spmside        .meta:nn = { spintent / spmside } { #1 },
5248     spmside        .value_required:n = true,
5249     unknown        .code:n =
5250     {
5251         \str_set:Nn \l__spintent_command_name_str { spcirc }
5252         \__spintent_unknown_keys_cmd:n {#1}
5253     },
5254 }

```

(End of definition for read-cmd and others.)

#### 11.46.4 Internal functions for \spcirc

`\l__spintent_spcirc_has_arg_bool` `\l__spintent_spcirc_intent_tl` `\l__spintent_spcirc_intent_arg_tl` `\l__spintent_spcirc_connector_tl` `\l__spintent_spcirc_connector_arg_tl` `\l__spintent_spcirc_radio_arg_tl`  
`\l__spintent_spcirc_has_arg_bool` records whether the `d()` argument was given at all (`\spcirc` with no `(...)` at all has neither a center nor a radius). `\l__spintent_spcirc_intent_tl` holds the word (circunferencia or círculo, with an optional `sub-index` suffix) paired with the symbol. `\l__spintent_spcirc_connector_tl` holds de-centro-en, paired with the (center) group; the center is read through `\__spintent_mathmlarg_point:nnn`, referenced as `$pt`, not concatenated into either word — it can be subscripted or primed (e.g. `O_{1}`, as in the Euler circle) without breaking anything. The `_arg_tl` variables hold each word's final wrapped form; a third, `\l__spintent_spcirc_radio_arg_tl`, is spoken directly on the , delimiter after the center, replacing its own reading rather than pairing with a group.

```

5255 \bool_new:N \l__spintent_spcirc_has_arg_bool
5256 \tl_new:N \l__spintent_spcirc_intent_tl
5257 \tl_new:N \l__spintent_spcirc_intent_arg_tl
5258 \tl_new:N \l__spintent_spcirc_connector_tl
5259 \tl_new:N \l__spintent_spcirc_connector_arg_tl
5260 \tl_new:N \l__spintent_spcirc_radio_arg_tl

```

(End of definition for `\l__spintent_spcirc_has_arg_bool` and others.)

`\__spintent_spcirc_sym:` Typesets `\bigodot` or the plain letter C, according to `print-sym`.

```

5261 \cs_new_protected:Nn \__spintent_spcirc_sym:
5262 {
5263     \str_if_eq:VnTF \l__spintent_spcirc_print_sym_str { odot }
5264     {
5265         \group_begin:
5266         \textstyle\bigodot
5267         \group_end:
5268     }
5269     { C }
5270 }

```

(End of definition for `\__spintent_spcirc_sym:`.)

`\__spintent_spcirc_word_and_article:` Sets `\l__spintent_spcirc_word_tl` to `read-arg` or circunferencia/círculo from `read-sty`, and `\l__spintent_spcirc_articulo_str` to the matching article, which always follows `read-sty` even when `read-arg` overrides the word.

```

5271 \cs_new_protected:Nn \__spintent_spcirc_word_and_article:
5272 {
5273     \tl_if_empty:NTF \l__spintent_spcirc_read_arg_tl
5274     {
5275         \str_if_eq:VnTF \l__spintent_spcirc_read_sty_str { circun }
5276         {
5277             \tl_set:Nn \l__spintent_spcirc_word_tl { circunferencia }

```

```

5278         \str_set:Nn \l__spintenc_spcirc_articulo_str { la }
5279     }
5280     {
5281         \tl_set:Nn \l__spintenc_spcirc_word_tl { círculo }
5282         \str_set:Nn \l__spintenc_spcirc_articulo_str { el }
5283     }
5284 }
5285 {
5286     \tl_set:Nn \l__spintenc_spcirc_word_tl { \l__spintenc_spcirc_read_arg_tl }
5287     \str_if_eq:VnTF \l__spintenc_spcirc_read_sty_str { circun }
5288     { \str_set:Nn \l__spintenc_spcirc_articulo_str { la } }
5289     { \str_set:Nn \l__spintenc_spcirc_articulo_str { el } }
5290 }
5291 }

```

(End of definition for `\l__spintenc_spcirc_word_and_article:`.)

`\l__spintenc_spcirc_radio_print:`

Dispatches on `\l__spintenc_geo_luaset_circ_radio_type_str`, reusing `\spnum/\spunit/\spside/\spmside` directly rather than reimplementing any parsing.

```

5292 \cs_new_protected:Nn \l__spintenc_spcirc_radio_print:
5293 {
5294     \str_case:Vn \l__spintenc_geo_luaset_circ_radio_type_str
5295     {
5296         { number } { \exp_args:Ne \spnum { \l__spintenc_geo_luaset_circ_radio_raw_str } }
5297         { measure } { \exp_args:Ne \spunit { \l__spintenc_geo_luaset_circ_radio_raw_str } }
5298         { label } { \l__spintenc_geo_luaset_circ_radio_raw_str }
5299         { segment }
5300         {
5301             \str_if_eq:VnTF \l__spintenc_spcirc_radio_sty_str { side }
5302             { \exp_args:Ne \spside { \l__spintenc_geo_luaset_circ_radio_raw_str } }
5303             { \exp_args:Ne \spmside { \l__spintenc_geo_luaset_circ_radio_raw_str } }
5304         }
5305     }
5306 }

```

(End of definition for `\l__spintenc_spcirc_radio_print:`.)

`\l__spintenc_spcirc_build_concept:`

Builds two words: `\l__spintenc_spcirc_intent_tl` (via `\l__spintenc_spcirc_word_and_article:`, plus an optional `sub-index` suffix), paired with the symbol, and `\l__spintenc_spcirc_connector_tl` (de-centro-en), paired with the (center) group — built even when `\l__spintenc_spcirc_has_arg_bool` is `false`, though `\l__spintenc_spcirc_mathml:` never uses it in that case. Neither carries the center's own text: it is read separately, by reference, through `\l__spintenc_mathmlarg_point:nnn` — so a subscripted or primed center (e.g.  $O_{1}$ ), as in the Euler circle) is never an issue.

```

5307 \cs_new_protected:Nn \l__spintenc_spcirc_build_concept:
5308 {
5309     \l__spintenc_spcirc_word_and_article:
5310     \tl_set:Ne \l__spintenc_spcirc_intent_tl { \l__spintenc_spcirc_word_tl }
5311     \str_if_empty:NF \l__spintenc_spcirc_sub_index_str
5312     {
5313         \tl_put_right:Ne \l__spintenc_spcirc_intent_tl
5314         {
5315             -
5316             \bool_if:NT \l__spintenc_spcirc_read_sub_bool { sub - }
5317             \l__spintenc_spcirc_sub_index_str
5318         }
5319     }
5320     \tl_set:Nn \l__spintenc_spcirc_connector_tl { de-centro-en }

```

`\silent-cmd true` overrides `\l__spintenc_spcirc_intent_arg_tl/\l__spintenc_spcirc_radio_arg_tl` with `_-` and `\l__spintenc_spcirc_connector_arg_tl` with `_($pt)` (the center is always read), regardless of `read-arg`.

```

5321     \bool_if:NTF \l__spintenc_spcirc_silent_cmd_bool
5322     {
5323         \tl_set:Nn \l__spintenc_spcirc_intent_arg_tl { _- }
5324         \tl_set:Ne \l__spintenc_spcirc_connector_arg_tl { _($ pt) }
5325         \tl_set:Nn \l__spintenc_spcirc_radio_arg_tl { _- }
5326     }

```

With `concept true`, `\l__spintenc_spcirc_intent_arg_tl` is `\l__spintenc_spcirc_intent_tl:prefix` (bare — the group it wraps has nothing to reference, only a silenced symbol), prefixed with `prefix` when given; `\l__spintenc_spcirc_connector_arg_tl` is `\l__spintenc_spcirc_connector_tl:prefix($pt)`; and `\l__spintenc_spcirc_radio_arg_tl` — spoken directly on the , delimiter, replacing its own reading

rather than pairing with any group — is the literal `_y-radio-`, or `_y-diámetro-` when `diameter` is `true`, read as its own phrase right before `\__spintent_spcirc_radio_print:`’s independent reading follows as the next sibling.

```

5327     {
5328         \bool_if:NTF \l__spintent_spcirc_concept_bool
5329         {
5330             \tl_set:Nx \l__spintent_spcirc_intent_arg_tl
5331             {
5332                 \tl_if_empty:NF \l__spintent_spcirc_prefix_tl
5333                 { \tl_use:N \l__spintent_spcirc_prefix_tl - }
5334                 \l__spintent_spcirc_intent_tl :prefix
5335             }
5336             \tl_set:Nx \l__spintent_spcirc_connector_arg_tl
5337             { \l__spintent_spcirc_connector_tl :prefix( $ pt ) }
5338             \bool_if:NTF \l__spintent_spcirc_diameter_bool
5339             { \tl_set:Nn \l__spintent_spcirc_radio_arg_tl { _y-diámetro- } }
5340             { \tl_set:Nn \l__spintent_spcirc_radio_arg_tl { _y-radio- } }
5341         }

```

With `concept` `false` and `read-arg` empty, the same `_/_($pt)` silencing applies as `silent-cmd`; with `read-arg` given, none of the three is silenced instead — the connector and `y-radio/y-diámetro` words are fixed structural pieces, not customized by `read-arg`, but once a real word is being spoken there is nothing left to silence around it (same convention as `\__spintent_spside_process:nn`).

```

5342     {
5343         \tl_if_empty:NTF \l__spintent_spcirc_read_arg_tl
5344         {
5345             \tl_set:Nn \l__spintent_spcirc_intent_arg_tl { _- }
5346             \tl_set:Nx \l__spintent_spcirc_connector_arg_tl { _ ( $ pt ) }
5347             \tl_set:Nn \l__spintent_spcirc_radio_arg_tl { _- }
5348         }
5349         {
5350             \tl_set:Nx \l__spintent_spcirc_intent_arg_tl { \l__spintent_spcirc_intent_tl }
5351             \tl_set:Nx \l__spintent_spcirc_connector_arg_tl
5352             { \l__spintent_spcirc_connector_tl :prefix( $ pt ) }
5353             \bool_if:NTF \l__spintent_spcirc_diameter_bool
5354             { \tl_set:Nn \l__spintent_spcirc_radio_arg_tl { _y-diámetro- } }
5355             { \tl_set:Nn \l__spintent_spcirc_radio_arg_tl { _y-radio- } }
5356         }
5357     }
5358 }
5359 }

```

(End of definition for `\__spintent_spcirc_build_concept:`)

`\__spintent_spcirc_process:n`

Receives the `d()` argument in `#1`, an explicit empty group when omitted (`\spcirc` itself converts `-NoValue-` to `{ }` before calling `\__spintent_spcirc_exec:nn`) — `\l__spintent_spcirc_has_arg_bool` set to `false` in that case, and `\__spintent_luafun_geo_circ_parse_and_set:n` never called at all — there is nothing to parse. Otherwise calls it, reports `spcirc-invalid-arg` on failure, then delegates to `\__spintent_spcirc_mathml:`, which calls `\__spintent_spcirc_build_concept:` itself.

```

5360 \cs_new_protected:Nn \__spintent_spcirc_process:n
5361 {
5362     \tl_if_empty:NTF {#1}
5363     {
5364         \bool_set_false:N \l__spintent_spcirc_has_arg_bool
5365     }
5366     {
5367         \bool_set_true:N \l__spintent_spcirc_has_arg_bool
5368         \__spintent_luafun_geo_circ_parse_and_set:n { \exp_not:n {#1} }
5369         \str_if_eq:VnT \l__spintent_geo_luasets_error_str { true }
5370         {
5371             \msg_error:nne { spintent } { spcirc-invalid-arg } { \exp_not:n {#1} }
5372         }
5373     }
5374     \__spintent_spcirc_mathml:
5375 }

```

(End of definition for `\__spintent_spcirc_process:n`)

`\__spintent_spcirc_mathml:`

Calls `\__spintent_spcirc_build_concept:`, then wraps the symbol (`\l__spintent_spcirc_intent_arg_tl`) in its own native `\MathMLintent` — this alone is the whole result when `\l__spintent_spcirc_has_arg_bool` is `false` (`\spcirc` with no `(...)` at all: no center, no radius).

Otherwise, the `(center)` group `(\__spintent_spcirc_connector_arg_tl)` follows as a separate sibling, never nested inside the symbol's own `\MathMLintent` — or MathCAT falls back to its own structural reading of the underlying `<mover>/<msub>` instead, the same problem already found for `\spangle`'s symbol and `\sparc`'s accent. When a radius is present, the `,` delimiter after the center carries `\__spintent_spcirc_radio_arg_tl` directly (replacing its own reading, not paired with any group), and `\__spintent_spcirc_radio_print:` follows as an ordinary sibling right after — wrapping it in another `\MathMLintent` would lose the reading of whatever it produces (itself a native `\MathMLintent`, from `\spnum/\spunit`, nested inside another one does not compose).

The symbol and, when `sub-index` is given, its subscript number are each silenced with `\__spintent_luafun_mrow_intent_arg:nn` *separately* — one queue entry per leaf. The two are reached differently by the shared `mathml_filter` callback's recursive lookup (the symbol via `nucleus`, the subscript via `sub`), so a single entry covering both leaves nothing registered for the second one, which the recursion then assigns whatever comes next in the queue instead — shifting every claim after it by one. The `(` delimiter is silenced the same way, but *without* the usual `{...}` around the character: `luamml` only keeps a delimiter's own math class (giving `<mo>` instead of `<mi>`, needed for MathCAT to read the group at all) when it is not itself sitting inside a brace group.

The final `)` — following `\__spintent_spcirc_radio_print:`, whose own output the shared queue does not reliably skip over — is typeset as `\rparen` inside its own native `\MathMLintent{:silent}` instead of a bare `)`: `\rparen` keeps its own math class through the wrapping, unlike a bare delimiter character, and this sits outside the shared queue entirely, so nothing in between can claim an entry meant for it.

The center itself is read through `\__spintent_mathmlarg_point:nnn`, referenced as `$pt`.

```

5376 \cs_new_protected:Nn \__spintent_spcirc_mathml:
5377 {
5378   \__spintent_invisible_sep: % (luamml issues #26)
5379   \__spintent_spcirc_build_concept:
5380   \MathMLintent { \__spintent_spcirc_intent_arg_tl }
5381   {
5382     \__spintent_inner_sep:
5383     \__spintent_luafun_mrow_intent_arg:nn
5384     { } { intent = ":silent" }
5385     { \__spintent_spcirc_sym: }
5386     \str_if_empty:NF \__spintent_spcirc_sub_index_str
5387     {
5388       \c_math_subscript_token
5389       {
5390         \__spintent_luafun_mrow_intent_arg:nn
5391         { } { intent = ":silent" }
5392         { \__spintent_spcirc_sub_index_str }
5393       }
5394     }
5395   }
5396   \bool_if:NT \__spintent_spcirc_has_arg_bool
5397   {
5398     \str_if_empty:NTF \__spintent_geo_luaset_circ_radio_type_str
5399     {
5400       \MathMLintent { \__spintent_spcirc_connector_arg_tl }
5401       {
5402         \__spintent_inner_sep:
5403         \__spintent_luafun_mrow_intent_arg:nn { } { intent = ":silent" } (
5404         \__spintent_mathmlarg_point:nnV { pt } { spcirc } \__spintent_geo_luaset_points:
5405         \__spintent_luafun_mrow_intent_arg:nn { } { intent = ":silent" } )
5406       }
5407     }
5408     {
5409       \MathMLintent { \__spintent_spcirc_connector_arg_tl }
5410       {
5411         \__spintent_inner_sep:
5412         \__spintent_luafun_mrow_intent_arg:nn { } { intent = ":silent" } (
5413         \__spintent_mathmlarg_point:nnV { pt } { spcirc } \__spintent_geo_luaset_points:
5414       }
5415       \MathMLintent { \__spintent_spcirc_radio_arg_tl } { , }
5416       \__spintent_spcirc_radio_print:
5417       \MathMLintent { :silent } { \rparen }
5418     }
5419   }
5420 }

```

(End of definition for `\__spintent_spcirc_mathml:`)

`\__spintent_spcirc_exec:nn` The function `\__spintent_spcirc_exec:nn` receives the optional keys in #1 and the argument in #2,

delegating the rendering to `\__spintent_spcirc_print:n`.

```
5421 \cs_new_protected:Nn \__spintent_spcirc_exec:nn
5422 {
5423   \keys_set:nn { spintent / spcirc } { #1 }
5424   \__spintent_spcirc_process:n { #2 }
5425 }
```

(End of definition for `\__spintent_spcirc_exec:nn`.)

`\spcirc` The public command `\spcirc` enforces math mode, opens a local group, and delegates execution to `\__spintent_spcirc_exec:nn`. It follows the same `d()`-argument pattern as `\spmcd/\spmcm` and `\spAfig/\spPfig`; an omitted radius forwards an explicit empty group rather than the `-NoValue-` marker.

```
5426 \NewDocumentCommand \spcirc { 0{} d() }
5427 {
5428   \mode_if_math:TF
5429   {
5430     \group_begin:
5431     \attribute \__spintent_scope_attr_int = 1
5432     \tl_if_novalue:nTF {#2}
5433     { \__spintent_spcirc_exec:nn { #1 } { } }
5434     { \__spintent_spcirc_exec:nn { #1 } { #2 } }
5435     \group_end:
5436   }
5437   { \msg_error:nnn { spintent } { need-math-mode } { \spcirc } }
5438 }
```

(End of definition for `\spcirc`. This function is documented on page 29.)

## 11.47 The commands `\spAcirc` and `\spPcirc`

The user commands `\spAcirc` and `\spPcirc` typeset the area and the perimeter of a circle or a circumference. The argument follows `\spcirc`'s own point-and-radius grammar for a real center (with an optional radius); when no comma is present and no valid center is found, it falls back to a bare number/letter shortcut, reusing the same Lua entry point as `\spAfig/\spPfig`. A table letter matched by that fallback is rejected, since it has no meaning for a circle.

### 11.47.1 Definition of variables for `\spAcirc` and `\spPcirc`

```
\__spintent_spAcirc_mode_str
\__spintent_spPcirc_mode_str
\__spintent_spAcirc_print_sym_str
\__spintent_spPcirc_print_sym_str
\__spintent_spAcirc_read_sty_str
\__spintent_spPcirc_read_sty_str
\__spintent_spAcirc_radio_sty_str
\__spintent_spPcirc_radio_sty_str
\__spintent_spAcirc_sub_index_str
\__spintent_spPcirc_sub_index_str
\__spintent_spAcirc_prefix_tl
\__spintent_spPcirc_prefix_tl
\__spintent_spAcirc_read_arg_tl
\__spintent_spPcirc_read_arg_tl
\__spintent_geo_circfig_mode_str
```

These variables store the value set by each corresponding key: `read-cmd`, `print-sym`, `read-sty`, `radio-sty`, `sub-index`, `prefix` and `read-arg`, respectively.

```
5439 \clist_map_inline:nn { Acirc , Pcirc }
5440 {
5441   \str_new:c { \__spintent_sp #1 _mode_str }
5442   \str_new:c { \__spintent_sp #1 _print_sym_str }
5443   \str_new:c { \__spintent_sp #1 _read_sty_str }
5444   \str_new:c { \__spintent_sp #1 _radio_sty_str }
5445   \str_new:c { \__spintent_sp #1 _sub_index_str }
5446   \tl_new:c { \__spintent_sp #1 _prefix_tl }
5447   \tl_new:c { \__spintent_sp #1 _read_arg_tl }
5448 }
```

(End of definition for `\__spintent_spAcirc_mode_str` and others.)

The variable `\__spintent_geo_circfig_mode_str` stores whether the argument was resolved as a real circle (`circle`) or as the bare number/letter shortcut (`bare`).

```
5449 \str_new:N \__spintent_geo_circfig_mode_str
```

(End of definition for `\__spintent_geo_circfig_mode_str`.)

### 11.47.2 Definition of error messages for `\spAcirc` and `\spPcirc`

```
spAcirc-invalid-arg
spPcirc-invalid-arg
```

The error message shown when the argument does not match any of the accepted forms: empty, a center point (optionally followed by a comma and a radius), or a bare number/letter.

```
5450 \clist_map_inline:nn { Acirc , Pcirc }
5451 {
5452   \msg_new:nnnn { spintent } { sp #1 -invalid-arg }
5453   {
5454     Invalid~argument~in~##1:~'##2'~\msg_line_context:.
5455   }
5456   {
5457     See~the~documentation~of~##1~for~the~accepted~argument.
5458   }
```

(End of definition for `spAcirc-invalid-arg` and `spPcirc-invalid-arg`.)

### 11.47.3 Definition of keys for \spAcirc and \spPcirc

We define the shared *(keys)* read-cmd, prefix, print-sym, read-sty, concept, sub-index, read-sub, read-arg and silent-cmd.

```

5459 \keys_define:nn { spintent / sp #1 }
5460 {
5461   read-cmd      .choices:nn = { school , mathcat }
5462   {
5463     \int_case:nn { \l_keys_choice_int }
5464     {
5465       {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school } }
5466       {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
5467     }
5468   },
5469   read-cmd      .initial:n = school,
5470   read-cmd      .value_required:n = true,
5471   read-cmd / unknown .code:n =
5472     \msg_error:nneee { spintent } { unknown-choice }
5473     { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
5474   prefix        .code:n =
5475     { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
5476   prefix        .initial:n = ,
5477   prefix        .value_required:n = true,
5478   print-sym     .choices:nn = { usual , odot }
5479   {
5480     \int_case:nn { \l_keys_choice_int }
5481     {
5482       {1} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { usual } }
5483       {2} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { odot } }
5484     }
5485   },
5486   print-sym     .initial:n = odot,
5487   print-sym     .value_required:n = true,
5488   print-sym / unknown .code:n =
5489     \msg_error:nneee { spintent } { unknown-choice }
5490     { print-sym } { usual,~odot } { \exp_not:n {##1} },
5491   read-sty      .choices:nn = { circun , circle }
5492   {
5493     \int_case:nn { \l_keys_choice_int }
5494     {
5495       {1} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { circun } }
5496       {2} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { circle } }
5497     }
5498   },
5499   read-sty      .initial:n = circun,
5500   read-sty      .value_required:n = true,
5501   concept       .bool_set:c = { l__spintent_sp #1 _concept_bool },
5502   concept       .initial:n = true,
5503   concept       .value_required:n = true,
5504   sub-index     .code:n =
5505     { \str_set:cn { l__spintent_sp #1 _sub_index_str } {##1} },
5506   sub-index     .initial:n = ,
5507   sub-index     .value_required:n = true,
5508   read-sub      .bool_set:c = { l__spintent_sp #1 _read_sub_bool },
5509   read-sub      .initial:n = false,
5510   read-sub      .value_required:n = true,
5511   read-arg      .code:n =
5512     { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _read_arg_tl } {##1} },
5513   read-arg      .initial:n = ,
5514   read-arg      .value_required:n = true,
5515   silent-cmd    .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
5516   silent-cmd    .initial:n = false,
5517   silent-cmd    .value_required:n = true,
5518   unknown      .code:n =
5519     {
5520       \str_set:Nn \l__spintent_command_name_str { sp #1 }
5521       \__spintent_unknown_keys_cmd:n {##1}
5522     },
5523 }
5524 }
```

(End of definition for read-cmd and others.)

```

\l__spintent_spcircfig_short_bool
\l__spintent_spcircfig_op_tl
\l__spintent_spcircfig_intent_tl
\l__spintent_spcircfig_intent_arg_tl

```

`\l__spintent_spcircfig_short_bool` records whether the argument resolved as the short number/letter form (via `\l__spintent_luafun_geo_afig_parse_and_set:n/\l__spintent_luafun_geo_pfig_parse_and_set:n` the same entry point `\spAfig/\spPfig` use) rather than a real circle. `\l__spintent_spcircfig_op_tl` holds *área* or *perímetro*. `\l__spintent_spcircfig_intent_tl` holds the full spoken phrase: `\l__spintent_spcircfig_del_word_tl` (via `\l__spintent_spcircfig_del_word:n`: the word, contracted with `del-`/`de-la-`, or `read-arg`), an optional `sub-index` suffix, and, when a center is given, `-con-centro-en` — the center itself is never concatenated into it: it is read separately, by reference, through `\l__spintent_mathmlarg_point:nnn`, so a subscripted or primed center (e.g.  $O_{\{1\}}$ , as in the Euler circle) is never an issue.

```

5525 \bool_new:N \l__spintent_spcircfig_short_bool
5526 \tl_new:N \l__spintent_spcircfig_op_tl
5527 \tl_new:N \l__spintent_spcircfig_del_word_tl
5528 \tl_new:N \l__spintent_spcircfig_intent_tl
5529 \tl_new:N \l__spintent_spcircfig_intent_arg_tl

```

(End of definition for `\l__spintent_spcircfig_short_bool` and others.)

```

\l__spintent_spcircfig_sym:n

```

Typesets `\bigodot` or the plain letter C, according to `print-sym`. Receives the short command name in ‘`#1`’.

```

5530 \cs_new_protected:Nn \l__spintent_spcircfig_sym:n
5531 {
5532   \str_if_eq:vnTF { \l__spintent_sp #1 _print_sym_str } { odot }
5533   { \bigodot }
5534   { C }
5535 }

```

(End of definition for `\l__spintent_spcircfig_sym:n`.)

```

\l__spintent_spcircfig_word_and_article:n

```

Sets `\l__spintent_spcirc_word_tl` to `read-arg` or `circunferencia/círculo` from `read-sty`, and `\l__spintent_spcirc_articulo_str` to the matching article, which always follows `read-sty` even when `read-arg` overrides the word. Receives the short command name in ‘`#1`’.

```

5536 \cs_new_protected:Nn \l__spintent_spcircfig_word_and_article:n
5537 {
5538   \str_if_eq:vnTF { \l__spintent_sp #1 _read_sty_str } { circun }
5539   { \str_set:Nn \l__spintent_spcirc_articulo_str { la } }
5540   { \str_set:Nn \l__spintent_spcirc_articulo_str { el } }
5541   \tl_if_empty:cTF { \l__spintent_sp #1 _read_arg_tl }
5542   {
5543     \str_if_eq:vnTF { \l__spintent_sp #1 _read_sty_str } { circun }
5544     { \tl_set:Nn \l__spintent_spcirc_word_tl { circunferencia } }
5545     { \tl_set:Nn \l__spintent_spcirc_word_tl { círculo } }
5546   }
5547   {
5548     \tl_set:Ne \l__spintent_spcirc_word_tl
5549     {
5550       \tl_use:c { \l__spintent_sp #1 _read_arg_tl }
5551     }
5552   }
5553 }

```

(End of definition for `\l__spintent_spcircfig_word_and_article:n`.)

```

\l__spintent_spcircfig_del_word:n

```

Sets `\l__spintent_spcircfig_del_word_tl` to `\l__spintent_spcircfig_word_and_article:n`’s word, prefixed with `del-` or `de-la-` — contracted, not `de-el-`, since a screen reader does not contract that on its own. Receives the short command name in ‘`#1`’.

```

5554 \cs_new_protected:Nn \l__spintent_spcircfig_del_word:n
5555 {
5556   \l__spintent_spcircfig_word_and_article:n {#1}
5557   \str_if_eq:VnTF \l__spintent_spcirc_articulo_str { el }
5558   {
5559     \tl_set:Ne \l__spintent_spcircfig_del_word_tl { del- \l__spintent_spcirc_word_tl }
5560   }
5561   {
5562     \tl_set:Ne \l__spintent_spcircfig_del_word_tl { de-la- \l__spintent_spcirc_word_tl }
5563   }
5564 }

```

(End of definition for `\l__spintent_spcircfig_del_word:n`.)

```

\l__spintent_spcircfig_build_short:n

```

Handles the short number/letter case: builds `\l__spintent_spcircfig_intent_arg_tl` as `\l__spintent_spcircfig_intent_arg_tl` (*ber word*), wrapped in bare `:prefix` (no reference — there is nothing to point to), or `_-` when `silent-cmd` is `true` or `concept` is `false`. Receives the short command name in ‘`#1`’.

```

5565 \cs_new_protected:Nn \l__spintent_spcircfig_build_short:n

```

```

5566 {
5567   \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
5568   {
5569     \tl_set:Nn \l__spintent_spcircfig_intent_arg_tl { _- }
5570   }
5571   {
5572     \bool_if:cTF { l__spintent_sp #1 _concept_bool }
5573     {
5574       \__spintent_spcircfig_del_word:n {#1}
5575       \tl_set:Ne \l__spintent_spcircfig_intent_arg_tl
5576       {
5577         \l__spintent_spcircfig_op_tl - \l__spintent_spcircfig_del_word_tl -
5578         \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub- }
5579         \l__spintent_geo_luaset_sub_spoken_str :prefix
5580       }
5581     }
5582     {
5583       \tl_set:Nn \l__spintent_spcircfig_intent_arg_tl { _- }
5584     }
5585   }
5586 }

```

(End of definition for `\__spintent_spcircfig_build_short:n`.)

`\__spintent_spcircfig_finish_intent:nn`

Appends an optional `sub-index` suffix and, when `#2` is non-empty, `-con-centro-en` to `\l__spintent_spcircfig_intent_arg_tl` — shared by both branches of `\__spintent_spcircfig_build_concept:nn`, after either one has set the word itself. Receives the short command name in ‘`#1`’ and the argument in ‘`#2`’.

```

5587 \cs_new_protected:Nn \__spintent_spcircfig_finish_intent:nn
5588 {
5589   \str_if_empty:cF { l__spintent_sp #1 _sub_index_str }
5590   {
5591     \tl_put_right:Ne \l__spintent_spcircfig_intent_tl
5592     {
5593       -
5594       \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub- }
5595       \str_use:c { l__spintent_sp #1 _sub_index_str }
5596     }
5597   }
5598   \tl_if_empty:nF {#2}
5599   {
5600     \tl_put_right:Nn \l__spintent_spcircfig_intent_tl { -con-centro-en }
5601   }
5602 }

```

(End of definition for `\__spintent_spcircfig_finish_intent:nn`.)

`\__spintent_spcircfig_build_concept:nn`

Builds `\l__spintent_spcircfig_intent_arg_tl`. `_-` when `silent-cmd` is `true`, or when `false` and `concept` is `false` with `read-arg` empty (`_( $\$pt$ )` instead when `#2` is non-empty — the center is still read either way). Otherwise `\l__spintent_spcircfig_intent_tl:prefix` — with ( `$\$pt$` ) appended when `#2` is non-empty — prefixed with `prefix` when given, where `\l__spintent_spcircfig_intent_tl` is: with `concept` `true`, `\l__spintent_spcircfig_op_tl- \l__spintent_spcircfig_del_word_tl` (via `\__spintent_spcircfig_del_word:n`: the word, contracted with `del-/de-la-`, or `read-arg` the same way); with `false` and `read-arg` given, `read-arg` verbatim instead, with `\l__spintent_spcircfig_op_tl` dropped entirely — `área/ perímetro` is itself the concept `concept` toggles, not the `-con-centro-en` that follows, which always stays (it identifies which figure, not the concept). No automatic article either — the word (and any article it needs) is the user’s to write, same convention as `\__spintent_spcirc_process:n`. Either way, an optional `sub-index` suffix and, when `#2` is non-empty, `-con-centro-en` follow. Receives the short command name in ‘`#1`’ and the argument in ‘`#2`’.

```

5603 \cs_new_protected:Nn \__spintent_spcircfig_build_concept:nn
5604 {
5605   \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
5606   {
5607     \tl_if_empty:nTF {#2}
5608     { \tl_set:Nn \l__spintent_spcircfig_intent_arg_tl { _- } }
5609     { \tl_set:Ne \l__spintent_spcircfig_intent_arg_tl { _( $\$pt$ ) } }
5610   }
5611   {
5612     \bool_if:cTF { l__spintent_sp #1 _concept_bool }
5613     {
5614       \__spintent_spcircfig_del_word:n {#1}
5615       \tl_set:Ne \l__spintent_spcircfig_intent_tl

```

```

5616         {
5617             \l__spintent_spcircfig_op_tl - \l__spintent_spcircfig_del_word_tl
5618         }
5619         \__spintent_spcircfig_finish_intent:nn {#1} {#2}
5620         \tl_set:Nc \l__spintent_spcircfig_intent_arg_tl
5621         {
5622             \tl_if_empty:cF { \l__spintent_sp #1 _prefix_tl }
5623             {
5624                 \tl_use:c { \l__spintent_sp #1 _prefix_tl } -
5625             }
5626             \l__spintent_spcircfig_intent_tl :prefix
5627             \tl_if_empty:nF {#2} { ( $ pt ) }
5628         }
5629     }
5630     {
5631         \tl_if_empty:cTF { \l__spintent_sp #1 _read_arg_tl }
5632         {
5633             \tl_if_empty:nTF {#2}
5634             { \tl_set:Nn \l__spintent_spcircfig_intent_arg_tl { _- } }
5635             { \tl_set:Nc \l__spintent_spcircfig_intent_arg_tl { _ ( $ pt ) } }
5636         }
5637         {
5638             \tl_set:Nc \l__spintent_spcircfig_intent_tl
5639             { \tl_use:c { \l__spintent_sp #1 _read_arg_tl } }
5640             \__spintent_spcircfig_finish_intent:nn {#1} {#2}
5641             \tl_set:Nc \l__spintent_spcircfig_intent_arg_tl
5642             {
5643                 \l__spintent_spcircfig_intent_tl :prefix
5644                 \tl_if_empty:nF {#2} { ( $ pt ) }
5645             }
5646         }
5647     }
5648 }
5649 }

```

(End of definition for `\__spintent_spcircfig_build_concept:nn`.)

`\__spintent_spcircfig_process:nn`

Receives the short command name in ‘#1’ and the argument in ‘#2’. Sets `\l__spintent_spcircfig_op_tl` to `área/perímetro`. With #2 empty, `\l__spintent_spcircfig_short_bool` is set `false` and nothing else is parsed. With a comma in #2, reports `sp#1-invalid-arg` immediately — `\@@_luafun_geo_circ_parse_and_set:` (shared with `\spcirc`) still accepts a second, comma-separated part as a generic `label`-type radius, matching almost any text, so it would never itself report an error for a comma-containing argument here, where a comma is never meaningful at all. Without a comma, calls `\__spintent_luafun_geo_circ_parse_and_set:n`; on error, retries through `\__spintent_luafun_geo_afig_parse_and_set:n/\__spintent_luafun_geo_pfig_parse_and_set:n` (the same entry point `\spAfig/\spPfig` use) for the short number/letter form, rejecting a table-letter match (not meaningful for a circle) back to an error. Reports `sp#1-invalid-arg` on failure either way, then delegates to `\__spintent_spcircfig_build_short:n/\__spintent_spcircfig_build_concept:nn` and `\__spintent_spcircfig_mathml:nn`.

- Unlike `\spcirc`, there is no radius/diameter here at all: a formula’s left-hand side (`A_0`) names the figure, it does not carry the values needed to compute it — those belong on the right-hand side, elsewhere in the document.

```

5650 \cs_new_protected:Nn \__spintent_spcircfig_process:nn
5651 {
5652     \str_if_eq:eeTF {#1} { Acirc }
5653     { \tl_set:Nn \l__spintent_spcircfig_op_tl { área } }
5654     { \tl_set:Nn \l__spintent_spcircfig_op_tl { perímetro } }
5655     \tl_if_empty:nTF {#2}
5656     {
5657         \bool_set_false:N \l__spintent_spcircfig_short_bool
5658     }
5659     {
5660         \bool_set_false:N \l__spintent_spcircfig_short_bool
5661         \tl_if_in:nnTF {#2} { , }
5662         {
5663             \msg_error:nnee { spintent } { sp #1 -invalid-arg }
5664             { \c_backslash_str sp #1 } { \exp_not:n {#2} }
5665         }
5666         {
5667             \__spintent_luafun_geo_circ_parse_and_set:n { \exp_not:n {#2} }
5668             \str_if_eq:VnT \l__spintent_geo_luaerror_str { true }
5669         }

```

```

5670         \str_if_eq:eeTF {#1} { Acirc }
5671         { \__spintent_luafun_geo_afig_parse_and_set:n { \exp_not:n {#2} } }
5672         { \__spintent_luafun_geo_pfig_parse_and_set:n { \exp_not:n {#2} } }
5673     \str_if_eq:VnF \__spintent_geo_luaset_error_str { true }
5674     {
5675         \str_if_empty:NTF \__spintent_geo_luaset_letra_word_str
5676         { \bool_set_true:N \__spintent_spcircfig_short_bool }
5677         { \str_set:Nn \__spintent_geo_luaset_error_str { true } }
5678     }
5679 }
5680 \bool_if:NF \__spintent_spcircfig_short_bool
5681 {
5682     \str_if_eq:VnT \__spintent_geo_luaset_error_str { true }
5683     {
5684         \msg_error:nnee { spintent } { sp #1 -invalid-arg }
5685         { \c_backslash_str sp #1 } { \exp_not:n {#2} }
5686     }
5687 }
5688 }
5689 }
5690 \bool_if:NTF \__spintent_spcircfig_short_bool
5691 { \__spintent_spcircfig_build_short:n {#1} }
5692 { \__spintent_spcircfig_build_concept:nn {#1} {#2} }
5693 \__spintent_spcircfig_mathml:nn { #1 } { #2 }
5694 }

```

(End of definition for `\__spintent_spcircfig_process:nn`.)

`\__spintent_spcircfig_mathml:nn`

Renders `\__spintent_spcircfig_intent_arg_tl`, already built by `\__spintent_spcircfig_process:nn`. Three cases: the short case prints the symbol with the short value as its own subscript; an empty `#2` prints the symbol alone; a real center prints the symbol (plus optional `sub-index` subscript) and the center between parentheses. In every case a single native `\MathMLintent` wraps `\__spintent_mathcal:n{⟨A|P⟩}` and the whole subscript together — splitting the symbol and the center into separate `\MathMLintent` siblings inside the same subscript was tried and reads as `<msub>`'s own native structural narration instead. The center is read through `\__spintent_mathmlarg_point:nnn`, referenced as `$pt` (same convention as `\spcirc`). Receives the short command name in `'#1'` and the argument in `'#2'`.

```

5695 \cs_new_protected:Nn \__spintent_spcircfig_mathml:nn
5696 {
5697     \__spintent_invisible_sep: % (luamml issues #26)
5698     \MathMLintent { \__spintent_spcircfig_intent_arg_tl }
5699     {
5700         \__spintent_inner_sep:
5701         \__spintent_luafun_mrow_intent_arg:nn
5702         { } { intent = ":silent" }
5703     }
5704     \str_if_eq:eeTF {#1} { Acirc } { \__spintent_mathcal:n { A } } { \__spintent_mathcal:n
5705     }
5706     \bool_if:NTF \__spintent_spcircfig_short_bool
5707     {
5708         \c_math_subscript_token
5709         {
5710             \__spintent_spcircfig_sym:n {#1}
5711             \c_math_subscript_token { \__spintent_geo_luaset_print_tl }
5712         }
5713     }
5714     {
5715         \c_math_subscript_token
5716         {
5717             \MathMLintent { :silent } { \__spintent_spcircfig_sym:n {#1} }
5718             \str_if_empty:cF { \__spintent_sp #1 _sub_index_str }
5719             {
5720                 \c_math_subscript_token
5721                 {
5722                     \MathMLintent { :silent }
5723                     { \str_use:c { \__spintent_sp #1 _sub_index_str } }
5724                 }
5725             }
5726         }
5727         \tl_if_empty:NF {#2}
5728         {
5729             \MathMLintent { :silent } { \lparen }
5730             \__spintent_mathmlarg_point:nnV { pt } { sp #1 } \__spintent_geo_luaset_poin

```

```

5730             \MathMLIntent { :silent } { \rparen }
5731         }
5732     }
5733 }
5734 }
5735 }

```

(End of definition for `\__spintenc_spcircfig_mathml:nn`.)

`\__spintenc_spcircfig_exec:nnn`

The function `\__spintenc_spcircfig_exec:nnn` receives the short name of the command in #1, the optional keys in #2, and the argument in #3, delegating the rendering to `\__spintenc_spcircfig_process:nn`.

```

5736 \cs_new_protected:Nn \__spintenc_spcircfig_exec:nnn
5737 {
5738     \keys_set:nn { spintenc / sp #1 } { #2 }
5739     \__spintenc_spcircfig_process:nn { #1 } { #3 }
5740 }

```

(End of definition for `\__spintenc_spcircfig_exec:nnn`.)

`\spAcirc`  
`\spPcirc`

Both public commands `\spAcirc` and `\spPcirc` enforce math mode, open a local group, and delegate execution to `\__spintenc_spcircfig_exec:nnn`.

```

5741 \NewDocumentCommand \spAcirc { 0{} d() }
5742 {
5743     \mode_if_math:TF
5744     {
5745         \group_begin:
5746         \attribute \l__spintenc_scope_attr_int = 1
5747         \tl_if_novalue:nTF {#2}
5748         { \__spintenc_spcircfig_exec:nnn { Acirc } { #1 } { } }
5749         { \__spintenc_spcircfig_exec:nnn { Acirc } { #1 } { #2 } }
5750     \group_end:
5751     }
5752     { \msg_error:nnn { spintenc } { need-math-mode } { \spAcirc } }
5753 }
5754 \NewDocumentCommand \spPcirc { 0{} d() }
5755 {
5756     \mode_if_math:TF
5757     {
5758         \group_begin:
5759         \attribute \l__spintenc_scope_attr_int = 1
5760         \tl_if_novalue:nTF {#2}
5761         { \__spintenc_spcircfig_exec:nnn { Pcirc } { #1 } { } }
5762         { \__spintenc_spcircfig_exec:nnn { Pcirc } { #1 } { #2 } }
5763     \group_end:
5764     }
5765     { \msg_error:nnn { spintenc } { need-math-mode } { \spPcirc } }
5766 }

```

(End of definition for `\spAcirc` and `\spPcirc`. These functions are documented on page 30.)

## 11.48 Finish package

Finish package implementation.

```

5767 \file_input_stop:
5768 </package>

```

12 Índice de la Implementación

The italic numbers denote the pages where the corresponding entry is described, the numbers underlined and all others indicate the line on which they are implemented in the package code.

|                                                                                                                                                                    |                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| A                                                                                                                                                                  |                                                                                |
| \angle .....                                                                                                                                                       | 120                                                                            |
| autofit .....                                                                                                                                                      | <u>3145</u>                                                                    |
| C                                                                                                                                                                  |                                                                                |
| cifras-sep .....                                                                                                                                                   | <u>284</u> , <u>569</u> , <u>842</u> , <u>1865</u> , <u>1936</u> , <u>2484</u> |
| Document class:                                                                                                                                                    |                                                                                |
| l3doc .....                                                                                                                                                        | 37                                                                             |
| cmd-key-unknown .....                                                                                                                                              | <u>203</u>                                                                     |
| cmd-key-value-unknown .....                                                                                                                                        | <u>203</u>                                                                     |
| Commands provide by <b>spintent</b> :                                                                                                                              |                                                                                |
| \spmoney .....                                                                                                                                                     | 55                                                                             |
| \spnum .....                                                                                                                                                       | 45                                                                             |
| \spsiglo .....                                                                                                                                                     | 64, 67                                                                         |
| \spunit .....                                                                                                                                                      | 50                                                                             |
| concept <u>2027</u> , <u>2198</u> , <u>3145</u> , <u>3744</u> , <u>3951</u> , <u>4236</u> , <u>4389</u> , <u>4573</u> , <u>4961</u> ,<br><u>5179</u> , <u>5459</u> |                                                                                |
| concept* .....                                                                                                                                                     | <u>4573</u>                                                                    |
| currency .....                                                                                                                                                     | <u>842</u>                                                                     |
| currency-no-subunits .....                                                                                                                                         | <u>811</u>                                                                     |
| D                                                                                                                                                                  |                                                                                |
| diameter .....                                                                                                                                                     | <u>5179</u>                                                                    |
| dolar-math .....                                                                                                                                                   | <u>842</u>                                                                     |
| I                                                                                                                                                                  |                                                                                |
| inline .....                                                                                                                                                       | <u>2346</u>                                                                    |
| inline-numeric-denominator .....                                                                                                                                   | <u>547</u>                                                                     |
| invalid-angle-format .....                                                                                                                                         | <u>2434</u>                                                                    |
| invalid-century-format .....                                                                                                                                       | <u>1231</u>                                                                    |
| invalid-currency .....                                                                                                                                             | <u>811</u>                                                                     |
| invalid-date-format .....                                                                                                                                          | <u>1461</u>                                                                    |
| invalid-time-format .....                                                                                                                                          | <u>1382</u>                                                                    |
| J                                                                                                                                                                  |                                                                                |
| join-word .....                                                                                                                                                    | <u>2562</u>                                                                    |
| K                                                                                                                                                                  |                                                                                |
| Keys for \spAcirc provide by <b>spintent</b> :                                                                                                                     |                                                                                |
| concept .....                                                                                                                                                      | 149                                                                            |
| prefix .....                                                                                                                                                       | 148, 149                                                                       |
| print-sym .....                                                                                                                                                    | 148–150                                                                        |
| radio-sty .....                                                                                                                                                    | 148                                                                            |
| read-arg .....                                                                                                                                                     | 148, 149                                                                       |
| read-cmd .....                                                                                                                                                     | 148, 149                                                                       |
| read-sty .....                                                                                                                                                     | 148, 149                                                                       |
| read-sub .....                                                                                                                                                     | 149                                                                            |
| silent-cmd .....                                                                                                                                                   | 149                                                                            |
| sub-index .....                                                                                                                                                    | 148, 149                                                                       |
| Keys for \spAfig provide by <b>spintent</b> :                                                                                                                      |                                                                                |
| prefix .....                                                                                                                                                       | 131                                                                            |
| print-sym .....                                                                                                                                                    | 130, 131                                                                       |
| read-arg .....                                                                                                                                                     | 131, 133                                                                       |
| read-cmd .....                                                                                                                                                     | 130, 131                                                                       |
| read-sub .....                                                                                                                                                     | 131                                                                            |
| Keys for \spangle provide by <b>spintent</b> :                                                                                                                     |                                                                                |
| concept .....                                                                                                                                                      | 119                                                                            |
| prefix .....                                                                                                                                                       | 119                                                                            |
| print-m .....                                                                                                                                                      | 119                                                                            |
| read-cmd .....                                                                                                                                                     | 119                                                                            |
| recto .....                                                                                                                                                        | 119                                                                            |
| silent-cmd .....                                                                                                                                                   | 119                                                                            |
| Keys for \sparc provide by <b>spintent</b> :                                                                                                                       |                                                                                |
| concept .....                                                                                                                                                      | 124                                                                            |
| prefix .....                                                                                                                                                       | 124                                                                            |
| read-cmd .....                                                                                                                                                     | 124                                                                            |
| silent-cmd .....                                                                                                                                                   | 124                                                                            |
| Keys for \spcirc provide by <b>spintent</b> :                                                                                                                      |                                                                                |
| prefix .....                                                                                                                                                       | 142                                                                            |
| print-sym .....                                                                                                                                                    | 142–144                                                                        |
| radio-sty .....                                                                                                                                                    | 142, 143                                                                       |
| read-arg .....                                                                                                                                                     | 142                                                                            |
| read-cmd .....                                                                                                                                                     | 142                                                                            |
| read-sty .....                                                                                                                                                     | 142, 143                                                                       |
| read-sub .....                                                                                                                                                     | 143                                                                            |
| sub-index .....                                                                                                                                                    | 142, 143                                                                       |
| Keys for \spcoord provide by <b>spintent</b> :                                                                                                                     |                                                                                |
| silent-cmd .....                                                                                                                                                   | 97                                                                             |
| Keys for \spdiam provide by <b>spintent</b> :                                                                                                                      |                                                                                |
| concept .....                                                                                                                                                      | 139                                                                            |
| read-arg .....                                                                                                                                                     | 139                                                                            |
| read-cmd .....                                                                                                                                                     | 139                                                                            |
| read-sub .....                                                                                                                                                     | 139                                                                            |
| silent-cmd .....                                                                                                                                                   | 139                                                                            |
| sub-index .....                                                                                                                                                    | 139                                                                            |
| upper-case .....                                                                                                                                                   | 139                                                                            |
| Keys for \spdiv provide by <b>spintent</b> :                                                                                                                       |                                                                                |
| cifras-sep .....                                                                                                                                                   | 77                                                                             |
| notation-sym .....                                                                                                                                                 | 77                                                                             |
| read-parens .....                                                                                                                                                  | 77                                                                             |
| read-period-sep .....                                                                                                                                              | 77                                                                             |
| read-sign .....                                                                                                                                                    | 77                                                                             |
| read-sym .....                                                                                                                                                     | 77                                                                             |
| set-sym .....                                                                                                                                                      | 77                                                                             |
| wrap-parens .....                                                                                                                                                  | 77                                                                             |
| Keys for \spdsym provide by <b>spintent</b> :                                                                                                                      |                                                                                |
| prefix .....                                                                                                                                                       | 74                                                                             |
| read-sym .....                                                                                                                                                     | 74                                                                             |
| set-sym .....                                                                                                                                                      | 74                                                                             |
| suffix .....                                                                                                                                                       | 74                                                                             |
| Keys for \spmangle provide by <b>spintent</b> :                                                                                                                    |                                                                                |
| concept .....                                                                                                                                                      | 119                                                                            |
| prefix .....                                                                                                                                                       | 119                                                                            |
| print-m .....                                                                                                                                                      | 119                                                                            |
| read-cmd .....                                                                                                                                                     | 119                                                                            |
| recto .....                                                                                                                                                        | 119                                                                            |
| silent-cmd .....                                                                                                                                                   | 119                                                                            |
| Keys for \spmarc provide by <b>spintent</b> :                                                                                                                      |                                                                                |
| concept .....                                                                                                                                                      | 124                                                                            |
| prefix .....                                                                                                                                                       | 124                                                                            |
| read-cmd .....                                                                                                                                                     | 124                                                                            |
| silent-cmd .....                                                                                                                                                   | 124                                                                            |
| Keys for \spmcd provide by <b>spintent</b> :                                                                                                                       |                                                                                |
| concept .....                                                                                                                                                      | 80                                                                             |
| prefix .....                                                                                                                                                       | 80                                                                             |
| read-cmd .....                                                                                                                                                     | 80                                                                             |
| result .....                                                                                                                                                       | 80                                                                             |
| silent-cmd .....                                                                                                                                                   | 80                                                                             |
| upper-case .....                                                                                                                                                   | 80                                                                             |

|                                                |               |                                                |               |
|------------------------------------------------|---------------|------------------------------------------------|---------------|
| Keys for \spmcmm provide by <b>spintent</b> :  |               | Keys for \spPcirc provide by <b>spintent</b> : |               |
| concept .....                                  | 80            | concept .....                                  | 149           |
| prefix .....                                   | 80            | prefix .....                                   | 148, 149      |
| read-cmd .....                                 | 80            | print-sym .....                                | 148–150       |
| result .....                                   | 80            | radio-sty .....                                | 148           |
| silent-cmd .....                               | 80            | read-arg .....                                 | 148, 149      |
| upper-case .....                               | 80            | read-cmd .....                                 | 148, 149      |
| Keys for \spmoney provide by <b>spintent</b> : |               | read-sty .....                                 | 148, 149      |
| cifras-sep .....                               | 56            | read-sub .....                                 | 149           |
| currency .....                                 | 56            | silent-cmd .....                               | 149           |
| dolar-math .....                               | 56            | sub-index .....                                | 148, 149      |
| sub-units .....                                | 56            | Keys for \spPfig provide by <b>spintent</b> :  |               |
| Keys for \spmQ provide by <b>spintent</b> :    |               | prefix .....                                   | 131           |
| join-word .....                                | 91, 93        | print-sym .....                                | 130, 131      |
| print-dfrac .....                              | 91–95         | read-arg .....                                 | 131, 133      |
| read-parens .....                              | 91, 93        | read-cmd .....                                 | 130, 131      |
| use-spnZ .....                                 | 91, 94, 95    | read-sub .....                                 | 131           |
| wrap-parens .....                              | 91, 92        | Keys for \spPoint provide by <b>spintent</b> : |               |
| Keys for \spmside provide by <b>spintent</b> : |               | concept .....                                  | 101           |
| concept .....                                  | 115           | prefix .....                                   | 101           |
| prefix .....                                   | 114, 115      | read-cmd .....                                 | 101           |
| print-m .....                                  | 114, 115      | silent-cmd .....                               | 101           |
| print-sym .....                                | 139           | Keys for \spPoly provide by <b>spintent</b> :  |               |
| read-arg .....                                 | 115           | concept .....                                  | 128           |
| read-cmd .....                                 | 114, 115, 117 | prefix .....                                   | 127, 128      |
| read-sty .....                                 | 114, 115      | print-sym .....                                | 128           |
| silent-cmd .....                               | 115, 117      | read-cmd .....                                 | 127, 128      |
| Keys for \spmsym provide by <b>spintent</b> :  |               | silent-cmd .....                               | 128           |
| not-print .....                                | 75            | Keys for \sprad provide by <b>spintent</b> :   |               |
| prefix .....                                   | 75            | concept .....                                  | 139           |
| read-sym .....                                 | 75            | read-arg .....                                 | 139           |
| set-sym .....                                  | 75            | read-cmd .....                                 | 139           |
| suffix .....                                   | 75            | read-sub .....                                 | 139           |
| Keys for \spmul provide by <b>spintent</b> :   |               | silent-cmd .....                               | 139           |
| cifras-sep .....                               | 78            | sub-index .....                                | 139           |
| notation-sym .....                             | 78            | upper-case .....                               | 139           |
| read-parens .....                              | 78            | Keys for \sprayo provide by <b>spintent</b> :  |               |
| read-period-sep .....                          | 78            | concept .....                                  | 112           |
| read-sign .....                                | 78            | prefix .....                                   | 112           |
| read-sym .....                                 | 78            | read-cmd .....                                 | 112, 113      |
| set-sym .....                                  | 78            | read-sty .....                                 | 112           |
| wrap-parens .....                              | 78            | silent-cmd .....                               | 112, 113      |
| Keys for \spnD provide by <b>spintent</b> :    |               | Keys for \sprazon provide by <b>spintent</b> : |               |
| concept .....                                  | 83            | inline .....                                   | 86            |
| prefix .....                                   | 83            | silent-cmd .....                               | 86            |
| result-num .....                               | 83            | use-dfrac .....                                | 86            |
| result .....                                   | 83            | Keys for \spread provide by <b>spintent</b> :  |               |
| silent-cmd .....                               | 83            | read-arg .....                                 | 73            |
| Keys for \spnM provide by <b>spintent</b> :    |               | silent-cmd .....                               | 73            |
| concept .....                                  | 83            | Keys for \sprecta provide by <b>spintent</b> : |               |
| prefix .....                                   | 83            | prefix .....                                   | 108           |
| result-num .....                               | 83            | read-cmd .....                                 | 108           |
| result .....                                   | 83            | silent-cmd .....                               | 108           |
| silent-cmd .....                               | 83            | Keys for \spshort provide by <b>spintent</b> : |               |
| Keys for \spnum provide by <b>spintent</b> :   |               | read-arg .....                                 | 62            |
| cifras-sep .....                               | 44–46         | small-caps .....                               | 62            |
| notation-sym .....                             | 45            | Keys for \spside provide by <b>spintent</b> :  |               |
| read-period-sep .....                          | 45, 47        | concept .....                                  | 115           |
| read-sign .....                                | 45            | prefix .....                                   | 114, 115      |
| Keys for \spnZ provide by <b>spintent</b> :    |               | read-arg .....                                 | 114, 115      |
| cifras-sep .....                               | 89            | read-cmd .....                                 | 114, 115, 117 |
| read-parens .....                              | 89, 90        | read-sty .....                                 | 114, 115      |
| read-sign .....                                | 89            | silent-cmd .....                               | 115, 117      |
| wrap-parens .....                              | 89, 90        | Keys for \spsiglo provide by <b>spintent</b> : |               |
|                                                |               | print-era .....                                | 65, 66        |

Keys for \spunit provide by [spintent](#):

|                       |    |
|-----------------------|----|
| cifras-sep .....      | 51 |
| notation-sym .....    | 51 |
| print-cdot .....      | 51 |
| read-cdot .....       | 51 |
| read-period-sep ..... | 51 |
| read-slash .....      | 51 |

Keys for \spusep provide by [spintent](#):

|              |    |
|--------------|----|
| silent ..... | 70 |
| size .....   | 70 |

Keys for \spvec provide by [spintent](#):

|                  |          |
|------------------|----------|
| autofit .....    | 104      |
| concept .....    | 104      |
| prefix .....     | 103, 104 |
| read-cmd .....   | 103, 104 |
| silent-cmd ..... | 104      |

## M

|                            |      |
|----------------------------|------|
| mathcal-fallback .....     | 80   |
| \MathMLintent .....        | 122  |
| mc-invalid-arguments ..... | 2020 |
| md-invalid-argument .....  | 2224 |

Lua module provide by [spintent](#):

|              |                                                                                    |
|--------------|------------------------------------------------------------------------------------|
| spintent.lua | 37, 38, 40, 43-45, 50, 54-57, 61, 62, 64, 66, 67, 69, 80, 83, 88, 91, 95, 101, 138 |
|--------------|------------------------------------------------------------------------------------|

## N

|                    |                      |
|--------------------|----------------------|
| not-print .....    | 1791                 |
| notation-sym ..... | 284, 569, 1865, 1936 |

## P

Packages:

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| expl3 .....            | 43, 45                                                                                   |
| lua-unicode-math ..... | 37                                                                                       |
| luamml .....           | 40, 91, 93                                                                               |
| tagpdf .....           | 61, 66                                                                                   |
| unicode-math .....     | 37, 71                                                                                   |
| prefix                 | 1721, 1791, 2027, 2198, 3024, 3145, 3387, 3607, 3744, 3951, 4236, 4389, 4573, 5179, 5459 |
| print-dfrac .....      | 2562                                                                                     |
| print-era .....        | 1239                                                                                     |
| print-m .....          | 3744, 3951                                                                               |
| print-sym .....        | 4389, 4573, 4961, 5179, 5459                                                             |

## R

|                                |                                                                              |
|--------------------------------|------------------------------------------------------------------------------|
| radio-sty .....                | 5179                                                                         |
| read-arg .....                 | 1109, 1648, 3744, 4573, 4961, 5179, 5459                                     |
| read-cdot .....                | 569                                                                          |
| read-cmd ..                    | 2027, 3024, 3145, 3387, 3607, 3744, 3951, 4236, 4389, 4573, 4961, 5179, 5459 |
| read-parens .....              | 1865, 1936, 2484, 2562                                                       |
| read-period-sep .....          | 284, 569, 1865, 1936                                                         |
| read-sign .....                | 284, 1865, 1936, 2484                                                        |
| read-slash .....               | 569                                                                          |
| read-sty .....                 | 3607, 3744, 5179, 5459                                                       |
| read-sub .....                 | 4573, 4961, 5179, 5459                                                       |
| read-sym .....                 | 1721, 1791, 1865, 1936                                                       |
| recto .....                    | 3951                                                                         |
| result .....                   | 2027, 2198                                                                   |
| result-num .....               | 2198                                                                         |
| right-angle-sqr-fallback ..... | 80                                                                           |
| \rightangle .....              | 120                                                                          |

## S

seq commands:

|                              |    |
|------------------------------|----|
| \seq_set_from_clist:Nn ..... | 62 |
|------------------------------|----|

|                                       |                                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------------------------|
| set-sym .....                         | 1721, 1791, 1865, 1936                                                                               |
| \setspintent .....                    | 3, 41, 191                                                                                           |
| silent .....                          | 1509                                                                                                 |
| silent-cmd                            | 1648, 2027, 2198, 2346, 2776, 3024, 3145, 3387, 3607, 3744, 3951, 4236, 4389, 4573, 4961, 5179, 5459 |
| size .....                            | 1509                                                                                                 |
| small-caps .....                      | 1109                                                                                                 |
| \spAcirc .....                        | 30, 148, 5741                                                                                        |
| spAcirc-invalid-arg .....             | 5450                                                                                                 |
| \spAfig .....                         | 25, 130, 4927                                                                                        |
| spAfig-invalid-arg .....              | 4535                                                                                                 |
| spAfig-invalid-points .....           | 4535                                                                                                 |
| spAfig-read-arg-key .....             | 4535                                                                                                 |
| spAfig-sym-cardinality-conflict ..... | 4535                                                                                                 |
| spAfig-sym-letra-conflict .....       | 4535                                                                                                 |
| \spang .....                          | 23, 88, 2474                                                                                         |
| \spangle .....                        | 22, 119, 4199                                                                                        |
| spangle-invalid-arg .....             | 3941                                                                                                 |
| \sparc .....                          | 24, 124, 4358                                                                                        |
| sparc-invalid-points .....            | 4226                                                                                                 |
| \spcirc .....                         | 29, 142, 5426                                                                                        |
| spcirc-invalid-arg .....              | 5172                                                                                                 |
| \spcoord .....                        | 18, 96, 2913                                                                                         |
| spcoord-invalid-arg .....             | 2772                                                                                                 |
| \spdate .....                         | 11, 69, 1485                                                                                         |
| \spdiam .....                         | 28, 138, 5141                                                                                        |
| \spdiv .....                          | 13, 77, 1926                                                                                         |
| spdiv-empty-argument .....            | 1885                                                                                                 |
| \spdsym .....                         | 12, 74, 1768                                                                                         |

spintent internal commands:

|                                                                                 |         |
|---------------------------------------------------------------------------------|---------|
| \_spintent_collapse_hyphens:N .....                                             | 128     |
| \l\_spintent_command_name_str .....                                             | 203     |
| \l\_spintent_geo_circfig_mode_str ....                                          | 5449    |
| \l\_spintent_geo_final_print_tl .....                                           | 4527    |
| \l\_spintent_geo_luaset_angulo_case_-str .....                                  | 2745    |
| \l\_spintent_geo_luaset_circ_radio_raw_-str .....                               | 5161    |
| \l\_spintent_geo_luaset_circ_radio_type_-str .....                              | 5161    |
| \l\_spintent_geo_luaset_concept_str ...                                         | 2745    |
| \l\_spintent_geo_luaset_error_str ....                                          | 2745    |
| \l\_spintent_geo_luaset_intent_str ...                                          | 2745    |
| \l\_spintent_geo_luaset_letra_str ....                                          | 2745    |
| \l\_spintent_geo_luaset_letra_word_str                                          | 2745    |
| \l\_spintent_geo_luaset_nombre_es_num_-str .....                                | 2745    |
| \l\_spintent_geo_luaset_points_-clist\l\_spintent_geo_luaset_is_-name_str ..... | 2745    |
| \l\_spintent_geo_luaset_poly_sym_str ..                                         | 2745    |
| \l\_spintent_geo_luaset_print_tl .....                                          | 2745    |
| \l\_spintent_geo_luaset_sub_spoken_str                                          | 2745    |
| \l\_spintent_geo_luaset_sub_str .....                                           | 2745    |
| \l\_spintent_geo_luaset_tipo_str .....                                          | 2745    |
| \l\_spintent_geo_print_sym_str .....                                            | 4643    |
| \_spintent_inner_sep: .....                                                     | 120     |
| \_spintent_invisible_sep: .....                                                 | 116     |
| \c\_spintent_invisible_sep_int .....                                            | 124     |
| \c\_spintent_invisible_sep_tl .....                                             | 93      |
| \_spintent_luafun_clean_split_and_set:n                                         | 76, 241 |

|                                                                                                      |                             |                                                           |                      |
|------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------|----------------------|
| <code>\__spintent_luafun_mrow_intent_arg:nn</code> .                                                 | <a href="#">116</a>         | <code>\__spintent_setspintent_parse_args:nn</code> .      | <a href="#">174</a>  |
| <code>\__spintent_luafun_set_debug_ghost:n</code> ...                                                | <a href="#">71</a>          | <code>\__spintent_setspintent_set_keys:nn</code> ...      | <a href="#">163</a>  |
| <code>\__spintent_luafun_spang_parse:n</code> . . . . .                                              | <a href="#">88</a>          | <code>\l__spintent_smparc_prefix_tl</code> . . . . .      | <a href="#">4221</a> |
| <code>\__spintent_luafun_sdate_parse:n</code> . . . . .                                              | <a href="#">69</a>          | <code>\__spintent_sp_rad_diam_exec:nn</code> . . . . .    | <a href="#">5136</a> |
| <code>\__spintent_luafun_spmoney_lookup_data:n</code>                                                | <a href="#">76</a>          | <code>\l__spintent_sp_rad_diam_letra_tl</code> . . . . .  | <a href="#">4959</a> |
| <code>\__spintent_luafun_spmQ_scale_int:nn</code> ...                                                | <a href="#">76</a>          | <code>\__spintent_sp_rad_diam_print:n</code> . . . . .    | <a href="#">5008</a> |
| <code>\__spintent_luafun_sptime_parse:n</code> . . . . .                                             | <a href="#">68</a>          | <code>\l__spintent_sp_rad_diam_word_tl</code> . . . . .   | <a href="#">4959</a> |
| <code>\__spintent_luafun_spunit_lookup_alias:n</code>                                                | <a href="#">76</a>          | <code>\l__spintent_spAcirc_mode_str</code> . . . . .      | <a href="#">5439</a> |
| <code>\l__spintent_luaset_arg_above_tl</code> . . . . .                                              | <a href="#">260</a>         | <code>\l__spintent_spAcirc_prefix_tl</code> . . . . .     | <a href="#">5439</a> |
| <code>\l__spintent_luaset_arg_below_tl</code> . . . . .                                              | <a href="#">260</a>         | <code>\l__spintent_spAcirc_print_sym_str</code> . . . . . | <a href="#">5439</a> |
| <code>\l__spintent_luaset_cal_fam_ok_str</code> . . . . .                                            | <a href="#">67</a>          | <code>\l__spintent_spAcirc_radio_sty_str</code> . . . . . | <a href="#">5439</a> |
| <code>\l__spintent_luaset_dec_is_all_zeros-</code><br><code>str</code> . . . . .                     | <a href="#">253</a>         | <code>\l__spintent_spAcirc_read_arg_tl</code> . . . . .   | <a href="#">5439</a> |
| <code>\l__spintent_luaset_dec_sep_tl</code> . . . . .                                                | <a href="#">241</a>         | <code>\l__spintent_spAcirc_read_sty_str</code> . . . . .  | <a href="#">5439</a> |
| <code>\l__spintent_luaset_decimal_period_str</code>                                                  | <a href="#">253</a>         | <code>\l__spintent_spAcirc_sub_index_str</code> . . . . . | <a href="#">5439</a> |
| <code>\l__spintent_luaset_denom_has_number-</code><br><code>str</code> . . . . .                     | <a href="#">260</a>         | <code>\l__spintent_spafig_mode_str</code> . . . . .       | <a href="#">4528</a> |
| <code>\l__spintent_luaset_exponent_tl</code> . . . . .                                               | <a href="#">253</a>         | <code>\l__spintent_spafig_prefix_tl</code> . . . . .      | <a href="#">4528</a> |
| <code>\l__spintent_luaset_format_part_dec_str</code>                                                 | <a href="#">248</a>         | <code>\l__spintent_spafig_print_sym_str</code> . . . . .  | <a href="#">4528</a> |
| <code>\l__spintent_luaset_format_part_int_str</code>                                                 | <a href="#">248</a>         | <code>\l__spintent_spafig_read_arg_tl</code> . . . . .    | <a href="#">4528</a> |
| <code>\l__spintent_luaset_has_exponent_str</code> ..                                                 | <a href="#">253</a>         | <code>\l__spintent_spang_luaset_error_str</code> ...      | <a href="#">2430</a> |
| <code>\l__spintent_luaset_has_integer_str</code> ..                                                  | <a href="#">94, 95, 241</a> | <code>\l__spintent_spang_luaset_grado_str</code> ...      | <a href="#">2430</a> |
| <code>\l__spintent_luaset_has_natural_str</code> ...                                                 | <a href="#">241</a>         | <code>\l__spintent_spang_luaset_minuto_str</code> ..      | <a href="#">2430</a> |
| <code>\l__spintent_luaset_has_units_str</code> . . . . .                                             | <a href="#">260</a>         | <code>\l__spintent_spang_luaset_segundo_str</code> .      | <a href="#">2430</a> |
| <code>\l__spintent_luaset_millions_str</code> . . . . .                                              | <a href="#">253</a>         | <code>\__spintent_spang_process:n</code> . . . . .        | <a href="#">2439</a> |
| <code>\l__spintent_luaset_not_decimal_not-</code><br><code>period_str</code> . . . . .               | <a href="#">253</a>         | <code>\__spintent_spangle_exec:nnn</code> . . . . .       | <a href="#">4194</a> |
| <code>\l__spintent_luaset_not_decimal_period-</code><br><code>str</code> . . . . .                   | <a href="#">253</a>         | <code>\l__spintent_spangle_intent_arg_tl</code> . . . . . | <a href="#">4010</a> |
| <code>\l__spintent_luaset_only_part_dec_str</code> .                                                 | <a href="#">248</a>         | <code>\l__spintent_spangle_intent_tl</code> . . . . .     | <a href="#">4010</a> |
| <code>\l__spintent_luaset_only_part_int_str</code> .                                                 | <a href="#">248</a>         | <code>\__spintent_spangle_mathml:n</code> . . . . .       | <a href="#">4052</a> |
| <code>\l__spintent_luaset_only_part_period-</code><br><code>str</code> . . . . .                     | <a href="#">248</a>         | <code>\l__spintent_spangle_mode_str</code> . . . . .      | <a href="#">3936</a> |
| <code>\l__spintent_luaset_part_dec_tl</code> . . . . .                                               | <a href="#">241</a>         | <code>\l__spintent_spangle_prefix_tl</code> . . . . .     | <a href="#">3936</a> |
| <code>\l__spintent_luaset_part_int_tl</code> . . . . .                                               | <a href="#">93, 241</a>     | <code>\l__spintent_spangle_print_m_now_bool</code> .      | <a href="#">4010</a> |
| <code>\l__spintent_luaset_part_period_tl</code> . . . . .                                            | <a href="#">241</a>         | <code>\__spintent_spangle_process:nn</code> . . . . .     | <a href="#">4017</a> |
| <code>\l__spintent_luaset_scope_attr_str</code> . . . . .                                            | <a href="#">67</a>          | <code>\l__spintent_spangle_pt_idx_int</code> . . . . .    | <a href="#">4010</a> |
| <code>\l__spintent_luaset_sign_tl</code> . . . . .                                                   | <a href="#">94, 241</a>     | <code>\l__spintent_spangle_pt_seq</code> . . . . .        | <a href="#">4010</a> |
| <code>\l__spintent_luaset_status_str</code> . . . . .                                                | <a href="#">260</a>         | <code>\l__spintent_spangle_refs_tl</code> . . . . .       | <a href="#">4010</a> |
| <code>\l__spintent_luaset_sym_fam_ok_str</code> . . . . .                                            | <a href="#">67</a>          | <code>\__spintent_spangle_sym:n</code> . . . . .          | <a href="#">3991</a> |
| <code>\__spintent_mathcal:n</code> . . . . .                                                         | <a href="#">94</a>          | <code>\__spintent_sparc_exec:nnn</code> . . . . .         | <a href="#">4353</a> |
| <code>\__spintent_mathmlarg_point:nnn</code> . . . . .                                               | <a href="#">2928</a>        | <code>\l__spintent_sparc_intent_arg_tl</code> . . . . .   | <a href="#">4280</a> |
| <code>\l__spintent_mathmlarg_point_kv_tl</code> . . . . .                                            | <a href="#">2924</a>        | <code>\l__spintent_sparc_intent_tl</code> . . . . .       | <a href="#">4280</a> |
| <code>\l__spintent_mathmlarg_point_mode_str</code> .                                                 | <a href="#">2924</a>        | <code>\__spintent_sparc_mathml:n</code> . . . . .         | <a href="#">4298</a> |
| <code>\l__spintent_mathmlarg_point_ref_tl</code> ...                                                 | <a href="#">2924</a>        | <code>\l__spintent_sparc_mode_str</code> . . . . .        | <a href="#">4221</a> |
| <code>\l__spintent_mathmlarg_point_silent-</code><br><code>bool</code> . . . . .                     | <a href="#">2924</a>        | <code>\l__spintent_sparc_prefix_tl</code> . . . . .       | <a href="#">4221</a> |
| <code>\l__spintent_mc_args_seq</code> . . . . .                                                      | <a href="#">2007</a>        | <code>\__spintent_sparc_print_arc:n</code> . . . . .      | <a href="#">4274</a> |
| <code>\l__spintent_mc_luaset_all_numeric_str</code>                                                  | <a href="#">2007</a>        | <code>\__spintent_sparc_process:nn</code> . . . . .       | <a href="#">4285</a> |
| <code>\l__spintent_mc_luaset_types_str</code> . . . . .                                              | <a href="#">2007</a>        | <code>\l__spintent_sparc_pt_idx_int</code> . . . . .      | <a href="#">4280</a> |
| <code>\__spintent_mc_process_args:nn</code> . . . . .                                                | <a href="#">2068</a>        | <code>\l__spintent_sparc_pt_seq</code> . . . . .          | <a href="#">4280</a> |
| <code>\l__spintent_mc_raw_args_seq</code> . . . . .                                                  | <a href="#">2007</a>        | <code>\l__spintent_sparc_refs_tl</code> . . . . .         | <a href="#">4280</a> |
| <code>\l__spintent_mc_types_seq\__spintent-</code><br><code>spmcd_print_result_bool</code> . . . . . | <a href="#">2007</a>        | <code>\__spintent_sparc_wide_arc:n</code> . . . . .       | <a href="#">4270</a> |
| <code>\l__spintent_md_out_seq</code> . . . . .                                                       | <a href="#">2189</a>        | <code>\l__spintent_spcirc_articulo_str</code> . . . . .   | <a href="#">5163</a> |
| <code>\__spintent_print_spunit_hierarchy:</code> ...                                                 | <a href="#">670</a>         | <code>\__spintent_spcirc_build_concept:</code> . . . . .  | <a href="#">5307</a> |
| <code>\__spintent_right_angle_sqr:</code> . . . . .                                                  | <a href="#">107</a>         | <code>\l__spintent_spcirc_connector_arg_tl</code> ..      | <a href="#">5255</a> |
| <code>\__spintent_sanitize_affix:Nn</code> . . . . .                                                 | <a href="#">136</a>         | <code>\l__spintent_spcirc_connector_tl</code> . . . . .   | <a href="#">5255</a> |
| <code>\l__spintent_saved_luamml_flag_int</code> . . . . .                                            | <a href="#">93, 2543</a>    | <code>\__spintent_spcirc_exec:nn</code> . . . . .         | <a href="#">5421</a> |
| <code>\l__spintent_scope_attr_int</code> . . . . .                                                   | <a href="#">74</a>          | <code>\l__spintent_spcirc_has_arg_bool</code> . . . . .   | <a href="#">5255</a> |
| <code>\l__spintent_setspintent_input_arg_cmd-</code><br><code>tl</code> . . . . .                    | <a href="#">154</a>         | <code>\l__spintent_spcirc_intent_arg_tl</code> . . . . .  | <a href="#">5255</a> |
|                                                                                                      |                             | <code>\l__spintent_spcirc_intent_tl</code> . . . . .      | <a href="#">5255</a> |
|                                                                                                      |                             | <code>\__spintent_spcirc_mathml:</code> . . . . .         | <a href="#">5376</a> |
|                                                                                                      |                             | <code>\l__spintent_spcirc_mode_str</code> . . . . .       | <a href="#">5163</a> |
|                                                                                                      |                             | <code>\l__spintent_spcirc_prefix_tl</code> . . . . .      | <a href="#">5163</a> |
|                                                                                                      |                             | <code>\l__spintent_spcirc_print_sym_str</code> . . . . .  | <a href="#">5163</a> |
|                                                                                                      |                             | <code>\__spintent_spcirc_process:n</code> . . . . .       | <a href="#">5360</a> |
|                                                                                                      |                             | <code>\l__spintent_spcirc_radio_arg_tl</code> . . . . .   | <a href="#">5255</a> |

|                                                                                  |                          |                                                                                     |                      |
|----------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------|----------------------|
| <code>\__spintrent_spcirc_radio_print:</code> .....                              | <a href="#">5292</a>     | <code>\__spintrent_spfig_build_refs:</code> .....                                   | <a href="#">4787</a> |
| <code>\l__spintrent_spcirc_radio_sty_str</code> ....                             | <a href="#">5163</a>     | <code>\__spintrent_spfig_build_simple:n</code> ....                                 | <a href="#">4676</a> |
| <code>\l__spintrent_spcirc_read_sty_str</code> ....                              | <a href="#">5163</a>     | <code>\__spintrent_spfig_exec:nnn</code> .....                                      | <a href="#">4922</a> |
| <code>\l__spintrent_spcirc_sub_index_str</code> ....                             | <a href="#">5163</a>     | <code>\l__spintrent_spfig_intent_arg_tl</code> ....                                 | <a href="#">4643</a> |
| <code>\__spintrent_spcirc_sym:</code> .....                                      | <a href="#">5261</a>     | <code>\l__spintrent_spfig_intent_tl</code> .....                                    | <a href="#">4643</a> |
| <code>\__spintrent_spcirc_word_and_article:</code> ..                            | <a href="#">5271</a>     | <code>\__spintrent_spfig_mathml:n</code> .....                                      | <a href="#">4882</a> |
| <code>\l__spintrent_spcirc_word_tl</code> .....                                  | <a href="#">5171</a>     | <code>\l__spintrent_spfig_op_tl</code> .....                                        | <a href="#">4643</a> |
| <code>\__spintrent_spcircfig_build_concept:nn</code>                             | <a href="#">5603</a>     | <code>\__spintrent_spfig_points_loop:n</code> .....                                 | <a href="#">4870</a> |
| <code>\__spintrent_spcircfig_build_short:n</code> ...                            | <a href="#">5565</a>     | <code>\__spintrent_spfig_process:nn</code> .....                                    | <a href="#">4652</a> |
| <code>\__spintrent_spcircfig_del_word:n</code> ....                              | <a href="#">5554</a>     | <code>\l__spintrent_spfig_pt_idx_int</code> .....                                   | <a href="#">4643</a> |
| <code>\__spintrent_spcircfig_exec:nnn</code> .....                               | <a href="#">5736</a>     | <code>\l__spintrent_spfig_pt_seq</code> .....                                       | <a href="#">4643</a> |
| <code>\__spintrent_spcircfig_finish_intent:nn</code>                             | <a href="#">5587</a>     | <code>\l__spintrent_spfig_refs_tl</code> .....                                      | <a href="#">4643</a> |
| <code>\l__spintrent_spcircfig_intent_arg_tl</code> ..                            | <a href="#">5525</a>     | <code>\__spintrent_spfig_sym:</code> .....                                          | <a href="#">4635</a> |
| <code>\l__spintrent_spcircfig_intent_tl</code> ....                              | <a href="#">5525</a>     | <code>\l__spintrent_spfig_symbol_tl</code> .....                                    | <a href="#">4643</a> |
| <code>\__spintrent_spcircfig_mathml:nn</code> .....                              | <a href="#">5695</a>     | <code>\l__spintrent_spmangle_mode_str</code> .....                                  | <a href="#">3936</a> |
| <code>\l__spintrent_spcircfig_op_tl</code> .....                                 | <a href="#">5525</a>     | <code>\l__spintrent_spmangle_prefix_tl</code> .....                                 | <a href="#">3936</a> |
| <code>\__spintrent_spcircfig_process:nn</code> ....                              | <a href="#">5650</a>     | <code>\l__spintrent_spmarc_mode_str</code> .....                                    | <a href="#">4221</a> |
| <code>\l__spintrent_spcircfig_short_bool</code> ....                             | <a href="#">5525</a>     | <code>\__spintrent_spmc_process:nnnn</code> .....                                   | <a href="#">2103</a> |
| <code>\__spintrent_spcircfig_sym:n</code> .....                                  | <a href="#">5530</a>     | <code>\l__spintrent_spmcd_prefix_tl</code> .....                                    | <a href="#">2007</a> |
| <code>\__spintrent_spcircfig_word_and_-</code><br><code>article:n</code> .....   | <a href="#">5536</a>     | <code>\l__spintrent_spmcd_print_upper_bool</code> ...                               | <a href="#">2007</a> |
| <code>\__spintrent_spcoord_exec:nn</code> .....                                  | <a href="#">2895</a>     | <code>\l__spintrent_spmcd_read_terse_bool</code> ...                                | <a href="#">2007</a> |
| <code>\__spintrent_spcoord_intent_infix:</code> ....                             | <a href="#">2866</a>     | <code>\l__spintrent_spmcd_suffix_tl</code> .....                                    | <a href="#">2007</a> |
| <code>\l__spintrent_spcoord_intent_str</code> .....                              | <a href="#">2768</a>     | <code>\l__spintrent_spmcm_prefix_tl</code> .....                                    | <a href="#">2007</a> |
| <code>\l__spintrent_spcoord_luaset_error_str</code> .                            | <a href="#">2759</a>     | <code>\l__spintrent_spmcm_print_result_bool</code> ..                               | <a href="#">2007</a> |
| <code>\l__spintrent_spcoord_luaset_sep_str</code> ...                            | <a href="#">2759</a>     | <code>\l__spintrent_spmcm_print_upper_bool</code> ...                               | <a href="#">2007</a> |
| <code>\l__spintrent_spcoord_luaset_x_has_frac_-</code><br><code>str</code> ..... | <a href="#">2759</a>     | <code>\l__spintrent_spmcm_read_terse_bool</code> ...                                | <a href="#">2007</a> |
| <code>\l__spintrent_spcoord_luaset_x_sign_str</code>                             | <a href="#">2759</a>     | <code>\l__spintrent_spmcm_spmcd_luaset_error_-</code><br><code>str</code> .....     | <a href="#">2007</a> |
| <code>\l__spintrent_spcoord_luaset_x_type_str</code>                             | <a href="#">2759</a>     | <code>\l__spintrent_spmcm_suffix_tl</code> .....                                    | <a href="#">2007</a> |
| <code>\l__spintrent_spcoord_luaset_x_value_-</code><br><code>str</code> .....    | <a href="#">2759</a>     | <code>\__spintrent_spmoney_build_tree:</code> .....                                 | <a href="#">1021</a> |
| <code>\l__spintrent_spcoord_luaset_y_has_frac_-</code><br><code>str</code> ..... | <a href="#">2759</a>     | <code>\__spintrent_spmoney_check_sub_units:</code> ..                               | <a href="#">974</a>  |
| <code>\l__spintrent_spcoord_luaset_y_sign_str</code>                             | <a href="#">2759</a>     | <code>\l__spintrent_spmoney_extracted_curr_str</code>                               | <a href="#">803</a>  |
| <code>\l__spintrent_spcoord_luaset_y_type_str</code>                             | <a href="#">2759</a>     | <code>\l__spintrent_spmoney_extracted_num_tl</code> .                               | <a href="#">803</a>  |
| <code>\l__spintrent_spcoord_luaset_y_value_-</code><br><code>str</code> .....    | <a href="#">2759</a>     | <code>\__spintrent_spmoney_intent_int:</code> .....                                 | <a href="#">939</a>  |
| <code>\__spintrent_spcoord_mathml:</code> .....                                  | <a href="#">2878</a>     | <code>\l__spintrent_spmoney_intent_money_str</code> .                               | <a href="#">803</a>  |
| <code>\__spintrent_spcoord_print:n</code> .....                                  | <a href="#">2843</a>     | <code>\l__spintrent_spmoney_intent_sign_str</code> ..                               | <a href="#">803</a>  |
| <code>\l__spintrent_spcoord_raw_arg_tl</code> ....                               | <a href="#">2768</a>     | <code>\l__spintrent_spmoney_intent_sub_unit_-</code><br><code>str</code> .....      | <a href="#">803</a>  |
| <code>\__spintrent_spcoord_set_wrap_parens_-</code><br><code>size:</code> .....  | <a href="#">2787</a>     | <code>\l__spintrent_spmoney_luaset_conde_str</code> .                               | <a href="#">793</a>  |
| <code>\__spintrent_spcoord_value:n</code> .....                                  | <a href="#">2834</a>     | <code>\l__spintrent_spmoney_luaset_currency_arg_-</code><br><code>str</code> .....  | <a href="#">793</a>  |
| <code>\l__spintrent_spcoord_wrap_parens_l_tl</code> .                            | <a href="#">2768</a>     | <code>\l__spintrent_spmoney_luaset_plur_str</code> ..                               | <a href="#">793</a>  |
| <code>\l__spintrent_spcoord_wrap_parens_r_tl</code> .                            | <a href="#">2768</a>     | <code>\l__spintrent_spmoney_luaset_position_-</code><br><code>str</code> .....      | <a href="#">793</a>  |
| <code>\l__spintrent_sptime_luaset_error_str</code> ..                            | <a href="#">1459</a>     | <code>\l__spintrent_spmoney_luaset_print_iso_-</code><br><code>str</code> .....     | <a href="#">793</a>  |
| <code>\l__spintrent_sptime_luaset_output_str</code> .                            | <a href="#">1459</a>     | <code>\l__spintrent_spmoney_luaset_sing_str</code> ..                               | <a href="#">793</a>  |
| <code>\__spintrent_sptime_process:n</code> .....                                 | <a href="#">70, 1470</a> | <code>\l__spintrent_spmoney_luaset_status_str</code>                                | <a href="#">793</a>  |
| <code>\l__spintrent_spdiam_mode_str</code> .....                                 | <a href="#">4953</a>     | <code>\l__spintrent_spmoney_luaset_subplur_str</code>                               | <a href="#">793</a>  |
| <code>\l__spintrent_spdiam_sub_index_str</code> ....                             | <a href="#">4953</a>     | <code>\l__spintrent_spmoney_luaset_subsing_str</code>                               | <a href="#">793</a>  |
| <code>\l__spintrent_spdiam_upper_case_bool</code> ...                            | <a href="#">4953</a>     | <code>\l__spintrent_spmoney_luaset_symbol_str</code>                                | <a href="#">793</a>  |
| <code>\__spintrent_spdiv_process:nnn</code> .....                                | <a href="#">1911</a>     | <code>\__spintrent_spmoney_normalize_key:n</code> ...                               | <a href="#">828</a>  |
| <code>\__spintrent_spdiv_render_op:nn</code> .....                               | <a href="#">1890</a>     | <code>\__spintrent_spmoney_parse_and_route:n</code> .                               | <a href="#">1062</a> |
| <code>\__spintrent_spdiv_render_parens:nn</code> ...                             | <a href="#">1899</a>     | <code>\__spintrent_spmoney_print_dec_sub_-</code><br><code>unit:</code> .....       | <a href="#">1021</a> |
| <code>\l__spintrent_spdsym_prefix_tl</code> .....                                | <a href="#">1717</a>     | <code>\__spintrent_spmoney_print_int:</code> .....                                  | <a href="#">910</a>  |
| <code>\l__spintrent_spdsym_read_sym_tl</code> .....                              | <a href="#">1717</a>     | <code>\__spintrent_spmoney_print_sep_con:</code> ...                                | <a href="#">1021</a> |
| <code>\l__spintrent_spdsym_suffix_tl</code> .....                                | <a href="#">1717</a>     | <code>\__spintrent_spmoney_print_symbol:</code> ....                                | <a href="#">876</a>  |
| <code>\l__spintrent_spdsym_symbol_tl</code> .....                                | <a href="#">1717</a>     | <code>\__spintrent_spmoney_render_layout:</code> ...                                | <a href="#">889</a>  |
| <code>\__spintrent_spfig_build_concept_-</code><br><code>vertices:n</code> ..... | <a href="#">4800</a>     | <code>\__spintrent_spmoney_render_number_node:</code>                               | <a href="#">901</a>  |
|                                                                                  |                          | <code>\__spintrent_spmoney_render_only_integer_-</code><br><code>node:</code> ..... | <a href="#">901</a>  |

|                                           |             |                                             |         |
|-------------------------------------------|-------------|---------------------------------------------|---------|
| \\__spintent_spmoney_res_main_voice_str   | 803         | \\__spintent_spmoney_print_num_start: ...   | 446     |
| \\__spintent_spmoney_set_and_print_sub-   |             | \\__spintent_spmoney_print_number_core: ..  | 446     |
| units: .....                              | 974         | \\__spintent_spmoney_print_part_dec: ....   | 425     |
| \\__spintent_spmoney_set_currency:n ...   | 828         | \\__spintent_spmoney_print_part_period: ..  | 425     |
| \\__spintent_spmoney_sub_units_bool ...   | 803         | \\__spintent_spmoney_print_period_bar: ...  | 354     |
| \\__spintent_spmoney_subnum_int .....     | 803         | \\__spintent_spmoney_read_period_sep_str .  | 45,     |
| \\__spintent_spmoney_validate_money: ...  | 860         | 279                                         |         |
| \\__spintent_spmQ_exec:nnnn .....         | 2723        | \\__spintent_spmoney_read_terse_bool .      | 45, 279 |
| \\__spintent_spmQ_join_word_str ...       | 93, 2543    | \\__spintent_spmoney_render_dec: .....      | 337     |
| \\__spintent_spmQ_print:nnn .....         | 2715        | \\__spintent_spmoney_render_decimal_sep: .  | 391     |
| \\__spintent_spmQ_print_dfrac_bool ....   | 94          | \\__spintent_spmoney_render_int: .....      | 33      |
| \\__spintent_spmQ_print_scale_int: ....   | 2661        | \\__spintent_spmoney_render_only_period: .  | 354     |
| \\__spintent_spmQ_print_scale_int:n ...   | 2670        | \\__spintent_spmoney_render_part:n .....    | 337     |
| \\__spintent_spmQ_print_spnZ_dfrac:nn ..  | 2698        | \\__spintent_spmoney_render_period_sep: ..  | 354     |
| \\__spintent_spmQ_set_join_word: .....    | 2652        | \\__spintent_spnz_process:nn .....          | 2517    |
| \\__spintent_spmQ_set_wrap_parens_size:   | 2597        | \\__spintent_spnz_render_parens: .....      | 2505    |
| \\__spintent_spmQ_start_wrap_parens: ...  | 2632        | \\__spintent_spnz_circ_mode_str .....       | 5439    |
| \\__spintent_spmQ_stop_wrap_parens: ...   | 2632        | \\__spintent_spnz_circ_prefix_tl .....      | 5439    |
| \\__spintent_spmQ_use_spnZ_bool .....     | 94          | \\__spintent_spnz_circ_print_sym_str ...    | 5439    |
| \\__spintent_spm_side_mode_str .....      | 3727        | \\__spintent_spnz_circ_radio_sty_str ...    | 5439    |
| \\__spintent_spm_side_prefix_tl .....     | 3727        | \\__spintent_spnz_circ_read_arg_tl ....     | 5439    |
| \\__spintent_spm_side_read_arg_tl ....    | 3727        | \\__spintent_spnz_circ_read_sty_str ....    | 5439    |
| \\__spintent_spm_side_read_sty_str ....   | 3727        | \\__spintent_spnz_circ_sub_index_str ...    | 5439    |
| \\__spintent_spm_sym_prefix_tl .....      | 1787        | \\__spintent_spnz_fig_mode_str .....        | 4528    |
| \\__spintent_spm_sym_read_sym_tl ....     | 1787        | \\__spintent_spnz_fig_prefix_tl .....       | 4528    |
| \\__spintent_spm_sym_suffix_tl .....      | 1787        | \\__spintent_spnz_fig_print_sym_str ....    | 4528    |
| \\__spintent_spm_sym_symbol_tl .....      | 1787        | \\__spintent_spnz_fig_read_arg_tl ....      | 4528    |
| \\__spintent_spmul_process:nnn .....      | 1982        | \\__spintent_spnz_point_exec:nnn .....      | 3099    |
| \\__spintent_spmul_render_op:nn .....     | 1961        | \\__spintent_spnz_point_intent_arg_tl ...   | 3013    |
| \\__spintent_spmul_render_parens:nn ...   | 1970        | \\__spintent_spnz_point_mathml:n .....      | 3090    |
| \\__spintent_spnD_luaset_is_truncated_-   |             | \\__spintent_spnz_point_mode_str .....      | 3013    |
| str .....                                 | 2189        | \\__spintent_spnz_point_prefix_tl .....     | 3013    |
| \\__spintent_spnD_luaset_result_tl ...    | 2189        | \\__spintent_spnz_point_print:n .....       | 3055    |
| \\__spintent_spnD_prefix_tl .....         | 2189        | \\__spintent_spnz_poly_build_concept: ....  | 4448    |
| \\__spintent_spnD_print_result_num_tl .   | 2189        | \\__spintent_spnz_poly_build_refs: .....    | 4435    |
| \\__spintent_spnM_luaset_is_truncated_-   |             | \\__spintent_spnz_poly_exec:nn .....        | 4511    |
| str .....                                 | 2189        | \\__spintent_spnz_poly_intent_arg_tl ....   | 4423    |
| \\__spintent_spnM_luaset_result_tl ...    | 2189        | \\__spintent_spnz_poly_mathml: .....        | 4491    |
| \\__spintent_spnM_prefix_tl .....         | 2189        | \\__spintent_spnz_poly_mode_str .....       | 4380    |
| \\__spintent_spnM_print_result_num_tl .   | 2189        | \\__spintent_spnz_poly_points_loop: .....   | 4469    |
| \\__spintent_spnM_spnD_exec:nnnn .....    | 2278        | \\__spintent_spnz_poly_prefix_tl .....      | 4380    |
| \\__spintent_spnM_spnD_intent:nn .....    | 2236        | \\__spintent_spnz_poly_process:n .....      | 4481    |
| \\__spintent_spnM_spnD_luaset_error_-     |             | \\__spintent_spnz_poly_pt_idx_int .....     | 4423    |
| str .....                                 | 2189        | \\__spintent_spnz_poly_pt_seq .....         | 4423    |
| \\__spintent_spnM_spnD_process:nnnn ...   | 2231        | \\__spintent_spnz_poly_refs_tl .....        | 4423    |
| \\__spintent_spnM_spnD_result:n .....     | 2253        | \\__spintent_spnz_poly_sym: .....           | 4427    |
| \\__spintent_spnM_spnD_silent:nn .....    | 2245        | \\__spintent_spnz_sprad_mode_str .....      | 4953    |
| \\__spintent_spnmd_luaset_is_number_-     |             | \\__spintent_spnz_sprad_sub_index_str ....  | 4953    |
| str .....                                 | 2189        | \\__spintent_spnz_sprad_upper_case_bool ... | 4953    |
| \\__spintent_spnQ_intent_read_sign_str    | 2543        | \\__spintent_spnz_sprayo_exec:nn .....      | 3711    |
| \\__spintent_spnQ_read_parens_bool ....   | 93          | \\__spintent_spnz_sprayo_intent_arg_tl .... | 3651    |
| \\__spintent_spnQ_wrap_parens_l_tl ..     | 92, 93,     | \\__spintent_spnz_sprayo_mathml: .....      | 3664    |
| 2543                                      |             | \\__spintent_spnz_sprayo_mode_str .....     | 3597    |
| \\__spintent_spnQ_wrap_parens_r_tl ..     | 92, 93,     | \\__spintent_spnz_sprayo_prefix_tl .....    | 3597    |
| 2543                                      |             | \\__spintent_spnz_sprayo_print:n .....      | 3655    |
| \\__spintent_spmnum__notation_sym_tl .... | 45          | \\__spintent_spnz_sprayo_pt_idx_int .....   | 3651    |
| \\__spintent_spmnum_intent_read_dec_sep_- |             | \\__spintent_spnz_sprayo_pt_seq .....       | 3651    |
| str .....                                 | 45, 47, 279 | \\__spintent_spnz_sprayo_read_sty_str ....  | 3597    |
| \\__spintent_spmnum_intent_read_sign_str  | 45,         | \\__spintent_spnz_sprayo_refs_tl .....      | 3651    |
| 279                                       |             | \\__spintent_spnz_sprazon_exec:nnn .....    | 2363    |
| \\__spintent_spmnum_notation_sym_tl ...   | 279         | \\__spintent_spnz_spread_exec:nn .....      | 1666    |
| \\__spintent_spmnum_print_integer: .....  | 446         |                                             |         |

|                                                          |                          |                                                            |                               |
|----------------------------------------------------------|--------------------------|------------------------------------------------------------|-------------------------------|
| <a href="#">\l__spintent_spread_luaset_reading_str</a>   | <a href="#">1645</a>     | <a href="#">\l__spintent_sptime_luaset_seconds_str</a>     | <a href="#">1377</a>          |
| <a href="#">\l__spintent_spread_luaset_status_str</a>    | <a href="#">1645</a>     | <a href="#">\l__spintent_spunit_exp_seq</a>                | <a href="#">540</a>           |
| <a href="#">\l__spintent_spread_read_arg_tl</a>          | <a href="#">1645</a>     | <a href="#">\l__spintent_spunit_exponent_tl</a>            | <a href="#">540</a>           |
| <a href="#">\__spintent_sprecta_exec:nn</a>              | <a href="#">3581</a>     | <a href="#">\__spintent_spunit_format_canonical:</a>       | <a href="#">603</a>           |
| <a href="#">\l__spintent_sprecta_intent_arg_tl</a>       | <a href="#">3373</a>     | <a href="#">\l__spintent_spunit_is_mult_bool</a>           | <a href="#">540</a>           |
| <a href="#">\l__spintent_sprecta_kv_tl</a>               | <a href="#">3373</a>     | <a href="#">\l__spintent_spunit_luaset_canonical_-</a>     |                               |
| <a href="#">\__spintent_sprecta_mathml:n</a>             | <a href="#">3427</a>     | <a href="#">str</a>                                        | <a href="#">533</a>           |
| <a href="#">\l__spintent_sprecta_mode_str</a>            | <a href="#">3373</a>     | <a href="#">\l__spintent_spunit_luaset_compact_exp_-</a>   |                               |
| <a href="#">\l__spintent_sprecta_prefix_tl</a>           | <a href="#">3373</a>     | <a href="#">str</a>                                        | <a href="#">533</a>           |
| <a href="#">\__spintent_sprecta_print:n</a>              | <a href="#">3418</a>     | <a href="#">\l__spintent_spunit_luaset_is_compact_-</a>    |                               |
| <a href="#">\l__spintent_sprecta_pt_idx_int</a>          | <a href="#">3373</a>     | <a href="#">str</a>                                        | <a href="#">533</a>           |
| <a href="#">\l__spintent_sprecta_pt_seq</a>              | <a href="#">3373</a>     | <a href="#">\l__spintent_spunit_luaset_is_-</a>            |                               |
| <a href="#">\l__spintent_sprecta_refs_tl</a>             | <a href="#">3373</a>     | <a href="#">sexagesimal_str</a>                            | <a href="#">533</a>           |
| <a href="#">\__spintent_spsshort_execute_layout:</a>     | <a href="#">1170</a>     | <a href="#">\l__spintent_spunit_luaset_lookup_status_-</a> |                               |
| <a href="#">\l__spintent_spsshort_luaset_base_str</a>    | <a href="#">1087</a>     | <a href="#">str</a>                                        | <a href="#">533</a>           |
| <a href="#">\l__spintent_spsshort_luaset_expanded_-</a>  |                          | <a href="#">\l__spintent_spunit_luaset_read_str</a>        | <a href="#">533</a>           |
| <a href="#">str</a>                                      | <a href="#">1087</a>     | <a href="#">\l__spintent_spunit_luaset_status_str</a>      | <a href="#">533</a>           |
| <a href="#">\l__spintent_spsshort_luaset_layout_str</a>  | <a href="#">1087</a>     | <a href="#">\l__spintent_spunit_mult_seq</a>               | <a href="#">540</a>           |
| <a href="#">\l__spintent_spsshort_luaset_output_str</a>  | <a href="#">1087</a>     | <a href="#">\__spintent_spunit_process_base_exp:n</a>      | <a href="#">611</a>           |
| <a href="#">\l__spintent_spsshort_luaset_status_str</a>  | <a href="#">1087</a>     | <a href="#">\__spintent_spunit_process_mult:n</a>          | <a href="#">663</a>           |
| <a href="#">\l__spintent_spsshort_luaset_suffix_str</a>  | <a href="#">1087</a>     | <a href="#">\l__spintent_spunit_read_cdot_bool</a>         | <a href="#">540</a>           |
| <a href="#">\__spintent_spsshort_print:nn</a>            | <a href="#">1123</a>     | <a href="#">\l__spintent_spunit_read_slash_str</a>         | <a href="#">540</a>           |
| <a href="#">\__spintent_spsshort_process_and_-</a>       |                          | <a href="#">\__spintent_spunit_safe_exec:n</a>             | <a href="#">682</a>           |
| <a href="#">render:n</a>                                 | <a href="#">1135</a>     | <a href="#">\l__spintent_spunit_unit_name_tl</a>           | <a href="#">540</a>           |
| <a href="#">\__spintent_spsshort_render_bypass:n</a>     | <a href="#">1146</a>     | <a href="#">\l__spintent_spusep_arg_tl</a>                 | <a href="#">70, 71, 1506</a>  |
| <a href="#">\__spintent_spsshort_render_error:n</a>      | <a href="#">1146</a>     | <a href="#">\__spintent_spusep_convert:n</a>               | <a href="#">1568</a>          |
| <a href="#">\__spintent_spsshort_render_fallback:</a>    | <a href="#">1146</a>     | <a href="#">\l__spintent_spusep_left_tl</a>                | <a href="#">70, 71, 1506</a>  |
| <a href="#">\__spintent_spsshort_render_success:</a>     | <a href="#">1146</a>     | <a href="#">\l__spintent_spusep_right_tl</a>               | <a href="#">70, 71, 1506</a>  |
| <a href="#">\__spintent_spsshort_tag_span:nn</a>         | <a href="#">1096</a>     | <a href="#">\__spintent_spusep_set_parens:nn</a>           | <a href="#">1557</a>          |
| <a href="#">\__spintent_spside_exec:nnn</a>              | <a href="#">3909</a>     | <a href="#">\l__spintent_spusep_silent_bool</a>            | <a href="#">71</a>            |
| <a href="#">\l__spintent_spside_intent_arg_tl</a>        | <a href="#">3801</a>     | <a href="#">\__spintent_sspec_exec:nnn</a>                 | <a href="#">3337</a>          |
| <a href="#">\l__spintent_spside_intent_tl</a>            | <a href="#">3801</a>     | <a href="#">\l__spintent_sspec_intent_arg_tl</a>           | <a href="#">3134</a>          |
| <a href="#">\__spintent_spside_mathml:n</a>              | <a href="#">3834</a>     | <a href="#">\l__spintent_sspec_kv_tl</a>                   | <a href="#">3134</a>          |
| <a href="#">\l__spintent_spside_mode_str</a>             | <a href="#">3727</a>     | <a href="#">\__spintent_sspec_mathml:n</a>                 | <a href="#">3196</a>          |
| <a href="#">\__spintent_spside_points_loop:n</a>         | <a href="#">3897</a>     | <a href="#">\l__spintent_sspec_mode_str</a>                | <a href="#">3134</a>          |
| <a href="#">\l__spintent_spside_prefix_tl</a>            | <a href="#">3727</a>     | <a href="#">\l__spintent_sspec_prefix_tl</a>               | <a href="#">3134</a>          |
| <a href="#">\__spintent_spside_process:nn</a>            | <a href="#">3806</a>     | <a href="#">\__spintent_sspec_print:n</a>                  | <a href="#">3179</a>          |
| <a href="#">\l__spintent_spside_pt_idx_int</a>           | <a href="#">3801</a>     | <a href="#">\l__spintent_sspec_pt_idx_int</a>              | <a href="#">3134</a>          |
| <a href="#">\l__spintent_spside_pt_seq</a>               | <a href="#">3801</a>     | <a href="#">\l__spintent_sspec_pt_seq</a>                  | <a href="#">3134</a>          |
| <a href="#">\l__spintent_spside_read_arg_tl</a>          | <a href="#">3727</a>     | <a href="#">\l__spintent_sspec_refs_tl</a>                 | <a href="#">3134</a>          |
| <a href="#">\l__spintent_spside_read_sty_str</a>         | <a href="#">3727</a>     | <a href="#">\__spintent_unknown_keys:n</a>                 | <a href="#">230</a>           |
| <a href="#">\l__spintent_spside_refs_tl</a>              | <a href="#">3801</a>     | <a href="#">\__spintent_unknown_keys:nn</a>                | <a href="#">230</a>           |
| <a href="#">\l__spintent_spsiglo_luaset_arabic_str</a>   | <a href="#">1227</a>     | <a href="#">spintent.lua</a>                               | <a href="#">70</a>            |
| <a href="#">\l__spintent_spsiglo_luaset_error_str</a>    | <a href="#">1227</a>     | <a href="#">\spmangle</a>                                  | <a href="#">23, 119, 4199</a> |
| <a href="#">\l__spintent_spsiglo_luaset_roman_min_-</a>  |                          | <a href="#">spmangle-invalid-arg</a>                       | <a href="#">3941</a>          |
| <a href="#">str</a>                                      | <a href="#">1227</a>     | <a href="#">\spmarc</a>                                    | <a href="#">24, 124, 4358</a> |
| <a href="#">\l__spintent_spsiglo_print_era_str</a>       | <a href="#">1227</a>     | <a href="#">spmarc-invalid-points</a>                      | <a href="#">4226</a>          |
| <a href="#">\__spintent_spsiglo_print_math:n</a>         | <a href="#">1311</a>     | <a href="#">\spmcd</a>                                     | <a href="#">15, 80, 2169</a>  |
| <a href="#">\__spintent_spsiglo_print_text:n</a>         | <a href="#">1311</a>     | <a href="#">\spmcm</a>                                     | <a href="#">16, 80, 2169</a>  |
| <a href="#">\__spintent_spsiglo_render_simple:nn</a>     | <a href="#">1262</a>     | <a href="#">\spmoney</a>                                   | <a href="#">6, 55, 1076</a>   |
| <a href="#">\__spintent_spsiglo_render_with_-</a>        |                          | <a href="#">\spmQ</a>                                      | <a href="#">17, 91, 2735</a>  |
| <a href="#">era:nnn</a>                                  | <a href="#">1262</a>     | <a href="#">spmQ-not-integer</a>                           | <a href="#">2548</a>          |
| <a href="#">\__spintent_spsiglo_tag_span:nn</a>          | <a href="#">1298</a>     | <a href="#">spmQ-zero-denominator</a>                      | <a href="#">2548</a>          |
| <a href="#">\__spintent_sptime_build_seconds:nn</a>      | <a href="#">1391</a>     | <a href="#">\spmside</a>                                   | <a href="#">21, 114, 3914</a> |
| <a href="#">\__spintent_sptime_build_tree:</a>           | <a href="#">69, 1409</a> | <a href="#">spmside</a>                                    | <a href="#">5179</a>          |
| <a href="#">\l__spintent_sptime_luaset_base_str</a>      | <a href="#">1377</a>     | <a href="#">spmside-invalid-points</a>                     | <a href="#">3734</a>          |
| <a href="#">\l__spintent_sptime_luaset_error_str</a>     | <a href="#">1377</a>     | <a href="#">\spmsym</a>                                    | <a href="#">13, 75, 1840</a>  |
| <a href="#">\l__spintent_sptime_luaset_has_seconds_-</a> |                          | <a href="#">\spmμl</a>                                     | <a href="#">14, 78, 1997</a>  |
| <a href="#">str</a>                                      | <a href="#">1377</a>     | <a href="#">spmμl-empty-argument</a>                       | <a href="#">1956</a>          |
| <a href="#">\l__spintent_sptime_luaset_is_fraction_-</a> |                          | <a href="#">\spnD</a>                                      | <a href="#">14, 83, 2326</a>  |
| <a href="#">str</a>                                      | <a href="#">1377</a>     | <a href="#">\spnewsym</a>                                  | <a href="#">11, 1699</a>      |

